

ក្រសួងអប់រំ យុវជន និងកីឡា
វិទ្យាល័យជាតិ

អ៊ុច ប៊ុនសាន
គ្រូបង្រៀនកម្រិតឧត្តម

លីមីតនៃអនុគមន៍
ថ្នាក់ទី ១២

$$\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \ln a$$

ត្រូវប្រើប្រាស់ក្របខណ្ឌប្រភេទអនុគមន៍
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លីមីតនៃអនុគមន៍

ផ្នែកទី ១២

អ៊ូច ប៊ុនថន
គ្រូបង្រៀនកម្រិតឧត្តម
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ការផ្ដកថា

សួស្ដីបួនៗសិស្សានុសិស្សថ្នាក់ទី១២ និងប្រិយមិត្តអ្នកអានជាទីមេត្រី!

សៀវភៅ **លីមីតនៃអនុគមន៍ ថ្នាក់ទី១២** ជាសៀវភៅដែលខ្ញុំបានរៀបចំចុងក្រុងឡើង យ៉ាងយកចិត្តទុកដាក់បំផុត ដោយផ្ដោតទៅលើគន្លឹះសំខាន់ៗនៃមេរៀន វិធីសាស្ត្រគណនា និង លំហាត់តាមចំណុចនីមួយៗភ្ជាប់ជាមួយដំណោះស្រាយដែលមានលក្ខណៈងាយយល់ ផងដែរ។ សៀវភៅនេះបានដកស្រង់ចេញពីប្រភពឯកសារជាភាសាខ្មែរ អង់គ្លេស បារាំង និងរុស្ស៊ី។

ខ្ញុំមានសង្ឃឹមថា សៀវភៅនេះនឹងក្លាយជាឯកសារស្រាវជ្រាវជាទីគាប់ចិត្តមួយសំរាប់ជួយដោះស្រាយបញ្ហាអ្នកដែលចង់ចេះចង់ដឹងផ្នែកលីមីតនៃអនុគមន៍ជាក់ជាមិនខាន។

វាពិតណាស់ថា អ្វីៗទាំងអស់មិនសុទ្ធតែល្អឥតខ្ចោះនោះទេ អាស្រ័យហេតុនេះ មេត្តាអធ្យាស្រ័យរាល់កំហុសឆ្គង ដោយអចេតនាក្នុងការផលិតសៀវភៅនេះ ហើយខ្ញុំក៏ រង់ចាំទទួលការរិះគន់ កែលំអ ក្នុងន័យស្ថាបនាពីសំណាក់ប្រិយមិត្តអ្នកអានពីគ្រប់មជ្ឈដ្ឋានទាំងអស់ ដើម្បីឲ្យការសរសេរលើកក្រោយៗទៀតកាន់តែមានភាពល្អប្រសើរ។

ជាចុងក្រោយខ្ញុំសូមជូនពរឲ្យ មានសុខភាពល្អ មានសំណាងល្អ មានសុភមង្គល និងជោគជ័យគ្រប់ភារកិច្ចជាពិសេសបួនៗសិស្សានុសិស្សថ្នាក់ទី១២ សូមឲ្យការប្រឡង នាពេលខាងមុខទទួលបានជោគជ័យគ្រប់ៗគ្នា។

ព្រៃវែង, ថ្ងៃទី០១ ខែមករា ឆ្នាំ២០១៦

អ្នករៀបរៀង

អ៊ុច ប៊ុនថង

គ្រូបង្រៀនកម្រិតឧត្តម

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លីមីតនៃអនុគមន៍

1. លីមីតនៃអនុគមន៍គ្រប់ចំនួនកំណត់ និងអនន្ត

ក. លីមីតនៃអនុគមន៍ត្រង់ចំនួនកំណត់

និយមន័យ

- ❶ $\lim_{x \rightarrow a} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0$ ដែល $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$
- ❷ $\lim_{x \rightarrow a} f(x) = +\infty \Leftrightarrow \forall M > 0, \exists \delta > 0$ ដែល $0 < |x - a| < \delta \Rightarrow f(x) > M$
- ❸ $\lim_{x \rightarrow a} f(x) = -\infty \Leftrightarrow \forall M > 0, \exists \delta > 0$ ដែល $0 < |x - a| < \delta \Rightarrow f(x) < -M$

លំហាត់

1) ប្រើនិយមន័យបង្ហាញថា:

a). $\lim_{x \rightarrow 1} (5x - 3) = 2$

b). $\lim_{x \rightarrow 2} (3x - 2) = 4$

c). $\lim_{x \rightarrow 1} (2x - 1) = 1$

d). $\lim_{x \rightarrow 2} (3x + 1) = 7$

e). $\lim_{x \rightarrow 4} (2x - 1) = 7$

f). $\lim_{x \rightarrow -3} (2x + 5) = -1$

g). $\lim_{x \rightarrow 2} x^2 = 4$

h). $\lim_{x \rightarrow 3} (2x - 1) = 5$

i). $\lim_{x \rightarrow 2} (3x + 2) = 8$

j). $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

k). រកចំនួនធំបំផុតនៃ $\delta > 0$ ដែល $0 < |x + 2| < \delta$ នោះ $0 < |3x + 6| < 0.03$

2) ប្រើនិយមន័យបង្ហាញថា:

a). $\lim_{x \rightarrow -2} \frac{x^2 - 1}{x + 1} = -3$

b). $\lim_{x \rightarrow 0} \sqrt[3]{x} = 0$

c). $\lim_{x \rightarrow 1} (x^2 + 1) = 2$

d). $\lim_{x \rightarrow 7} \frac{8}{x - 3} = 2$

e). $\lim_{x \rightarrow a} (mx + b) = ma + b$

f). $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$

$$g). \lim_{x \rightarrow 2^-} \sqrt{(x-2)(x-3)} = 0$$

$$h). \lim_{x \rightarrow 3} (5x-2) = 13$$

$$i). \lim_{x \rightarrow 2} \frac{x^2-1}{x} = \frac{3}{2}$$

$$j). \text{គេឲ្យ } \lim_{x \rightarrow 3} (2x+4) = 10 \text{ រក } \delta \text{ បើ } \varepsilon = 0.01$$

3) គេមានអនុគមន៍ $f(x)$ និងចំនួនពិត L ដែល $\lim_{x \rightarrow a} f(x) = L$ ។ រកចំនួន $\alpha > 0$

ដោយស្គាល់ចំនួន ε ដែល $|f(x) - L| < \varepsilon$ កាលណា $0 < |x - a| < \alpha$:

$$a). \lim_{x \rightarrow -1} (3-4x) = 7, \varepsilon = 0.02$$

$$b). \lim_{x \rightarrow 3} (2x+4) = 10, \varepsilon = 0.01$$

$$c). \lim_{x \rightarrow 3} x^2 = 9, \varepsilon = 0.005$$

$$d). \lim_{x \rightarrow 1} \sqrt{x} = 1, \varepsilon = 0.002$$

$$e). \lim_{x \rightarrow -2} \frac{x^2-4}{x+2} = -4, \varepsilon = 0.01$$

$$f). \lim_{x \rightarrow 2} (4x-5) = 3, \varepsilon = 0.001$$

$$g). \lim_{x \rightarrow -2} (2+5x) = -8, \varepsilon = 0.002$$

$$h). \lim_{x \rightarrow 2} (x-1)^2 = 1, \varepsilon = 0.001$$

4) ប្រើនិយមន័យបង្ហាញថា:

$$a). \lim_{x \rightarrow 2^-} \frac{3x-1}{x-2} = -\infty$$

$$b). \lim_{x \rightarrow 1^+} \frac{7x+2}{x-1} = +\infty$$

$$c). \lim_{x \rightarrow 1^-} \frac{x+8}{1-x} = +\infty$$

$$d). \lim_{x \rightarrow 3^+} \frac{x}{x-3} = +\infty$$

$$e). \lim_{x \rightarrow -1^+} \frac{x+5}{x+1} = +\infty$$

$$f). \lim_{x \rightarrow -2^+} \frac{x-9}{2+x} = -\infty$$

$$g). \lim_{x \rightarrow 0^-} \frac{x+1}{x} = -\infty$$

$$h). \lim_{x \rightarrow 2^+} \frac{3x+3}{2-x} = -\infty$$

$$i). \lim_{x \rightarrow 1^+} \frac{x}{1-x} = -\infty$$

$$j). \lim_{x \rightarrow 2^+} \frac{4x+1}{x-2} = +\infty$$

ជំនោះស្រាវជ្រាវ

1) ប្រើនិយមន័យបង្ហាញថា:

$$a). \lim_{x \rightarrow 1} (5x-3) = 2$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(5x-3)-2|<\varepsilon$$

$$|5x-5|<\varepsilon$$

$$5|x-1|<\varepsilon$$

$$|x-1|<\frac{\varepsilon}{5} \text{ យក } \delta=\frac{\varepsilon}{5}>0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{5}$ ដែល $|x-1| < \frac{\varepsilon}{5}$ នាំឲ្យ $|(5x-3)-2| < \varepsilon$

ដូចនេះ បញ្ជាក់ថា $\lim_{x \rightarrow 1} (5x-3) = 2$ ។

$$\text{b). } \lim_{x \rightarrow 2} (3x-2) = 4$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x)-L| < \varepsilon$

$$|(3x-2)-4|<\varepsilon$$

$$|3x-6|<\varepsilon$$

$$3|x-2|<\varepsilon$$

$$|x-2|<\frac{\varepsilon}{3} \text{ យក } \delta=\frac{\varepsilon}{3}>0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3}$ ដែល $|x-2| < \frac{\varepsilon}{3}$ នាំឲ្យ $|(3x-2)-4| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow 2} (3x-2) = 4$ ។

$$\text{c). } \lim_{x \rightarrow 1} (2x-1) = 1$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x)-L| < \varepsilon$

$$|(2x-1)-1|<\varepsilon$$

$$|2x-2|<\varepsilon$$

$$2|x-1|<\varepsilon$$

$$|x-1|<\frac{\varepsilon}{2} \text{ យក } \delta=\frac{\varepsilon}{2}>0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{2}$ ដែល $|x-1| < \frac{\varepsilon}{2}$ នាំឲ្យ $|(2x-1)-1| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow 1} (2x-1) = 1$ ។

$$d). \lim_{x \rightarrow 2} (3x+1) = 7$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(3x+1) - 7| < \varepsilon$$

$$|3x - 6| < \varepsilon$$

$$3|x - 2| < \varepsilon$$

$$|x - 2| < \frac{\varepsilon}{3} \text{ យក } \delta = \frac{\varepsilon}{3} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3}$ ដែល $|x - 2| < \frac{\varepsilon}{3}$ នាំឲ្យ $|(3x+1) - 7| < \varepsilon$

ដូចនេះ: $\lim_{x \rightarrow 2} (3x+1) = 7$ ។

$$e). \lim_{x \rightarrow 4} (2x-1) = 7$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(2x-1) - 7| < \varepsilon$$

$$|2x - 8| < \varepsilon$$

$$2|x - 4| < \varepsilon$$

$$|x - 4| < \frac{\varepsilon}{2} \text{ យក } \delta = \frac{\varepsilon}{2} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{2}$ ដែល $|x - 4| < \frac{\varepsilon}{2}$ នាំឲ្យ $|(2x-1) - 7| < \varepsilon$

ដូចនេះ: $\lim_{x \rightarrow 4} (2x-1) = 7$ ។

$$f). \lim_{x \rightarrow -3} (2x+5) = -1$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(2x+5) - (-1)| < \varepsilon$$

$$|2x + 6| < \varepsilon$$

$$2|x + 3| < \varepsilon$$

$$|x - (-3)| < \frac{\varepsilon}{2} \text{ យក } \delta = \frac{\varepsilon}{2} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{2}$ ដែល $|x - (-3)| < \frac{\varepsilon}{2}$ នាំឱ្យ $|(2x+5) - (-1)| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow -3} (2x+5) = -1$ ។

g). $\lim_{x \rightarrow 2} x^2 = 4$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|x^2 - 4| < \varepsilon$$

$$\text{គេមាន } |x^2 - 4| = |(x-2)(x+2)| = |x-2| \cdot |x+2|$$

$$\text{នាំឱ្យ } |x-2| = \frac{|x^2 - 4|}{|x+2|}$$

ដោយ $x \rightarrow 2$ នោះ $1 < x < 3$ នោះ $3 < x+2 < 5$ នាំឱ្យ $3 < |x+2| < 5$

$$\text{គេបាន } |x-2| = \frac{|x^2 - 4|}{|x+2|} < \frac{\varepsilon}{5} \text{ យក } \delta = \frac{\varepsilon}{5} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{5}$ ដែល $|x-2| < \frac{\varepsilon}{5}$ នាំឱ្យ $|x^2 - 4| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow 2} x^2 = 4$ ។

h). $\lim_{x \rightarrow 3} (2x-1) = 5$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(2x-1) - 5| < \varepsilon$$

$$|2x-6| < \varepsilon$$

$$2|x-3| < \varepsilon \text{ ឬ } |x-3| < \frac{\varepsilon}{2} \text{ យក } \delta = \frac{\varepsilon}{2} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{2}$ ដែល $|x-3| < \frac{\varepsilon}{2}$ នាំឱ្យ $|(2x-1) - 5| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow 3} (2x-1) = 5$ ។

i). $\lim_{x \rightarrow 2} (3x+2) = 8$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(3x+2) - 8| < \varepsilon$$

$$|3x-6|<\varepsilon$$

$$3|x-2|<\varepsilon \text{ ឬ } |x-2|<\frac{\varepsilon}{3} \text{ យក } \delta=\frac{\varepsilon}{3}>0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3} \text{ ដែល } |x-2|<\frac{\varepsilon}{3} \text{ នាំឲ្យ } |(3x+2)-8|<\varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 2} (3x+2) = 8 \text{ ។}$$

$$\text{j). } \lim_{x \rightarrow -1} (x^2 - 1) = 0$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$|(x^2 - 1) - 0| < \varepsilon \text{ ឬ } |x^2 - 1| < \varepsilon$$

$$\text{គេមាន } |x^2 - 1| = |(x-1)(x+1)| = |x-1| |x+1|$$

$$\text{នាំឲ្យ } |x+1| = \frac{|x^2 - 1|}{|x-1|}$$

$$\text{ដោយ } x \rightarrow -1 \text{ នោះ: } -2 < x < 0 \text{ ឬ } -3 < x-1 < -1$$

$$\text{នាំឲ្យ } |x-1| < 3 \text{ នាំឲ្យ } |x+1| = \frac{|x^2 - 1|}{|x-1|} < \frac{\varepsilon}{3} \text{ យក } \delta = \frac{\varepsilon}{3} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3} \text{ ដែល } |x+1| < \frac{\varepsilon}{3} \text{ នាំឲ្យ } |(x^2 - 1) - 0| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow -1} (x^2 - 1) = 0 \text{ ។}$$

$$\text{k). រកចំនួនធំបំផុតនៃ } \delta > 0 \text{ ដែល } 0 < |x+2| < \delta \text{ នោះ: } 0 < |3x+6| < 0.03$$

$$\text{គេមាន } 0 < |3x+6| < 0.03$$

$$0 < 3|x+2| < 0.03 \text{ ឬ } 0 < |x+2| < 0.01$$

$$\text{គេទាញបាន } \delta = 0.01$$

$$\text{ដូចនេះ: } \boxed{\delta = 0.01}$$

2) ប្រើនិយមន័យបង្ហាញថា:

$$\text{a). } \lim_{x \rightarrow -2} \frac{x^2 - 1}{x + 1} = -3$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$\left| \frac{x^2-1}{x+1} - (-3) \right| < \varepsilon$$

$$\left| \frac{x^2-1}{x+1} + 3 \right| < \varepsilon$$

$$\left| \frac{x^2-1+3(x+1)}{x+1} \right| < \varepsilon$$

$$\left| \frac{x^2+3x+2}{x+1} \right| < \varepsilon$$

$$\left| \frac{(x+2)(x+1)}{x+1} \right| < \varepsilon$$

$$|x+2| < \varepsilon \text{ យក } \delta = \varepsilon > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \varepsilon$ ដែល $|x+1| < \delta$ នាំឲ្យ $\left| \frac{x^2-1}{x+1} - (-3) \right| < \varepsilon$

ដូចនេះ: $\lim_{x \rightarrow -2} \frac{x^2-1}{x+1} = -3$

b). $\lim_{x \rightarrow 0} \sqrt[3]{x} = 0$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|\sqrt[3]{x} - 0| < \varepsilon \text{ ឬ } |\sqrt[3]{x}| < \varepsilon$$

$$(|\sqrt[3]{x}|)^3 < \varepsilon^3 \text{ នោះ } |x| < \varepsilon^3 \text{ យក } \delta = \varepsilon^3 > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \varepsilon^3$ ដែល $|x| < \delta$ នាំឲ្យ $|\sqrt[3]{x} - 0| < \varepsilon$

ដូចនេះ: $\lim_{x \rightarrow 0} \sqrt[3]{x} = 0$ ។

c). $\lim_{x \rightarrow 1} (x^2 + 1) = 2$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(x^2 + 1) - 2| < \varepsilon \text{ ឬ } |x^2 - 1| < \varepsilon$$

គេមាន $x^2 - 1 = (x-1)(x+1)$ នោះ $x-1 = \frac{x^2-1}{x+1}$

ដោយ $x \rightarrow 1$ នោះ $0 < x < 2$ នាំឲ្យ $1 < x+1 < 3$ នាំឲ្យ $|x+1| < 3$

$$\text{គេបាន } |x-1| = \frac{|x^2-1|}{|x+1|} < \frac{\varepsilon}{3} \text{ យក } \delta = \frac{\varepsilon}{3} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{3} \text{ ដែល } |x-1| < \frac{\varepsilon}{3} \text{ នាំឲ្យ } |(x^2+1)-2| < \varepsilon$$

$$\text{ដូចនេះ } \lim_{x \rightarrow 1} (x^2+1) = 2 \text{ ។}$$

$$d). \lim_{x \rightarrow 7} \frac{8}{x-3} = 2$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{8}{x-3} - 2 \right| < \varepsilon$$

$$\left| \frac{8-2(x-3)}{x-3} \right| < \varepsilon$$

$$\left| \frac{8-2x+6}{x-3} \right| < \varepsilon$$

$$\left| \frac{14-2x}{x-3} \right| < \varepsilon$$

$$\left| \frac{-2(x-7)}{x-3} \right| < \varepsilon$$

$$\frac{|-2||x-7|}{|x-3|} < \varepsilon$$

ដោយ $x \rightarrow 7$ នោះ $6 < x < 8$ ឬ $3 < x-3 < 5$ នាំឲ្យ $|x-3| < 5$

$$\text{នាំឲ្យ } \left| \frac{8}{x-3} - 2 \right| = \frac{|-2||x-7|}{|x-3|} = \frac{2|x-7|}{|x-3|} \Rightarrow |x-7| = \frac{|x-3|}{2} \left| \frac{8}{x-3} - 2 \right|$$

$$\Rightarrow |x-7| = \frac{|x-3|}{2} \left| \frac{8}{x-3} - 2 \right| < \frac{5}{2} \varepsilon$$

$$\Rightarrow |x-7| < \frac{5}{2} \varepsilon \text{ យក } \delta = \frac{5}{2} \varepsilon > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{5}{2} \varepsilon \text{ ដែល } |x-7| < \frac{5}{2} \varepsilon \text{ នាំឲ្យ } \left| \frac{8}{x-3} - 2 \right| < \varepsilon$$

ដូចនេះ $\lim_{x \rightarrow 7} \frac{8}{x-3} = 2$ ។

e). $\lim_{x \rightarrow a} (mx + b) = ma + b$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(mx + b) - (ma + b)| < \varepsilon$$

$$|mx - ma| < \varepsilon \quad \text{ឬ} \quad |m||x - a| < \varepsilon$$

$$|x - a| < \frac{\varepsilon}{|m|} \quad \text{យក} \quad \delta = \frac{\varepsilon}{|m|} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{|m|}$ ដែល $|x - a| < \frac{\varepsilon}{|m|}$

នាំឲ្យ $|(mx + b) - (ma + b)| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow a} (mx + b) = ma + b$ ។

f). $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|\sqrt{x} - \sqrt{2}| < \varepsilon$$

$$\text{គេមាន } |x - 2| = |(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})| = |\sqrt{x} - \sqrt{2}| |\sqrt{x} + \sqrt{2}|$$

ដោយ $x \rightarrow 2$ នោះ $1 < x < 3$

$$\Rightarrow 1 < \sqrt{x} < \sqrt{3}$$

$$\Rightarrow 1 + \sqrt{2} < \sqrt{x} + \sqrt{2} < \sqrt{3} + \sqrt{2}$$

$$\Rightarrow |\sqrt{x} + \sqrt{2}| < \sqrt{3} + \sqrt{2}$$

នាំឲ្យ $|x - 2| = |\sqrt{x} - \sqrt{2}| |\sqrt{x} + \sqrt{2}| < (\sqrt{3} + \sqrt{2})\varepsilon = \delta$

គេបាន $\forall \varepsilon > 0, \exists \delta = (\sqrt{3} + \sqrt{2})\varepsilon$ ដែល $|x - 2| < (\sqrt{3} + \sqrt{2})\varepsilon$

នាំឲ្យ $|\sqrt{x} - \sqrt{2}| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow 2} \sqrt{x} = \sqrt{2}$ ។

g). $\lim_{x \rightarrow 2^-} \sqrt{(x-2)(x-3)} = 0$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|\sqrt{(x-2)(x-3)}-0|<\varepsilon$$

$$|(x-2)(x-3)|<\varepsilon^2 \text{ ឬ } |x-2||x-3|<\varepsilon^2$$

$$|x-2|<\frac{\varepsilon^2}{|x-3|}$$

ដោយ $x \rightarrow 2^-$ នោះ $1 < x < 2$ នោះ $-2 < x-3 < -1 \Rightarrow |x-3| < 2$

$$\text{នាំឲ្យ } |x-2| < \frac{\varepsilon^2}{2} = \delta$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon^2}{2} \text{ ដែល } |x-2| < \frac{\varepsilon^2}{2}$$

$$\text{នាំឲ្យ } |\sqrt{(x-2)(x-3)}-0|<\varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 2^-} \sqrt{(x-2)(x-3)} = 0 \text{ ។}$$

$$\text{h). } \lim_{x \rightarrow 3} (5x-2) = 13$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(5x-2)-13| < \varepsilon$$

$$|5x-15| < \varepsilon$$

$$5|x-3| < \varepsilon$$

$$|x-3| < \frac{\varepsilon}{5} \text{ យក } \delta = \frac{\varepsilon}{5} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{5} \text{ ដែល } |x-2| < \frac{\varepsilon}{5} \text{ នាំឲ្យ } |(5x-2)-13| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 3} (5x-2) = 13 \text{ ។}$$

$$\text{i). } \lim_{x \rightarrow 2} \frac{x^2-1}{x} = \frac{3}{2}$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{x^2-1}{x} - \frac{3}{2} \right| < \varepsilon$$

$$\left| \frac{2(x^2-1)-3x}{2x} \right| < \varepsilon$$

$$\left| \frac{2x^2 - 3x - 2}{2x} \right| < \varepsilon \quad \text{ឬ} \quad \left| \frac{(x-2)(2x+1)}{2x} \right| < \varepsilon$$

$$\left| \frac{2x^2 - 3x - 2}{2x} \right| = \left| \frac{(x-2)(2x+1)}{2x} \right| = \frac{|x-2| |2x+1|}{2|x|}$$

$$\Rightarrow |x-2| = \frac{\left| \frac{2x^2 - 3x - 2}{2x} \right|}{|2x+1|} \times 2|x|$$

ដោយ $x \rightarrow 2$ នោះ $1 < x < 3$ នាំឲ្យ $|x| < 3$

$$2 < 2x < 6$$

$$3 < 2x+1 < 7 \quad \text{នាំឲ្យ} \quad |2x+1| < 7$$

$$\Rightarrow |x-2| = \frac{\left| \frac{2x^2 - 3x - 2}{2x} \right|}{|2x+1|} \times 2|x| < \frac{\varepsilon}{7} \times 2 \times 3$$

$$\Rightarrow |x-2| < \frac{6\varepsilon}{7} \quad \text{យក} \quad \delta = \frac{6\varepsilon}{7} > 0$$

គេបាន $\forall \varepsilon > 0, \exists \delta = \frac{6\varepsilon}{7}$ ដែល $|x-2| < \frac{6\varepsilon}{7}$ នាំឲ្យ $\left| \frac{x^2-1}{x} - \frac{3}{2} \right| < \varepsilon$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 2} \frac{x^2-1}{x} = \frac{3}{2} \quad \text{។}$$

j). រក δ បើ $\varepsilon = 0.01$

$$\text{គេមាន } \lim_{x \rightarrow 3} (2x+4) = 10$$

តាមនិយមន័យគេបាន: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(2x+4) - 10| < \varepsilon$$

$$|2x-6| < \varepsilon \quad \text{ឬ} \quad 2|x-3| < \varepsilon$$

$$|x-3| < \frac{\varepsilon}{2} \quad \text{គេបាន} \quad \delta = \frac{\varepsilon}{2} = \frac{0.01}{2} = 0.005$$

$$\text{ដូចនេះ: } \boxed{\delta = 0.005}$$

3) រកចំនួន $\alpha > 0$ ដែល $0 < |x - a| < \alpha$ នាំឱ្យ $|f(x) - L| < \varepsilon$:

a). $\lim_{x \rightarrow -1} (3 - 4x) = 7, \varepsilon = 0.02$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(3 - 4x) - 7| < \varepsilon$$

$$|-4x - 4| < \varepsilon$$

$$|-4||x + 1| < \varepsilon$$

$$4|x + 1| < \varepsilon \Rightarrow |x + 1| < \frac{\varepsilon}{4}$$

$$\Rightarrow \alpha = \frac{\varepsilon}{4} = \frac{0.02}{4} = 0.005$$

ដូចនេះ: $\boxed{\alpha = 0.005}$

b). $\lim_{x \rightarrow 3} (2x + 4) = 10, \varepsilon = 0.01$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|(2x + 4) - 10| < \varepsilon$$

$$|2x - 6| < \varepsilon$$

$$2|x - 3| < \varepsilon$$

$$\Rightarrow |x - 3| < \frac{\varepsilon}{2}$$

$$\Rightarrow \alpha = \frac{\varepsilon}{2} = \frac{0.01}{2} = 0.005$$

ដូចនេះ: $\boxed{\alpha = 0.005}$

c). $\lim_{x \rightarrow 3} x^2 = 9, \varepsilon = 0.005$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$|x^2 - 9| < \varepsilon$$

គេមាន $|x^2 - 9| = |(x - 3)(x + 3)| = |x - 3||x + 3|$

$$\Rightarrow |x - 3| = \frac{|x^2 - 9|}{|x + 3|}$$

តែ $x \rightarrow 3$ នោះ $2 < x < 4$ នាំឱ្យ $5 < x + 3 < 7$ នាំឱ្យ $|x + 3| < 7$

$$\Rightarrow |x-3| = \frac{|x^2-9|}{|x+3|} < \frac{\varepsilon}{7}$$

$$\text{គេបាន } \alpha = \frac{\varepsilon}{7} = \frac{0.003}{7} = 0.00043$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.00043}$$

$$\text{d). } \lim_{x \rightarrow 1} \sqrt{x} = 1, \varepsilon = 0.002$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$|\sqrt{x} - 1| < \varepsilon$$

$$\text{គេមាន } |x-1| = |\sqrt{x}-1||\sqrt{x}+1|$$

$$\text{តែ } x \rightarrow 1 \text{ នោះ: } 0 < x < 2 \text{ នោះ: } 0 < \sqrt{x} < \sqrt{2}$$

$$\text{នាំឲ្យ } 1 < \sqrt{x} + 1 < \sqrt{2} + 1 \text{ នាំឲ្យ } |\sqrt{x} + 1| < \sqrt{2} + 1$$

$$\text{នាំឲ្យ } |x-1| = |\sqrt{x}-1||\sqrt{x}+1| < \varepsilon(\sqrt{2}+1)$$

$$\Rightarrow |x-1| < (\sqrt{2}+1)\varepsilon$$

$$\text{គេបាន } \alpha = (\sqrt{2}+1)\varepsilon = (\sqrt{2}+1)(0.002) = 0.0048$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.0048}$$

$$\text{e). } \lim_{x \rightarrow -2} \frac{x^2-4}{x+2} = -4, \varepsilon = 0.01$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$\left| \frac{x^2-4}{x+2} - (-4) \right| < \varepsilon$$

$$\left| \frac{(x-2)(x+2)}{x+2} + 4 \right| < \varepsilon$$

$$|x-2+4| < \varepsilon$$

$$|x+2| < \varepsilon \text{ នាំឲ្យ } \alpha = \varepsilon = 0.01$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.01}$$

$$\text{f). } \lim_{x \rightarrow 2} (4x-5) = 3, \varepsilon = 0.001$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$|(4x-5)-3| < \varepsilon$$

$$|4x-8|<\varepsilon$$

$$4|x-2|<\varepsilon$$

$$|x-2|<\frac{\varepsilon}{4}$$

$$\text{គេបាន } \alpha = \frac{\varepsilon}{4} = \frac{0.001}{4} = 0.00025$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.00025}$$

$$\text{g). } \lim_{x \rightarrow -2} (2+5x) = -8, \varepsilon = 0.002$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$|(2+5x) - (-8)| < \varepsilon$$

$$|(2+5x) + 8| < \varepsilon$$

$$|5x+10| < \varepsilon$$

$$5|x+2| < \varepsilon$$

$$|x+2| < \frac{\varepsilon}{5}$$

$$\text{គេបាន } \alpha = \frac{\varepsilon}{5} = \frac{0.002}{5} = 0.0004$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.0004}$$

$$\text{h). } \lim_{x \rightarrow 2} (x-1)^2 = 1, \varepsilon = 0.001$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$|(x-1)^2 - 1| < \varepsilon$$

$$|(x-1+1)(x-1-1)| < \varepsilon$$

$$|x(x-2)| < \varepsilon$$

$$\text{គេមាន } |x(x-2)| = |x||x-2| \text{ នាំឲ្យ } |x-2| = \frac{|x(x-2)|}{|x|}$$

$$\text{ដោយ } 1 < x < 3 \text{ នោះ: } |x| < 3$$

$$\text{នាំឲ្យ } |x-2| = \frac{|x(x-2)|}{|x|} < \frac{\varepsilon}{3} \text{ គេបាន } \alpha = \frac{\varepsilon}{3} = \frac{0.001}{3} = 0.0003$$

$$\text{ដូចនេះ: } \boxed{\alpha = 0.0003}$$

4) ប្រើនិយមន័យបង្ហាញថា:

$$a). \lim_{x \rightarrow 2^-} \frac{3x-1}{x-2} = -\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$$\frac{3x-1}{x-2} < -M \text{ ឬ } 3 + \frac{5}{x-2} < -M$$

$$\frac{5}{x-2} < -M-3 \text{ នោះ: } \frac{5}{|x-2|} > |-M-3|$$

$$\frac{5}{|x-2|} > M+3$$

$$|x-2| < \frac{5}{M+3} \text{ តាង } \delta = \frac{5}{M+3} > 0$$

គេបាន $\forall M > 0, \exists \delta = \frac{5}{M+3}$ ដែល $|x-2| < \delta$ នាំឲ្យ $\frac{3x-1}{x-2} < -M$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 2^-} \frac{3x-1}{x-2} = -\infty \text{ ។}$$

$$b). \lim_{x \rightarrow 1^+} \frac{7x+2}{x-1} = +\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) > M$

$$\text{គេមាន } f(x) = \frac{7x+2}{x-1} = 7 + \frac{9}{x-1}$$

$$\text{ដើម្បីឲ្យ } f(x) > M \text{ គេឲ្យ } \frac{9}{x-1} > M \Rightarrow x-1 < \frac{9}{M}$$

$$\text{ដោយ } x \rightarrow 1^+ \text{ នោះ: } x-1 > 0 \text{ នាំឲ្យ } 0 < x-1 < \frac{9}{M}$$

$$\text{នាំឲ្យ } |x-1| < \frac{9}{M} \text{ យក } \delta = \frac{9}{M} > 0$$

គេបាន $\forall M > 0, \exists \delta = \frac{9}{M}$ ដែល $|x-1| < \delta$ នាំឲ្យ $f(x) > M$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 1^+} \frac{7x+2}{x-1} = +\infty \text{ ។}$$

$$c). \lim_{x \rightarrow 1^-} \frac{x+8}{1-x} = +\infty$$

$$\text{គេមាន } f(x) = \frac{x+8}{1-x} = -1 + \frac{9}{1-x}$$

$$\text{ចំពោះ } \forall M > 0, f(x) > M$$

$$-1 + \frac{9}{1-x} > M \text{ ឬ } \frac{9}{1-x} > M+1 \text{ ឬ } 1-x < \frac{9}{M+1}$$

$$\text{ដោយ } x < 1 \text{ នោះ } 1-x > 0 \text{ នាំឲ្យ } 0 < 1-x < \frac{9}{M+1}$$

$$\text{នាំឲ្យ } |1-x| < \frac{9}{M+1} \text{ ឬ } |x-1| < \frac{9}{M+1} \text{ យក } \delta = \frac{9}{M+1} > 0$$

$$\text{គេបាន } \forall M > 0, \exists \delta = \frac{9}{M+1} \text{ ដែល } |x-1| < \delta \text{ នាំឲ្យ } f(x) > M$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 1^-} \frac{x+8}{1-x} = +\infty \text{ ។}$$

$$\text{d). } \lim_{x \rightarrow 3^+} \frac{x}{x-3} = +\infty$$

$$\text{គេមាន } f(x) = \frac{x}{x-3} = 1 + \frac{3}{x-3}$$

$$\text{ដើម្បីឲ្យ } f(x) > M, \forall M > 0 \text{ គេគ្រាន់តែឲ្យ } \frac{3}{x-3} > M \text{ នាំឲ្យ } x-3 < \frac{3}{M}$$

$$\text{តែ } x > 3 \text{ នាំឲ្យ } x-3 > 0 \text{ គេបាន } 0 < x-3 < \frac{3}{M}$$

$$\text{នាំឲ្យ } |x-3| < \frac{3}{M} \text{ យក } \delta = \frac{3}{M} > 0$$

$$\text{គេបាន } \forall M > 0, \exists \delta = \frac{3}{M} \text{ ដែល } |x-3| < \delta \text{ នាំឲ្យ } f(x) > M$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow 3^+} \frac{x}{x-3} = +\infty \text{ ។}$$

$$\text{e). } \lim_{x \rightarrow -1^+} \frac{x+5}{x+1} = +\infty$$

$$\text{គេមាន } f(x) = \frac{x+5}{x+1} = 1 + \frac{4}{x+1}$$

$$\text{ដើម្បីឲ្យ } f(x) > M, \forall M > 0 \text{ គេត្រូវឲ្យ } \frac{4}{x+1} > M \Rightarrow x+1 < \frac{4}{M}$$

តែ $x > -1$ នោះ $x+1 > 0$ នាំឲ្យ $0 < x+1 < \frac{4}{M}$

នាំឲ្យ $|x+1| < \frac{4}{M}$ គេយក $\delta = \frac{4}{M} > 0$

គេបាន $\forall M > 0, \exists \delta = \frac{4}{M}$ ដែល $|x+1| < \delta$ នាំឲ្យ $f(x) > M$

ដូចនេះ $\lim_{x \rightarrow -1^+} \frac{x+5}{x+1} = +\infty$ ។

f). $\lim_{x \rightarrow -2^+} \frac{x-9}{2+x} = -\infty$

គេមាន $f(x) = \frac{x-9}{2+x} = 1 - \frac{11}{x+2}$

ចំពោះ $\forall M > 0, f(x) < -M$

គេបាន $1 - \frac{11}{x+2} < -M$ ឬ $M+1 < \frac{11}{x+2} \Rightarrow x+2 > \frac{11}{M+1}$

នាំឲ្យ $|x+2| > \frac{11}{M+1}$ គេយក $\delta = \frac{11}{M+1} > 0$

គេបាន $\forall M > 0, \exists \delta = \frac{11}{M+1}$ ដែល $|x+2| > \delta$ នាំឲ្យ $f(x) < -M$

ដូចនេះ $\lim_{x \rightarrow -2^+} \frac{x-9}{2+x} = -\infty$ ។

g). $\lim_{x \rightarrow 0^-} \frac{x+1}{x} = -\infty$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$\frac{x+1}{x} < -M$ ឬ $1 + \frac{1}{x} < -M$ ឬ $\frac{1}{x} < -M-1$

$\frac{1}{|x|} > |-M-1|$ នាំឲ្យ $|x| < \frac{1}{M+1}$ គេយក $\delta = \frac{1}{M+1} > 0$

គេបាន $\forall M > 0, \exists \delta = \frac{1}{M+1}$ ដែល $|x| < \delta$ នាំឲ្យ $f(x) < -M$

ដូចនេះ $\lim_{x \rightarrow 0^-} \frac{x+1}{x} = -\infty$ ។

$$h). \lim_{x \rightarrow 2^+} \frac{3x+3}{2-x} = -\infty$$

$$\text{គេមាន } f(x) = \frac{3x+3}{2-x} = -3 - \frac{9}{x-2}$$

$$\text{ដើម្បីឲ្យ } f(x) < -M, \forall M > 0 \text{ គេគ្រាន់តែឲ្យ } -\frac{9}{x-2} < -M$$

$$\left| -\frac{9}{x-2} \right| > |-M| \text{ ឬ } \frac{9}{|x-2|} > M \text{ នាំឲ្យ } |x-2| < \frac{9}{M} \text{ យក } \delta = \frac{9}{M} > 0$$

$$\text{គេបាន } \forall M > 0, \exists \delta = \frac{9}{M} \text{ ដែល } |x-2| < \delta \text{ នាំឲ្យ } f(x) < -M$$

$$\text{ដូចនេះ } \lim_{x \rightarrow 2^+} \frac{3x+3}{2-x} = -\infty \text{ ។}$$

$$i). \lim_{x \rightarrow 1^+} \frac{x}{1-x} = -\infty$$

$$\text{គេមាន } f(x) = \frac{x}{1-x} = -1 - \frac{1}{x-1}$$

$$\text{ដើម្បីឲ្យ } f(x) < -M, \forall M > 0 \text{ គេគ្រាន់តែឲ្យ } -\frac{1}{x-1} < -M$$

$$\left| -\frac{1}{x-1} \right| > |-M| \text{ ឬ } \frac{1}{|x-1|} > M \text{ នាំឲ្យ } |x-1| < \frac{1}{M} \text{ គេយក } \delta = \frac{1}{M} > 0$$

$$\text{គេបាន } \forall M > 0, \exists \delta = \frac{1}{M} \text{ ដែល } |x-1| < \delta \text{ នាំឲ្យ } f(x) < -M$$

$$\text{ដូចនេះ } \lim_{x \rightarrow 1^+} \frac{x}{1-x} = -\infty \text{ ។}$$

$$j). \lim_{x \rightarrow 2^+} \frac{4x+1}{x-2} = +\infty$$

$$\text{គេមាន } f(x) = \frac{4x+1}{x-2} = 4 + \frac{9}{x-2}$$

$$\text{ដើម្បីឲ្យ } f(x) > M, \forall M > 0 \text{ គេគ្រាន់តែឲ្យ } \frac{9}{x-2} > M$$

$$\text{នាំឲ្យ } \frac{9}{|x-2|} > M \text{ នាំឲ្យ } |x-2| < \frac{9}{M} \text{ គេយក } \delta = \frac{9}{M} > 0$$

គេបាន $\forall M > 0, \exists \delta = \frac{9}{M}$ ដែល $|x-2| < \delta$ នាំឱ្យ $f(x) > M$

ដូចនេះ $\lim_{x \rightarrow 2^+} \frac{4x+1}{x-2} = +\infty$ ។

ខ. លីមីតនៃអនុគមន៍ត្រង់អនន្ត

និយមន័យ

- ❶ $\lim_{x \rightarrow +\infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists N > 0$ ដែល $x > N \Rightarrow |f(x) - L| < \varepsilon$
- ❷ $\lim_{x \rightarrow -\infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists N > 0$ ដែល $x < -N \Rightarrow |f(x) - L| < \varepsilon$
- ❸ $\lim_{x \rightarrow +\infty} f(x) = +\infty \Leftrightarrow \forall M > 0, \exists N > 0$ ដែល $x > N \Rightarrow f(x) > M$
- ❹ $\lim_{x \rightarrow +\infty} f(x) = -\infty \Leftrightarrow \forall M > 0, \exists N > 0$ ដែល $x > N \Rightarrow f(x) < -M$
- ❺ $\lim_{x \rightarrow -\infty} f(x) = +\infty \Leftrightarrow \forall M > 0, \exists N > 0$ ដែល $x < -N \Rightarrow f(x) > M$
- ❻ $\lim_{x \rightarrow -\infty} f(x) = -\infty \Leftrightarrow \forall M > 0, \exists N > 0$ ដែល $x < -N \Rightarrow f(x) < -M$

លំហាត់

1) ប្រើនិយមន័យបង្ហាញថា:

a). $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$

b). $\lim_{x \rightarrow +\infty} \frac{x+1}{x} = 1$

c). $\lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = 1$

d). $\lim_{x \rightarrow +\infty} \frac{1}{x^2} = 0$

e). $\lim_{x \rightarrow -\infty} \frac{x^2 - 2}{x^2 + 2} = 1$

f). $\lim_{x \rightarrow +\infty} \frac{2x+4}{x} = 2$

g). $\lim_{x \rightarrow +\infty} \frac{3x+1}{x} = 3$

h). $\lim_{x \rightarrow -\infty} \frac{3x-1}{x-1} = 3$

i). $\lim_{x \rightarrow +\infty} \frac{x^2 - 2}{x^2 + 2} = 1$

2) ប្រើនិយមន័យបង្ហាញថា:

a). $\lim_{x \rightarrow +\infty} (-x^2) = -\infty$

b). $\lim_{x \rightarrow -\infty} (x^2 - x - 1) = +\infty$

$$c). \lim_{x \rightarrow -\infty} (1 - x^2) = -\infty$$

$$d). \lim_{x \rightarrow +\infty} \frac{x + 2x^2}{3x + 5} = +\infty$$

$$e). \lim_{x \rightarrow -\infty} \frac{x^2 + 3}{x} = -\infty$$

$$f). \lim_{x \rightarrow +\infty} \frac{3 - x + x^3}{1 - x^2} = -\infty$$

$$g). \lim_{x \rightarrow -\infty} \frac{6x - x^2}{2x + 1} = +\infty$$

$$h). \lim_{x \rightarrow +\infty} \frac{x^2 + 3}{2 - x} = -\infty$$

$$i). \lim_{x \rightarrow -\infty} (x^2 + 2x + 3) = +\infty$$

$$j). \lim_{x \rightarrow +\infty} (1 + 6x - x^2) = -\infty$$

ដំណោះស្រាយ

1) ប្រើនិយមន័យបង្ហាញថា:

$$a). \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{1}{x} - 0 \right| < \varepsilon \quad \text{ឬ} \quad \left| \frac{1}{x} \right| < \varepsilon$$

$$\frac{1}{|x|} < \varepsilon \quad \text{នាំឱ្យ} \quad |x| > \frac{1}{\varepsilon}$$

$$\text{តែ } x < 0 \quad x < -\frac{1}{\varepsilon} \quad \text{តាំង } N = \frac{1}{\varepsilon} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists N = \frac{1}{\varepsilon} > 0 \text{ ដែល } x < -N \text{ នាំឱ្យ } \left| \frac{1}{x} - 0 \right| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow -\infty} \frac{1}{x} = 0 \quad \forall$$

$$b). \lim_{x \rightarrow +\infty} \frac{x+1}{x} = 1$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{x+1}{x} - 1 \right| < \varepsilon \quad \text{ឬ} \quad \left| 1 + \frac{1}{x} - 1 \right| < \varepsilon \quad \text{ឬ} \quad \left| \frac{1}{x} \right| < \varepsilon$$

$$\frac{1}{|x|} < \varepsilon \quad \text{ឬ} \quad |x| > \frac{1}{\varepsilon}$$

ដោយ $x > 0$ នាំឲ្យ $x > \frac{1}{\varepsilon}$ យក $N = \frac{1}{\varepsilon} > 0$

គេបាន $\forall \varepsilon > 0, \exists N = \frac{1}{\varepsilon}$ ដែល $x > N$ នាំឲ្យ $\left| \frac{x+1}{x} - 1 \right| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow +\infty} \frac{x+1}{x} = 1$ ។

$$c). \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = 1$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{x^2}{x^2 - 1} - 1 \right| < \varepsilon$$

$$\left| 1 + \frac{1}{x^2 - 1} - 1 \right| < \varepsilon$$

$$\left| \frac{1}{x^2 - 1} \right| < \varepsilon \text{ នាំឲ្យ } |x^2 - 1| > \frac{1}{\varepsilon}$$

$$(|x^2 - 1|)^2 > \left(\frac{1}{\varepsilon} \right)^2 \text{ នាំឲ្យ } x^2 - 1 > \frac{1}{\varepsilon^2}$$

$$\text{នាំឲ្យ } x^2 > \frac{1}{\varepsilon^2} + 1 \text{ នាំឲ្យ } \sqrt{x^2} > \sqrt{\frac{1}{\varepsilon^2} + 1} \text{ ឬ } |x| > \sqrt{\frac{1}{\varepsilon^2} + 1}$$

$$\text{ដោយ } x < 0 \text{ នាំឲ្យ } x < -\sqrt{\frac{1}{\varepsilon^2} + 1} \text{ គេយក } N = \sqrt{\frac{1}{\varepsilon^2} + 1} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists N = \sqrt{\frac{1}{\varepsilon^2} + 1} \text{ ដែល } x < -N \text{ នាំឲ្យ } \left| \frac{x^2}{x^2 - 1} - 1 \right| < \varepsilon$$

$$\text{ដូចនេះ } \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = 1 \text{ ។}$$

$$d). \lim_{x \rightarrow +\infty} \frac{1}{x^2} = 0$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{1}{x^2} - 0 \right| < \varepsilon \text{ ឬ } \left| \frac{1}{x^2} \right| < \varepsilon \text{ ឬ } \frac{1}{x^2} < \varepsilon$$

$$\text{នាំឲ្យ } x^2 > \frac{1}{\varepsilon} \text{ នាំឲ្យ } \sqrt{x^2} > \sqrt{\frac{1}{\varepsilon}} \text{ ឬ } |x| > \sqrt{\frac{1}{\varepsilon}}$$

$$\text{ដោយ } x \rightarrow +\infty \text{ នោះ } |x| = x \text{ នាំឲ្យ } x > \sqrt{\frac{1}{\varepsilon}} \text{ គេយក } N = \sqrt{\frac{1}{\varepsilon}} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists N = \sqrt{\frac{1}{\varepsilon}} \text{ ដែល } x > N \text{ នាំឲ្យ } \left| \frac{1}{x^2} - 0 \right| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow +\infty} \frac{1}{x^2} = 0 \text{ ។}$$

$$\text{e). } \lim_{x \rightarrow -\infty} \frac{x^2 - 2}{x^2 + 2} = 1$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$\left| \frac{x^2 - 2}{x^2 + 2} - 1 \right| < \varepsilon \text{ ឬ } \left| 1 - \frac{4}{x^2 + 2} - 1 \right| < \varepsilon \text{ នាំឲ្យ } \left| \frac{4}{x^2 + 2} \right| < \varepsilon$$

$$\frac{4}{x^2 + 2} < \varepsilon \text{ នាំឲ្យ } x^2 + 2 > \frac{4}{\varepsilon} \text{ នាំឲ្យ } x^2 > \frac{4}{\varepsilon}$$

$$\text{នាំឲ្យ } \sqrt{x^2} > \sqrt{\frac{4}{\varepsilon}} \text{ នាំឲ្យ } |x| > \frac{2}{\sqrt{\varepsilon}}$$

$$\text{ដោយ } x \rightarrow -\infty \text{ នាំឲ្យ } |x| = -x$$

$$\text{នាំឲ្យ } x < -\frac{2}{\sqrt{\varepsilon}} \text{ គេយក } N = \frac{2}{\sqrt{\varepsilon}} > 0$$

$$\text{គេបាន } \forall \varepsilon > 0, \exists N = \frac{2}{\sqrt{\varepsilon}} \text{ ដែល } x < -N \text{ នាំឲ្យ } \left| \frac{x^2 - 2}{x^2 + 2} - 1 \right| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow -\infty} \frac{x^2 - 2}{x^2 + 2} = 1 \text{ ។}$$

$$\text{f). } \lim_{x \rightarrow +\infty} \frac{2x + 4}{x} = 2$$

$$\text{តាមនិយមន័យ: } \forall \varepsilon > 0, |f(x) - L| < \varepsilon$$

$$\left| \frac{2x + 4}{x} - 2 \right| < \varepsilon \text{ ឬ } \left| 2 + \frac{4}{x} - 2 \right| < \varepsilon$$

$$\left| \frac{4}{x} \right| < \varepsilon \quad \text{ឬ} \quad \frac{4}{|x|} < \varepsilon \quad \text{នាំឲ្យ} \quad |x| > \frac{4}{\varepsilon}$$

ដោយ $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{នាំឲ្យ} \quad x > \frac{4}{\varepsilon} \quad \text{គេយក} \quad N = \frac{4}{\varepsilon} > 0$$

$$\text{គេបាន} \quad \forall \varepsilon > 0 \text{ មាន } N = \frac{4}{\varepsilon} \text{ ដែល } x > N \text{ នាំឲ្យ} \quad \left| \frac{2x+4}{x} - 2 \right| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow +\infty} \frac{2x+4}{x} = 2 \quad \text{។}$$

$$\text{g). } \lim_{x \rightarrow +\infty} \frac{3x+1}{x} = 3$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{3x+1}{x} - 3 \right| < \varepsilon \quad \text{ឬ} \quad \left| 3 + \frac{1}{x} - 3 \right| < \varepsilon$$

$$\left| \frac{1}{x} \right| < \varepsilon \quad \text{ឬ} \quad \frac{1}{|x|} < \varepsilon \quad \text{នាំឲ្យ} \quad |x| > \frac{1}{\varepsilon}$$

ដោយ $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{នាំឲ្យ} \quad x > \frac{1}{\varepsilon} \quad \text{គេយក} \quad N = \frac{1}{\varepsilon} > 0$$

$$\text{គេបាន} \quad \forall \varepsilon > 0, \exists N = \frac{1}{\varepsilon} \text{ ដែល } x > N \text{ នាំឲ្យ} \quad \left| \frac{3x+1}{x} - 3 \right| < \varepsilon$$

$$\text{ដូចនេះ: } \lim_{x \rightarrow +\infty} \frac{3x+1}{x} = 3 \quad \text{។}$$

$$\text{h). } \lim_{x \rightarrow -\infty} \frac{3x-1}{x-1} = 3$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{3x-1}{x-1} - 3 \right| < \varepsilon \quad \text{ឬ} \quad \left| 3 + \frac{2}{x-1} - 3 \right| < \varepsilon$$

$$\left| \frac{2}{x-1} \right| < \varepsilon \quad \text{ឬ} \quad \frac{2}{|x-1|} < \varepsilon \Rightarrow |x-1| > \frac{2}{\varepsilon}$$

ដោយ $x \rightarrow -\infty$ នោះ $|x-1| = -(x-1)$

នាំឲ្យ $x-1 < -\frac{2}{\varepsilon}$ នាំឲ្យ $x-1 < -\frac{2}{\varepsilon}$ នាំឲ្យ $x < -\frac{2}{\varepsilon}$ គេយក $N = \frac{2}{\varepsilon} > 0$

គេបាន $\forall \varepsilon > 0, \exists N = \frac{2}{\varepsilon}$ ដែល $x < -N$ នាំឲ្យ $\left| \frac{3x-1}{x-1} - 3 \right| < \varepsilon$

ដូចនេះ $\lim_{x \rightarrow -\infty} \frac{3x-1}{x-1} = 3$ ។

$$i). \lim_{x \rightarrow +\infty} \frac{x^2-2}{x^2+2} = 1$$

តាមនិយមន័យ: $\forall \varepsilon > 0, |f(x) - L| < \varepsilon$

$$\left| \frac{x^2-2}{x^2+2} - 1 \right| < \varepsilon \text{ ឬ } \left| 1 - \frac{4}{x^2+2} - 1 \right| < \varepsilon \text{ នាំឲ្យ } \left| \frac{4}{x^2+2} \right| < \varepsilon$$

$$\frac{4}{x^2+2} < \varepsilon \text{ នាំឲ្យ } x^2+2 > \frac{4}{\varepsilon} \text{ នាំឲ្យ } x^2 > \frac{4}{\varepsilon}$$

$$\text{នាំឲ្យ } \sqrt{x^2} > \sqrt{\frac{4}{\varepsilon}} \text{ នាំឲ្យ } |x| > \frac{2}{\sqrt{\varepsilon}}$$

ដោយ $x \rightarrow +\infty$ នាំឲ្យ $|x| = x$

$$\text{នាំឲ្យ } x > \frac{2}{\sqrt{\varepsilon}} \text{ គេយក } N = \frac{2}{\sqrt{\varepsilon}} > 0$$

គេបាន $\forall \varepsilon > 0, \exists N = \frac{2}{\sqrt{\varepsilon}}$ ដែល $x > N$ នាំឲ្យ $\left| \frac{x^2-2}{x^2+2} - 1 \right| < \varepsilon$

$$\text{ដូចនេះ } \lim_{x \rightarrow +\infty} \frac{x^2-2}{x^2+2} = 1 \text{ ។}$$

2) ប្រើនិយមន័យបង្ហាញថា:

$$a). \lim_{x \rightarrow +\infty} (-x^2) = -\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$$-x^2 < -M \text{ ឬ } x^2 > M$$

$$\text{នាំឲ្យ } \sqrt{x^2} > \sqrt{M} \text{ នាំឲ្យ } |x| > \sqrt{M}$$

ដោយ $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{នាំឲ្យ } x > \sqrt{M} \text{ គេយក } N = \sqrt{M} > 0$$

គេបាន $\forall M > 0, \exists N = \sqrt{M}$ ដែល $x > N$ នាំឲ្យ $f(x) < -M$

ដូចនេះ: $\lim_{x \rightarrow +\infty} (-x^2) = -\infty$ ។

b). $\lim_{x \rightarrow -\infty} (x^2 - x - 1) = +\infty$

តាមនិយមន័យ: $\forall M > 0, f(x) > M$

$$x^2 - x - 1 > M \text{ ឬ } (x - \frac{1}{2})^2 - \frac{1}{4} - 1 > M$$

$$\text{ឬ } (x - \frac{1}{2})^2 > M + \frac{5}{4} \text{ នាំឲ្យ } \sqrt{(x - \frac{1}{2})^2} > \sqrt{M + \frac{5}{4}}$$

$$|x - \frac{1}{2}| > \sqrt{M + \frac{5}{4}} \text{ តែ } x \rightarrow -\infty \text{ នាំឲ្យ } x - \frac{1}{2} < -\sqrt{M + \frac{5}{4}}$$

$$\text{នាំឲ្យ } x < -\sqrt{M + \frac{5}{4}} \text{ គេយក } N = \sqrt{M + \frac{5}{4}} > 0$$

គេបាន $\forall M > 0, \exists N = \sqrt{M + \frac{5}{4}}$ ដែល $x < -N$ នាំឲ្យ $f(x) > M$

ដូចនេះ: $\lim_{x \rightarrow -\infty} (x^2 - x - 1) = +\infty$ ។

c). $\lim_{x \rightarrow -\infty} (1 - x^2) = -\infty$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$$1 - x^2 < -M \text{ ឬ } -x^2 < -M - 1 \text{ ឬ } x^2 > M + 1$$

$$\sqrt{x^2} > \sqrt{M + 1} \text{ ឬ } |x| > \sqrt{M + 1}$$

ដោយ $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\text{នាំឲ្យ } x < -\sqrt{M + 1} \text{ គេយក } N = \sqrt{M + 1} > 0$$

គេបាន $\forall M > 0, \exists N = \sqrt{M + 1}$ ដែល $x < -N$ នាំឲ្យ $f(x) < -M$

ដូចនេះ: $\lim_{x \rightarrow -\infty} (1 - x^2) = -\infty$ ។

d). $\lim_{x \rightarrow +\infty} \frac{x + 2x^2}{3x + 5} = +\infty$

តាមនិយមន័យ: $\forall M > 0, f(x) > M$

$$\frac{x+2x^2}{3x+5} > M \text{ ឬ } \frac{2}{3}x - \frac{7}{9} + \frac{35/9}{3x+5} > M$$

$$\text{ដោយ } x \rightarrow +\infty \text{ នោះ } \frac{35/9}{3x+5} \rightarrow 0$$

$$\text{នាំឲ្យ } \frac{2}{3}x - \frac{7}{9} > M \text{ នាំឲ្យ } \frac{2}{3}x > M + \frac{7}{9} \text{ ឬ } x > \frac{3}{2}\left(M + \frac{7}{9}\right)$$

$$\text{តេយក } N = \frac{3}{2}\left(M + \frac{7}{9}\right) > 0$$

$$\text{តេបាន } \forall M > 0, \exists N = \frac{3}{2}\left(M + \frac{7}{9}\right) \text{ ដែល } x > N \text{ នាំឲ្យ } f(x) > M$$

$$\text{ដូចនេះ } \lim_{x \rightarrow +\infty} \frac{x+2x^2}{3x+5} = +\infty \text{ ។}$$

$$\text{e). } \lim_{x \rightarrow -\infty} \frac{x^2+3}{x} = -\infty$$

$$\text{តាមនិយមន័យ: } \forall M > 0, f(x) < -M$$

$$\frac{x^2+3}{x} < -M \text{ ឬ } x + \frac{3}{x} < -M$$

$$\text{ដោយ } x \rightarrow -\infty \text{ នោះ } \frac{3}{x} \rightarrow 0 \text{ នាំឲ្យ } x < -M \text{ តេយក } N = M > 0$$

$$\text{តេបាន } \forall M > 0, \exists N = M \text{ ដែល } x < -N \text{ នាំឲ្យ } \lim_{x \rightarrow -\infty} \frac{x^2+3}{x} = -\infty$$

$$\text{ដូចនេះ } \lim_{x \rightarrow -\infty} \frac{x^2+3}{x} = -\infty \text{ ។}$$

$$\text{f). } \lim_{x \rightarrow +\infty} \frac{3-x+x^3}{1-x^2} = -\infty$$

$$\text{តាមនិយមន័យ: } \forall M > 0, f(x) < -M$$

$$\frac{3-x+x^3}{1-x^2} < -M \text{ ឬ } -x + \frac{3}{1-x^2} < -M$$

$$\text{ដោយ } x \rightarrow +\infty \text{ នោះ } \frac{3}{1-x^2} \rightarrow 0 \text{ នាំឲ្យ } -x < -M$$

$$\text{នាំឲ្យ } x > M \text{ តេយក } N = M > 0$$

គេបាន $\forall M > 0, \exists N = M$ ដែល $x > N$ នាំឲ្យ $\frac{3-x+x^3}{1-x^2} < -M$

ដូចនេះ $\lim_{x \rightarrow +\infty} \frac{3-x+x^3}{1-x^2} = -\infty$ ។

$$g). \lim_{x \rightarrow -\infty} \frac{6x-x^2}{2x+1} = +\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) > M$

$$\text{គេមាន } f(x) = \frac{6x-x^2}{2x+1} = -\frac{1}{2}x + \frac{13}{4} - \frac{13/4}{2x+1}$$

ដើម្បីឲ្យ $f(x) > M$ គេគ្រាន់តែឲ្យ $-\frac{1}{2}x > M$

$$(\text{ព្រោះ } \frac{13/4}{2x+1} \rightarrow 0 \text{ ពេល } x \rightarrow -\infty)$$

នាំឲ្យ $x < -2M$ គេយក $N = 2M > 0$

គេបាន $\forall M > 0, \exists N = 2M$ ដែល $x < -N$ នាំឲ្យ $f(x) > M$

ដូចនេះ $\lim_{x \rightarrow -\infty} \frac{6x-x^2}{2x+1} = +\infty$ ។

$$h). \lim_{x \rightarrow +\infty} \frac{x^2+3}{2-x} = -\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$$\frac{x^2+3}{2-x} < -M$$

$$-x-2 + \frac{7}{2-x} < -M$$

នាំឲ្យ $-x-2 < -M$ (ព្រោះ $\frac{7}{2-x} \rightarrow 0$ ពេល $x \rightarrow +\infty$)

នាំឲ្យ $-x < -M$ នាំឲ្យ $x > M$ គេយក $N = M > 0$

គេបាន $\forall M > 0, \exists N = M$ ដែល $x > N$ នាំឲ្យ $\frac{x^2+3}{2-x} < -M$

ដូចនេះ $\lim_{x \rightarrow +\infty} \frac{x^2+3}{2-x} = -\infty$ ។

$$i). \lim_{x \rightarrow -\infty} (x^2 + 2x + 3) = +\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) > M$

$$x^2 + 2x + 3 > M \text{ ឬ } (x^2 + 2x + 1) + 2 > M$$

$$(x+1)^2 > M \text{ ឬ } |x+1| > \sqrt{M}$$

$$\text{នាំឲ្យ } x+1 < -\sqrt{M} \text{ (ព្រោះ } x \rightarrow -\infty)$$

$$\text{នាំឲ្យ } x < -\sqrt{M} - 1 \text{ នាំឲ្យ } x < -(\sqrt{M} + 1)$$

$$\text{គេយក } N = \sqrt{M} + 1 > 0$$

គេបាន $\forall M > 0, \exists N = \sqrt{M} + 1$ ដែល $x < -N$ នាំឲ្យ $x^2 + 2x + 3 > M$

$$\text{ដូចនេះ: } \lim_{x \rightarrow -\infty} (x^2 + 2x + 3) = +\infty$$

$$j). \lim_{x \rightarrow +\infty} (1 + 6x - x^2) = -\infty$$

តាមនិយមន័យ: $\forall M > 0, f(x) < -M$

$$1 + 6x - x^2 < -M$$

$$10 + (-9 + 6x - x^2) < -M$$

$$10 - (9 - 6x + x^2) < -M$$

$$10 - (3 - x)^2 < -M$$

$$-(3 - x)^2 < -M$$

$$(3 - x)^2 > M \text{ នាំឲ្យ } |3 - x| > \sqrt{M}$$

$$3 - x < -\sqrt{M} \text{ ឬ } -x < -\sqrt{M} - 3$$

$$\text{នាំឲ្យ } x > \sqrt{M} + 3 \text{ គេយក } N = \sqrt{M} + 3 > 0$$

គេបាន $\forall M > 0, \exists N = \sqrt{M} + 3$ ដែល $x > N$ នាំឲ្យ $1 + 6x - x^2 < -M$

$$\text{ដូចនេះ: } \lim_{x \rightarrow +\infty} (1 + 6x - x^2) = -\infty$$

2. ប្រមាណវិធីលីមីត

បើ $\lim_{x \rightarrow a} f(x) = \ell$, $\lim_{x \rightarrow a} g(x) = m$ និង $\lim_{x \rightarrow a} h(x) = n$ ដែល $(\ell, m, n) \in \mathbb{R}$

នោះគេបាន៖

- ❶ $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \ell \pm m$ ដែល $(a$ អាចជាចំនួនកំណត់ ឬអនន្ត)
- ❷ $\lim_{x \rightarrow a} [f(x) + g(x) - h(x)] = \ell + m - n$
- ❸ $\lim_{x \rightarrow a} k \cdot f(x) = k\ell$
- ❹ $\lim_{x \rightarrow a} f(x)g(x)h(x) = \ell \cdot m \cdot n$
- ❺ $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\ell}{m}$ ដែល $m \neq 0$, $g(a) \neq 0$
- ❻ $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n = \ell^n$
- ❼ $\lim_{x \rightarrow a} [f(x)]^{g(x)} = [\lim_{x \rightarrow a} f(x)]^{\lim_{x \rightarrow a} g(x)} = \ell^m$
- ❽ $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{\ell}$ ដែល $n \in \mathbb{N}$, $n \geq 2$ ។

លំហាត់

- 1) គេឲ្យ $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -1$ និង $\lim_{x \rightarrow c} h(x) = 0$ ។

គណនាលីមីតខាងក្រោម៖

ក/. $\lim_{x \rightarrow c} [f(x) - g(x)]$

ខ/. $\lim_{x \rightarrow c} [f(x)]^2$

គ/. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$

ឃ/. $\lim_{x \rightarrow c} \frac{1}{f(x) - g(x)}$

ង/. $\lim_{x \rightarrow c} \frac{h(x)}{f(x)}$

ច/. $\lim_{x \rightarrow c} \frac{f(x)}{h(x)}$

- 2) គេឲ្យ $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 0$ និង $\lim_{x \rightarrow c} h(x) = -2$ ។

គណនាលីមីតខាងក្រោម៖

ក/. $\lim_{x \rightarrow c} [3f(x) + 2h(x)]$

ខ/. $\lim_{x \rightarrow c} [h(x)]^3$

$$\text{គ/} \lim_{x \rightarrow c} \frac{h(x)}{x - c}$$

$$\text{ឃ/} \lim_{x \rightarrow c} \frac{g(x)}{h(x)}$$

$$\text{ង/} \lim_{x \rightarrow c} \frac{4}{f(x) - h(x)}$$

$$\text{ច/} \lim_{x \rightarrow c} [3 + g(x)]^2$$

3) គណនាលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow 0} (2x - 1)$$

$$\text{ខ/} \lim_{x \rightarrow 1} (2 - 5x)$$

$$\text{គ/} \lim_{x \rightarrow 2} x^2$$

$$\text{ឃ/} \lim_{x \rightarrow 4} \sqrt{x}$$

$$\text{ង/} \lim_{x \rightarrow -3} (|x| - 2)$$

$$\text{ច/} \lim_{x \rightarrow 0} |x|$$

$$\text{ឆ/} \lim_{x \rightarrow 1} \frac{3}{x + 1}$$

$$\text{ជ/} \lim_{x \rightarrow 0} \frac{4}{x + 1}$$

$$\text{ណ/} \lim_{x \rightarrow -1} \frac{2}{x + 1}$$

$$\text{ញ/} \lim_{x \rightarrow 2} \frac{1}{3x - 6}$$

$$\text{ដ/} \lim_{x \rightarrow 1} \frac{x}{x + 1}$$

$$\text{ប/} \lim_{x \rightarrow 4} \frac{x}{\sqrt{x} + 1}$$

$$\text{ខ/} \lim_{x \rightarrow 4} \frac{x}{\sqrt{x} + 1}$$

$$\text{ឈ/} \lim_{x \rightarrow -1} \frac{1 - x}{x + 1}$$

$$\text{ណ/} \lim_{x \rightarrow -2} \frac{|x|}{x}$$

4) គណនាតម្លៃលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow -4} (x^2 + 3x - 7)$$

$$\text{ខ/} \lim_{x \rightarrow 3} (5 - 4x)^2$$

$$\text{គ/} \lim_{x \rightarrow 2} (3)$$

$$\text{ឃ/} \lim_{x \rightarrow 2} \frac{x^2 + x + 1}{x^2 + 2x}$$

$$\text{ង/} \lim_{x \rightarrow \sqrt{3}} |x^2 - 8|$$

$$\text{ច/} \lim_{x \rightarrow 5} \frac{2 - x^2}{4x}$$

$$\text{ឆ/} \lim_{x \rightarrow -1} \frac{x^2 + 1}{3x^5 + 4}$$

$$\text{ជ/} \lim_{x \rightarrow -2} 3|x - 1|$$

$$\text{ណ/} \lim_{x \rightarrow 0} (x - \frac{4}{x})$$

$$\text{ញ/} \lim_{x \rightarrow 1} \frac{1 + x - x^2}{(3x - 1)(x^4 - 2)}$$

ដំណោះស្រាយ

1) គេឱ្យ $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -1$ និង $\lim_{x \rightarrow c} h(x) = 0$

$$\text{ក/} \lim_{x \rightarrow c} [f(x) - g(x)] = 2 - (-1) = 2 + 1 = 3$$

$$\text{ខ/} \lim_{x \rightarrow c} [f(x)]^2 = (2)^2 = 4$$

$$\text{គ/} \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{2}{-1} = -2$$

$$\text{ឃ/} \lim_{x \rightarrow c} \frac{1}{f(x) - g(x)} = \frac{1}{2 - (-1)} = \frac{1}{2+1} = \frac{1}{3}$$

$$\text{ង/} \lim_{x \rightarrow c} \frac{h(x)}{f(x)} = \frac{0}{2} = 0$$

$$\text{ច/} \lim_{x \rightarrow c} \frac{f(x)}{h(x)} = \frac{2}{0} = \pm \infty$$

$$2) \text{ តេឡី} \lim_{x \rightarrow c} f(x) = 3, \lim_{x \rightarrow c} g(x) = 0 \text{ និង } \lim_{x \rightarrow c} h(x) = -2$$

$$\text{ក/} \lim_{x \rightarrow c} [3f(x) + 2h(x)] = 3(3) + 2(-2) = 9 - 4 = 5$$

$$\text{ខ/} \lim_{x \rightarrow c} [h(x)]^3 = (-2)^3 = -8$$

$$\text{គ/} \lim_{x \rightarrow c} \frac{h(x)}{x - c} = \frac{-2}{c - c} = \frac{-2}{0} = \pm \infty$$

$$\text{ឃ/} \lim_{x \rightarrow c} \frac{g(x)}{h(x)} = \frac{0}{-2} = 0$$

$$\text{ង/} \lim_{x \rightarrow c} \frac{4}{f(x) - h(x)} = \frac{4}{3 - (-2)} = \frac{4}{3+2} = \frac{4}{5}$$

$$\text{ច/} \lim_{x \rightarrow c} [3 + g(x)]^2 = (3 + 0)^2 = 3^2 = 9$$

3) គណនាលីមីតខាងក្រោម៖

$$\text{ក/} \lim_{x \rightarrow 0} (2x - 1) = 2(0) - 1 = 0 - 1 = -1$$

$$\text{ខ/} \lim_{x \rightarrow 1} (2 - 5x) = 2 - 5(1) = 2 - 5 = -3$$

$$\text{គ/} \lim_{x \rightarrow 2} x^2 = 2^2 = 4$$

$$\text{ឃ/} \lim_{x \rightarrow 4} \sqrt{x} = \sqrt{4} = 2$$

$$\text{ង/} \lim_{x \rightarrow -3} (|x| - 2) = |-3| - 2 = 3 - 2 = 1$$

$$\text{ច/} \lim_{x \rightarrow 0} |x| = |0| = 0$$

$$\text{ឆ/} \lim_{x \rightarrow 1} \frac{3}{x+1} = \frac{3}{1+1} = \frac{3}{2}$$

$$\text{ជ.} \quad \lim_{x \rightarrow 0} \frac{4}{x+1} = \frac{4}{0+1} = \frac{4}{1} = 4$$

$$\text{ឈ.} \quad \lim_{x \rightarrow -1} \frac{2}{x+1} = \frac{2}{-1+1} = \frac{2}{0} = \pm\infty$$

$$\text{ញ.} \quad \lim_{x \rightarrow 2} \frac{1}{3x-6} = \frac{1}{3(2)-6} = \frac{1}{0} = \pm\infty$$

$$\text{ដ.} \quad \lim_{x \rightarrow 1} \frac{x}{x+1} = \frac{1}{1+1} = \frac{1}{2}$$

$$\text{ប.} \quad \lim_{x \rightarrow 4} \frac{x}{\sqrt{x}+1} = \frac{4}{\sqrt{4}+1} = \frac{4}{2+1} = \frac{4}{3}$$

$$\text{ខ.} \quad \lim_{x \rightarrow 4} \frac{x}{\sqrt{x+1}} = \frac{4}{\sqrt{4+1}} = \frac{4}{\sqrt{5}} = \frac{4\sqrt{5}}{5}$$

$$\text{ឈ.} \quad \lim_{x \rightarrow -1} \frac{1-x}{x+1} = \frac{1-(-1)}{-1+1} = \frac{1+1}{0} = \frac{2}{0} = \pm\infty$$

$$\text{ណ.} \quad \lim_{x \rightarrow -2} \frac{|x|}{x} = \frac{|-2|}{-2} = \frac{2}{-2} = -1$$

4) គណនាតម្លៃលីមីតខាងក្រោម៖

$$\text{ក.} \quad \lim_{x \rightarrow -4} (x^2 + 3x - 7) = 16 - 12 - 7 = -3$$

$$\text{ខ.} \quad \lim_{x \rightarrow 3} (5 - 4x)^2 = (5 - 12)^2 = (-7)^2 = 49$$

$$\text{គ.} \quad \lim_{x \rightarrow 2} (3) = 3$$

$$\text{ឃ.} \quad \lim_{x \rightarrow 2} \frac{x^2 + x + 1}{x^2 + 2x} = \frac{4 + 2 + 1}{4 + 4} = \frac{7}{8}$$

$$\text{ង.} \quad \lim_{x \rightarrow \sqrt{3}} |x^2 - 8| = |(\sqrt{3})^2 - 8| = |3 - 8| = |-5| = 5$$

$$\text{ច.} \quad \lim_{x \rightarrow 5} \frac{2 - x^2}{4x} = \frac{2 - 25}{20} = \frac{-23}{20} = -\frac{23}{20}$$

$$\text{ឆ.} \quad \lim_{x \rightarrow -1} \frac{x^2 + 1}{3x^5 + 4} = \frac{1 + 1}{-3 + 4} = \frac{2}{1} = 2$$

$$\text{ជ.} \lim_{x \rightarrow -2} 3|x-1| = 3|-2-1| = 3|-3| = 3 \cdot 3 = 9$$

$$\text{ឈ.} \lim_{x \rightarrow 0} (x - \frac{4}{x}) = 0 - \frac{4}{0} = -\frac{4}{0} = \pm \infty$$

$$\text{ញ.} \lim_{x \rightarrow 1} \frac{1+x-x^2}{(3x-1)(x^4-2)} = \frac{1+1-1}{(3-1)(1-2)} = \frac{1}{(2)(-1)} = -\frac{1}{2}$$

3. លីមីតរាងមិនកំណត់

ក. លីមីតរាងមិនកំណត់ $(\frac{0}{0})$

វិធាន: ដើម្បីគណនាលីមីតរាងមិនកំណត់ $(\frac{0}{0})$ គេត្រូវ៖

- ❶ បំបែកភាគយក និងភាគបែងជាផលគុណកត្តា
- ❷ សម្រួលកត្តារួមចោល
- ❸ គណនាលីមីតនៃកន្សោមថ្មី

លំហាត់

1) គណនាលីមីតខាងក្រោម៖

$$\text{ក.} \lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1}$$

$$\text{ខ.} \lim_{x \rightarrow 0} \frac{x}{|x|}$$

$$\text{គ.} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1}$$

$$\text{ឃ.} \lim_{x \rightarrow 2} \frac{x - 2}{x^2 - 4}$$

$$\text{ង.} \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$\text{ច.} \lim_{x \rightarrow -2} \frac{x^2 - x - 6}{(x + 2)^2}$$

$$\text{ឆ.} \lim_{x \rightarrow 0} \frac{x^4}{3x^3 + 2x^2 + x}$$

$$\text{ជ.} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1}$$

$$\text{ឈ.} \lim_{x \rightarrow 1} \frac{x^5 - 1}{x^4 - 1}$$

$$\text{ញ.} \lim_{x \rightarrow -2} \frac{(x^2 - x - 6)^2}{(x + 2)^2}$$

$$\text{ដ.} \lim_{x \rightarrow 0} \frac{x^2 + 2x}{x^2 + x}$$

$$\text{ប.} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - x}$$

$$\text{ខ.} \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}$$

$$\text{ឈ.} \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$$

2) គណនាលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow 1} \frac{x^3 - 1}{\sqrt{x} - 1}$$

$$\text{គ/} \lim_{x \rightarrow 3^+} \frac{\sqrt{x-3}}{|x-3|}$$

$$\text{ង/} \lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x-3}$$

$$\text{ឆ/} \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{\sqrt{x^2 - 1}}$$

$$\text{ឈ/} \lim_{x \rightarrow 5} \frac{\sqrt{x^2 + 5} - \sqrt{30}}{x-5}$$

$$\text{ជ/} \lim_{x \rightarrow 5} \frac{\sqrt{2x-1} - 3}{5-x}$$

$$\text{ខ/} \lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x^2 - 4}$$

$$\text{ឃ/} \lim_{x \rightarrow 4} \frac{\sqrt{x+5} - 3}{x-4}$$

$$\text{ច/} \lim_{x \rightarrow 7} \frac{7-x}{\sqrt{x+2} - 3}$$

$$\text{ឝ/} \lim_{x \rightarrow 1} \frac{\sqrt{x^2 + 1} - \sqrt{2}}{x-1}$$

$$\text{ញ/} \lim_{x \rightarrow 5} \frac{\sqrt{x+4} - 3}{x^2 - 6x + 5}$$

$$\text{ប/} \lim_{x \rightarrow 1} \frac{1 - \sqrt{2x-1}}{1-x}$$

3) គណនាលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{\sqrt{x+7} - 3}$$

$$\text{គ/} \lim_{x \rightarrow 2} \frac{x - \sqrt{x+2}}{\sqrt{4x+1} - 3}$$

$$\text{ង/} \lim_{x \rightarrow 0} \frac{\sqrt{x+1} - \sqrt{1-2x}}{1 - \sqrt{1-3x}}$$

$$\text{ខ/} \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - \sqrt{3x+4}}{\sqrt{x+1} - 1}$$

$$\text{ឃ/} \lim_{x \rightarrow 5} \frac{\sqrt{x^2 - 7x + 16} - \sqrt{6}}{(x-5)\sqrt{x+1}}$$

$$\text{ច/} \lim_{x \rightarrow -1} \frac{1 - \sqrt{x+2}}{1 - \sqrt{2x^2 - 1}}$$

4) គណនាលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x-1}$$

$$\text{គ/} \lim_{x \rightarrow 1} \frac{1 - \sqrt[1987]{x}}{1-x}$$

$$\text{ង/} \lim_{x \rightarrow 1} \frac{1 - \sqrt[3]{2-x}}{1-x}$$

$$\text{ខ/} \lim_{x \rightarrow 0} \frac{\sqrt[n]{a+x} - \sqrt[n]{a}}{x}$$

$$\text{ឃ/} \lim_{x \rightarrow 0} \frac{\sqrt[5]{(x+1)^3} - 1}{x}$$

$$\text{ច/} \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - 1}{x(x+2)}$$

$$\text{ឆ/} \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{\sqrt[3]{x+6} - 2}$$

$$\text{ជ/} \lim_{x \rightarrow 1} \frac{\sqrt[5]{x^2 + x - 1} - 1}{1 + \sqrt[3]{1 - 2x}}$$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

$$\text{ក/} \lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^3 + x^2 + x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x^3 + x^2 + x + 1)$$

$$= 1 + 1 + 1 + 1 = 4$$

$$\text{ខ/} \lim_{x \rightarrow 0} \frac{x}{|x|} \text{ មានរាង } \left(\frac{0}{0} \right)$$

❖ ករណី $x \rightarrow 0^+$ នោះ $|x| = x$ គេបាន:

$$\lim_{x \rightarrow 0} \frac{x}{|x|} = \lim_{x \rightarrow 0} \frac{x}{x} = \lim_{x \rightarrow 0} (1) = 1$$

❖ ករណី $x \rightarrow 0^-$ នោះ $|x| = -x$ គេបាន:

$$\lim_{x \rightarrow 0} \frac{x}{|x|} = \lim_{x \rightarrow 0} \frac{x}{-x} = \lim_{x \rightarrow 0} (-1) = -1$$

$$\text{គ/} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)^2} = \lim_{x \rightarrow 1} \frac{x+1}{x-1} = \frac{2}{0} = \pm\infty$$

$$\text{ឃ/} \lim_{x \rightarrow 2} \frac{x-2}{x^2-4} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{1}{x+2} = \frac{1}{2+2} = \frac{1}{4}$$

$$\text{ង/} \lim_{x \rightarrow 2} \frac{x^2-4}{x-2} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} (x+2) = 2+2 = 4$$

$$\text{ច/} \lim_{x \rightarrow -2} \frac{x^2 - x - 6}{(x+2)^2} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow -2} \frac{(x+2)(x-3)}{(x+2)^2} = \lim_{x \rightarrow -2} \frac{x-3}{x+2} = \frac{-5}{0} = \pm\infty$$

$$\text{ឆ/} \lim_{x \rightarrow 0} \frac{x^4}{3x^3 + 2x^2 + x} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{x \cdot x^3}{x(3x^2 + 2x + 1)} = \lim_{x \rightarrow 0} \frac{x^3}{3x^2 + 2x + 1} = \frac{0}{1} = 0$$

$$\text{ជ/} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x^2 + x + 1)} = \lim_{x \rightarrow 1} \frac{x+1}{x^2 + x + 1} = \frac{1+1}{1+1+1} = \frac{2}{3}$$

$$\text{ឈ/} \lim_{x \rightarrow 1} \frac{x^5 - 1}{x^4 - 1} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^4 + x^3 + x^2 + x + 1)}{(x-1)(x^3 + x^2 + x + 1)} = \lim_{x \rightarrow 1} \frac{x^4 + x^3 + x^2 + x + 1}{x^3 + x^2 + x + 1} = \frac{5}{4}$$

$$\text{ញ/} \lim_{x \rightarrow -2} \frac{(x^2 - x - 6)^2}{(x+2)^2} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow -2} \frac{[(x+2)(x-3)]^2}{(x+2)^2} = \lim_{x \rightarrow -2} \frac{(x+2)^2(x-3)^2}{(x+2)^2}$$

$$= \lim_{x \rightarrow -2} (x-3)^2 = (-2-3)^2 = (-5)^2 = 25$$

$$\text{ដ/} \lim_{x \rightarrow 0} \frac{x^2 + 2x}{x^2 + x} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{x(x+2)}{x(x+1)} = \lim_{x \rightarrow 0} \frac{x+2}{x+1} = \frac{0+2}{0+1} = 2$$

$$\text{ប/} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - x} \text{មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x(x-1)} = \lim_{x \rightarrow 1} \frac{x+1}{x} = \frac{1+1}{1} = 2$$

$$ឧ/. \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x(x-2)} = \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x} = \frac{4 + 4 + 4}{2} = \frac{12}{2} = 6$$

$$ឆ/. \lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9)}{(x-3)(x+3)} = \lim_{x \rightarrow 3} \frac{x^2 + 3x + 9}{x+3} = \frac{9 + 9 + 9}{3 + 3} = \frac{27}{6} = \frac{9}{2}$$

2) គណនាលីមីតខាងក្រោម:

$$ក/. \lim_{x \rightarrow 1} \frac{x^3 - 1}{\sqrt{x} - 1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x^3 - 1)(\sqrt{x} + 1)}{(\sqrt{x} - 1)(\sqrt{x} + 1)} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 1)(\sqrt{x} + 1)}{(x-1)}$$

$$= \lim_{x \rightarrow 1} (x^2 + x + 1)(\sqrt{x} + 1) = (1 + 1 + 1)(1 + 1) = 3 \times 2 = 6$$

$$ខ/. \lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x^2 - 4} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{(\sqrt{x+2} - 2)(\sqrt{x+2} + 2)}{(x^2 - 4)(\sqrt{x+2} + 2)} = \lim_{x \rightarrow 2} \frac{x + 2 - 4}{(x-2)(x+2)(\sqrt{x+2} + 2)}$$

$$= \lim_{x \rightarrow 2} \frac{x - 2}{(x-2)(x+2)(\sqrt{x+2} + 2)} = \lim_{x \rightarrow 2} \frac{1}{(x+2)(\sqrt{x+2} + 2)}$$

$$= \frac{1}{4(2+2)} = \frac{1}{16}$$

$$គ/. \lim_{x \rightarrow 3^+} \frac{\sqrt{x-3}}{|x-3|} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 3^+} \frac{(\sqrt{x-3})(\sqrt{x-3})}{(x-3)\sqrt{x-3}} = \lim_{x \rightarrow 3^+} \frac{x-3}{(x-3)\sqrt{x-3}}$$

$$= \lim_{x \rightarrow 3^+} \frac{1}{\sqrt{x-3}} = \frac{1}{0^+} = +\infty$$

$$\text{ឃ/} \lim_{x \rightarrow 4} \frac{\sqrt{x+5}-3}{x-4} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 4} \frac{(\sqrt{x+5}-3)(\sqrt{x+5}+3)}{(x-4)(\sqrt{x+5}+3)} = \lim_{x \rightarrow 4} \frac{x+5-9}{(x-4)(\sqrt{x+5}+3)}$$

$$= \lim_{x \rightarrow 4} \frac{x-4}{(x-4)(\sqrt{x+5}+3)} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x+5}+3} = \frac{1}{3+3} = \frac{1}{6}$$

$$\text{ង/} \lim_{x \rightarrow 3} \frac{\sqrt{x+1}-2}{x-3} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{x+1}-2)(\sqrt{x+1}+2)}{(x-3)(\sqrt{x+1}+2)} = \lim_{x \rightarrow 3} \frac{x+1-4}{(x-3)(\sqrt{x+1}+2)}$$

$$= \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(\sqrt{x+1}+2)} = \lim_{x \rightarrow 3} \frac{1}{\sqrt{x+1}+2} = \frac{1}{2+2} = \frac{1}{4}$$

$$\text{ច/} \lim_{x \rightarrow 7} \frac{7-x}{\sqrt{x+2}-3} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 7} \frac{-(x-7)(\sqrt{x+2}+3)}{(\sqrt{x+2}-3)(\sqrt{x+2}+3)} = \lim_{x \rightarrow 7} \frac{-(x-7)(\sqrt{x+2}+3)}{x+2-9}$$

$$= \lim_{x \rightarrow 7} \frac{-(x-7)(\sqrt{x+2}+3)}{x-7} = \lim_{x \rightarrow 7} [-(\sqrt{x+2}+3)] = -(3+3) = -6$$

$$\text{ឆ/} \lim_{x \rightarrow 1} \frac{x^2+x-2}{\sqrt{x^2-1}} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{\sqrt{(x-1)(x+1)}} = \lim_{x \rightarrow 1} \frac{(x+2)}{\sqrt{x+1}} \sqrt{\frac{(x-1)^2}{x-1}}$$

$$= \lim_{x \rightarrow 1} \frac{(x+2)\sqrt{x-1}}{\sqrt{x+1}} = \frac{(1+2)(0)}{\sqrt{1+1}} = \frac{0}{\sqrt{2}} = 0$$

$$\text{ជ/} \lim_{x \rightarrow 1} \frac{\sqrt{x^2+1}-\sqrt{2}}{x-1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(\sqrt{x^2+1}-\sqrt{2})(\sqrt{x^2+1}+\sqrt{2})}{(x-1)(\sqrt{x^2+1}+\sqrt{2})} = \lim_{x \rightarrow 1} \frac{x^2+1-2}{(x-1)(\sqrt{x^2+1}+\sqrt{2})}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 1}{(x-1)(\sqrt{x^2 + 1} + \sqrt{2})} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(\sqrt{x^2 + 1} + \sqrt{2})}$$

$$= \lim_{x \rightarrow 1} \frac{x+1}{\sqrt{x^2 + 1} + \sqrt{2}} = \frac{1+1}{\sqrt{2} + \sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

ឈ/. $\lim_{x \rightarrow 5} \frac{\sqrt{x^2 + 5} - \sqrt{30}}{x - 5}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 5} \frac{(\sqrt{x^2 + 5} - \sqrt{30})(\sqrt{x^2 + 5} + \sqrt{30})}{(x-5)(\sqrt{x^2 + 5} + \sqrt{30})}$$

$$= \lim_{x \rightarrow 5} \frac{x^2 + 5 - 30}{(x-5)(\sqrt{x^2 + 5} + \sqrt{30})} = \lim_{x \rightarrow 5} \frac{x^2 - 25}{(x-5)(\sqrt{x^2 + 5} + \sqrt{30})}$$

$$= \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{(x-5)(\sqrt{x^2 + 5} + \sqrt{30})} = \lim_{x \rightarrow 5} \frac{x+5}{\sqrt{x^2 + 5} + \sqrt{30}}$$

$$= \frac{5+5}{\sqrt{25+5} + \sqrt{30}} = \frac{10}{\sqrt{30} + \sqrt{30}} = \frac{10}{2\sqrt{30}} = \frac{10\sqrt{30}}{2 \cdot 30} = \frac{\sqrt{30}}{6}$$

ញ/. $\lim_{x \rightarrow 5} \frac{\sqrt{x+4} - 3}{x^2 - 6x + 5}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 5} \frac{(\sqrt{x+4} - 3)(\sqrt{x+4} + 3)}{(x^2 - 6x + 5)(\sqrt{x+4} + 3)} = \lim_{x \rightarrow 5} \frac{x+4-9}{(x-5)(x-1)(\sqrt{x+4} + 3)}$$

$$= \lim_{x \rightarrow 5} \frac{x-5}{(x-5)(x-1)(\sqrt{x+4} + 3)} = \lim_{x \rightarrow 5} \frac{1}{(x-1)(\sqrt{x+4} + 3)}$$

$$= \frac{1}{4(3+3)} = \frac{1}{4 \times 6} = \frac{1}{24}$$

ដ/. $\lim_{x \rightarrow 5} \frac{\sqrt{2x-1} - 3}{5-x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 5} \frac{(\sqrt{2x-1} - 3)(\sqrt{2x-1} + 3)}{(5-x)(\sqrt{2x-1} + 3)} = \lim_{x \rightarrow 5} \frac{2x-1-9}{-(x-5)(\sqrt{2x-1} + 3)}$$

$$= \lim_{x \rightarrow 5} \frac{2x-10}{-(x-5)(\sqrt{2x-1} + 3)} = \lim_{x \rightarrow 5} \frac{2(x-5)}{-(x-5)(\sqrt{2x-1} + 3)}$$

$$= \lim_{x \rightarrow 5} \frac{2}{-(\sqrt{2x-1}+3)} = \frac{2}{-(3+3)} = -\frac{2}{6} = -\frac{1}{3}$$

ប្រ. $\lim_{x \rightarrow 1} \frac{1-\sqrt{2x-1}}{1-x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 1} \frac{(1-\sqrt{2x-1})(1+\sqrt{2x-1})}{(1-x)(1+\sqrt{2x-1})} = \lim_{x \rightarrow 1} \frac{1-(2x-1)}{(1-x)(1+\sqrt{2x-1})}$$

$$= \lim_{x \rightarrow 1} \frac{1-2x+1}{(1-x)(1+\sqrt{2x-1})} = \lim_{x \rightarrow 1} \frac{2-2x}{(1-x)(1+\sqrt{2x-1})}$$

$$= \lim_{x \rightarrow 1} \frac{2(1-x)}{(1-x)(1+\sqrt{2x-1})} = \lim_{x \rightarrow 1} \frac{2}{1+\sqrt{2x-1}} = \frac{2}{1+1} = \frac{2}{2} = 1$$

3) គណនាលីមីតខាងក្រោម:

ក/. $\lim_{x \rightarrow 2} \frac{\sqrt{x+2}-2}{\sqrt{x+7}-3}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 2} \frac{(x+2-4)(\sqrt{x+7}+3)}{(x+7-9)(\sqrt{x+2}+2)} = \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{x+7}+3)}{(x-2)(\sqrt{x+2}+2)}$$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{x+7}+3}{\sqrt{x+2}+2} = \frac{3+3}{2+2} = \frac{6}{4} = \frac{3}{2}$$

ខ/. $\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-\sqrt{3x+4}}{\sqrt{x+1}-1}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{[(x+4)-(3x+4)](\sqrt{x+1}+1)}{(x+1-1)(\sqrt{x+4}+\sqrt{3x+4})}$$

$$= \lim_{x \rightarrow 0} \frac{(-2x)(\sqrt{x+1}+1)}{(x)(\sqrt{x+4}+\sqrt{3x+4})} = \lim_{x \rightarrow 0} \frac{-2(\sqrt{x+1}+1)}{\sqrt{x+4}+\sqrt{3x+4}}$$

$$= \frac{-2(1+1)}{(2+2)} = -\frac{4}{4} = -1$$

គ/. $\lim_{x \rightarrow 2} \frac{x-\sqrt{x+2}}{\sqrt{4x+1}-3}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 2} \frac{[x^2-(x+2)](\sqrt{4x+1}+3)}{(4x+1-9)(x+\sqrt{x+2})} = \lim_{x \rightarrow 2} \frac{(x^2-x-2)(\sqrt{4x+1}+3)}{(4x-8)(x+\sqrt{x+2})}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 2} \frac{(x-2)(x+1)(\sqrt{4x+1}+3)}{4(x-2)(x+\sqrt{x+2})} = \lim_{x \rightarrow 2} \frac{(x+1)(\sqrt{4x+1}+3)}{4(x+\sqrt{x+2})} \\
 &= \frac{3(3+3)}{4(2+2)} = \frac{18}{16} = \frac{9}{8}
 \end{aligned}$$

ឃ/. $\lim_{x \rightarrow 5} \frac{\sqrt{x^2 - 7x + 16} - \sqrt{6}}{(x-5)\sqrt{x+1}}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 5} \frac{[(x^2 - 7x + 16) - 6]}{(x-5)\sqrt{x+1}(\sqrt{x^2 - 7x + 16} + \sqrt{6})}$$

$$= \lim_{x \rightarrow 5} \frac{(x^2 - 7x + 10)}{(x-5)(\sqrt{x^2 - 7x + 16} + \sqrt{6})\sqrt{x+1}}$$

$$= \lim_{x \rightarrow 5} \frac{(x-5)(x-2)}{(x-5)(\sqrt{x^2 - 7x + 16} + \sqrt{6})\sqrt{x+1}}$$

$$= \lim_{x \rightarrow 5} \frac{(x-2)}{(\sqrt{x^2 - 7x + 16} + \sqrt{6})\sqrt{x+1}} = \frac{3}{(\sqrt{6} + \sqrt{6})\sqrt{6}} = \frac{3}{2\sqrt{6}\sqrt{6}} = \frac{1}{4}$$

ង/. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - \sqrt{1-2x}}{1 - \sqrt{1-3x}}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{[(x+1) - (1-2x)](1 + \sqrt{1-3x})}{x \rightarrow 0 [1 - (1-3x)](\sqrt{x+1} + \sqrt{1-2x})} = \lim_{x \rightarrow 0} \frac{(3x)(1 + \sqrt{1-3x})}{(3x)(\sqrt{x+1} + \sqrt{1-2x})}$$

$$= \lim_{x \rightarrow 0} \frac{(1 + \sqrt{1-3x})}{(\sqrt{x+1} + \sqrt{1-2x})} = \frac{1+1}{1+1} = \frac{2}{2} = 1$$

ច/. $\lim_{x \rightarrow -1} \frac{1 - \sqrt{x+2}}{1 - \sqrt{2x^2 - 1}}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow -1} \frac{[1 - (x+2)](1 + \sqrt{2x^2 - 1})}{x \rightarrow -1 [1 - (2x^2 - 1)](1 + \sqrt{x+2})} = \lim_{x \rightarrow -1} \frac{(-x-1)(1 + \sqrt{2x^2 - 1})}{(2 - 2x^2)(1 + \sqrt{x+2})}$$

$$= \lim_{x \rightarrow -1} \frac{-(x+1)(1 + \sqrt{2x^2 - 1})}{2(1-x)(1+x)(1 + \sqrt{x+2})} = \lim_{x \rightarrow -1} \frac{-(1 + \sqrt{2x^2 - 1})}{2(1-x)(1 + \sqrt{x+2})}$$

$$= \frac{-(1+1)}{2(1+1)(1+1)} = -\frac{2}{2 \cdot 2 \cdot 2} = -\frac{1}{4}$$

4) គណនាលីមីតខាងក្រោម:

$$\begin{aligned} \text{ក/} \quad \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1} & \text{មានរាង} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 1} \frac{(x-1)}{(x-1)(\sqrt[3]{x^2} + \sqrt[3]{x} + 1)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1} = \frac{1}{1+1+1} = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{ខ/} \quad \lim_{x \rightarrow 0} \frac{\sqrt[n]{a+x} - \sqrt[n]{a}}{x} & \text{មានរាង} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{(a+x) - a}{x(\sqrt[n]{(a+x)^{n-1}} + \sqrt[n]{(a+x)^{n-2}} \times \sqrt[n]{a} + \dots + \sqrt[n]{a^{n-1}})} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt[n]{(a+x)^{n-1}} + \sqrt[n]{(a+x)^{n-2}} \times \sqrt[n]{a} + \dots + \sqrt[n]{a^{n-1}})} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt[n]{(a+x)^{n-1}} + \sqrt[n]{(a+x)^{n-2}} \times \sqrt[n]{a} + \dots + \sqrt[n]{a^{n-1}}} \\ &= \frac{1}{\sqrt[n]{a^{n-1}} + \sqrt[n]{a^{n-1}} + \dots + \sqrt[n]{a^{n-1}}} = \frac{1}{n\sqrt[n]{a^{n-1}}} \\ &= \frac{\sqrt[n]{a}}{n\sqrt[n]{a^{n-1}} \times \sqrt[n]{a}} = \frac{\sqrt[n]{a}}{n\sqrt[n]{a^n}} = \frac{\sqrt[n]{a}}{na} \end{aligned}$$

$$\begin{aligned} \text{គ/} \quad \lim_{x \rightarrow 1} \frac{1 - \sqrt[1987]{x}}{1 - x} & \text{មានរាង} \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 1} \frac{(1-x)}{(1-x)(1 + \sqrt[1987]{x} + \sqrt[1987]{x^2} + \dots + \sqrt[1987]{x^{1986}})} \\ &= \lim_{x \rightarrow 1} \frac{1}{(1 + \sqrt[1987]{x} + \sqrt[1987]{x^2} + \dots + \sqrt[1987]{x^{1986}})} \\ &= \frac{1}{1+1+1+\dots+1} = \frac{1}{1987} \end{aligned}$$

$$\text{ឃ/} \quad \lim_{x \rightarrow 0} \frac{\sqrt[5]{(x+1)^3} - 1}{x} \text{មានរាង} \left(\frac{0}{0} \right)$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{(\sqrt[5]{x+1})^3 - 1}{x} = \lim_{x \rightarrow 0} \frac{(\sqrt[5]{x+1} - 1)[\sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1]}{x} \\
&= \lim_{x \rightarrow 0} \frac{(x+1-1)[\sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1]}{x[\sqrt[5]{(x+1)^4} + \sqrt[5]{(x+1)^3} + \sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1]} \\
&= \lim_{x \rightarrow 0} \frac{x[\sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1]}{x[\sqrt[5]{(x+1)^4} + \sqrt[5]{(x+1)^3} + \sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1]} \\
&= \lim_{x \rightarrow 0} \frac{\sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1}{\sqrt[5]{(x+1)^4} + \sqrt[5]{(x+1)^3} + \sqrt[5]{(x+1)^2} + \sqrt[5]{x+1} + 1} \\
&= \frac{1+1+1}{1+1+1+1+1} = \frac{3}{5}
\end{aligned}$$

$$\begin{aligned}
&\text{ឯ/} \lim_{x \rightarrow 1} \frac{1 - \sqrt[3]{2-x}}{1-x} \text{ មានរាង } \left(\frac{0}{0} \right) \\
&= \lim_{x \rightarrow 1} \frac{1 - (2-x)}{(1-x)[1 + \sqrt[3]{2-x} + \sqrt[3]{(2-x)^2}]} \\
&= \lim_{x \rightarrow 1} \frac{(-1+x)}{(1-x)[1 + \sqrt[3]{2-x} + \sqrt[3]{(2-x)^2}]} \\
&= \lim_{x \rightarrow 1} \frac{-(1-x)}{(1-x)[1 + \sqrt[3]{2-x} + \sqrt[3]{(2-x)^2}]} \\
&= \lim_{x \rightarrow 1} \frac{-1}{1 + \sqrt[3]{2-x} + \sqrt[3]{(2-x)^2}} = \frac{-1}{1+1+1} = -\frac{1}{3}
\end{aligned}$$

$$\begin{aligned}
&\text{ច/} \lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - 1}{x(x+2)} \text{ មានរាង } \left(\frac{0}{0} \right) \\
&= \lim_{x \rightarrow 0} \frac{x+1-1}{x(x+2)[\sqrt[3]{(x+1)^2} + \sqrt[3]{x+1} + 1]} \\
&= \lim_{x \rightarrow 0} \frac{x}{x(x+2)[\sqrt[3]{(x+1)^2} + \sqrt[3]{x+1} + 1]}
\end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{1}{(x+2)[\sqrt[3]{(x+1)^2} + \sqrt[3]{x+1} + 1]} = \frac{1}{2(1+1+1)} = \frac{1}{6}$$

$$\text{ឆ/} \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{\sqrt[3]{x+6} - 2} \text{មានរាង} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - x - 2)[\sqrt[3]{(x+6)^2} + \sqrt[3]{x+6} \cdot 2 + 4]}{x + 6 - 8}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+1)[\sqrt[3]{(x+6)^2} + \sqrt[3]{x+6} \cdot 2 + 4]}{x - 2}$$

$$= \lim_{x \rightarrow 2} (x+1)[\sqrt[3]{(x+6)^2} + \sqrt[3]{x+6} \cdot 2 + 4]$$

$$= (2+1)(2^2 + 2 \cdot 2 + 4) = 3(12) = 36$$

$$\text{ជ/} \lim_{x \rightarrow 1} \frac{\sqrt[5]{x^2 + x - 1} - 1}{1 + \sqrt[3]{1 - 2x}} \text{មានរាង} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 1} \frac{(x^2 + x - 1 - 1)[1 - \sqrt[3]{1 - 2x} + \sqrt[3]{(1 - 2x)^2}]}{(1 + 1 - 2x)[\sqrt[5]{(x^2 + x - 1)^4} + \sqrt[5]{(x^2 + x - 1)^3} + \dots + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{(x^2 + x - 2)[1 - \sqrt[3]{1 - 2x} + \sqrt[3]{(1 - 2x)^2}]}{(2 - 2x)[\sqrt[5]{(x^2 + x - 1)^4} + \sqrt[5]{(x^2 + x - 1)^3} + \dots + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{(x+2)(x-1)[1 - \sqrt[3]{1 - 2x} + \sqrt[3]{(1 - 2x)^2}]}{-2(x-1)[\sqrt[5]{(x^2 + x - 1)^4} + \sqrt[5]{(x^2 + x - 1)^3} + \dots + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{(x+2)[1 - \sqrt[3]{1 - 2x} + \sqrt[3]{(1 - 2x)^2}]}{-2[\sqrt[5]{(x^2 + x - 1)^4} + \sqrt[5]{(x^2 + x - 1)^3} + \dots + 1]}$$

$$= \frac{(3)(1+1+1)}{-2(1+1+\dots+1)} = \frac{(3)(3)}{(-2)(5)} = -\frac{9}{10}$$

ខ. លីមីតរាងមិនកំណត់ $\left(\frac{\infty}{\infty}\right)$

វិធាន: ដើម្បីគណនាលីមីតរាងមិនកំណត់ $\frac{\infty}{\infty}$ គេត្រូវ៖

- ❶ ដាក់តួដែលមានដឺក្រេធំជាងគេនៅភាគយក និងភាគបែងជាកត្តារួមសិន
- ❷ សម្រួលកត្តារួមចោល
- ❸ គណនាលីមីតនៃកន្សោមថ្មី

លំហាត់

1) គណនាលីមីតត្រង់ $-\infty$ និង $+\infty$ នៃអនុគមន៍ខាងក្រោម៖

a). $f(x) = \frac{2x-7}{3x}$

b). $f(x) = \frac{2x+3}{x+1}$

c). $f(x) = \frac{2x+1}{x}$

d). $f(x) = \frac{3x-1}{2-x}$

e). $f(x) = \frac{7+x^2}{-3+x^2}$

f). $f(x) = \frac{8x+3}{-x^2-5}$

g). $f(x) = \frac{x^2-x+3}{x^2+x+1}$

h). $f(x) = \frac{2x-1}{x^2-3x+2}$

i). $f(x) = \frac{5x^2-2x+1}{x^2-4}$

j). $f(x) = \frac{x^2+3x-2}{x^2+x+2}$

k). $f(x) = \frac{5x^2+x-3}{x+2}$

l). $f(x) = \frac{x^2+4x+7}{x^2+x-2}$

m). $f(x) = \frac{2x^3-3x}{x^3-x^2}$

n). $f(x) = \frac{2x^2-4x+3}{(x-1)(x-2)}$

o). $f(x) = \frac{2x}{3+(x+1)^2}$

p). $f(x) = \frac{(x-2)^2}{-x^2+2x+3}$

q). $f(x) = \frac{3x^2-4x+4}{(x-1)^2}$

r). $f(x) = \frac{1-x^2+x^3}{(1-x)^2(1+x)}$

2) គណនាលីមីតខាងក្រោម៖

a). $\lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{x}}$

b). $\lim_{x \rightarrow +\infty} \frac{\sqrt{x} + 3x}{2\sqrt{x} - 4x}$

$$\text{c). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{\sqrt{x}}$$

$$\text{d). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x}}{\sqrt{2x+3}}$$

$$\text{e). } \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2x - 3} + 2x}{\sqrt{x^2 + 4} + x}$$

$$\text{f). } \lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x} + 1}{\sqrt[3]{x^2}}$$

$$\text{g). } \lim_{x \rightarrow \infty} \frac{\sqrt[4]{2+x^4}}{\sqrt[3]{5+27x^3}}$$

$$\text{h). } \lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2+1} - x + 2}{x+3}$$

$$\text{i). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{x+1}$$

$$\text{j). } \lim_{x \rightarrow +\infty} \frac{x - \sqrt{x^2 - 1}}{\sqrt{x^2 - 4}}$$

$$\text{k). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x} - \sqrt{x+1}}{x+1}$$

$$\text{l). } \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 3} + x}{x+2}$$

ជំនោះស្រាយ

1) គណនាលីមីតក្រុង $-\infty$ និង $+\infty$ នៃអនុគមន៍ខាងក្រោម:

$$\text{a). } f(x) = \frac{2x-7}{3x}$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{2x-7}{3x} = \lim_{x \rightarrow \pm\infty} \frac{x(2-7/x)}{3x} = \lim_{x \rightarrow \pm\infty} \frac{(2-7/x)}{3} \\ &= \frac{2-0}{3} = \frac{2}{3} \end{aligned}$$

$$\text{b). } f(x) = \frac{2x+3}{x+1}$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{2x+3}{x+1} = \lim_{x \rightarrow \pm\infty} \frac{x(2+3/x)}{x(1+1/x)} = \lim_{x \rightarrow \pm\infty} \frac{2+3/x}{1+1/x} \\ &= \frac{2+0}{1+0} = \frac{2}{1} = 2 \end{aligned}$$

$$\text{c). } f(x) = \frac{2x+1}{x}$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{2x+1}{x} = \lim_{x \rightarrow \pm\infty} \frac{x(2+1/x)}{x} = \lim_{x \rightarrow \pm\infty} (2+1/x) \\ &= 2+0 = 2 \end{aligned}$$

$$d). f(x) = \frac{3x-1}{2-x}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{3x-1}{2-x} = \lim_{x \rightarrow \pm\infty} \frac{x(3-1/x)}{x(2/x-1)} = \lim_{x \rightarrow \pm\infty} \frac{(3-1/x)}{(2/x-1)} \\ &= \frac{3-0}{0-1} = \frac{3}{-1} = -3\end{aligned}$$

$$e). f(x) = \frac{7+x^2}{-3+x^2}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{7+x^2}{-3+x^2} = \lim_{x \rightarrow \pm\infty} \frac{x^2(7/x^2+1)}{x^2(-3/x^2+1)} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(7/x^2+1)}{(-3/x^2+1)} = \frac{0+1}{0+1} = \frac{1}{1} = 1\end{aligned}$$

$$f). f(x) = \frac{8x+3}{-x^2-5}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{8x+3}{-x^2-5} = \lim_{x \rightarrow \pm\infty} \frac{x(8+3/x)}{x^2(-1-5/x^2)} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(8+3/x)}{x(-1-5/x^2)} = \frac{8+0}{(\pm\infty)(-1+0)} = \frac{8}{\pm\infty} = 0\end{aligned}$$

$$g). f(x) = \frac{x^2-x+3}{x^2+x+1}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{x^2-x+3}{x^2+x+1} = \lim_{x \rightarrow \pm\infty} \frac{x^2(1-\frac{1}{x}+\frac{3}{x^2})}{x^2(1+\frac{1}{x}+\frac{1}{x^2})} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(1-\frac{1}{x}+\frac{3}{x^2})}{(1+\frac{1}{x}+\frac{1}{x^2})} = \frac{1-0+0}{1+0+0} = \frac{1}{1} = 1\end{aligned}$$

$$h). f(x) = \frac{2x-1}{x^2-3x+2}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{2x-1}{x^2-3x+2} = \lim_{x \rightarrow \pm\infty} \frac{x(2-\frac{1}{x})}{x^2(1-\frac{3}{x}+\frac{2}{x^2})} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(2-\frac{1}{x})}{x(1-\frac{3}{x}+\frac{2}{x^2})} = \frac{2-0}{(\pm\infty)(1-0+0)} = \frac{2}{\pm\infty} = 0\end{aligned}$$

$$\text{i). } f(x) = \frac{5x^2 - 2x + 1}{x^2 - 4}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{5x^2 - 2x + 1}{x^2 - 4} = \lim_{x \rightarrow \pm\infty} \frac{x^2(5-\frac{2}{x}+\frac{1}{x^2})}{x^2(1-\frac{4}{x^2})} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(5-\frac{2}{x}+\frac{1}{x^2})}{(1-\frac{4}{x^2})} = \frac{5-0+0}{1-0} = \frac{5}{1} = 5\end{aligned}$$

$$\text{j). } f(x) = \frac{x^2 + 3x - 2}{x^2 + x + 2}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{x^2 + 3x - 2}{x^2 + x + 2} = \lim_{x \rightarrow \pm\infty} \frac{x^2(1+\frac{3}{x}-\frac{2}{x^2})}{x^2(1+\frac{1}{x}+\frac{2}{x^2})} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(1+\frac{3}{x}-\frac{2}{x^2})}{(1+\frac{1}{x}+\frac{2}{x^2})} = \frac{1+0-0}{1+0+0} = \frac{1}{1} = 1\end{aligned}$$

$$\text{k). } f(x) = \frac{5x^2 + x - 3}{x + 2}$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{5x^2 + x - 3}{x + 2} = \lim_{x \rightarrow \pm\infty} \frac{x^2(5+\frac{1}{x}-\frac{3}{x^2})}{x(1+\frac{2}{x})}$$

$$= \lim_{x \rightarrow \pm\infty} \frac{x(5 + \frac{1}{x} - \frac{3}{x^2})}{(1 + \frac{2}{x})} = \frac{(\pm\infty)(5 + 0 - 0)}{1 + 0} = \frac{\pm\infty}{1} = \pm\infty$$

$$l). f(x) = \frac{x^2 + 4x + 7}{x^2 + x - 2}$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x^2 + 4x + 7}{x^2 + x - 2} = \lim_{x \rightarrow \pm\infty} \frac{x^2(1 + \frac{4}{x} + \frac{7}{x^2})}{x^2(1 + \frac{1}{x} - \frac{2}{x^2})}$$

$$= \lim_{x \rightarrow \pm\infty} \frac{(1 + \frac{4}{x} + \frac{7}{x^2})}{(1 + \frac{1}{x} - \frac{2}{x^2})} = \frac{1 + 0 + 0}{1 + 0 - 0} = \frac{1}{1} = 1$$

$$m). f(x) = \frac{2x^3 - 3x}{x^3 - x^2}$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{2x^3 - 3x}{x^3 - x^2} = \lim_{x \rightarrow \pm\infty} \frac{x^3(2 - \frac{3}{x^2})}{x^3(1 - \frac{1}{x})}$$

$$= \lim_{x \rightarrow \pm\infty} \frac{(2 - \frac{3}{x^2})}{(1 - \frac{1}{x})} = \frac{2 - 0}{1 - 0} = \frac{2}{1} = 2$$

$$n). f(x) = \frac{2x^2 - 4x + 3}{(x-1)(x-2)}$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{2x^2 - 4x + 3}{(x-1)(x-2)} = \lim_{x \rightarrow \pm\infty} \frac{x^2(2 - \frac{4}{x} + \frac{3}{x^2})}{x^2(1 - \frac{1}{x})(1 - \frac{2}{x})}$$

$$= \lim_{x \rightarrow \pm\infty} \frac{(2 - \frac{4}{x} + \frac{3}{x^2})}{(1 - \frac{1}{x})(1 - \frac{2}{x})} = \frac{2 - 0 + 0}{(1 - 0)(1 - 0)} = \frac{2}{1} = 2$$

$$\text{o). } f(x) = \frac{2x}{3+(x+1)^2}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{2x}{3+(x+1)^2} = \lim_{x \rightarrow \pm\infty} \frac{2x}{x^2[\frac{3}{x^2} + (1+\frac{1}{x})^2]} \\ &= \lim_{x \rightarrow \pm\infty} \frac{2}{x[\frac{3}{x^2} + (1+\frac{1}{x})^2]} = \frac{2}{(\pm\infty)[0 + (1+0)^2]} = \frac{2}{\pm\infty} = 0\end{aligned}$$

$$\text{p). } f(x) = \frac{(x-2)^2}{-x^2+2x+3}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{(x-2)^2}{-x^2+2x+3} = \lim_{x \rightarrow \pm\infty} \frac{x^2(1-\frac{2}{x})^2}{x^2(-1+\frac{2}{x}+\frac{3}{x^2})} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(1-\frac{2}{x})^2}{(-1+\frac{2}{x}+\frac{3}{x^2})} = \frac{(1-0)^2}{(-1+0+0)} = \frac{1}{-1} = -1\end{aligned}$$

$$\text{q). } f(x) = \frac{3x^2-4x+4}{(x-1)^2}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{3x^2-4x+4}{(x-1)^2} = \lim_{x \rightarrow \pm\infty} \frac{x^2(3-\frac{4}{x}+\frac{4}{x^2})}{x^2(1-\frac{1}{x})^2} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(3-\frac{4}{x}+\frac{4}{x^2})}{(1-\frac{1}{x})^2} = \frac{3-0+0}{(1-0)^2} = \frac{3}{1} = 3\end{aligned}$$

$$\text{r). } f(x) = \frac{1-x^2+x^3}{(1-x)^2(1+x)}$$

$$\begin{aligned}\lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} \frac{1 - x^2 + x^3}{(1-x)^2(1+x)} = \lim_{x \rightarrow \pm\infty} \frac{x^3(\frac{1}{x^3} - \frac{1}{x} + 1)}{x^3(\frac{1}{x} - 1)^2(\frac{1}{x} + 1)} \\ &= \lim_{x \rightarrow \pm\infty} \frac{(\frac{1}{x^3} - \frac{1}{x} + 1)}{(\frac{1}{x} - 1)^2(\frac{1}{x} + 1)} = \frac{0 - 0 + 1}{(0 - 1)^2(0 + 1)} = \frac{1}{(1)(1)} = 1\end{aligned}$$

2) គណនាលីមីតខាងក្រោម:

$$\begin{aligned}\text{a). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x}}} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{x}}{\sqrt{x} \cdot \sqrt{1 + \frac{\sqrt{x}}{x}}} = \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{1 + \sqrt{\frac{1}{x}}}} = \frac{1}{\sqrt{1 + \sqrt{0}}} = \frac{1}{1} = 1\end{aligned}$$

$$\begin{aligned}\text{b). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + 3x}{2\sqrt{x} - 4x} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\ &= \lim_{x \rightarrow +\infty} \frac{x(\frac{\sqrt{x}}{x} + 3)}{x(\frac{2\sqrt{x}}{x} - 4)} = \lim_{x \rightarrow +\infty} \frac{\sqrt{\frac{1}{x}} + 3}{2\sqrt{\frac{1}{x}} - 4} = \frac{0 + 3}{0 - 4} = -\frac{3}{4}\end{aligned}$$

$$\begin{aligned}\text{c). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x + \sqrt{x + \sqrt{x}}}}{\sqrt{x}} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \cdot \sqrt{1 + \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x^3}}}}}{\sqrt{x}} = \lim_{x \rightarrow +\infty} \sqrt{1 + \sqrt{\frac{1}{x} + \sqrt{\frac{1}{x^3}}}} = 1\end{aligned}$$

$$\text{d). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x}}{\sqrt{2x + 3}} \text{ មានរាង } \left(\frac{\infty}{\infty}\right)$$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} \frac{\sqrt{x}(1 + \frac{\sqrt[3]{x}}{\sqrt{x}} + \frac{\sqrt[4]{x}}{\sqrt{x}})}{\sqrt{x} \cdot \sqrt{2 + \frac{3}{x}}} = \lim_{x \rightarrow +\infty} \frac{1 + \sqrt[6]{\frac{x^2}{x^3} + 4\sqrt{\frac{x}{x^2}}}}{\sqrt{2 + \frac{3}{x}}} \\
&= \lim_{x \rightarrow +\infty} \frac{1 + \sqrt[6]{\frac{1}{x} + 4\sqrt{\frac{1}{x}}}}{\sqrt{2 + \frac{3}{x}}} = \frac{1 + 0 + 0}{\sqrt{2 + 0}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}
\end{aligned}$$

e). $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2x - 3} + 2x}{\sqrt{x^2 + 4} + x}$ មានរាង $(\frac{\infty}{\infty})$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{\sqrt{x^2(1 + \frac{4}{x^2})} + x} = \lim_{x \rightarrow \infty} \frac{|x| \sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{|x| \sqrt{(1 + \frac{4}{x^2})} + x}$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

គេបាន $\lim_{x \rightarrow +\infty} \frac{|x| \sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{|x| \sqrt{(1 + \frac{4}{x^2})} + x} = \lim_{x \rightarrow +\infty} \frac{x \sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{x \sqrt{(1 + \frac{4}{x^2})} + x}$

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2}{\sqrt{(1 + \frac{4}{x^2})} + 1} = \frac{\sqrt{1 + 0 - 0} + 2}{\sqrt{1 + 0} + 1} = \frac{1 + 2}{1 + 1} = \frac{3}{2}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

គេបាន $\lim_{x \rightarrow -\infty} \frac{|x| \sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{|x| \sqrt{(1 + \frac{4}{x^2})} + x} = \lim_{x \rightarrow -\infty} \frac{-x \sqrt{(1 + \frac{2}{x} - \frac{3}{x^2})} + 2x}{-x \sqrt{(1 + \frac{4}{x^2})} + x}$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 + \frac{2}{x} - \frac{3}{x^2}} + 2}{-\sqrt{1 + \frac{4}{x^2}} + 1} = \frac{-\sqrt{1+0-0} + 2}{-\sqrt{1+0} + 1} = \frac{-1+2}{-1+1} = \frac{1}{0} = \pm\infty$$

f). $\lim_{x \rightarrow +\infty} \frac{\sqrt[4]{x} + 1}{\sqrt[3]{x^2}}$ មានរាង $(\frac{\infty}{\infty})$

$$= \lim_{x \rightarrow +\infty} \left(\frac{\sqrt[4]{x}}{\sqrt[3]{x^2}} + \frac{1}{\sqrt[3]{x^2}} \right) = \lim_{x \rightarrow +\infty} \left(12 \frac{\sqrt[3]{x^3}}{\sqrt[3]{x^8}} + \frac{1}{\sqrt[3]{x^2}} \right)$$

$$= \lim_{x \rightarrow +\infty} \left(12 \frac{\sqrt[3]{1}}{\sqrt[3]{x^5}} + \frac{1}{\sqrt[3]{x^2}} \right) = 0 + 0 = 0$$

g). $\lim_{x \rightarrow \infty} \frac{\sqrt[4]{2+x^4}}{\sqrt[3]{5+27x^3}}$ មានរាង $(\frac{\infty}{\infty})$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt[4]{x^4(\frac{2}{x^4} + 1)}}{\sqrt[3]{x^3(\frac{5}{x^3} + 27)}} = \lim_{x \rightarrow \infty} \frac{|x| \sqrt[4]{(\frac{2}{x^4} + 1)}}{x \sqrt[3]{(\frac{5}{x^3} + 27)}}$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

គេបាន $\lim_{x \rightarrow +\infty} \frac{|x| \sqrt[4]{(\frac{2}{x^4} + 1)}}{x \sqrt[3]{(\frac{5}{x^3} + 27)}} = \lim_{x \rightarrow +\infty} \frac{x \sqrt[4]{(\frac{2}{x^4} + 1)}}{x \sqrt[3]{(\frac{5}{x^3} + 27)}}$

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt[4]{(\frac{2}{x^4} + 1)}}{\sqrt[3]{(\frac{5}{x^3} + 27)}} = \frac{\sqrt[4]{0+1}}{\sqrt[3]{0+27}} = \frac{\sqrt[4]{1}}{\sqrt[3]{27}} = \frac{1}{3}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow -\infty} \frac{|x| \sqrt[4]{\left(\frac{2}{x^4} + 1\right)}}{x^3 \sqrt[3]{\left(\frac{5}{x^3} + 27\right)}} &= \lim_{x \rightarrow -\infty} \frac{-x \sqrt[4]{\left(\frac{2}{x^4} + 1\right)}}{x^3 \sqrt[3]{\left(\frac{5}{x^3} + 27\right)}} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt[4]{\left(\frac{2}{x^4} + 1\right)}}{\sqrt[3]{\left(\frac{5}{x^3} + 27\right)}} = \frac{-\sqrt[4]{0+1}}{\sqrt[3]{0+27}} = \frac{-\sqrt[4]{1}}{\sqrt[3]{27}} = -\frac{1}{3} \end{aligned}$$

$$\text{h). } \lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2+1}-x+2}{x+3} \text{ មានរាង } \left(\frac{\infty}{\infty}\right)$$

$$\text{គេមាន } \frac{\sqrt{x^2+1}-x+2}{x+3} = \frac{|x| \sqrt{1+\frac{1}{x^2}}-x+2}{x+3}$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+1}-x+2}{x+3} &= \lim_{x \rightarrow +\infty} \frac{x \sqrt{1+\frac{1}{x^2}}-x+2}{x+3} \\ &= \lim_{x \rightarrow +\infty} \frac{\sqrt{1+\frac{1}{x^2}}-1+\frac{2}{x}}{1+\frac{3}{x}} = \frac{\sqrt{1+0}-1+0}{1+0} = \frac{0}{1} = 0 \end{aligned}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+1}-x+2}{x+3} &= \lim_{x \rightarrow -\infty} \frac{-x \sqrt{1+\frac{1}{x^2}}-x+2}{x+3} \\ &= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1+\frac{1}{x^2}}-1+\frac{2}{x}}{1+\frac{3}{x}} = \frac{-\sqrt{1+0}-1+0}{1+0} = \frac{-2}{1} = -2 \end{aligned}$$

$$\text{i). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x+\sqrt{x+\sqrt{x}}}}{x+1} \text{ មានរាង } \left(\frac{\infty}{\infty}\right)$$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \cdot \sqrt{1 + \sqrt{\frac{1}{x}} + \sqrt{\frac{1}{x^3}}}}{x(1 + \frac{1}{x})} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \cdot \sqrt{1 + \sqrt{\frac{1}{x}} + \sqrt{\frac{1}{x^3}}}}{\sqrt{x} \cdot \sqrt{x}(1 + \frac{1}{x})} \\
&= \lim_{x \rightarrow +\infty} \frac{\sqrt{1 + \sqrt{\frac{1}{x}} + \sqrt{\frac{1}{x^3}}}}{\sqrt{x}(1 + \frac{1}{x})} = \frac{\sqrt{1 + \sqrt{0} + \sqrt{0}}}{(+\infty)(1 + 0)} = \frac{1}{+\infty} = 0
\end{aligned}$$

$$\begin{aligned}
\text{j). } \lim_{x \rightarrow +\infty} \frac{x - \sqrt{x^2 - 1}}{\sqrt{x^2 - 4}} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\
&= \lim_{x \rightarrow +\infty} \frac{x - x\sqrt{1 - \frac{1}{x^2}}}{x\sqrt{1 - \frac{4}{x^2}}} = \lim_{x \rightarrow +\infty} \frac{1 - \sqrt{1 - \frac{1}{x^2}}}{\sqrt{1 - \frac{4}{x^2}}} = \frac{1 - \sqrt{1 - 0}}{\sqrt{1 - 0}} = 0
\end{aligned}$$

$$\begin{aligned}
\text{k). } \lim_{x \rightarrow +\infty} \frac{\sqrt{x} - \sqrt{x+1}}{x+1} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\
&= \lim_{x \rightarrow +\infty} \frac{x - (x+1)}{(x+1)(\sqrt{x} + \sqrt{x+1})} = \lim_{x \rightarrow +\infty} \frac{-1}{(x+1)(\sqrt{x} + \sqrt{x+1})} = 0
\end{aligned}$$

$$\begin{aligned}
\text{l). } \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 3} + x}{x+2} &\text{ មានរាង } \left(\frac{\infty}{\infty}\right) \\
&= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2(1 + \frac{3}{x^2})} + x}{x+2} = \lim_{x \rightarrow -\infty} \frac{|x|\sqrt{1 + \frac{3}{x^2}} + x}{x+2} \\
&= \lim_{x \rightarrow -\infty} \frac{-x\sqrt{1 + \frac{3}{x^2}} + x}{x+2} = \lim_{x \rightarrow -\infty} \frac{x(-\sqrt{1 + \frac{3}{x^2}} + 1)}{x(1 + \frac{2}{x})} \\
&= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 + (3/x^2)} + 1}{1 + (2/x)} = \frac{-\sqrt{1 + 0} + 1}{1 + 0} = \frac{-1 + 1}{1} = \frac{0}{1} = 0
\end{aligned}$$

គ. លីមីតរាងមិនកំណត់ ($+\infty - \infty$)

វិធាន: ដើម្បីគណនាលីមីតរាងមិនកំណត់ ($+\infty - \infty$) គេត្រូវ៖

❖ **ចំពោះអនុគមន៍ទូទៅ:**

- ❶ ដាក់តួដែលមានដឺក្រេខ្ពស់ជាងគេជាកត្តារួម
- ❷ គណនាលីមីតនៃកន្សោមថ្មី

❖ **ចំពោះអនុគមន៍អសនិទាន:**

- ❶ គេត្រូវគុណនិងចែកកន្សោមនោះជាមួយកន្សោមផ្លាស់របស់វា
- ❷ គណនាលីមីតនៃកន្សោមថ្មី

លំហាត់

1) គណនាលីមីតក្រុង $-\infty$ និង $+\infty$ នៃអនុគមន៍ខាងក្រោម:

- | | |
|----------------------------------|---------------------------------|
| a). $f(x) = -3x^2 + 5x + 2$ | b). $f(x) = x^3 - 5x^2 + 1$ |
| c). $f(x) = -2 + x - x^5$ | d). $f(x) = x^4 - 2x^3 + x - 2$ |
| e). $f(x) = -x^3 + x^2 - 4x + 5$ | f). $f(x) = 1 - 3x + 3x^4$ |

2) គណនាលីមីតខាងក្រោម:

- | | |
|--|---|
| a). $\lim_{x \rightarrow +\infty} (1 + x - x^5)$ | b). $\lim_{x \rightarrow -\infty} (1 - 3x - x^3)$ |
| c). $\lim_{x \rightarrow +\infty} (1 + x^3 - x^4)$ | d). $\lim_{x \rightarrow -\infty} (3x^3 - x^2 + 9)$ |

3) គណនាលីមីតខាងក្រោម:

- | | |
|--|---|
| a). $\lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x^2+3})$ | b). $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - x^2)$ |
| c). $\lim_{x \rightarrow \pm\infty} (x\sqrt{x^2+1} - x)$ | d). $\lim_{x \rightarrow +\infty} (\sqrt{2x^2+1} - \sqrt{x+x^2})$ |
| e). $\lim_{x \rightarrow -\infty} (\sqrt{2x^2+x} + 3x - 1)$ | f). $\lim_{x \rightarrow +\infty} (\sqrt{x^2+x} - \sqrt[3]{x^2-x})$ |

4) គណនាលីមីតខាងក្រោម:

- | | |
|--|--|
| a). $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - x)$ | b). $\lim_{x \rightarrow +\infty} (\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x})$ |
|--|--|

$$c). \lim_{x \rightarrow \infty} (\sqrt{x^2 - 1} - \sqrt{x^2 + 3x})$$

$$d). \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 2x} - \sqrt{x(x - 4)})$$

$$e). \lim_{x \rightarrow \infty} (\sqrt[4]{x^4 + 4} - x)$$

$$f). \lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + 3x^2} - \sqrt{x^2 - 2x})$$

ជំនោះស្រាយ

1) គណនាលីមីតត្រង់ $-\infty$ និង $+\infty$ នៃអនុគមន៍ខាងក្រោម៖

$$a). f(x) = -3x^2 + 5x + 2$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} (-3x^2 + 5x + 2) = \lim_{x \rightarrow \pm\infty} x^2 \left(-3 + \frac{5}{x} + \frac{2}{x^2}\right) \\ &= (+\infty)(-3 + 0 + 0) = (+\infty)(-3) = -\infty \end{aligned}$$

$$b). f(x) = x^3 - 5x^2 + 1$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} (x^3 - 5x^2 + 1) = \lim_{x \rightarrow \pm\infty} x^3 \left(1 - \frac{5}{x} + \frac{1}{x^2}\right) \\ &= (\pm\infty)(1 - 0 + 0) = (\pm\infty)(1) = \pm\infty \end{aligned}$$

$$c). f(x) = -2 + x - x^5$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} (-2 + x - x^5) = \lim_{x \rightarrow \pm\infty} x^5 \left(-\frac{2}{x^5} + \frac{1}{x^4} - 1\right) \\ &= (\pm\infty)(-0 + 0 - 1) = (\pm\infty)(-1) = \pm\infty \end{aligned}$$

$$d). f(x) = x^4 - 2x^3 + x - 2$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} (x^4 - 2x^3 + x - 2) = \lim_{x \rightarrow \pm\infty} x^4 \left(1 - \frac{2}{x} + \frac{1}{x^3} - \frac{2}{x^4}\right) \\ &= (+\infty)(1 - 0 + 0 - 0) = (+\infty)(1) = +\infty \end{aligned}$$

$$e). f(x) = -x^3 + x^2 - 4x + 5$$

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} f(x) &= \lim_{x \rightarrow \pm\infty} (-x^3 + x^2 - 4x + 5) \\ &= \lim_{x \rightarrow \pm\infty} x^3 \left(-1 + \frac{1}{x} - \frac{4}{x^2} + \frac{5}{x^3}\right) = (\pm\infty)(-1 + 0 - 0 + 0) = \pm\infty \end{aligned}$$

$$f). f(x) = 1 - 3x + 3x^4$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} (1 - 3x + 3x^4) = \lim_{x \rightarrow \pm\infty} x^4 \left(\frac{1}{x^4} - \frac{3}{x^3} + 3\right)$$

$$= (+\infty)(0 - 0 + 3) = (+\infty)(3) = +\infty$$

2) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow +\infty} (1 + x - x^5)$$

$$= \lim_{x \rightarrow +\infty} x^5 \left(\frac{1}{x^5} + \frac{1}{x^4} - 1 \right) = (+\infty)(0 + 0 - 1) = (+\infty)(-1) = -\infty$$

$$\text{b). } \lim_{x \rightarrow -\infty} (1 - 3x - x^3)$$

$$= \lim_{x \rightarrow -\infty} x^3 \left(\frac{1}{x^3} - \frac{3}{x^2} - 1 \right) = (-\infty)(0 - 0 - 1) = (-\infty)(-1) = +\infty$$

$$\text{c). } \lim_{x \rightarrow +\infty} (1 + x^3 - x^4)$$

$$= \lim_{x \rightarrow +\infty} x^4 \left(\frac{1}{x^4} + \frac{1}{x} - 1 \right) = (+\infty)(0 + 0 - 1) = -\infty$$

$$\text{d). } \lim_{x \rightarrow -\infty} (3x^3 - x^2 + 9)$$

$$= \lim_{x \rightarrow -\infty} x^3 \left(3 - \frac{1}{x} + \frac{9}{x^3} \right) = (-\infty)(3 - 0 + 0) = (-\infty)(3) = -\infty$$

3) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow +\infty} (\sqrt{x+1} - \sqrt{x^2+3})$$

$$= \lim_{x \rightarrow +\infty} \left(\sqrt{x+1} - x \sqrt{1 + \frac{3}{x^2}} \right) = \lim_{x \rightarrow +\infty} x \left(\frac{\sqrt{x+1}}{x} - \sqrt{1 + \frac{3}{x^2}} \right)$$

$$= \lim_{x \rightarrow +\infty} x \left(\sqrt{\frac{1}{x} + \frac{1}{x^2}} - \sqrt{1 + \frac{3}{x^2}} \right) = +\infty(0 - 1) = -\infty$$

$$\text{b). } \lim_{x \rightarrow -\infty} (\sqrt{x^2+x} - x^2) = \lim_{x \rightarrow -\infty} \left(|x| \sqrt{1 + \frac{1}{x}} - x^2 \right)$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\text{គេបាន } \lim_{x \rightarrow -\infty} \left(|x| \sqrt{1 + \frac{1}{x}} - x^2 \right) = \lim_{x \rightarrow -\infty} \left(-x \sqrt{1 + \frac{1}{x}} - x^2 \right)$$

$$= \lim_{x \rightarrow -\infty} x^2 \left(-\frac{1}{x} \sqrt{1 + \frac{1}{x}} - 1 \right) = (+\infty)(0 - 1) = -\infty$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{គេបាន } \lim_{x \rightarrow +\infty} \left(|x| \sqrt{1 + \frac{1}{x}} - x^2 \right) = \lim_{x \rightarrow +\infty} \left(x \sqrt{1 + \frac{1}{x}} - x^2 \right)$$

$$= \lim_{x \rightarrow +\infty} x^2 \left(\frac{1}{x} \sqrt{1 + \frac{1}{x}} - 1 \right) = (+\infty)(0 - 1) = -\infty$$

$$\text{c). } \lim_{x \rightarrow \pm\infty} (x\sqrt{x^2 + 1} - x) = \lim_{x \rightarrow \pm\infty} \left(x|x| \sqrt{1 + \frac{1}{x^2}} - x \right)$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{គេបាន } \lim_{x \rightarrow +\infty} \left(x|x| \sqrt{1 + \frac{1}{x^2}} - x \right) = \lim_{x \rightarrow +\infty} \left(x \cdot x \sqrt{1 + \frac{1}{x^2}} - x \right)$$

$$= \lim_{x \rightarrow +\infty} x^2 \left(\sqrt{1 + \frac{1}{x^2}} - \frac{1}{x} \right) = (+\infty)(1 - 0) = +\infty$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\text{គេបាន } \lim_{x \rightarrow -\infty} \left(x|x| \sqrt{1 + \frac{1}{x^2}} - x \right) = \lim_{x \rightarrow -\infty} \left(-x^2 \sqrt{1 + \frac{1}{x^2}} - x \right)$$

$$= \lim_{x \rightarrow -\infty} x^2 \left(-\sqrt{1 + \frac{1}{x^2}} - \frac{1}{x} \right) = (+\infty)(-1 - 0) = -\infty$$

$$\text{d). } \lim_{x \rightarrow +\infty} (\sqrt{2x^2 + 1} - \sqrt{x + x^2}) = \lim_{x \rightarrow +\infty} \left(x\sqrt{2 + \frac{1}{x^2}} - x\sqrt{\frac{1}{x} + 1} \right)$$

$$= \lim_{x \rightarrow +\infty} x \left(\sqrt{2 + \frac{1}{x^2}} - \sqrt{\frac{1}{x} + 1} \right) = (+\infty)(\sqrt{2} - 1) = +\infty$$

$$\text{e). } \lim_{x \rightarrow -\infty} (\sqrt{2x^2 + x + 3x - 1}) = \lim_{x \rightarrow -\infty} (|x| \sqrt{2 + \frac{1}{x}} + 3x - 1)$$

$$= \lim_{x \rightarrow -\infty} \left(-x\sqrt{2 + \frac{1}{x}} + 3x - 1 \right) = \lim_{x \rightarrow -\infty} x \left(-\sqrt{2 + \frac{1}{x}} + 3 - \frac{1}{x} \right)$$

$$= (-\infty)(-\sqrt{2} + 3 - 0) = (-\infty)(-\sqrt{2} + 3) = -\infty$$

$$\begin{aligned}
 \text{f). } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + x} - \sqrt[3]{x^2 - x}) &= \lim_{x \rightarrow +\infty} x \left(\frac{\sqrt{x^2 + x}}{x} - \frac{\sqrt[3]{x^2 - x}}{x} \right) \\
 &= \lim_{x \rightarrow +\infty} x \left(\sqrt{\frac{x^2 + x}{x^2}} - \sqrt[3]{\frac{x^2 - x}{x^3}} \right) = \lim_{x \rightarrow +\infty} x \left(\sqrt{1 + \frac{1}{x}} - \sqrt[3]{\frac{1}{x} - \frac{1}{x^2}} \right) \\
 &= (+\infty)(1 - 0) = +\infty
 \end{aligned}$$

4) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow \infty} \frac{(x^2 + x) - x^2}{(\sqrt{x^2 + x} + x)} = \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x} + x} = \lim_{x \rightarrow \infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x}} + x}$$

❖ បើ $x \rightarrow +\infty$ នោះ $|x| = x$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow +\infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x}} + x} &= \lim_{x \rightarrow +\infty} \frac{x}{x \sqrt{1 + \frac{1}{x}} + x} \\
 &= \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{1 + 1} = \frac{1}{2}
 \end{aligned}$$

❖ បើ $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow -\infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x}} + x} &= \lim_{x \rightarrow -\infty} \frac{x}{-x \sqrt{1 + \frac{1}{x}} + x} \\
 &= \lim_{x \rightarrow -\infty} \frac{1}{-\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{0^+} = +\infty
 \end{aligned}$$

b). $\lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x})$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow +\infty} \frac{x + \sqrt{x + \sqrt{x}} - x}{\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x}} = \lim_{x \rightarrow +\infty} \frac{\sqrt{x + \sqrt{x}}}{\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x}}$$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \left(\sqrt{1 + \sqrt{\frac{1}{x}}} \right)}{\sqrt{x} \left(\sqrt{1 + \sqrt{\frac{1}{x}} + \sqrt{\frac{1}{x^3}}} + 1 \right)} = \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{1 + \sqrt{\frac{1}{x}}} \right)}{\left(\sqrt{1 + \sqrt{\frac{1}{x}} + \sqrt{\frac{1}{x^3}}} + 1 \right)} \\
&= \frac{\sqrt{1 + \sqrt{0}}}{\sqrt{1 + \sqrt{0} + \sqrt{0}} + 1} = \frac{1}{1 + 1} = \frac{1}{2}
\end{aligned}$$

c). $\lim_{x \rightarrow \infty} (\sqrt{x^2 - 1} - \sqrt{x^2 + 3x})$ មានរាង $(\infty - \infty)$

$$\begin{aligned}
&= \lim_{x \rightarrow \infty} \frac{(x^2 - 1) - (x^2 + 3x)}{\sqrt{x^2 - 1} + \sqrt{x^2 + 3x}} = \lim_{x \rightarrow \infty} \frac{-1 - 3x}{|x| \sqrt{1 - \frac{1}{x^2}} + |x| \sqrt{1 + \frac{3}{x}}} \\
&= \lim_{x \rightarrow \infty} \frac{x(-\frac{1}{x} - 3)}{|x| \left(\sqrt{1 - \frac{1}{x^2}} + \sqrt{1 + \frac{3}{x}} \right)}
\end{aligned}$$

❖ បើ $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\text{គេបាន } \lim_{x \rightarrow -\infty} \frac{x(-\frac{1}{x} - 3)}{|x| \left(\sqrt{1 - \frac{1}{x^2}} + \sqrt{1 + \frac{3}{x}} \right)}$$

$$\begin{aligned}
&= \lim_{x \rightarrow -\infty} \frac{x(-\frac{1}{x} - 3)}{-x \left(\sqrt{1 - \frac{1}{x^2}} + \sqrt{1 + \frac{3}{x}} \right)} = \lim_{x \rightarrow -\infty} \frac{(-\frac{1}{x} - 3)}{- \left(\sqrt{1 - \frac{1}{x^2}} + \sqrt{1 + \frac{3}{x}} \right)} \\
&= \frac{-0 - 3}{-(1 + 1)} = \frac{-3}{-2} = \frac{3}{2}
\end{aligned}$$

❖ បើ $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{គេបាន } \lim_{x \rightarrow +\infty} \frac{x(-\frac{1}{x}-3)}{|x| \left(\sqrt{1-\frac{1}{x^2}} + \sqrt{1+\frac{3}{x}} \right)}$$

$$= \lim_{x \rightarrow +\infty} \frac{x(-\frac{1}{x}-3)}{x \left(\sqrt{1-\frac{1}{x^2}} + \sqrt{1+\frac{3}{x}} \right)} = \lim_{x \rightarrow +\infty} \frac{(-\frac{1}{x}-3)}{\left(\sqrt{1-\frac{1}{x^2}} + \sqrt{1+\frac{3}{x}} \right)}$$

$$= \frac{-0-3}{(1+1)} = \frac{-3}{2} = -\frac{3}{2}$$

d). $\lim_{x \rightarrow +\infty} (\sqrt{x^2+2x} - \sqrt{x(x-4)})$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow +\infty} \frac{(x^2+2x) - x(x-4)}{\sqrt{x^2+2x} + \sqrt{x(x-4)}} = \lim_{x \rightarrow +\infty} \frac{2x+4x}{x\sqrt{1+\frac{2}{x}} + x\sqrt{1-\frac{4}{x}}}$$

$$= \lim_{x \rightarrow +\infty} \frac{6x}{x \left(\sqrt{1+\frac{2}{x}} + \sqrt{1-\frac{4}{x}} \right)} = \lim_{x \rightarrow +\infty} \frac{6}{\sqrt{1+\frac{2}{x}} + \sqrt{1-\frac{4}{x}}}$$

$$= \frac{6}{\sqrt{1+0} + \sqrt{1-0}} = \frac{6}{1+1} = \frac{6}{2} = 3$$

e). $\lim_{x \rightarrow \infty} (\sqrt[4]{x^4+4} - x)$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow \infty} \frac{x^4+4-x^4}{\sqrt[4]{(x^4+4)^3} + \sqrt[4]{(x^4+4)^2} \cdot x + \sqrt[4]{(x^4+4)} \cdot x^2 + x^3}$$

$$= \lim_{x \rightarrow \infty} \frac{4}{\sqrt[4]{(x^4+4)^3} + \sqrt[4]{(x^4+4)^2} \cdot x + \sqrt[4]{(x^4+4)} \cdot x^2 + x^3} = 0$$

f). $\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3+3x^2} - \sqrt{x^2-2x})$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow +\infty} [(\sqrt[3]{x^3+3x^2} - x) - (\sqrt{x^2-2x} - x)]$$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} \left[\frac{x^3 + 3x^2 - x^3}{\sqrt[3]{(x^3 + 3x^2)^2} + \sqrt[3]{x^3 + 3x^2} \cdot x + x^2} - \frac{x^2 - 2x - x^2}{\sqrt{x^2 - 2x + x}} \right] \\
&= \lim_{x \rightarrow +\infty} \left[\frac{3x^2}{\sqrt[3]{(x^3 + 3x^2)^2} + \sqrt[3]{x^3 + 3x^2} \cdot x + x^2} - \frac{-2x}{\sqrt{x^2 - 2x + x}} \right] \\
&= \lim_{x \rightarrow +\infty} \left[\frac{3x^2}{x^2 \left(\sqrt[3]{\left(1 + \frac{3}{x}\right)^2} + \sqrt[3]{1 + \frac{3}{x}} + 1 \right)} + \frac{2x}{x \left(\sqrt{1 - \frac{2}{x}} + 1 \right)} \right] \\
&= \lim_{x \rightarrow +\infty} \left[\frac{3}{\sqrt[3]{\left(1 + \frac{3}{x}\right)^2} + \sqrt[3]{1 + \frac{3}{x}} + 1} + \frac{2}{\sqrt{1 - \frac{2}{x}} + 1} \right] \\
&= \frac{3}{1+1+1} + \frac{2}{1+1} = \frac{3}{3} + \frac{2}{2} = 1+1=2
\end{aligned}$$

4. លីមីតនៃអនុគមន៍ត្រីកោណមាត្រ

ក. ទ្រឹស្តីបទ

លីមីតនៃអនុគមន៍ត្រីកោណមាត្រ៖

- ❶ $\lim_{x \rightarrow a} \sin x = \sin a$
- ❷ $\lim_{x \rightarrow a} \cos x = \cos a$
- ❸ $\lim_{x \rightarrow a} \tan x = \tan a, (a \neq \pi/2 + 2k\pi, k \in \mathbb{Z})$
- ❹ $\lim_{x \rightarrow a} \cot x = \cot a, (a \neq k\pi, k \in \mathbb{Z})$

ឧទាហរណ៍

គណនាលីមីតខាងក្រោម៖

a). $\lim_{x \rightarrow \frac{\pi}{6}} (\sin x - 1)^2$

b). $\lim_{x \rightarrow \pi} \tan x$

c). $\lim_{x \rightarrow 7\pi} \cos \frac{x}{3}$

d). $\lim_{x \rightarrow 13\pi/4} (\cot x - \tan x)$

$$e). \lim_{x \rightarrow 0} \frac{\sin x + \cos^2 x}{x^2 + 1}$$

$$f). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \tan x}{\sin^2 x}$$

$$g). \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan x - \sin x}{\cos x - 1}$$

$$h). \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin^2 x + \cot^2 x}{2 \sin x}$$

$$i). \lim_{x \rightarrow 0} \frac{\cos x + \tan x}{\tan^2 x + 3}$$

$$j). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x + \sin x}{\cos^2 2x}$$

ដំណោះស្រាយ

$$a). \lim_{x \rightarrow \frac{\pi}{6}} (\sin x - 1)^2 = (\sin \frac{\pi}{6} - 1)^2 = (\frac{1}{2} - 1)^2 = (-\frac{1}{2})^2 = \frac{1}{4}$$

$$b). \lim_{x \rightarrow \pi} \tan x = \tan \pi = 0$$

$$c). \lim_{x \rightarrow 7\pi} \cos \frac{x}{3} = \cos \frac{7\pi}{3} = \cos(2\pi + \frac{\pi}{3}) = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$d). \lim_{x \rightarrow 13\pi/4} (\cot x - \tan x) = \cot \frac{13\pi}{4} - \tan \frac{13\pi}{4} \\ = \cot(3\pi + \frac{\pi}{4}) - \tan(3\pi + \frac{\pi}{4}) = \cot \frac{\pi}{4} - \tan \frac{\pi}{4} = 1 - 1 = 0$$

$$e). \lim_{x \rightarrow 0} \frac{\sin x + \cos^2 x}{x^2 + 1} = \frac{\sin 0 + \cos^2 0}{0^2 + 1} = \frac{1}{1} = 1$$

$$f). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \tan x}{\sin^2 x} = \frac{\cos \frac{\pi}{4} - \tan \frac{\pi}{4}}{\sin^2 \frac{\pi}{4}} = \frac{\frac{\sqrt{2}}{2} - 1}{(\frac{\sqrt{2}}{2})^2} = \frac{\frac{\sqrt{2}-2}{2}}{\frac{2}{4}} = \sqrt{2} - 2$$

$$g). \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan x - \sin x}{\cos x - 1} = \frac{\tan \frac{\pi}{3} - \sin \frac{\pi}{3}}{\cos \frac{\pi}{3} - 1} = \frac{\sqrt{3} - \frac{\sqrt{3}}{2}}{\frac{1}{2} - 1} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$$

$$h). \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin^2 x + \cot^2 x}{2 \sin x} = \frac{(1/2)^2 + (\sqrt{3})^2}{2 \cdot \frac{1}{2}} = \frac{(1/4) + 3}{1} = \frac{13}{4}$$

$$i). \lim_{x \rightarrow 0} \frac{\cos x + \tan x}{\tan^2 x + 3} = \frac{\cos 0 + \tan 0}{\tan^2 0 + 3} = \frac{1 + 0}{0 + 3} = \frac{1}{3}$$

$$j). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x + \sin x}{\cos^2 2x} = \frac{\cos \frac{\pi}{4} + \sin \frac{\pi}{4}}{\cos^2 2 \cdot \frac{\pi}{4}} = \frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}}{\cos^2 \frac{\pi}{2}} = \frac{\sqrt{2}}{0^+} = +\infty$$

ខ. ត្រីស្វីបទ

បើ x ជារង្វាស់មុំ ឬឆ្លុះគិតជាដឺក្រេនោះគេបាន៖

$$① \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$② \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

❖ ចំណាំ

$$① \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$② \lim_{x \rightarrow 0} \frac{\sin ax}{ax} = \lim_{x \rightarrow 0} \frac{ax}{\sin ax} = 1$$

$$③ \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$④ \lim_{x \rightarrow 0} \frac{\tan ax}{ax} = \lim_{x \rightarrow 0} \frac{ax}{\tan ax} = 1$$

$$⑤ \lim_{x \rightarrow 0} \frac{\sin^n x}{x^n} = \lim_{x \rightarrow 0} \frac{x^n}{\sin^n x} = 1$$

$$⑥ \lim_{x \rightarrow 0} \frac{\sin^n ax}{(ax)^n} = \lim_{x \rightarrow 0} \frac{(ax)^n}{\sin^n ax} = 1$$

$$⑦ \lim_{x \rightarrow 0} \frac{\tan^n x}{x^n} = \lim_{x \rightarrow 0} \frac{x^n}{\tan^n x} = 1$$

$$⑧ \lim_{x \rightarrow 0} \frac{\tan^n ax}{(ax)^n} = \lim_{x \rightarrow 0} \frac{(ax)^n}{\tan^n ax} = 1$$

លំហាត់

1) គណនាលីមីតខាងក្រោម៖

$$a). \lim_{x \rightarrow 0} \frac{2 \sin x}{3x}$$

$$b). \lim_{x \rightarrow 0} \frac{1 - \cos x}{2x}$$

$$c). \lim_{x \rightarrow 0} \frac{2 \tan x}{x}$$

$$d). \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

$$e). \lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 3x}$$

$$f). \lim_{x \rightarrow 0} \frac{\tan mx}{\sin nx}$$

$$g). \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$$

$$h). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{\cos 2x}$$

$$i). \lim_{x \rightarrow 0} \frac{\sin 3x}{x}$$

$$j). \lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$$

$$k). \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 5x}$$

$$l). \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

2) គណនាលីមីតខាងក្រោម៖

$$\text{a). } \lim_{x \rightarrow 0} \frac{\cos x - \cos^2 x}{x}$$

$$\text{c). } \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$$

$$\text{e). } \lim_{x \rightarrow 0} \frac{\sin 2x - 3x}{\sin 3x - 2x}$$

$$\text{g). } \lim_{x \rightarrow 0} \frac{x + \sin x}{2x - 3\sin 2x}$$

$$\text{i). } \lim_{x \rightarrow 0} \frac{1 - \cos 5x}{x^2}$$

$$\text{k). } \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$$

$$\text{b). } \lim_{x \rightarrow 0} \frac{\cos 3x - \cos^2 3x}{6x}$$

$$\text{d). } \lim_{x \rightarrow 0} \frac{\sin x - \sin 3x}{x}$$

$$\text{f). } \lim_{x \rightarrow 0} \frac{\sin x - 2x}{x - 2\sin x}$$

$$\text{h). } \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{3x^2}$$

$$\text{j). } \lim_{x \rightarrow 0} \frac{\sin(3x + 1989\pi)}{\sin(4x + 1987\pi)}$$

$$\text{l). } \lim_{x \rightarrow 0} \left(\frac{2}{\sin^2 x} - \frac{1}{1 - \cos x} \right)$$

3) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow 0} \frac{\sin x + 1 - \cos x}{x}$$

$$\text{c). } \lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin^2 x - 3\sin x + 1}{4\sin^2 x - 1}$$

$$\text{e). } \lim_{x \rightarrow 0} \frac{3x + \sin 2x}{3x - \sin 2x}$$

$$\text{g). } \lim_{x \rightarrow 0} \frac{\sin(a + x) - \sin(a - x)}{x}$$

$$\text{b). } \lim_{x \rightarrow 0} \frac{x \sin x - 2 \sin 4x}{\tan x}$$

$$\text{d). } \lim_{x \rightarrow 0} \frac{1 - \cos nx}{1 - \cos mx}$$

$$\text{f). } \lim_{x \rightarrow 0} \frac{(1 + x^2) - \cos x}{\tan^2 x}$$

$$\text{h). } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x}$$

4) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x}$$

$$\text{c). } \lim_{x \rightarrow 0} \frac{x(a - b)}{\sin ax - \sin bx}, (a \neq 0, b \neq 0, a \neq b)$$

$$\text{d). } \lim_{x \rightarrow 0} \frac{\sin(\sin(\sin x))}{x}$$

$$\text{f). } \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 + \sin nx - \cos nx}$$

$$\text{b). } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$\text{e). } \lim_{x \rightarrow 0} \frac{\sin 5x + x \sin 10x}{\sin 5x - x \sin 10x}$$

$$\text{g). } \lim_{x \rightarrow 0} \frac{(1 - \cos x) \sin x}{\tan^3 x}$$

5) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sin 3x}$

b). $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sin 5x}$

c). $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{\sin 7x}$

d). $\lim_{x \rightarrow 0} \frac{2 \sin 2x}{\sqrt{x+2} - \sqrt{2}}$

e). $\lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$

f). $\lim_{x \rightarrow 0} \frac{1+x \sin x - \cos 2x}{\sin^2 x}$

g). $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{1+x \sin x} - \cos x}$

h). $\lim_{x \rightarrow 0} \frac{2x - \sin x}{\sqrt{1 - \cos x}}$

i). $\lim_{x \rightarrow 0} \frac{\sqrt{3x+1} - \sqrt{2x+1}}{\sin x}$

j). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x \sin x} - \sqrt{\cos 2x}}{\cot^2(\frac{\pi}{2} - x)}$

6) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{(\sqrt[3]{x-1} - \sqrt[3]{x+1}) \sin x}{1 - \cos \pi x}$

b). $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$

c). $\lim_{x \rightarrow 0} \frac{\sin^2 x}{1 - \sqrt{\cos x}}$

d). $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{x+1} - 1}$

e). $\lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$

f). $\lim_{x \rightarrow 0} \frac{\sin x}{\sqrt[3]{x}}$

g). $\lim_{x \rightarrow 0^+} \frac{1 - \cos \sqrt{x}}{\sin x}$

h). $\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{x^2}$

i). $\lim_{x \rightarrow 0} \frac{\cos 6x - \cos x}{\sin^2 x}$

j). $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin^2 x} - \cos x}{\sin^2 x}$

k). $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$

l). $\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2}$

m). $\lim_{x \rightarrow 0} \frac{2012x + \sin 2013x - \tan 2014x}{2014x - \sin 2015x + \tan 2016x}$

7) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{\tan^3 x - \sin^3 x}$

b). $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{2x \tan 2x}$

$$c). \lim_{x \rightarrow 0} \frac{2\sin^2 x - 2(1 - \cos x)}{5x^2}$$

$$d). \lim_{x \rightarrow 0} \frac{\cos 3x - \cos x}{\sin 5x + \sin 3x}$$

$$e). \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2(1 + \sqrt{\cos x})}$$

$$f). \lim_{x \rightarrow 0} \frac{\cos x - \sqrt{\cos 2x}}{\sin^2 x}$$

$$g). \lim_{x \rightarrow 0} \frac{\tan x}{\sqrt[3]{(1 - \cos x)^2}}$$

$$h). \lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{x^4}$$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 0} \frac{2\sin x}{3x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{2}{3} = 1 \times \frac{2}{3} = \frac{2}{3}$$

$$b). \lim_{x \rightarrow 0} \frac{1 - \cos x}{2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \times \frac{1}{2} = 0 \times \frac{1}{2} = 0$$

$$c). \lim_{x \rightarrow 0} \frac{2\tan x}{x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{x} \times 2 = 1 \times 2 = 2$$

$$d). \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin ax}{ax} \times \frac{bx}{\sin bx} \times \frac{a}{b} = 1 \times 1 \times \frac{a}{b} = \frac{a}{b}$$

$$e). \lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 3x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{3x}{\sin 3x} \times \frac{4}{3} = 1 \times 1 \times \frac{4}{3} = \frac{4}{3}$$

$$f). \lim_{x \rightarrow 0} \frac{\tan mx}{\sin nx} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\tan mx}{mx} \times \frac{nx}{\sin nx} \times \frac{m}{n} = 1 \times 1 \times \frac{m}{n} = \frac{m}{n}$$

$$g). \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x(2 \sin \frac{x}{2} \cos \frac{x}{2})} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{x \cos \frac{x}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \times \frac{1}{2 \cos \frac{x}{2}} = 1 \times \frac{1}{2} = \frac{1}{2}$$

$$h). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{\cos 2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{\cos^2 x - \sin^2 x} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{(\cos x - \sin x)(\cos x + \sin x)}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{1}{(\cos x + \sin x)} = \frac{1}{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}} = \frac{1}{2 \cdot \frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$i). \lim_{x \rightarrow 0} \frac{\sin 3x}{x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times 3 = 1 \times 3 = 3$$

$$j). \lim_{x \rightarrow 0} \frac{\tan 3x}{2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times \frac{3}{2} = 1 \times \frac{3}{2} = \frac{3}{2}$$

$$k). \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 5x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times \frac{5x}{\sin 5x} \times \frac{2}{5} = 1 \times 1 \times \frac{2}{5} = \frac{2}{5}$$

$$\begin{aligned} \text{I). } \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2} = 2 \times 1 = 2 \end{aligned}$$

2) គណនាលីមីតខាងក្រោម:

$$\begin{aligned} \text{a). } \lim_{x \rightarrow 0} \frac{\cos x - \cos^2 x}{x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{\cos x(1 - \cos x)}{x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \times \cos x = 0 \times 1 = 0 \end{aligned}$$

$$\begin{aligned} \text{b). } \lim_{x \rightarrow 0} \frac{\cos 3x - \cos^2 3x}{6x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{(1 - \cos 3x)\cos 3x}{6x} = \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{3x} \times \frac{\cos 3x}{2} = 0 \times \frac{1}{2} = 0 \end{aligned}$$

$$\begin{aligned} \text{c). } \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{2\sin^2 2x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 2x}{(2x)^2} \times 4 \times 2 = 1 \times 4 \times 2 = 8 \end{aligned}$$

$$\begin{aligned} \text{d). } \lim_{x \rightarrow 0} \frac{\sin x - \sin 3x}{x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} - \frac{\sin 3x}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} - \frac{\sin 3x}{3x} \times 3 \right) = 1 - 1 \times 3 = -2 \end{aligned}$$

$$\begin{aligned} \text{e). } \lim_{x \rightarrow 0} \frac{\sin 2x - 3x}{\sin 3x - 2x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin 2x - 3x}{x}}{\frac{\sin 3x - 2x}{x}} = \lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{x} - 3}{\frac{\sin 3x}{x} - 2} = \lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{2x} \times 2 - 3}{\frac{\sin 3x}{3x} \times 3 - 2} \\ &= \frac{1 \times 2 - 3}{1 \times 3 - 2} = \frac{2 - 3}{3 - 2} = \frac{-1}{1} = -1 \end{aligned}$$

$$\text{f). } \lim_{x \rightarrow 0} \frac{\sin x - 2x}{x - 2\sin x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin x - 2x}{x}}{\frac{x - 2 \sin x}{x}} = \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - 2}{1 - 2 \times \frac{\sin x}{x}} = \frac{1 - 2}{1 - 2 \times 1} = \frac{-1}{-1} = 1$$

g). $\lim_{x \rightarrow 0} \frac{x + \sin x}{2x - 3 \sin 2x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\frac{x + \sin x}{x}}{\frac{2x - 3 \sin 2x}{x}} = \lim_{x \rightarrow 0} \frac{1 + \frac{\sin x}{x}}{2 - \frac{3 \sin 2x}{x}} = \lim_{x \rightarrow 0} \frac{1 + \frac{\sin x}{x}}{2 - \frac{\sin 2x}{\frac{x}{2}} \times 6}$$

$$= \frac{1 + 1}{2 - 1 \times 6} = \frac{2}{-4} = -\frac{1}{2}$$

h). $\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{3x^2}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{3x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \times \frac{1}{3} = 1 \times \frac{1}{3} = \frac{1}{3}$$

i). $\lim_{x \rightarrow 0} \frac{1 - \cos 5x}{x^2}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{5x}{2}}{x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 \frac{5x}{2}}{\left(\frac{5x}{2}\right)^2} \times \frac{25}{4} \times 2 = 1 \times \frac{25}{4} \times 2 = \frac{25}{2}$$

j). $\lim_{x \rightarrow 0} \frac{\sin(3x + 1989\pi)}{\sin(4x + 1987\pi)}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\sin(3x + \pi)}{\sin(4x + \pi)} = \lim_{x \rightarrow 0} \frac{-\sin 3x}{-\sin 4x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{4x}{\sin 4x} \times \frac{3}{4}$$

$$= 1 \times 1 \times \frac{3}{4} = \frac{3}{4}$$

k). $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2}} = \lim_{x \rightarrow 0} \frac{1}{2 \cos^2 \frac{x}{2}} = \frac{1}{2 \times 1} = \frac{1}{2}$$

l). $\lim_{x \rightarrow 0} \left(\frac{2}{\sin^2 x} - \frac{1}{1 - \cos x} \right)$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow 0} \left(\frac{2}{1 - \cos^2 x} - \frac{1}{1 - \cos x} \right) = \lim_{x \rightarrow 0} \frac{2 - (1 + \cos x)}{1 - \cos^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{(1 - \cos x)(1 + \cos x)} = \lim_{x \rightarrow 0} \frac{1}{1 + \cos x} = \frac{1}{1 + 1} = \frac{1}{2}$$

3) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{\sin x + 1 - \cos x}{x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} + \frac{1 - \cos x}{x} \right) = 1 + 0 = 1$$

b). $\lim_{x \rightarrow 0} \frac{x \sin x - 2 \sin 4x}{\tan x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{\frac{x \sin x - 2 \sin 4x}{x}}{\frac{\tan x}{x}} = \lim_{x \rightarrow 0} \frac{\sin x - \frac{2 \sin 4x}{x}}{\frac{\tan x}{x}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - 2 \times \frac{\sin 4x}{4x} \times 4}{\frac{\tan x}{x}} = \frac{0 - 2 \times 1 \times 4}{1} = -8$$

c). $\lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin^2 x - 3 \sin x + 1}{4 \sin^2 x - 1}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow \frac{\pi}{6}} \frac{(\sin x - 1)(2 \sin x - 1)}{(2 \sin x - 1)(2 \sin x + 1)} = \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin x - 1}{2 \sin x + 1}$$

$$= \frac{\frac{1}{2} - 1}{2 \cdot \frac{1}{2} + 1} = \frac{-\frac{1}{2}}{1 + 1} = \frac{-1}{2 \times 2} = -\frac{1}{4}$$

d). $\lim_{x \rightarrow 0} \frac{1 - \cos nx}{1 - \cos mx}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{nx}{2}}{2 \sin^2 \frac{mx}{2}} = \lim_{x \rightarrow 0} \frac{\sin^2 \frac{nx}{2}}{\left(\frac{nx}{2}\right)^2} \times \frac{\left(\frac{mx}{2}\right)^2}{\sin^2 \frac{mx}{2}} \times \frac{n^2}{m^2} = 1 \times 1 \times \frac{n^2}{m^2} = \frac{n^2}{m^2}$$

e). $\lim_{x \rightarrow 0} \frac{3x + \sin 2x}{3x - \sin 2x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\frac{3x + \sin 2x}{x}}{\frac{3x - \sin 2x}{x}} = \lim_{x \rightarrow 0} \frac{3 + \frac{\sin 2x}{x}}{3 - \frac{\sin 2x}{x}} = \lim_{x \rightarrow 0} \frac{3 + \frac{\sin 2x}{2x} \times 2}{3 - \frac{\sin 2x}{2x} \times 2}$$

$$= \frac{3 + 1 \times 2}{3 - 1 \times 2} = \frac{3 + 2}{3 - 2} = \frac{5}{1} = 5$$

f). $\lim_{x \rightarrow 0} \frac{(1 + x^2) - \cos x}{\tan^2 x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{x^2 + 1 - \cos x}{\tan^2 x} = \lim_{x \rightarrow 0} \left(\frac{x^2}{\tan^2 x} + \frac{2 \sin^2 \frac{x}{2}}{\frac{\sin^2 x}{\cos^2 x}} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{x^2}{\tan^2 x} + \frac{2 \sin^2 \frac{x}{2} \times \cos^2 x}{4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2}} \right) = \lim_{x \rightarrow 0} \left(\frac{x^2}{\tan^2 x} + \frac{\cos^2 x}{2 \cos^2 \frac{x}{2}} \right)$$

$$= 1 + \frac{1}{2 \times 1} = 1 + \frac{1}{2} = \frac{3}{2}$$

g). $\lim_{x \rightarrow 0} \frac{\sin(a+x) - \sin(a-x)}{x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{(\sin a \cos x + \cos a \sin x) - (\sin a \cos x - \cos a \sin x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin a \cos x + \cos a \sin x - \sin a \cos x + \cos a \sin x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos a \sin x + \cos a \sin x}{x} = \lim_{x \rightarrow 0} \frac{2 \cos a \sin x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times 2 \cos a = 1 \times 2 \cos a = 2 \cos a$$

h). $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\tan x(1 - \cos x)}{x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} \times (1 - \cos x) = 1 \times (1 - 1) = 0$$

4) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x + \sin x}{1 - \cos x - \sin x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{x}{2} - 2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2} + \cos \frac{x}{2}}{\sin \frac{x}{2} - \cos \frac{x}{2}} = \frac{0 + 1}{0 - 1} = \frac{1}{-1} = -1$$

b). $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\tan x(1 - \cos x)}{x^3} = \lim_{x \rightarrow 0} \frac{\tan x}{x} \times \frac{2 \sin^2 \frac{x}{2}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{x} \times \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{2}{4} = 1 \times 1 \times \frac{2}{4} = \frac{1}{2}$$

$$c). \lim_{x \rightarrow 0} \frac{x(a-b)}{\sin ax - \sin bx}, (a \neq 0, b \neq 0, a \neq b) \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\begin{aligned} & \frac{x(a-b)}{\sin ax - \sin bx} \\ &= \lim_{x \rightarrow 0} \frac{x}{\frac{\sin ax - \sin bx}{x}} = \lim_{x \rightarrow 0} \frac{a-b}{\frac{\sin ax}{x} - \frac{\sin bx}{x}} \\ &= \lim_{x \rightarrow 0} \frac{a-b}{\frac{\sin ax}{ax} \times a - \frac{\sin bx}{bx} \times b} = \frac{a-b}{1 \times a - 1 \times b} = \frac{a-b}{a-b} = 1 \end{aligned}$$

$$d). \lim_{x \rightarrow 0} \frac{\sin(\sin(\sin x))}{x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin(\sin x))}{\sin(\sin)x} \times \frac{\sin(\sin x)}{\sin x} \times \frac{\sin x}{x} = 1 \times 1 \times 1 = 1$$

$$e). \lim_{x \rightarrow 0} \frac{\sin 5x + x \sin 10x}{\sin 5x - x \sin 10x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\begin{aligned} & \frac{\sin 5x + x \sin 10x}{\sin 5x - x \sin 10x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin 5x}{x} + \sin 10x}{\frac{\sin 5x}{x} - \sin 10x} \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 5x}{5x} \times 5 + \sin 10x}{\frac{\sin 5x}{5x} \times 5 - \sin 10x} = \frac{1 \times 5 + 0}{1 \times 5 - 0} = \frac{5}{5} = 1$$

$$f). \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 + \sin nx - \cos nx} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{(1 - \cos x) + \sin x}{(1 - \cos nx) + \sin nx} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{nx}{2} + 2 \sin \frac{nx}{2} \cos \frac{nx}{2}} \\ &= \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2} + \sin \frac{x}{2} \cos \frac{x}{2}}{\sin^2 \frac{nx}{2} + \sin \frac{nx}{2} \cos \frac{nx}{2}} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2} \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)}{\sin \frac{nx}{2} \left(\sin \frac{nx}{2} + \cos \frac{nx}{2} \right)} \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\left(\frac{x}{2}\right)} \times \frac{\left(\frac{nx}{2}\right)}{\sin \frac{nx}{2}} \times \frac{1}{n} \times \frac{\sin \frac{x}{2} + \cos \frac{x}{2}}{\sin \frac{nx}{2} + \cos \frac{nx}{2}} = 1 \times 1 \times \frac{1}{n} \times \frac{0+1}{0+1} = \frac{1}{n}$$

g). $\lim_{x \rightarrow 0} \frac{(1 - \cos x) \sin x}{\tan^3 x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x) \sin x \cos^3 x}{\sin^3 x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} \cos^3 x}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} \cos^3 x}{4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2}} = \lim_{x \rightarrow 0} \frac{\cos^3 x}{2 \cos^2 \frac{x}{2}} = \frac{1}{2 \times 1} = \frac{1}{2}$$

5) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sin 3x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{(\sqrt{1+x} + \sqrt{1-x}) \sin 3x} = \lim_{x \rightarrow 0} \frac{2x}{(\sqrt{1+x} + \sqrt{1-x}) \sin 3x}$$

$$= \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} \times \frac{2}{3(\sqrt{1+x} + \sqrt{1-x})} = 1 \times \frac{2}{3(1+1)} = \frac{2}{3 \times 2} = \frac{1}{3}$$

b). $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sin 5x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{x+4-4}{(\sqrt{x+4}+2) \sin 5x} = \lim_{x \rightarrow 0} \frac{x}{(\sqrt{x+4}+2) \sin 5x}$$

$$= \lim_{x \rightarrow 0} \frac{5x}{\sin 5x} \times \frac{1}{5(\sqrt{x+4}+2)} = 1 \times \frac{1}{5(2+2)} = \frac{1}{5 \times 4} = \frac{1}{20}$$

c). $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{\sin 7x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{x+9-9}{(\sqrt{x+9}+3) \sin 7x} = \lim_{x \rightarrow 0} \frac{x}{\sin 7x (\sqrt{x+9}+3)}$$

$$= \lim_{x \rightarrow 0} \frac{7x}{\sin 7x} \times \frac{1}{7(\sqrt{x+9}+3)} = 1 \times \frac{1}{7(3+3)} = \frac{1}{7 \times 6} = \frac{1}{42}$$

$$\begin{aligned}
 \text{d). } \lim_{x \rightarrow 0} \frac{2 \sin 2x}{\sqrt{x+2} - \sqrt{2}} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{2 \sin 2x(\sqrt{x+2} + \sqrt{2})}{x+2-2} = \lim_{x \rightarrow 0} \frac{2 \sin 2x(\sqrt{x+2} + \sqrt{2})}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times 4(\sqrt{x+2} + \sqrt{2}) = 1 \times 4(\sqrt{2} + \sqrt{2}) = 4 \times 2\sqrt{2} = 8\sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{e). } \lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{2 - (1 + \cos x)}{(\sqrt{2} + \sqrt{1+\cos x}) \sin^2 x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{(\sqrt{2} + \sqrt{1+\cos x}) \sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{(\sqrt{2} + \sqrt{1+\cos x})(4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2})} \\
 &= \lim_{x \rightarrow 0} \frac{1}{(\sqrt{2} + \sqrt{1+\cos x})(2 \cos^2 \frac{x}{2})} = \frac{1}{(\sqrt{2} + \sqrt{2})(2)} \\
 &= \frac{1}{2\sqrt{2} \times 2} = \frac{1}{4\sqrt{2}} = \frac{\sqrt{2}}{4 \times 2} = \frac{\sqrt{2}}{8}
 \end{aligned}$$

$$\begin{aligned}
 \text{f). } \lim_{x \rightarrow 0} \frac{1 + x \sin x - \cos 2x}{\sin^2 x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{(1 - \cos 2x) + x \sin x}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x + x \sin x}{\sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{2 \sin x + x}{\sin x} = \lim_{x \rightarrow 0} \left(2 + \frac{x}{\sin x} \right) = 2 + 1 = 3
 \end{aligned}$$

$$\begin{aligned}
 \text{g). } \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{1+x \sin x} - \cos x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{\sin^2 x(\sqrt{1+x \sin x} + \cos x)}{1 + x \sin x - \cos^2 x} = \lim_{x \rightarrow 0} \frac{\sin^2 x(\sqrt{1+x \sin x} + \cos x)}{x \sin x + 1 - \cos^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin^2 x(\sqrt{1+x \sin x} + \cos x)}{x \sin x + \sin^2 x} = \lim_{x \rightarrow 0} \frac{\sin x(\sqrt{1+x \sin x} + \cos x)}{x + \sin x}
 \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x \sin x} + \cos x)}{\frac{x}{\sin x} + 1} = \frac{1+1}{1+1} = \frac{2}{2} = 1$$

h). $\lim_{x \rightarrow 0} \frac{2x - \sin x}{\sqrt{1 - \cos x}}$ មានរាង $\left(\frac{0}{0}\right)$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{2x - \sin x}{\sqrt{2 \sin^2 \frac{x}{2}}} = \lim_{x \rightarrow 0} \frac{2x - \sin x}{|\sin \frac{x}{2}| \sqrt{2}} = \lim_{x \rightarrow 0} \frac{2x - 2 \sin \frac{x}{2} \cos \frac{x}{2}}{|\sin \frac{x}{2}| \sqrt{2}} \\ &= \lim_{x \rightarrow 0} \frac{2(x - \sin \frac{x}{2} \cos \frac{x}{2})}{\sqrt{2} |\sin \frac{x}{2}|} = \lim_{x \rightarrow 0} \frac{\sqrt{2}(x - \sin \frac{x}{2} \cos \frac{x}{2})}{|\sin \frac{x}{2}|} \end{aligned}$$

❖ ករណី $x \rightarrow 0^+$ នោះ $|\sin \frac{x}{2}| = \sin \frac{x}{2}$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 0^+} \frac{2x - \sin x}{\sqrt{1 - \cos x}} &= \lim_{x \rightarrow 0^+} \frac{\sqrt{2}(x - \sin \frac{x}{2} \cos \frac{x}{2})}{\sin \frac{x}{2}} \\ &= \sqrt{2} \lim_{x \rightarrow 0^+} \left(\frac{x}{\sin \frac{x}{2}} - \cos \frac{x}{2} \right) = \sqrt{2} \lim_{x \rightarrow 0^+} \left(\frac{\frac{x}{2}}{\sin \frac{x}{2}} \times 2 - \cos \frac{x}{2} \right) \\ &= \sqrt{2}(1 \times 2 - 1) = \sqrt{2}(2 - 1) = \sqrt{2} \end{aligned}$$

❖ ករណី $x \rightarrow 0^-$ នោះ $|\sin \frac{x}{2}| = -\sin \frac{x}{2}$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 0^-} \frac{2x - \sin x}{\sqrt{1 - \cos x}} &= \lim_{x \rightarrow 0^-} \frac{\sqrt{2}(x - \sin \frac{x}{2} \cos \frac{x}{2})}{-\sin \frac{x}{2}} \\ &= \sqrt{2} \lim_{x \rightarrow 0^-} \left(-\frac{x}{\sin \frac{x}{2}} + \cos \frac{x}{2} \right) = \sqrt{2} \lim_{x \rightarrow 0^-} \left(-\frac{\frac{x}{2}}{\sin \frac{x}{2}} \times 2 + \cos \frac{x}{2} \right) \\ &= \sqrt{2}(-1 \times 2 + 1) = \sqrt{2}(-2 + 1) = -\sqrt{2} \end{aligned}$$

$$\begin{aligned}
 \text{i). } \lim_{x \rightarrow 0} \frac{\sqrt{3x+1} - \sqrt{2x+1}}{\sin x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{(3x+1) - (2x+1)}{(\sqrt{3x+1} + \sqrt{2x+1}) \sin x} = \lim_{x \rightarrow 0} \frac{x}{\sin x} \times \frac{1}{(\sqrt{3x+1} + \sqrt{2x+1})} \\
 &= 1 \times \frac{1}{1+1} = \frac{1}{2}
 \end{aligned}$$

$$\text{j). } \lim_{x \rightarrow 0} \frac{\sqrt{1+x \sin x} - \sqrt{\cos 2x}}{\cot^2\left(\frac{\pi}{2} - x\right)} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{1 + x \sin x - \cos 2x}{(\sqrt{1+x \sin x} + \sqrt{\cos 2x}) \tan^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{x \sin x + (1 - \cos 2x)}{(\sqrt{1+x \sin x} + \sqrt{\cos 2x}) \tan^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{(x \sin x + 2 \sin^2 x) \cos^2 x}{(\sqrt{1+x \sin x} + \sqrt{\cos 2x}) \sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{\left(\frac{x}{\sin x} + 2\right) \cos^2 x}{(\sqrt{1+x \sin x} + \sqrt{\cos 2x})} = \frac{(1+2)(1)}{(1+1)} = \frac{3}{2}
 \end{aligned}$$

6) គណនាលីមីតខាងក្រោម:

$$\begin{aligned}
 \text{a). } \lim_{x \rightarrow 0} \frac{(\sqrt[3]{x-1} + \sqrt[3]{x+1}) \sin x}{1 - \cos \pi x} & \text{ មានរាង } \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{(x-1+x+1) \sin x}{2 \sin^2 \frac{\pi}{2} x \left[\sqrt[3]{(x-1)^2} - \sqrt[3]{x-1} \sqrt[3]{x+1} + \sqrt[3]{(x+1)^2} \right]} \\
 &= \lim_{x \rightarrow 0} \frac{2x \sin x}{2 \sin^2 \frac{\pi}{2} x \left[\sqrt[3]{(x-1)^2} - \sqrt[3]{x-1} \sqrt[3]{x+1} + \sqrt[3]{(x+1)^2} \right]} \\
 &= \lim_{x \rightarrow 0} \frac{x \sin x}{\sin^2 \frac{\pi}{2} x \left[\sqrt[3]{(x-1)^2} - \sqrt[3]{x-1} \sqrt[3]{x+1} + \sqrt[3]{(x+1)^2} \right]}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\frac{x \sin x}{x^2}}{\frac{\sin^2 \frac{\pi}{2} x}{x^2} \times \left[\sqrt[3]{(x-1)^2} - \sqrt[3]{x-1} \sqrt[3]{x+1} + \sqrt[3]{(x+1)^2} \right]} \\
&= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x}}{\frac{\sin^2 \frac{\pi}{2} x}{\left(\frac{\pi}{2} x\right)^2} \times \frac{\pi^2}{4} \left[\sqrt[3]{(x-1)^2} - \sqrt[3]{x-1} \sqrt[3]{x+1} + \sqrt[3]{(x+1)^2} \right]} \\
&= \frac{1}{1 \times \frac{\pi^2}{4} (1+1+1)} = \frac{4}{3\pi^2}
\end{aligned}$$

b). $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$ មានរាង $\left(\frac{0}{0}\right)$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{(1+\sin x) - (1-\sin x)}{\tan x (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \\
&= \lim_{x \rightarrow 0} \frac{2\sin x}{\tan x (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \\
&= \lim_{x \rightarrow 0} \frac{2\sin x \cos x}{\sin x (\sqrt{1+\sin x} + \sqrt{1-\sin x})} \\
&= \lim_{x \rightarrow 0} \frac{2\cos x}{(\sqrt{1+\sin x} + \sqrt{1-\sin x})} = \frac{2}{1+1} = \frac{2}{2} = 1
\end{aligned}$$

c). $\lim_{x \rightarrow 0} \frac{\sin^2 x}{1 - \sqrt{\cos x}}$ មានរាង $\left(\frac{0}{0}\right)$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\sin^2 x (1 + \sqrt{\cos x})}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{4\sin^2 \frac{x}{2} \cos^2 \frac{x}{2} (1 + \sqrt{\cos x})}{2\sin^2 \frac{x}{2}} \\
&= \lim_{x \rightarrow 0} 2\cos^2 \frac{x}{2} (1 + \sqrt{\cos x}) = 2 \times 1 \times (1+1) = 2 \times 2 = 4
\end{aligned}$$

$$d). \lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{x+1}-1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin x \cos x (\sqrt{x+1}+1)}{x+1-1} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x (\sqrt{x+1}+1)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times 2 \cos x (\sqrt{x+1}+1) = 1 \times 2 \times 1(1+1) = 2 \times 2 = 4$$

$$e). \lim_{x \rightarrow 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{(1+\tan x) - (1+\sin x)}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})} = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x(1 - \cos x)}{x^3(\sqrt{1+\tan x} + \sqrt{1+\sin x})}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{x} \times \frac{2 \sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{1}{4} \times \frac{1}{(\sqrt{1+\tan x} + \sqrt{1+\sin x})}$$

$$= 1 \times 2 \times \frac{1}{4} \times \frac{1}{1+1} = \frac{2}{4} \times \frac{1}{2} = \frac{1}{4}$$

$$f). \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt[3]{x}} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt[3]{x}} \times \frac{\sqrt[3]{x^2}}{\sqrt[3]{x^2}} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \sqrt[3]{x^2} = 1 \times 0 = 0$$

$$g). \lim_{x \rightarrow 0^+} \frac{1 - \cos \sqrt{x}}{\sin x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0^+} \frac{2 \sin^2 \frac{\sqrt{x}}{2}}{\sin x} = 2 \lim_{x \rightarrow 0^+} \frac{\sin^2 \frac{\sqrt{x}}{2}}{\left(\frac{\sqrt{x}}{2}\right)^2} \times \frac{x}{\sin x} \times \frac{1}{4} = 2 \times 1 \times 1 \times \frac{1}{4} = \frac{1}{2}$$

$$h). \lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{x^2} \text{ មានរាង } \left(\frac{0}{0} \right)$$

❖ រូបថតទី១

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{-2 \sin\left(\frac{ax+bx}{2}\right) \sin\left(\frac{ax-bx}{2}\right)}{x^2} \\
 &= -2 \lim_{x \rightarrow 0} \frac{\sin\left(\frac{a+b}{2}\right)x}{x} \times \frac{\sin\left(\frac{a-b}{2}\right)x}{x} \\
 &= -2 \lim_{x \rightarrow 0} \frac{\sin\left(\frac{a+b}{2}\right)x}{\left(\frac{a+b}{2}\right)x} \times \left(\frac{a+b}{2}\right) \times \frac{\sin\left(\frac{a-b}{2}\right)x}{\left(\frac{a-b}{2}\right)x} \times \left(\frac{a-b}{2}\right) \\
 &= -2 \times \left(\frac{a+b}{2}\right) \times \left(\frac{a-b}{2}\right) = -\frac{(a+b)(a-b)}{2} = -\frac{a^2-b^2}{2} = \frac{b^2-a^2}{2}
 \end{aligned}$$

❖ រូបថតទី២

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{\cos ax - 1 + 1 - \cos bx}{x^2} = \lim_{x \rightarrow 0} \frac{(1 - \cos bx) - (1 - \cos ax)}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{bx}{2} - 2 \sin^2 \frac{ax}{2}}{x^2} = 2 \lim_{x \rightarrow 0} \left(\frac{\sin^2 \frac{bx}{2}}{x^2} - \frac{\sin^2 \frac{ax}{2}}{x^2} \right) \\
 &= 2 \lim_{x \rightarrow 0} \left(\frac{\sin^2 \frac{bx}{2}}{\left(\frac{bx}{2}\right)^2} \times \frac{b^2}{4} - \frac{\sin^2 \frac{ax}{2}}{\left(\frac{ax}{2}\right)^2} \times \frac{a^2}{4} \right) = 2 \left(\frac{b^2}{4} - \frac{a^2}{4} \right) \\
 &= \frac{2(b^2 - a^2)}{4} = \frac{b^2 - a^2}{2}
 \end{aligned}$$

i). $\lim_{x \rightarrow 0} \frac{\cos 6x - \cos x}{\sin^2 x}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\cos 6x - 1 + 1 - \cos x}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{(1 - \cos x) - (1 - \cos 6x)}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 \frac{x}{2} - 2\sin^2 3x}{\sin^2 x} = 2 \lim_{x \rightarrow 0} \left(\frac{\sin^2 \frac{x}{2}}{\sin^2 x} - \frac{\sin^2 3x}{\sin^2 x} \right)$$

$$= 2 \lim_{x \rightarrow 0} \left(\frac{\sin^2 \frac{x}{2}}{4\sin^2 \frac{x}{2} \cos^2 \frac{x}{2}} - \frac{\sin^2 3x}{(3x)^2} \times \frac{x^2}{\sin^2 x} \times 9 \right)$$

$$= 2 \lim_{x \rightarrow 0} \left(\frac{1}{4\cos^2 \frac{x}{2}} - \frac{\sin^2 3x}{(3x)^2} \times \frac{x^2}{\sin^2 x} \times 9 \right)$$

$$= 2 \left(\frac{1}{4} - 1 \times 1 \times 9 \right) = 2 \left(\frac{1}{4} - 9 \right) = 2 \times \frac{1-36}{4} = -\frac{35}{2}$$

j). $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin^2 x} - \cos x}{\sin^2 x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{1+\sin^2 x - \cos^2 x}{\sin^2 x(\sqrt{1+\sin^2 x} + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x + (1 - \cos^2 x)}{\sin^2 x(\sqrt{1+\sin^2 x} + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x + \sin^2 x}{\sin^2 x(\sqrt{1+\sin^2 x} + \cos x)} = \lim_{x \rightarrow 0} \frac{2\sin^2 x}{\sin^2 x(\sqrt{1+\sin^2 x} + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{2}{\sqrt{1+\sin^2 x} + \cos x} = \frac{2}{1+1} = \frac{2}{2} = 1$$

k). $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x (\cos^2 x - \sin^2 x)}{x^2} = \lim_{x \rightarrow 0} \frac{(1 - \cos^3 x) + \cos x \sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x + \cos^2 x) + \cos x \sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 \frac{x}{2} (1 + \cos x + \cos^2 x) + \cos x (4\sin^2 \frac{x}{2} \cos^2 \frac{x}{2})}{x^2}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{x^2} \times \left[2(1 + \cos x + \cos^2 x) + 4 \cos x \cos^2 \frac{x}{2} \right] \\
&= \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{1}{4} \times \left[2(1 + \cos x + \cos^2 x) + 4 \cos x \cos^2 \frac{x}{2} \right]
\end{aligned}$$

$$= 1 \times \frac{1}{4} [2(1+1+1) + 4] = \frac{1}{4} (6+4) = \frac{10}{4} = \frac{5}{2}$$

$$l). \lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x \cos x}{x^2 (1 + \cos x \sqrt{\cos 2x})} = \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x^2 (1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x + \cos^2 x)}{x^2 (1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} \times \frac{(1 + \cos x + \cos^2 x)}{(1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{1}{4} \times \frac{(1 + \cos x + \cos^2 x)}{(1 + \cos x \sqrt{\cos 2x})} = 2 \times \frac{1}{4} \times \frac{1+1+1}{1+1} = \frac{2}{4} \times \frac{3}{2} = \frac{3}{4}$$

$$m). \lim_{x \rightarrow 0} \frac{2012x + \sin 2013x - \tan 2014x}{2014x - \sin 2015x + \tan 2016x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\frac{2012x + \sin 2013x - \tan 2014x}{x}}{\frac{2014x - \sin 2015x + \tan 2016x}{x}}$$

$$= \lim_{x \rightarrow 0} \frac{2012 + \frac{\sin 2013x}{x} - \frac{\tan 2014x}{x}}{2014 - \frac{\sin 2015x}{x} + \frac{\tan 2016x}{x}}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{2012 + \frac{\sin 2013x}{2013x} \times 2013 - \frac{\tan 2014x}{2014x} \times 2014}{2014 - \frac{\sin 2015x}{2015x} \times 2015 + \frac{\tan 2016x}{2016x} \times 2016} \\
&= \frac{2012 + 2013 - 2014}{2014 - 2015 + 2016} = \frac{2012 - 1}{2014 + 1} = \frac{2011}{2015}
\end{aligned}$$

7) គណនាលីមីតខាងក្រោម:

$$\begin{aligned}
\text{a). } & \lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{\tan^3 x - \sin^3 x} \text{ មានរាង } \left(\frac{0}{0} \right) \\
&= \lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{\tan^3 x (1 - \cos^3 x)} = \lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{\frac{\sin^3 x}{\cos^3 x} (1 - \cos x)(1 + \cos x + \cos^2 x)} \\
&= \lim_{x \rightarrow 0} \frac{(1 - \cos x) \cos^3 x}{\sin^3 x (1 + \cos x + \cos^2 x)} \\
&= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} \cos^3 x}{8 \sin^3 \frac{x}{2} \cos^3 \frac{x}{2} (1 + \cos x + \cos^2 x)} \\
&= \lim_{x \rightarrow 0} \frac{\cos^3 x}{4 \sin \frac{x}{2} \cos^3 \frac{x}{2} (1 + \cos x + \cos^2 x)} = \frac{1}{4(0)(1)(1+1+1)} = \frac{1}{0} = \pm \infty \\
\text{b). } & \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{2x \tan 2x} \text{ មានរាង } \left(\frac{0}{0} \right) \\
&= \lim_{x \rightarrow 0} \frac{2 \sin^2 2x}{2x \frac{\sin 2x}{\cos 2x}} = \lim_{x \rightarrow 0} \frac{\sin 2x}{x \times \frac{1}{\cos x}} = \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times 2 \cos x = 1 \times 2 = 2 \\
\text{c). } & \lim_{x \rightarrow 0} \frac{2 \sin^2 x - 2(1 - \cos x)}{5x^2} \text{ មានរាង } \left(\frac{0}{0} \right) \\
&= \lim_{x \rightarrow 0} \frac{2 \cdot 4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2} - 2 \cdot 2 \sin^2 \frac{x}{2}}{5x^2}
\end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2} \left(8 \cos^2 \frac{x}{2} - 4 \right)}{5x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2} \right)^2} \times \frac{1}{4 \cdot 5} \times \left(8 \cos^2 \frac{x}{2} - 4 \right)$$

$$= 1 \times \frac{1}{20} \times (8 - 4) = \frac{4}{20} = \frac{1}{5}$$

d). $\lim_{x \rightarrow 0} \frac{\cos 3x - \cos x}{\sin 5x + \sin 3x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{\cos 3x - 1 + 1 - \cos x}{\sin 5x + \sin 3x} = \lim_{x \rightarrow 0} \frac{(1 - \cos x) - (1 - \cos 3x)}{\sin 5x + \sin 3x}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} - 2 \sin^2 \frac{3x}{2}}{\sin 5x + \sin 3x} = 2 \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2} - \sin^2 \frac{3x}{2}}{\sin 5x + \sin 3x}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\frac{\sin^2 \frac{x}{2}}{\frac{x^2}{2}} - \frac{\sin^2 \frac{3x}{2}}{\frac{(3x)^2}{2}}}{\frac{1}{x} \left(\frac{\sin 5x}{x} + \frac{\sin 3x}{x} \right)} = 2 \lim_{x \rightarrow 0} \frac{x \left(\frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2} \right)^2} \times \frac{1}{4} - \frac{\sin^2 \frac{3x}{2}}{\left(\frac{3x}{2} \right)^2} \times \frac{9}{4} \right)}{\left(\frac{\sin 5x}{5x} \times 5 + \frac{\sin 3x}{3x} \times 3 \right)}$$

$$= 2 \times \frac{0 \left(\frac{1}{4} - \frac{9}{4} \right)}{(5 + 3)} = \frac{0}{8} = 0$$

e). $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 (1 + \sqrt{\cos x})}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2 (1 + \sqrt{\cos x})} = \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2} \right)^2} \times \frac{2}{4} \times \frac{1}{1 + \sqrt{\cos x}}$$

$$= 1 \times \frac{2}{4} \times \frac{1}{1 + 1} = \frac{2}{4} \times \frac{1}{2} = \frac{1}{4}$$

f). $\lim_{x \rightarrow 0} \frac{\cos x - \sqrt{\cos 2x}}{\sin^2 x}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - \cos 2x}{\sin^2 x (\cos x + \sqrt{\cos 2x})} = \lim_{x \rightarrow 0} \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin^2 x (\cos x + \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sin^2 x (\cos x + \sqrt{\cos 2x})} = \lim_{x \rightarrow 0} \frac{1}{\cos x + \sqrt{\cos 2x}} = \frac{1}{2}$$

g). $\lim_{x \rightarrow 0} \frac{\tan x}{\sqrt[3]{(1 - \cos x)^2}}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\sqrt[3]{\tan^3 x}}{\sqrt[3]{(1 - \cos x)^2}} = \lim_{x \rightarrow 0} \sqrt[3]{\frac{\sin^3 x}{\cos^3 x (2 \sin^2 \frac{x}{2})^2}}$$

$$= \lim_{x \rightarrow 0} \sqrt[3]{\frac{8 \sin^3 \frac{x}{2} \cos^3 \frac{x}{2}}{\cos^3 x 4 \sin^4 \frac{x}{2}}} = \lim_{x \rightarrow 0} \sqrt[3]{\frac{2 \cos^3 \frac{x}{2}}{\cos^3 x \sin \frac{x}{2}}} = \sqrt[3]{\frac{2}{0}} = \pm \infty$$

h). $\lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{x^4}$ មានរាង $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos(2 \sin^2 \frac{x}{2})}{x^4} = \lim_{x \rightarrow 0} \frac{2 \sin^2(\sin^2 \frac{x}{2})}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2(\sin^2 \frac{x}{2})}{(\sin^2 \frac{x}{2})^2} \times \frac{\sin^4 \frac{x}{2}}{\left(\frac{x}{2}\right)^4} \times \frac{2}{16} = 1 \times 1 \times \frac{2}{16} = \frac{1}{8}$$

គ. ទ្រឹស្តីបទ

បើ $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ មានរាងមិនកំណត់ $\frac{0}{0}$ (អនុគមន៍ត្រីកោណមាត្រ)

គេតាង $u = x - a$ នោះ $x = u + a$

ពេល $x \rightarrow a$ នោះ $u \rightarrow 0$

រួចអនុវត្តន៍រូបមន្ត $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$ ឬ $\lim_{u \rightarrow 0} \frac{\tan u}{u} = 1$

លំហាត់

1) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}}$$

$$\text{b). } \lim_{x \rightarrow 1} \frac{\sin \pi x}{x - 1}$$

$$\text{c). } \lim_{x \rightarrow 2} \frac{4 - x^2}{\sin \pi x}$$

$$\text{d). } \lim_{x \rightarrow 1} \frac{\tan \pi x}{1 - x}$$

$$\text{e). } \lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{(\pi - x)^2}$$

$$\text{f). } \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x}$$

$$\text{g). } \lim_{x \rightarrow 1} \frac{1 - x^2}{\sin \pi x}$$

$$\text{h). } \lim_{x \rightarrow \pi} \frac{1 + \cos x}{(x - \pi)^2}$$

$$\text{i). } \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{(1 - \sqrt{x})}$$

$$\text{j). } \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$$

2) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin x - \sqrt{3} \cos x}{\sin 3x}$$

$$\text{b). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan \frac{x}{2} - 1}{x - \frac{\pi}{2}}$$

$$\text{c). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\pi - 4x}$$

$$\text{d). } \lim_{x \rightarrow a} \frac{\sin^2 x - \sin^2 a}{x^2 - a^2}$$

$$\text{e). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin^2 x - 1}{4x - \pi}$$

$$\text{f). } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sqrt{3} \cos x - \sin x}{x - \frac{\pi}{3}}$$

$$\text{g). } \lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{x^2 + 4x - 5}$$

$$\text{h). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt{(1 - \sin x)^2}}$$

$$\text{i). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x + \cos 2x + 1}{\cos 2x + \sin x}$$

$$\text{j). } \lim_{x \rightarrow \frac{\pi}{12}} \frac{\tan 3x - 1}{2 \sin 2x - 1}$$

3) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx}, (m, n \in \mathbb{N})$$

$$\text{b). } \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x}$$

$$\text{c). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sqrt{2} \cos x}{1 - \sqrt{2} \sin x}$$

$$\text{d). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan x - 1}{(2 \cos x - \sqrt{2})}$$

$$\text{e). } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(x - \frac{\pi}{3}\right)}{1 - 2 \cos x}$$

$$\text{f). } \lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin x - 1}{4 \cos^2 x - 3}$$

$$\text{g). } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan^3 x - 3 \tan x}{\cos\left(x + \frac{\pi}{6}\right)}$$

$$\text{h). } \lim_{x \rightarrow a} \frac{\tan x - \tan a}{\cos x - \cos a}$$

$$\text{i). } \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin\left(x - \frac{\pi}{6}\right)}{\frac{\sqrt{3}}{2} - \cos x}$$

$$\text{j). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x^2 - \left(\frac{\pi}{4}\right)^2}$$

$$\text{k). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x + \cos x}{1 - \sin x - \cos x}$$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} \text{ មានរាង } \left(\frac{0}{0}\right)$$

$$\text{តាង } u = x - \frac{\pi}{2} \text{ នោះ } x = \frac{\pi}{2} + u$$

$$\text{ពេល } x \rightarrow \frac{\pi}{2} \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} = \lim_{u \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} + u\right)}{u} = \lim_{u \rightarrow 0} \frac{-\sin u}{u} = -1$$

$$\text{ដូចនេះ } \boxed{\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} = -1}$$

$$\text{b). } \lim_{x \rightarrow 1} \frac{\sin \pi x}{x-1} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\text{តាង } u = x-1 \text{ នោះ } x = 1+u$$

$$\text{ពេល } x \rightarrow 1 \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow 1} \frac{\sin \pi x}{x-1} = \lim_{u \rightarrow 0} \frac{\sin \pi(1+u)}{u} = \lim_{u \rightarrow 0} \frac{\sin(\pi + \pi u)}{u}$$

$$= \lim_{u \rightarrow 0} \frac{-\sin \pi u}{u} = - \lim_{u \rightarrow 0} \frac{\sin \pi u}{\pi u} \times \pi = -\pi$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{\sin \pi x}{x-1} = -\pi}$$

$$\text{c). } \lim_{x \rightarrow 2} \frac{4-x^2}{\sin \pi x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\text{តាង } u = x-2 \text{ នោះ } x = 2+u$$

$$\text{ពេល } x \rightarrow 2 \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow 2} \frac{4-x^2}{\sin \pi x} = \lim_{u \rightarrow 0} \frac{4-(2+u)^2}{\sin \pi(2+u)} = \lim_{u \rightarrow 0} \frac{4-(4+4u+u^2)}{\sin(2\pi + \pi u)}$$

$$= \lim_{u \rightarrow 0} \frac{-u(4+u)}{\sin \pi u} = \lim_{u \rightarrow 0} \frac{\pi u}{\sin \pi u} \times \frac{-(4+u)}{\pi} = 1 \times \frac{-4}{\pi} = -\frac{4}{\pi}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 2} \frac{4-x^2}{\sin \pi x} = -\frac{4}{\pi}}$$

$$\text{d). } \lim_{x \rightarrow 1} \frac{\tan \pi x}{1-x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

$$\text{តាង } u = x-1 \text{ នោះ } x = 1+u$$

$$\text{ពេល } x \rightarrow 1 \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow 1} \frac{\tan \pi x}{1-x} = \lim_{u \rightarrow 0} \frac{\tan \pi(1+u)}{1-(1+u)} = \lim_{u \rightarrow 0} \frac{\tan(\pi + \pi u)}{-u}$$

$$= \lim_{u \rightarrow 0} \frac{\tan \pi u}{-u} = - \lim_{u \rightarrow 0} \frac{\tan \pi u}{\pi u} \times \pi = -1 \times \pi = -\pi$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{\tan \pi x}{1-x} = -\pi}$$

$$e). \lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{(\pi - x)^2} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាំង $u = \pi - x$ នោះ $x = \pi - u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{(\pi - x)^2} &= \lim_{u \rightarrow 0} \frac{1 - \sin \left(\frac{\pi - u}{2} \right)}{u^2} \\ &= \lim_{u \rightarrow 0} \frac{1 - \sin \left(\frac{\pi}{2} - \frac{u}{2} \right)}{u^2} = \lim_{u \rightarrow 0} \frac{1 - \cos \frac{u}{2}}{u^2} = \lim_{u \rightarrow 0} \frac{2 \sin^2 \frac{u}{4}}{u^2} \\ &= \lim_{u \rightarrow 0} \frac{\sin^2 \frac{u}{4}}{\left(\frac{u}{4} \right)^2} \times \frac{2}{16} = 1 \times \frac{2}{16} = \frac{1}{8} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{1 - \sin \frac{x}{2}}{(\pi - x)^2} = \frac{1}{8}}$$

$$f). \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាំង $u = x - \pi$ នោះ $x = \pi + u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} &= \lim_{u \rightarrow 0} \frac{\sin 3(\pi + u)}{\sin 2(\pi + u)} = \lim_{u \rightarrow 0} \frac{\sin(3\pi + 3u)}{\sin(2\pi + 2u)} \\ &= \lim_{u \rightarrow 0} \frac{-\sin 3u}{\sin 2u} = - \lim_{u \rightarrow 0} \frac{\sin 3u}{3u} \times \frac{2u}{\sin 2u} \times \frac{3}{2} = -1 \times 1 \times \frac{3}{2} = -\frac{3}{2} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} = -\frac{3}{2}}$$

$$g). \lim_{x \rightarrow 1} \frac{1 - x^2}{\sin \pi x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាំង $u = 1 - x$ នោះ $x = 1 - u$

ពេល $x \rightarrow 1$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 1} \frac{1-x^2}{\sin \pi x} &= \lim_{u \rightarrow 0} \frac{1-(1-u)^2}{\sin \pi(1-u)} = \lim_{u \rightarrow 0} \frac{1-(1-2u+u^2)}{\sin(\pi - \pi u)} \\ &= \lim_{u \rightarrow 0} \frac{(2u-u^2)}{\sin(\pi u)} = \lim_{u \rightarrow 0} \frac{\pi u}{\sin \pi u} \times \frac{2-u}{\pi} = 1 \times \frac{2}{\pi} = \frac{2}{\pi} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{1-x^2}{\sin \pi x} = \frac{2}{\pi}}$$

$$\text{h). } \lim_{x \rightarrow \pi} \frac{1+\cos x}{(x-\pi)^2} \text{ មានរាង } \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

តាង $u = x - \pi$ នោះ $x = \pi + u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{1+\cos x}{(x-\pi)^2} &= \lim_{u \rightarrow 0} \frac{1+\cos(\pi+u)}{u^2} \\ &= \lim_{u \rightarrow 0} \frac{1-\cos u}{u^2} = \lim_{u \rightarrow 0} \frac{2\sin^2 \frac{u}{2}}{u^2} = \lim_{u \rightarrow 0} \frac{\sin^2 \frac{u}{2}}{\left(\frac{u}{2}\right)^2} \times \frac{2}{4} = 1 \times \frac{2}{4} = \frac{1}{2} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{1+\cos x}{(x-\pi)^2} = \frac{1}{2}}$$

$$\text{i). } \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{(1-\sqrt{x})} \text{ មានរាង } \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

តាង $u = 1 - x$ នោះ $x = 1 - u$

ពេល $x \rightarrow 1$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{(1-\sqrt{x})} &= \lim_{x \rightarrow 1} \frac{(1+\sqrt{x}) \cos \frac{\pi}{2} x}{1-x} \\ &= \lim_{u \rightarrow 0} \frac{(1+\sqrt{1-u}) \cos \frac{\pi}{2} (1-u)}{u} = \lim_{u \rightarrow 0} \frac{(1+\sqrt{1-u}) \cos \left(\frac{\pi}{2} - \frac{\pi u}{2} \right)}{u} \end{aligned}$$

$$= \lim_{u \rightarrow 0} \frac{(1 + \sqrt{1-u}) \sin\left(\frac{\pi u}{2}\right)}{u} = \lim_{u \rightarrow 0} \frac{\sin\left(\frac{\pi u}{2}\right)}{\left(\frac{\pi u}{2}\right)} \times \frac{\pi}{2} (1 + \sqrt{1-u})$$

$$= 1 \times \frac{\pi}{2} (1+1) = \frac{\pi}{2} \times 2 = \pi$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{(1 - \sqrt{x})} = \pi$

j). $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - a$ នោះ $x = a + u$

ពេល $x \rightarrow a$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \lim_{u \rightarrow 0} \frac{\sin(a+u) - \sin a}{u}$

$$= \lim_{u \rightarrow 0} \frac{\sin a \cos u + \cos a \sin u - \sin a}{u}$$

$$= \lim_{u \rightarrow 0} \frac{\sin a(\cos u - 1) + \cos a \sin u}{u}$$

$$= \lim_{u \rightarrow 0} \frac{\sin a(-2 \sin^2 \frac{u}{2}) + \cos a(2 \sin \frac{u}{2} \cos \frac{u}{2})}{u}$$

$$= \lim_{u \rightarrow 0} \frac{\sin \frac{u}{2}}{u} \times (-2 \sin a \sin \frac{u}{2} + 2 \cos a \cos \frac{u}{2})$$

$$= \lim_{u \rightarrow 0} \left[\frac{\sin \frac{u}{2}}{\frac{u}{2}} \times \frac{1}{2} \times (-2 \sin a \sin \frac{u}{2} + 2 \cos a \cos \frac{u}{2}) \right]$$

$$= 1 \times \frac{1}{2} \times (0 + 2 \cos a) = \frac{1}{2} \times (2 \cos a) = \cos a$$

ដូចនេះ: $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \cos a$

2) គណនាលីមីតខាងក្រោម:

$$\begin{aligned} \text{a). } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin x - \sqrt{3} \cos x}{\sin 3x} &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \left(\frac{1}{2} \sin x - \frac{\sqrt{3}}{2} \cos x \right)}{\sin 3x} \\ &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \left(\sin x \cos \frac{\pi}{3} - \cos x \sin \frac{\pi}{3} \right)}{\sin 3x} = \lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \sin \left(x - \frac{\pi}{3} \right)}{\sin 3x} \end{aligned}$$

តាង $u = x - \frac{\pi}{3}$ នោះ $x = \frac{\pi}{3} + u$

ពេល $x \rightarrow \frac{\pi}{3}$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \sin \left(x - \frac{\pi}{3} \right)}{\sin 3x} &= \lim_{u \rightarrow 0} \frac{2 \sin u}{\sin 3 \left(\frac{\pi}{3} + u \right)} = \lim_{u \rightarrow 0} \frac{2 \sin u}{\sin(\pi + 3u)} \\ &= \lim_{u \rightarrow 0} \frac{2 \sin u}{-\sin(3u)} = -2 \lim_{u \rightarrow 0} \frac{\sin u}{u} \times \frac{3u}{\sin 3u} \times \frac{1}{3} = -2 \times 1 \times 1 \times \frac{1}{3} = -\frac{2}{3} \end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin x - \sqrt{3} \cos x}{\sin 3x} = -\frac{2}{3}}$

$$\text{b). } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{\tan \frac{x}{2} - 1}{2}}{x - \frac{\pi}{2}} \text{ មានរាង } \left(\frac{0}{0} \right)$$

(ប្រើរូបមន្ត $\tan(a-b) = \frac{\tan a - \tan b}{1 + \tan a \tan b}$)

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{\tan \frac{x}{2} - \tan \frac{\pi}{4}}{2}}{x - \frac{\pi}{2}} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan \left(\frac{x}{2} - \frac{\pi}{4} \right) (1 + \tan \frac{x}{2})}{x - \frac{\pi}{2}}$$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan\left(\frac{x}{2} - \frac{\pi}{4}\right)(1 + \tan\frac{x}{2})}{x - \frac{\pi}{2}} \\
 = \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{4} + \frac{u}{2} - \frac{\pi}{4}\right)\left[1 + \tan\left(\frac{\pi}{4} + \frac{u}{2}\right)\right]}{u} \\
 = \lim_{u \rightarrow 0} \frac{\tan\left(\frac{u}{2}\right)\left[1 + \tan\left(\frac{\pi}{4} + \frac{u}{2}\right)\right]}{u} = \lim_{u \rightarrow 0} \frac{\tan\left(\frac{u}{2}\right)}{\left(\frac{u}{2}\right)} \times \frac{1 + \tan\left(\frac{\pi}{4} + \frac{u}{2}\right)}{2} \\
 = 1 \times \frac{1+1}{2} = \frac{2}{2} = 1
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan\frac{x}{2} - 1}{x - \frac{\pi}{2}} = 1$

c). $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\pi - 4x}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{4}$ នោះ $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\pi - 4x} &= \lim_{u \rightarrow 0} \frac{\sin\left(\frac{\pi}{4} + u\right) - \cos\left(\frac{\pi}{4} + u\right)}{\pi - 4\left(\frac{\pi}{4} + u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\left(\sin\frac{\pi}{4}\cos u + \cos\frac{\pi}{4}\sin u\right) - \left(\cos\frac{\pi}{4}\cos u - \sin\frac{\pi}{4}\sin u\right)}{-4u}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{\sin \frac{\pi}{4} \cos u + \cos \frac{\pi}{4} \sin u - \cos \frac{\pi}{4} \cos u + \sin \frac{\pi}{4} \sin u}{-4u} \\
&= \lim_{u \rightarrow 0} \frac{\frac{\sqrt{2}}{2} \cos u + \frac{\sqrt{2}}{2} \sin u - \frac{\sqrt{2}}{2} \cos u + \frac{\sqrt{2}}{2} \sin u}{-4u} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{2} \sin u}{-4u} = \lim_{u \rightarrow 0} \frac{\sin u}{u} \times \frac{\sqrt{2}}{-4} = 1 \times \frac{\sqrt{2}}{-4} = -\frac{\sqrt{2}}{4}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\pi - 4x} = -\frac{\sqrt{2}}{4}$

d). $\lim_{x \rightarrow a} \frac{\sin^2 x - \sin^2 a}{x^2 - a^2}$ មានរាង $\left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow a} \frac{(\sin x - \sin a)(\sin x + \sin a)}{(x - a)(x + a)}$$

តាង $u = x - a$ នោះ $x = a + u$

ពេល $x \rightarrow a$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow a} \frac{(\sin x - \sin a)(\sin x + \sin a)}{(x - a)(x + a)}$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{[\sin(a + u) - \sin a][\sin(a + u) + \sin a]}{u(a + u + a)} \\
&= \lim_{u \rightarrow 0} \frac{(\sin a \cos u + \cos a \sin u - \sin a)[\sin(a + u) + \sin a]}{u(2a + u)} \\
&= \lim_{u \rightarrow 0} \frac{(\sin a \cos u - \sin a + \cos a \sin u)[\sin(a + u) + \sin a]}{u(2a + u)} \\
&= \lim_{u \rightarrow 0} \frac{\sin a(\cos u - 1) + \cos a \sin u}{u} \times \frac{\sin(a + u) + \sin a}{2a + u} \\
&= \lim_{u \rightarrow 0} \left(\frac{\sin a(\cos u - 1)}{u} + \frac{\cos a \sin u}{u} \right) \times \frac{\sin(a + u) + \sin a}{2a + u} \\
&= \lim_{u \rightarrow 0} \left(\sin a \times \frac{\cos u - 1}{u} + \cos a \times \frac{\sin u}{u} \right) \times \frac{\sin(a + u) + \sin a}{2a + u}
\end{aligned}$$

$$= (0 + \cos a) \times \frac{\sin a + \sin a}{2a} = \frac{\cos a(2 \sin a)}{2a} = \frac{\sin 2a}{2a}$$

ដូចនេះ: $\lim_{x \rightarrow a} \frac{\sin^2 x - \sin^2 a}{x^2 - a^2} = \frac{\sin 2a}{2a}$

e). $\lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin^2 x - 1}{4x - \pi}$ មានរាង $\left(\frac{0}{0} \right)$

តាង $u = x - \frac{\pi}{4}$ នោះ $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin^2 x - 1}{4x - \pi} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\sqrt{2} \sin x - 1)(\sqrt{2} \sin x + 1)}{4x - \pi}$

$$= \lim_{u \rightarrow 0} \frac{[\sqrt{2} \sin(\frac{\pi}{4} + u) - 1][\sqrt{2} \sin(\frac{\pi}{4} + u) + 1]}{4(\frac{\pi}{4} + u) - \pi}$$

$$= \lim_{u \rightarrow 0} \frac{[\sqrt{2}(\sin \frac{\pi}{4} \cos u + \cos \frac{\pi}{4} \sin u) - 1][\sqrt{2} \sin(\frac{\pi}{4} + u) + 1]}{4u}$$

$$= \lim_{u \rightarrow 0} \frac{[\sqrt{2}(\frac{\sqrt{2}}{2} \cos u + \frac{\sqrt{2}}{2} \sin u) - 1][\sqrt{2} \sin(\frac{\pi}{4} + u) + 1]}{4u}$$

$$= \lim_{u \rightarrow 0} \frac{(\cos u + \sin u) - 1}{u} \times \frac{\sqrt{2} \sin(\frac{\pi}{4} + u) + 1}{4}$$

$$= \lim_{u \rightarrow 0} \left(\frac{\cos u - 1}{u} + \frac{\sin u}{u} \right) \times \frac{\sqrt{2} \sin(\frac{\pi}{4} + u) + 1}{4}$$

$$= (0 + 1) \times \frac{\sqrt{2} \cdot \frac{\sqrt{2}}{2} + 1}{4} = 1 \times \frac{1 + 1}{4} = \frac{2}{4} = \frac{1}{2}$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{2\sin^2 x - 1}{4x - \pi} = \frac{1}{2}$$

f). $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sqrt{3}\cos x - \sin x}{x - \frac{\pi}{3}}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{3}$ នោះ $x = \frac{\pi}{3} + u$

ពេល $x \rightarrow \frac{\pi}{3}$ នោះ $u \rightarrow 0$

គេបាន
$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sqrt{3}\cos x - \sin x}{x - \frac{\pi}{3}} &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{2\left(\frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x\right)}{x - \frac{\pi}{3}} \\ &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{2\left(\sin \frac{\pi}{3}\cos x - \cos \frac{\pi}{3}\sin x\right)}{x - \frac{\pi}{3}} = \lim_{x \rightarrow \frac{\pi}{3}} \frac{2\sin\left(\frac{\pi}{3} - x\right)}{x - \frac{\pi}{3}} \\ &= \lim_{u \rightarrow 0} \frac{2\sin(-u)}{u} = -2 \lim_{u \rightarrow 0} \frac{\sin u}{u} = -2 \times 1 = -2 \end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sqrt{3}\cos x - \sin x}{x - \frac{\pi}{3}} = -2$$

g). $\lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{x^2 + 4x - 5}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - 1$ នោះ $x = 1 + u$ ឬ $\sqrt{x} = \sqrt{1 + u}$

ពេល $x \rightarrow 1$ នោះ $u \rightarrow 0$

គេបាន
$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{x^2 + 4x - 5} &= \lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{(x - 1)(x + 5)} \\ &= \lim_{u \rightarrow 0} \frac{\sin(\sqrt{1 + u} - 1)}{u(1 + u + 5)} = \lim_{u \rightarrow 0} \frac{\sin(\sqrt{1 + u} - 1)}{(\sqrt{1 + u} - 1)} \times \frac{(\sqrt{1 + u} - 1)}{u(u + 6)} \end{aligned}$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{\sin(\sqrt{1+u}-1)}{(\sqrt{1+u}-1)} \times \frac{(1+u-1)}{u(u+6)(\sqrt{1+u}+1)} \\
&= \lim_{u \rightarrow 0} \frac{\sin(\sqrt{1+u}-1)}{(\sqrt{1+u}-1)} \times \frac{u}{u(u+6)(\sqrt{1+u}+1)} \\
&= \lim_{u \rightarrow 0} \frac{\sin(\sqrt{1+u}-1)}{(\sqrt{1+u}-1)} \times \frac{1}{(u+6)(\sqrt{1+u}+1)} = 1 \times \frac{1}{(6)(1+1)} = \frac{1}{12}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{\sin(\sqrt{x}-1)}{x^2+4x-5} = \frac{1}{12}$

h). $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt{(1-\sin x)^2}}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt{(1-\sin x)^2}} = \lim_{u \rightarrow 0} \frac{\cos(\frac{\pi}{2} + u)}{\sqrt{\left(1 - \sin(\frac{\pi}{2} + u)\right)^2}}$

$$= \lim_{u \rightarrow 0} \frac{-\sin u}{\sqrt{(1-\cos u)^2}} = \lim_{u \rightarrow 0} \frac{-\sin u}{\sqrt{\left(2\sin^2 \frac{u}{2}\right)^2}}$$

$$= \lim_{u \rightarrow 0} \frac{-2\sin \frac{u}{2} \cos \frac{u}{2}}{2\sin^2 \frac{u}{2}} = \lim_{u \rightarrow 0} \frac{-2\cos \frac{u}{2}}{2\sin \frac{u}{2}} = \frac{-2}{0} = \pm \infty$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sqrt{(1-\sin x)^2}} = \pm \infty$

i). $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x + \cos 2x + 1}{\cos 2x + \sin x}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x + \cos 2x + 1}{\cos 2x + \sin x} &= \lim_{u \rightarrow 0} \frac{\cos^2\left(\frac{\pi}{2} + u\right) + \cos 2\left(\frac{\pi}{2} + u\right) + 1}{\cos 2\left(\frac{\pi}{2} + u\right) + \sin\left(\frac{\pi}{2} + u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\sin^2 u + \cos(\pi + 2u) + 1}{\cos(\pi + 2u) + \cos u} = \lim_{u \rightarrow 0} \frac{\sin^2 u - \cos 2u + 1}{-\cos 2u + \cos u} \\
 &= \lim_{u \rightarrow 0} \frac{\sin^2 u + 1 - \cos 2u}{\cos u - \cos 2u} = \lim_{u \rightarrow 0} \frac{\sin^2 u + 2\sin^2 u}{\cos u - (2\cos^2 u - 1)} \\
 &= \lim_{u \rightarrow 0} \frac{3\sin^2 u}{\cos u - 2\cos^2 u + 1} = \lim_{u \rightarrow 0} \frac{3\sin^2 u}{(\cos u - \cos^2 u) + (1 - \cos^2 u)} \\
 &= \lim_{u \rightarrow 0} \frac{3(4\sin^2 \frac{u}{2} \cos^2 \frac{u}{2})}{\cos u(1 - \cos u) + (1 - \cos u)(1 + \cos u)} \\
 &= \lim_{u \rightarrow 0} \frac{12\sin^2 \frac{u}{2} \cos^2 \frac{u}{2}}{(1 - \cos u)(\cos u + 1 + \cos u)} = \lim_{u \rightarrow 0} \frac{12\sin^2 \frac{u}{2} \cos^2 \frac{u}{2}}{2\sin^2 \frac{u}{2} (2\cos u + 1)} \\
 &= \lim_{u \rightarrow 0} \frac{6\cos^2 \frac{u}{2}}{(2\cos u + 1)} = \frac{6}{(2+1)} = \frac{6}{3} = 2
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x + \cos 2x + 1}{\cos 2x + \sin x} = 2$

j). $\lim_{x \rightarrow \frac{\pi}{12}} \frac{\tan 3x - 1}{2\sin 2x - 1}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{12}$ នោះ $x = \frac{\pi}{12} + u$

ពេល $x \rightarrow \frac{\pi}{12}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{12}} \frac{\tan 3x - 1}{2 \sin 2x - 1} &= \lim_{u \rightarrow 0} \frac{\tan 3\left(\frac{\pi}{12} + u\right) - 1}{2 \sin 2\left(\frac{\pi}{12} + u\right) - 1} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{4} + 3u\right) - 1}{2\left(\sin \frac{\pi}{6} \cos 2u + \cos \frac{\pi}{6} \sin 2u\right) - 1} \\
 &= \lim_{u \rightarrow 0} \frac{\frac{\tan \frac{\pi}{4} + \tan 3u}{1 - \tan \frac{\pi}{4} \tan 3u} - 1}{2\left(\sin \frac{\pi}{6} \cos 2u + \cos \frac{\pi}{6} \sin 2u\right) - 1} \\
 &= \lim_{u \rightarrow 0} \frac{\frac{1 + \tan 3u}{1 - \tan 3u} - 1}{\cos 2u + \sqrt{3} \sin 2u - 1} \\
 &= \lim_{u \rightarrow 0} \frac{(1 + \tan 3u) - (1 - \tan 3u)}{(1 - \tan 3u)[\sqrt{3} \sin 2u - (1 - \cos 2u)]} \\
 &= \lim_{u \rightarrow 0} \frac{2 \tan 3u}{(1 - \tan 3u)(2\sqrt{3} \sin u \cos u - 2 \sin^2 u)} \\
 &= \lim_{u \rightarrow 0} \frac{\tan 3u}{\sin u} \times \frac{2}{(1 - \tan 3u)(2\sqrt{3} \cos u - 2 \sin u)} \\
 &= \lim_{u \rightarrow 0} \frac{\tan 3u}{3u} \times \frac{u}{\sin u} \times 3 \times \frac{2}{(1 - \tan 3u)(2\sqrt{3} \cos u - 2 \sin u)} \\
 &= 1 \times 1 \times 3 \times \frac{2}{(1 - 0)(2\sqrt{3} - 0)} = \frac{6}{2\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{12}} \frac{\tan 3x - 1}{2 \sin 2x - 1} = \sqrt{3}$

3) គណនាលីមីតខាងក្រោម:

a) $\lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx}, (m, n \in \mathbb{N})$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \pi$ នោះ $x = \pi + u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} = \lim_{u \rightarrow 0} \frac{\sin m(\pi + u)}{\sin n(\pi + u)} = \lim_{u \rightarrow 0} \frac{\sin(m\pi + mu)}{\sin(n\pi + nu)}$$

❖ ករណី m, n ជាចំនួនគត់ ឬ ជាចំនួនសេស

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} &= \lim_{u \rightarrow 0} \frac{\sin(m\pi + mu)}{\sin(n\pi + nu)} = \lim_{u \rightarrow 0} \frac{\sin(mu)}{\sin(nu)} \\ &= \lim_{u \rightarrow 0} \frac{\sin(mu)}{mu} \times \frac{nu}{\sin(nu)} \times \frac{m}{n} = 1 \times 1 \times \frac{m}{n} = \frac{m}{n} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} = \frac{m}{n}}$$

❖ ករណី (m គូ និង n សេស) ឬ (m សេស និង n គូ)

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} &= \lim_{u \rightarrow 0} \frac{\sin(m\pi + mu)}{\sin(n\pi + nu)} = - \lim_{u \rightarrow 0} \frac{\sin(mu)}{\sin(nu)} \\ &= - \lim_{u \rightarrow 0} \frac{\sin(mu)}{mu} \times \frac{nu}{\sin(nu)} \times \frac{m}{n} = -1 \times 1 \times \frac{m}{n} = -\frac{m}{n} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{\sin mx}{\sin nx} = -\frac{m}{n}}$$

$$\text{b). } \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាង $u = x - \pi$ នោះ $x = \pi + u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} &= \lim_{u \rightarrow 0} \frac{\sin 3(\pi + u)}{\sin 2(\pi + u)} = \lim_{u \rightarrow 0} \frac{\sin(3\pi + 3u)}{\sin(2\pi + 2u)} \\ &= \lim_{u \rightarrow 0} \frac{-\sin(3u)}{\sin(2u)} = - \lim_{u \rightarrow 0} \frac{\sin(3u)}{(3u)} \times \frac{(2u)}{\sin(2u)} \times \frac{3}{2} = -1 \times 1 \times \frac{3}{2} = -\frac{3}{2} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{\sin 3x}{\sin 2x} = -\frac{3}{2}}$$

$$\text{c). } \lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sqrt{2} \cos x}{1 - \sqrt{2} \sin x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាង $u = x - \frac{\pi}{4}$ នោះ $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sqrt{2} \cos x}{1 - \sqrt{2} \sin x} &= \lim_{u \rightarrow 0} \frac{1 - \sqrt{2} \cos\left(\frac{\pi}{4} + u\right)}{1 - \sqrt{2} \sin\left(\frac{\pi}{4} + u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{1 - \sqrt{2} \left(\cos \frac{\pi}{4} \cos u - \sin \frac{\pi}{4} \sin u \right)}{1 - \sqrt{2} \left(\sin \frac{\pi}{4} \cos u + \cos \frac{\pi}{4} \sin u \right)} \\
 &= \lim_{u \rightarrow 0} \frac{1 - \sqrt{2} \left(\frac{\sqrt{2}}{2} \cos u - \frac{\sqrt{2}}{2} \sin u \right)}{1 - \sqrt{2} \left(\frac{\sqrt{2}}{2} \cos u + \frac{\sqrt{2}}{2} \sin u \right)} \\
 &= \lim_{u \rightarrow 0} \frac{1 - \cos u + \sin u}{1 - \cos u - \sin u} = \lim_{u \rightarrow 0} \frac{2 \sin^2 \frac{u}{2} + 2 \sin \frac{u}{2} \cos \frac{u}{2}}{2 \sin^2 \frac{u}{2} - 2 \sin \frac{u}{2} \cos \frac{u}{2}} \\
 &= \lim_{u \rightarrow 0} \frac{\sin \frac{u}{2} + \cos \frac{u}{2}}{\sin \frac{u}{2} - \cos \frac{u}{2}} = \frac{0 + 1}{0 - 1} = \frac{1}{-1} = -1
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sqrt{2} \cos x}{1 - \sqrt{2} \sin x} = -1$

d). $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan x - 1}{(2 \cos x - \sqrt{2})}$ មានរាង $\left(\frac{0}{0} \right)$

តាង $u = x - \frac{\pi}{4}$ នោះ $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan x - 1}{(2 \cos x - \sqrt{2})} &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{4} + u\right) - 1}{2 \cos\left(\frac{\pi}{4} + u\right) - \sqrt{2}} \\
&= \lim_{u \rightarrow 0} \frac{\frac{1 + \tan u}{1 - \tan u} - 1}{2 \left(\cos \frac{\pi}{4} \cos u - \sin \frac{\pi}{4} \sin u \right) - \sqrt{2}} \\
&= \lim_{u \rightarrow 0} \frac{\frac{1 + \tan u}{1 - \tan u} - 1}{2 \left(\frac{\sqrt{2}}{2} \cos u - \frac{\sqrt{2}}{2} \sin u \right) - \sqrt{2}} \\
&= \lim_{u \rightarrow 0} \frac{(1 + \tan u) - (1 - \tan u)}{(1 - \tan u)(\sqrt{2} \cos u - \sqrt{2} \sin u - \sqrt{2})} \\
&= \lim_{u \rightarrow 0} \frac{2 \tan u}{(1 - \tan u)[\sqrt{2}(\cos u - 1) - \sqrt{2} \sin u]} \\
&= \lim_{u \rightarrow 0} \frac{\frac{2 \tan u}{u}}{(1 - \tan u) \left[\frac{\sqrt{2}(\cos u - 1) - \sqrt{2} \sin u}{u} \right]} \\
&= \lim_{u \rightarrow 0} \frac{2 \cdot \frac{\tan u}{u}}{(1 - \tan u) \sqrt{2} \left(\frac{\cos u - 1}{u} - \frac{\sin u}{u} \right)} = \frac{2}{\sqrt{2}(0 - 1)} = -\frac{2}{\sqrt{2}} = -\sqrt{2}
\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan x - 1}{(2 \cos x - \sqrt{2})} = -\sqrt{2}}$

e). $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(x - \frac{\pi}{3}\right)}{1 - 2 \cos x}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - \frac{\pi}{3}$ នោះ $x = \frac{\pi}{3} + u$

ពេល $x \rightarrow \frac{\pi}{3}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(x - \frac{\pi}{3}\right)}{1 - 2\cos x} &= \lim_{u \rightarrow 0} \frac{\sin u}{1 - 2\cos\left(\frac{\pi}{3} + u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\sin u}{1 - 2\left(\cos \frac{\pi}{3} \cos u - \sin \frac{\pi}{3} \sin u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\sin u}{1 - 2\left(\frac{1}{2}\cos u - \frac{\sqrt{3}}{2}\sin u\right)} = \lim_{u \rightarrow 0} \frac{\sin u}{1 - \cos u + \sqrt{3}\sin u} \\
 &= \lim_{u \rightarrow 0} \frac{\frac{\sin u}{u}}{\frac{1 - \cos u + \sqrt{3}\sin u}{u}} = \lim_{u \rightarrow 0} \frac{\frac{\sin u}{u}}{\frac{1 - \cos u}{u} + \sqrt{3} \cdot \frac{\sin u}{u}} \\
 &= \frac{1}{0 + \sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(x - \frac{\pi}{3}\right)}{1 - 2\cos x} = \frac{\sqrt{3}}{3}$

f). $\lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin x - 1}{4\cos^2 x - 3}$ មានរាង $\left(\frac{0}{0}\right)$

តាំង $u = x - \frac{\pi}{6}$ នោះ $x = \frac{\pi}{6} + u$

ពេល $x \rightarrow \frac{\pi}{6}$ នោះ $u \rightarrow 0$

$$\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin x - 1}{4\cos^2 x - 3} = \lim_{u \rightarrow 0} \frac{2\sin\left(\frac{\pi}{6} + u\right) - 1}{4\cos^2\left(\frac{\pi}{6} + u\right) - 3}$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{2 \left(\sin \frac{\pi}{6} \cos u + \cos \frac{\pi}{6} \sin u \right) - 1}{4 \left(\cos \frac{\pi}{6} \cos u - \sin \frac{\pi}{6} \sin u \right)^2 - 3} \\
&= \lim_{u \rightarrow 0} \frac{2 \left(\frac{1}{2} \cos u + \frac{\sqrt{3}}{2} \sin u \right) - 1}{4 \left(\frac{\sqrt{3}}{2} \cos u - \frac{1}{2} \sin u \right)^2 - 3} \\
&= \lim_{u \rightarrow 0} \frac{\cos u + \sqrt{3} \sin u - 1}{(\sqrt{3} \cos u - \sin u)^2 - 3} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{3} \sin u - 1 + \cos u}{(3 \cos^2 u - 2\sqrt{3} \cos u \sin u + \sin^2 u) - 3} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{3} \sin u - (1 - \cos u)}{3(\cos^2 u - 1) - 2\sqrt{3} \cos u \sin u + \sin^2 u} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{3} \sin u - (1 - \cos u)}{-3 \sin^2 u - 2\sqrt{3} \cos u \sin u + \sin^2 u} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{3} \sin u - (1 - \cos u)}{-2 \sin^2 u - 2\sqrt{3} \cos u \sin u} = \lim_{u \rightarrow 0} \frac{\sqrt{3} \sin u - (1 - \cos u)}{\sin u (-2 \sin u - 2\sqrt{3} \cos u)} \\
&= \lim_{u \rightarrow 0} \frac{\sqrt{3} \cdot \frac{\sin u}{u} - \frac{1 - \cos u}{u}}{\frac{\sin u}{u} \times (-2 \sin u - 2\sqrt{3} \cos u)} = \frac{\sqrt{3} - 0}{1 \times (0 - 2\sqrt{3})} = \frac{\sqrt{3}}{-2\sqrt{3}} = -\frac{1}{2}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin x - 1}{4 \cos^2 x - 3} = -\frac{1}{2}$

g). $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan^3 x - 3 \tan x}{\cos \left(x + \frac{\pi}{6} \right)}$ មានរាង $\left(\frac{0}{0} \right)$

តាង $u = x - \frac{\pi}{3}$ នោះ $x = \frac{\pi}{3} + u$

ពេល $x \rightarrow \frac{\pi}{3}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan^3 x - 3 \tan x}{\cos\left(x + \frac{\pi}{6}\right)} &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan x (\tan^2 x - 3)}{\cos\left(x + \frac{\pi}{6}\right)} \\
 &= \lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan x (\tan x + \sqrt{3})(\tan x - \sqrt{3})}{\cos\left(x + \frac{\pi}{6}\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] \left[\tan\left(\frac{\pi}{3} + u\right) - \sqrt{3} \right]}{\cos\left(\frac{\pi}{3} + u + \frac{\pi}{6}\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] \left[\frac{\tan \frac{\pi}{3} + \tan u}{1 - \tan \frac{\pi}{3} \tan u} - \sqrt{3} \right]}{\cos\left(\frac{\pi}{2} + u\right)} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] \left[\frac{\sqrt{3} + \tan u}{1 - \sqrt{3} \tan u} - \sqrt{3} \right]}{-\sin u} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] \left[\frac{(\sqrt{3} + \tan u) - \sqrt{3}(1 - \sqrt{3} \tan u)}{1 - \sqrt{3} \tan u} \right]}{-\sin u} \\
 &= \lim_{u \rightarrow 0} \frac{\tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] (4 \tan u)}{-\sin u (1 - \sqrt{3} \tan u)} \\
 &= \lim_{u \rightarrow 0} \frac{4 \tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right] (\sin u)}{-\sin u (1 - \sqrt{3} \tan u) (\cos u)}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{4 \tan\left(\frac{\pi}{3} + u\right) \left[\tan\left(\frac{\pi}{3} + u\right) + \sqrt{3} \right]}{-(1 - \sqrt{3} \tan u)(\cos u)} \\
&= \frac{4(\sqrt{3})(\sqrt{3} + \sqrt{3})}{-(1-0)(1)} = \frac{4\sqrt{3} \times 2\sqrt{3}}{-1} = -8 \times 3 = -24
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\tan^3 x - 3 \tan x}{\cos\left(x + \frac{\pi}{6}\right)} = -24$

h). $\lim_{x \rightarrow a} \frac{\tan x - \tan a}{\cos x - \cos a}$ មានរាង $\left(\frac{0}{0}\right)$

តាង $u = x - a$ នោះ: $x = u + a$

ពេល $x \rightarrow a$ នោះ: $u \rightarrow 0$

$$\begin{aligned}
\text{គេបាន } \lim_{x \rightarrow a} \frac{\tan x - \tan a}{\cos x - \cos a} &= \lim_{x \rightarrow a} \frac{\frac{\sin(x-a)}{\cos x \cos a}}{-2 \sin\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right)} \\
&= \lim_{x \rightarrow a} \frac{\sin(x-a)}{-2 \sin\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right) \cos x \cos a} \\
&= \lim_{u \rightarrow 0} \frac{\sin u}{-2 \sin\left(\frac{u+2a}{2}\right) \sin\left(\frac{u}{2}\right) \cos(u+a) \cos a} \\
&= \lim_{u \rightarrow 0} \frac{2 \sin\left(\frac{u}{2}\right) \cos\left(\frac{u}{2}\right)}{-2 \sin\left(\frac{u+2a}{2}\right) \sin\left(\frac{u}{2}\right) \cos(u+a) \cos a} \\
&= \lim_{u \rightarrow 0} \frac{\cos\left(\frac{u}{2}\right)}{-\sin\left(\frac{u+2a}{2}\right) \cos(u+a) \cos a} = \frac{1}{-\sin a \cos a \cos a} \\
&= -\frac{2}{2 \sin a \cos a \cos a} = -\frac{2}{\sin 2a \cos a}
\end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow a} \frac{\tan x - \tan a}{\cos x - \cos a} = -\frac{2}{\sin 2a \cos a}$$

i).
$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin\left(x - \frac{\pi}{6}\right)}{\frac{\sqrt{3}}{2} - \cos x} \text{ មានរាង } \left(\frac{0}{0}\right)$$

តាង $u = x - \frac{\pi}{6}$ នោះ $x = \frac{\pi}{6} + u$

ពេល $x \rightarrow \frac{\pi}{6}$ នោះ $u \rightarrow 0$

គេបាន
$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin\left(x - \frac{\pi}{6}\right)}{\frac{\sqrt{3}}{2} - \cos x} &= \lim_{u \rightarrow 0} \frac{\sin u}{\frac{\sqrt{3}}{2} - \cos\left(\frac{\pi}{6} + u\right)} \\ &= \lim_{u \rightarrow 0} \frac{\sin u}{\frac{\sqrt{3}}{2} - \cos \frac{\pi}{6} \cos u + \sin \frac{\pi}{6} \sin u} \\ &= \lim_{u \rightarrow 0} \frac{\sin u}{\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \cos u + \frac{1}{2} \sin u} = \lim_{u \rightarrow 0} \frac{\sin u}{\frac{\sqrt{3}}{2} (1 - \cos u) + \frac{1}{2} \sin u} \\ &= \lim_{u \rightarrow 0} \frac{\frac{\sin u}{u}}{\frac{\sqrt{3}}{2} \times \frac{1 - \cos u}{u} + \frac{1}{2} \times \frac{\sin u}{u}} = \frac{1}{\frac{\sqrt{3}}{2} \times 0 + \frac{1}{2}} = \frac{1}{0 + \frac{1}{2}} = 2 \end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin\left(x - \frac{\pi}{6}\right)}{\frac{\sqrt{3}}{2} - \cos x} = 2$$

j).
$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x^2 - \left(\frac{\pi}{4}\right)^2} \text{ មានរាង } \left(\frac{0}{0}\right)$$

តាំង $u = x - \frac{\pi}{4}$ នោះ $x = u + \frac{\pi}{4}$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x^2 - \left(\frac{\pi}{4}\right)^2} &= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(\sin x - \frac{\sqrt{2}}{2})(\sin x + \frac{\sqrt{2}}{2})}{(x - \frac{\pi}{4})(x + \frac{\pi}{4})} \\
 &= \lim_{u \rightarrow 0} \frac{\left[\sin\left(\frac{\pi}{4} + u\right) - \frac{\sqrt{2}}{2} \right] \left[\sin\left(\frac{\pi}{4} + u\right) + \frac{\sqrt{2}}{2} \right]}{u \left(\frac{\pi}{4} + u + \frac{\pi}{4} \right)} \\
 &= \lim_{u \rightarrow 0} \frac{\left[\frac{\sqrt{2}}{2} \cos u + \frac{\sqrt{2}}{2} \sin u - \frac{\sqrt{2}}{2} \right] \left[\sin\left(\frac{\pi}{4} + u\right) + \frac{\sqrt{2}}{2} \right]}{u \left(\frac{\pi}{2} + u \right)} \\
 &= \lim_{u \rightarrow 0} \frac{\left[\frac{\sqrt{2}}{2} (\cos u - 1) + \frac{\sqrt{2}}{2} \sin u \right] \left[\sin\left(\frac{\pi}{4} + u\right) + \frac{\sqrt{2}}{2} \right]}{u \left(\frac{\pi}{2} + u \right)} \\
 &= \lim_{u \rightarrow 0} \frac{\frac{\sqrt{2}}{2} [(\cos u - 1) + \sin u]}{u} \times \frac{\sin\left(\frac{\pi}{4} + u\right) + \frac{\sqrt{2}}{2}}{\frac{\pi}{2} + u} \\
 &= \lim_{u \rightarrow 0} \frac{\sqrt{2}}{2} \left(\frac{\cos u - 1}{u} + \frac{\sin u}{u} \right) \times \frac{\sin\left(\frac{\pi}{4} + u\right) + \frac{\sqrt{2}}{2}}{\frac{\pi}{2} + u} \\
 &= \frac{\sqrt{2}}{2} (0 + 1) \times \frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}}{\frac{\pi}{2}} = \frac{\frac{\sqrt{2}}{2} \times \sqrt{2}}{\frac{\pi}{2}} = \frac{2}{\pi}
 \end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x - \frac{1}{2}}{x^2 - \left(\frac{\pi}{4}\right)^2} = \frac{2}{\pi}$$

k).
$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x + \cos x}{1 - \sin x - \cos x} \text{ មានរាង } \left(\frac{0}{0} \right)$$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

គេបាន
$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x + \cos x}{1 - \sin x - \cos x} = \lim_{u \rightarrow 0} \frac{1 - \sin\left(\frac{\pi}{2} + u\right) + \cos\left(\frac{\pi}{2} + u\right)}{1 - \sin\left(\frac{\pi}{2} + u\right) - \cos\left(\frac{\pi}{2} + u\right)}$$

$$= \lim_{u \rightarrow 0} \frac{1 - \cos u - \sin u}{1 - \cos u + \sin u} = \lim_{u \rightarrow 0} \frac{2 \sin^2 \frac{u}{2} - 2 \sin \frac{u}{2} \cos \frac{u}{2}}{2 \sin^2 \frac{u}{2} + 2 \sin \frac{u}{2} \cos \frac{u}{2}}$$

$$= \lim_{u \rightarrow 0} \frac{\frac{\sin \frac{u}{2} - \cos \frac{u}{2}}{2}}{\frac{\sin \frac{u}{2} + \cos \frac{u}{2}}{2}} = \frac{0 - 1}{0 + 1} = \frac{-1}{1} = -1$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x + \cos x}{1 - \sin x - \cos x} = -1$$

២. លីមីតរាងមិនកំណត់ ($0 \times \infty$)

លំហាត់

1) គណនាលីមីតខាងក្រោម:

a).
$$\lim_{x \rightarrow 0} x \cot x$$

b).
$$\lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2}{\cos x} + \cos x - 3 \right)$$

c).
$$\lim_{x \rightarrow \frac{\pi}{2}} (\pi - 2x) \tan x$$

d).
$$\lim_{x \rightarrow \frac{\pi}{3}} \left(\frac{x}{2} - \frac{\pi}{3} \cos x \right) \cdot \frac{1}{x - \frac{\pi}{3}}$$

$$e). \lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos 2x) \tan x$$

$$f). \lim_{x \rightarrow \frac{\pi}{2}} \tan x \cot \left(\frac{x}{2} + \frac{\pi}{4} \right)$$

$$g). \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \tan x$$

$$h). \lim_{x \rightarrow a} \sin \frac{(x-a)}{2} \cdot \tan \frac{\pi x}{2a}$$

$$i). \lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$$

$$j). \lim_{x \rightarrow \infty} x \sin \frac{\pi}{x}$$

2) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow +\infty} x^2 \left(1 - \cos \frac{1}{x} \right)$$

$$b). \lim_{x \rightarrow \infty} (5x+1) \tan \frac{2}{x}$$

$$c). \lim_{x \rightarrow \infty} (x+4) \sin \frac{3}{x}$$

$$d). \lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan^2 x$$

$$e). \lim_{x \rightarrow -\frac{\pi}{2}} (1 + \sin x) \tan^2 x$$

$$f). \lim_{x \rightarrow +\infty} x \sin \frac{1}{x}$$

$$g). \lim_{x \rightarrow \pi} \tan x \tan \frac{x}{2}$$

$$h). \lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \tan 2x$$

$$i). \lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \frac{\tan 2x}{\tan 4x}$$

$$j). \lim_{x \rightarrow 2} (x-2) \tan \frac{\pi}{x}$$

ជំនោះស្រាវជ្រាវ

1) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 0} x \cot x \text{ មានរាង } (0 \times \infty)$$

$$= \lim_{x \rightarrow 0} x \cdot \frac{\cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{x}{\sin x} \times \cos x = 1 \times 1 = 1$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} x \cot x = 1}$$

$$b). \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2}{\cos x} + \cos x - 3 \right) \text{ មានរាង } (\infty \times 0)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2(\sin^2 x + \cos^2 x)}{\cos x} + \cos x - 3 \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2\sin^2 x}{\cos x} + \frac{2\cos^2 x}{\cos x} + \cos x - 3 \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2\sin^2 x}{\cos x} + 2\cos x + \cos x - 3 \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2\sin^2 x}{\cos x} + 3(\cos x - 1) \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2\sin^2 x}{\cos x} - 3 \cdot 2\sin^2 \frac{x}{2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \times \frac{2}{\cos x} - \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{1}{4} \times 6 \right)$$

$$= 1 \times 2 - 1 \times \frac{6}{4} = 2 - \frac{3}{2} = \frac{4-3}{2} = \frac{1}{2}$$

$$\boxed{\text{ដូច្នេះ: } \lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{2}{\cos x} + \cos x - 3 \right) = \frac{1}{2}}$$

c). $\lim_{x \rightarrow \frac{\pi}{2}} (\pi - 2x) \tan x$ មានរាង $(0 \times \infty)$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{2}} (\pi - 2x) \tan x = \lim_{u \rightarrow 0} (-2u) \tan \left(\frac{\pi}{2} + u \right)$

$$= \lim_{u \rightarrow 0} (-2u)(-\cot u) = \lim_{u \rightarrow 0} 2u \cdot \frac{\cos u}{\sin u}$$

$$= \lim_{u \rightarrow 0} \frac{u}{\sin u} \times 2 \cos u = 1 \times 2 = 2$$

ដូច្នេះ: $\boxed{\lim_{x \rightarrow \frac{\pi}{2}} (\pi - 2x) \tan x = 2}$

d). $\lim_{x \rightarrow \frac{\pi}{3}} \left(\frac{x}{2} - \frac{\pi}{3} \cos x \right) \cdot \frac{1}{x - \frac{\pi}{3}}$ មានរាង $(0 \times \infty)$

តាង $u = x - \frac{\pi}{3}$ នោះ $x = \frac{\pi}{3} + u$

ពេល $x \rightarrow \frac{\pi}{3}$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{3}} \left(\frac{x}{2} - \frac{\pi}{3} \cos x \right) \cdot \frac{1}{x - \frac{\pi}{3}}$

$$= \lim_{u \rightarrow 0} \left(\frac{\frac{\pi}{3} + u}{2} - \frac{\pi}{3} \cos \left(\frac{\pi}{3} + u \right) \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\left(\frac{\pi}{6} + \frac{u}{2} \right) - \frac{\pi}{3} \cos \left(\frac{\pi}{3} + u \right) \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\left(\frac{\pi}{6} + \frac{u}{2} \right) - \frac{\pi}{3} \left(\cos \frac{\pi}{3} \cos u - \sin \frac{\pi}{3} \sin u \right) \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\left(\frac{\pi}{6} + \frac{u}{2} \right) - \frac{\pi}{3} \left(\frac{1}{2} \cos u - \frac{\sqrt{3}}{2} \sin u \right) \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\frac{\pi}{6} + \frac{u}{2} - \frac{\pi}{6} \cos u + \frac{\pi\sqrt{3}}{6} \sin u \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\frac{\pi}{6} (1 - \cos u) + \frac{u}{2} + \frac{\pi\sqrt{3}}{6} \sin u \right) \cdot \frac{1}{u}$$

$$= \lim_{u \rightarrow 0} \left(\frac{\pi}{6} \cdot \frac{1 - \cos u}{u} + \frac{1}{2} + \frac{\pi\sqrt{3}}{6} \cdot \frac{\sin u}{u} \right) = 0 + \frac{1}{2} + \frac{\pi\sqrt{3}}{2} = \frac{1 + \sqrt{3}\pi}{2}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \frac{\pi}{3}} \left(\frac{x}{2} - \frac{\pi}{3} \cos x \right) \cdot \frac{1}{x - \frac{\pi}{3}} = \frac{1 + \sqrt{3} \pi}{2}}$$

e). $\lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos 2x) \tan x$ មានរាង $(0 \times \infty)$

$$= \lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos 2x) \tan x = \lim_{x \rightarrow \frac{\pi}{2}} 2 \cos^2 x \times \frac{\sin x}{\cos x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} 2 \cos x \times \sin x = 2(0)(1) = 0$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \frac{\pi}{2}} (1 + \cos 2x) \tan x = 0}$

f). $\lim_{x \rightarrow \frac{\pi}{2}} \tan x \cot \left(\frac{x}{2} + \frac{\pi}{4} \right)$ មានរាង $(\infty \times 0)$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

$$\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{2}} \tan x \cot \left(\frac{x}{2} + \frac{\pi}{4} \right) = \lim_{u \rightarrow 0} \tan \left(\frac{\pi}{2} + u \right) \cot \left(\frac{\frac{\pi}{2} + u}{2} + \frac{\pi}{4} \right)$$

$$= \lim_{u \rightarrow 0} (-\cot u) \cot \left(\frac{\pi}{4} + \frac{u}{2} + \frac{\pi}{4} \right) = \lim_{u \rightarrow 0} (-\cot u) \cot \left(\frac{\pi}{2} + \frac{u}{2} \right)$$

$$= \lim_{u \rightarrow 0} (-\cot u) \left(-\tan \frac{u}{2} \right) = \lim_{u \rightarrow 0} \frac{\cos u}{\sin u} \cdot \frac{\sin \frac{u}{2}}{\cos \frac{u}{2}}$$

$$= \lim_{u \rightarrow 0} \frac{\cos u}{2 \sin \frac{u}{2} \cos \frac{u}{2}} \cdot \frac{\sin \frac{u}{2}}{\cos \frac{u}{2}} = \lim_{u \rightarrow 0} \frac{\cos u}{2 \cos \frac{u}{2}} \cdot \frac{1}{\cos \frac{u}{2}} = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \frac{\pi}{2}} \tan x \cot \left(\frac{x}{2} + \frac{\pi}{4} \right) = \frac{1}{2}}$$

$$\text{g). } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \tan x \text{ មានរាង } (0 \times \infty)$$

$$\text{តាង } u = x - \frac{\pi}{2} \text{ នោះ } x = \frac{\pi}{2} + u$$

$$\text{ពេល } x \rightarrow \frac{\pi}{2} \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \tan x = \lim_{u \rightarrow 0} u \tan \left(\frac{\pi}{2} + u \right)$$

$$= \lim_{u \rightarrow 0} u(-\cot u) = \lim_{u \rightarrow 0} \frac{u}{\sin u} \cdot (-\cos u) = 1(-1) = -1$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \tan x = -1}$$

$$\text{h). } \lim_{x \rightarrow a} \sin \frac{(x-a)}{2} \cdot \tan \frac{\pi x}{2a} \text{ មានរាង } (0 \times \infty)$$

$$\text{តាង } u = x - a \text{ នោះ } x = a + u$$

$$\text{ពេល } x \rightarrow a \text{ នោះ } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow a} \sin \frac{(x-a)}{2} \cdot \tan \frac{\pi x}{2a} = \lim_{u \rightarrow 0} \sin \frac{u}{2} \cdot \tan \frac{\pi(a+u)}{2a}$$

$$= \lim_{u \rightarrow 0} \sin \frac{u}{2} \cdot \tan \left(\frac{\pi}{2} + \frac{\pi u}{2a} \right) = \lim_{u \rightarrow 0} \sin \frac{u}{2} \cdot (-\cot \frac{\pi u}{2a})$$

$$= - \lim_{u \rightarrow 0} \sin \frac{u}{2} \cdot \frac{\cos \frac{\pi u}{2a}}{\sin \frac{\pi u}{2a}} = - \lim_{u \rightarrow 0} \frac{\sin \frac{u}{2}}{\frac{u}{2}} \times \frac{\frac{\pi u}{2a}}{\sin \frac{\pi u}{2a}} \times \frac{a}{\pi} \times \cos \frac{\pi u}{2a}$$

$$= -1 \times 1 \times \frac{a}{\pi} \times 1 = -\frac{a}{\pi}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow a} \sin \frac{(x-a)}{2} \cdot \tan \frac{\pi x}{2a} = -\frac{a}{\pi}}$

i). $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$ មានរាង $(0 \times \infty)$

តាង $u = 1-x$ នោះ: $x = 1-u$

ពេល $x \rightarrow 1$ នោះ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} = \lim_{u \rightarrow 0} u \tan \frac{\pi(1-u)}{2}$

$$= \lim_{u \rightarrow 0} u \tan \left(\frac{\pi}{2} - \frac{\pi}{2} u \right) = \lim_{u \rightarrow 0} u \cot \frac{\pi}{2} u = \lim_{u \rightarrow 0} u \cdot \frac{\cos \frac{\pi}{2} u}{\sin \frac{\pi}{2} u}$$

$$= \lim_{u \rightarrow 0} \frac{\frac{\pi}{2} u}{\sin \frac{\pi}{2} u} \times \frac{2}{\pi} \times \cos \frac{\pi}{2} u = 1 \times \frac{2}{\pi} \times 1 = \frac{2}{\pi}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} = \frac{2}{\pi}}$

j). $\lim_{x \rightarrow \infty} x \sin \frac{\pi}{x}$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ: $x = \frac{1}{u}$

ពេល $x \rightarrow \infty$ នោះ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \infty} x \sin \frac{\pi}{x} = \lim_{u \rightarrow 0} \frac{1}{u} \sin \pi u = \lim_{u \rightarrow 0} \frac{\sin \pi u}{\pi u} \times \pi = 1 \times \pi = \pi$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} x \sin \frac{\pi}{x} = \pi}$

2) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow +\infty} x^2 (1 - \cos \frac{1}{x})$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ: $x = \frac{1}{u}$

ពេល $x \rightarrow +\infty$ នោះ $u \rightarrow 0$

$$\begin{aligned}\text{គេបាន } \lim_{x \rightarrow +\infty} x^2(1 - \cos \frac{1}{x}) &= \lim_{u \rightarrow 0} \left(\frac{1}{u}\right)^2 (1 - \cos u) \\ &= \lim_{u \rightarrow 0} \frac{2 \sin^2 \frac{u}{2}}{u^2} = \lim_{u \rightarrow 0} \frac{\sin^2 \frac{u}{2}}{\left(\frac{u}{2}\right)^2} \times \frac{1}{4} \times 2 = 1 \times \frac{2}{4} = \frac{1}{2}\end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} x^2(1 - \cos \frac{1}{x}) = \frac{1}{2}}$$

b). $\lim_{x \rightarrow \infty} (5x+1) \tan \frac{2}{x}$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ $x = \frac{1}{u}$

ពេល $x \rightarrow \infty$ នោះ $u \rightarrow 0$

$$\begin{aligned}\text{គេបាន } \lim_{x \rightarrow \infty} (5x+1) \tan \frac{2}{x} &= \lim_{u \rightarrow 0} \left(5 \cdot \frac{1}{u} + 1\right) \tan 2u \\ &= \lim_{u \rightarrow 0} \left(\frac{5 \tan 2u}{u} + \tan 2u\right) = \lim_{u \rightarrow 0} \left(\frac{\tan 2u}{2u} \times 5 \times 2 + \tan 2u\right) \\ &= 1 \times 5 \times 2 + 0 = 10\end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \infty} (5x+1) \tan \frac{2}{x} = 10}$$

c). $\lim_{x \rightarrow \infty} (x+4) \sin \frac{3}{x}$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ $x = \frac{1}{u}$

ពេល $x \rightarrow \infty$ នោះ $u \rightarrow 0$

$$\begin{aligned}\text{គេបាន } \lim_{x \rightarrow \infty} (x+4) \sin \frac{3}{x} &= \lim_{u \rightarrow 0} \left(\frac{1}{u} + 4\right) \sin 3u \\ &= \lim_{u \rightarrow 0} \left(\frac{\sin 3u}{u} + 4 \sin 3u\right) = \lim_{u \rightarrow 0} \left(\frac{\sin 3u}{3u} \times 3 + 4 \sin 3u\right) \\ &= 1 \times 3 + 0 = 3\end{aligned}$$

ដូច្នេះ: $\lim_{x \rightarrow \infty} (x+4) \sin \frac{3}{x} = 3$

d). $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan^2 x$ មានរាង $(0 \times \infty)$

តាង $u = x - \frac{\pi}{2}$ នោះ $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ $u \rightarrow 0$

$$\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan^2 x = \lim_{u \rightarrow 0} \left(1 - \sin \left(\frac{\pi}{2} + u \right) \right) \tan^2 \left(\frac{\pi}{2} + u \right)$$

$$= \lim_{u \rightarrow 0} (1 - \cos u) \cot^2 u = \lim_{u \rightarrow 0} 2 \sin^2 \frac{u}{2} \times \frac{\cos^2 u}{\sin^2 u}$$

$$= \lim_{u \rightarrow 0} 2 \sin^2 \frac{u}{2} \times \frac{\cos^2 u}{4 \sin^2 \frac{u}{2} \cos^2 \frac{u}{2}} = \lim_{u \rightarrow 0} \frac{\cos^2 u}{2 \cos^2 \frac{u}{2}} = \frac{1}{2}$$

ដូច្នេះ: $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x) \tan^2 x = \frac{1}{2}$

e). $\lim_{x \rightarrow -\frac{\pi}{2}} (1 + \sin x) \tan^2 x$ មានរាង $(0 \times \infty)$

តាង $u = x + \frac{\pi}{2}$ នោះ $x = u - \frac{\pi}{2}$

ពេល $x \rightarrow -\frac{\pi}{2}$ នោះ $u \rightarrow 0$

$$\lim_{x \rightarrow -\frac{\pi}{2}} (1 + \sin x) \tan^2 x = \lim_{u \rightarrow 0} \left[1 + \sin \left(u - \frac{\pi}{2} \right) \tan^2 \left(u - \frac{\pi}{2} \right) \right]$$

$$= \lim_{u \rightarrow 0} \left[1 - \sin \left(\frac{\pi}{2} - u \right) \tan^2 \left(\frac{\pi}{2} - u \right) \right] = \lim_{u \rightarrow 0} (1 - \cos u) \cot^2 u$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} 2 \sin^2 \frac{u}{2} \times \frac{\cos^2 u}{\sin^2 u} = \lim_{u \rightarrow 0} 2 \sin^2 \frac{u}{2} \times \frac{\cos^2 u}{4 \sin^2 \frac{u}{2} \cos^2 \frac{u}{2}} \\
&= \lim_{u \rightarrow 0} \frac{\cos^2 u}{2 \cos^2 \frac{u}{2}} = \frac{1}{2 \times 1} = \frac{1}{2}
\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow -\frac{\pi}{2}} (1 + \sin x) \tan^2 x = \frac{1}{2}}$

f). $\lim_{x \rightarrow +\infty} x \sin \frac{1}{x}$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ $x = \frac{1}{u}$

ពេល $x \rightarrow +\infty$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow +\infty} x \sin \frac{1}{x} = \lim_{u \rightarrow 0} \frac{1}{u} \sin u = \lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$

ដូចនេះ: $\boxed{\lim_{x \rightarrow +\infty} x \sin \frac{1}{x} = 1}$

g). $\lim_{x \rightarrow \pi} \tan x \tan \frac{x}{2}$ មានរាង $(\infty \times 0)$

តាង $u = x - \pi$ នោះ $x = \pi + u$

ពេល $x \rightarrow \pi$ នោះ $u \rightarrow 0$

$$\begin{aligned}
&\text{គេបាន } \lim_{x \rightarrow \pi} \tan x \tan \frac{x}{2} = \lim_{u \rightarrow 0} \tan(\pi + u) \tan \frac{\pi + u}{2} \\
&= \lim_{u \rightarrow 0} \tan(\pi + u) \tan \left(\frac{\pi}{2} + \frac{u}{2} \right) = \lim_{u \rightarrow 0} \tan u \times (-\cot \frac{u}{2}) \\
&= - \lim_{u \rightarrow 0} \frac{\sin u}{\cos u} \times \frac{\cos \frac{u}{2}}{\sin \frac{u}{2}} = - \lim_{u \rightarrow 0} \frac{2 \sin \frac{u}{2} \cos \frac{u}{2}}{\cos u} \times \frac{\cos \frac{u}{2}}{\sin \frac{u}{2}}
\end{aligned}$$

$$= - \lim_{u \rightarrow 0} \frac{2 \cos \frac{u}{2}}{\cos u} \times \cos \frac{u}{2} = - \frac{2}{1} \times 1 = -2$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \pi} \tan x \tan \frac{x}{2} = -2}$

h). $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \tan 2x$ មានរាង $(0 \times \infty)$

តាង $u = x - \frac{\pi}{4}$ នោះ: $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \tan 2x$

$$= \lim_{u \rightarrow 0} \left[1 - \sin 2 \left(\frac{\pi}{4} + u \right) \right] \tan 2 \left(\frac{\pi}{4} + u \right)$$

$$= \lim_{u \rightarrow 0} \left[1 - \sin \left(\frac{\pi}{2} + 2u \right) \right] \tan \left(\frac{\pi}{2} + 2u \right)$$

$$= \lim_{u \rightarrow 0} (1 - \cos 2u)(-\cot 2u) = - \lim_{u \rightarrow 0} 2 \sin^2 u \times \frac{\cos 2u}{\sin 2u}$$

$$= - \lim_{u \rightarrow 0} 2 \sin^2 u \times \frac{\cos 2u}{2 \sin u \cos u} = - \lim_{u \rightarrow 0} \sin u \times \frac{\cos 2u}{\cos u} = -0 \times 1 = 0$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \tan 2x = 0}$

i). $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \frac{\tan 2x}{\tan 4x}$ មានរាង $(0 \times \infty)$

តាង $u = x - \frac{\pi}{4}$ នោះ: $x = \frac{\pi}{4} + u$

ពេល $x \rightarrow \frac{\pi}{4}$ នោះ: $u \rightarrow 0$

$$\text{គេបាន } \lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \frac{\tan 2x}{\tan 4x}$$

$$= \lim_{u \rightarrow 0} \left[1 - \sin 2\left(\frac{\pi}{4} + u\right) \right] \frac{\tan 2\left(\frac{\pi}{4} + u\right)}{\tan 4\left(\frac{\pi}{4} + u\right)}$$

$$= \lim_{u \rightarrow 0} \left[1 - \sin\left(\frac{\pi}{2} + 2u\right) \right] \frac{\tan\left(\frac{\pi}{2} + 2u\right)}{\tan(\pi + 4u)}$$

$$= \lim_{u \rightarrow 0} (1 - \cos 2u) \frac{-\cot 2u}{\tan 4u} = - \lim_{u \rightarrow 0} 2\sin^2 u \times \frac{\cos 2u}{\sin 2u} \times \frac{\cos 4u}{\sin 4u}$$

$$= - \lim_{u \rightarrow 0} 2\sin^2 u \times \frac{\cos 2u}{2\sin u \cos u} \times \frac{\cos 4u}{2\sin 2u \cos 2u}$$

$$= - \lim_{u \rightarrow 0} 2\sin^2 u \times \frac{\cos 2u}{2\sin u \cos u} \times \frac{\cos 4u}{4\sin u \cos u \cos 2u}$$

$$= - \lim_{u \rightarrow 0} \frac{\cos 2u}{\cos u} \times \frac{\cos 4u}{4\cos u \cos 2u} = -1 \times \frac{1}{4} = -\frac{1}{4}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \frac{\pi}{4}} (1 - \sin 2x) \frac{\tan 2x}{\tan 4x} = -\frac{1}{4}}$$

$$\text{j). } \lim_{x \rightarrow 2} (x - 2) \tan \frac{\pi}{x} \text{ មានរាង } (0 \times \infty)$$

$$\text{តាង } u = x - 2 \text{ នោះ: } x = u + 2$$

$$\text{ពេល } x \rightarrow 2 \text{ នោះ: } u \rightarrow 0$$

$$\text{គេបាន } \lim_{x \rightarrow 2} (x - 2) \tan \frac{\pi}{x} = \lim_{u \rightarrow 0} u \tan \frac{\pi}{u + 2} = \lim_{u \rightarrow 0} u \times \frac{\sin\left(\frac{\pi}{u + 2}\right)}{\cos\left(\frac{\pi}{u + 2}\right)}$$

$$= \lim_{u \rightarrow 0} u \times \frac{\sin\left(\frac{\pi}{u + 2}\right)}{\sin\left(\frac{\pi}{2} - \frac{\pi}{u + 2}\right)} = \lim_{u \rightarrow 0} \frac{u \sin\left(\frac{\pi}{u + 2}\right)}{\sin\left(\frac{\pi(u + 2) - 2\pi}{2(u + 2)}\right)}$$

$$\begin{aligned}
&= \lim_{u \rightarrow 0} \frac{u \sin\left(\frac{\pi}{u+2}\right)}{\sin\left(\frac{\pi u}{2(u+2)}\right)} = \lim_{u \rightarrow 0} \frac{\left(\frac{\pi u}{2(u+2)}\right)}{\sin\left(\frac{\pi u}{2(u+2)}\right)} \times \frac{u \sin\left(\frac{\pi}{u+2}\right)}{\left(\frac{\pi u}{2(u+2)}\right)} \\
&= \lim_{u \rightarrow 0} \frac{\left(\frac{\pi u}{2(u+2)}\right)}{\sin\left(\frac{\pi u}{2(u+2)}\right)} \times \frac{\sin\left(\frac{\pi}{u+2}\right)}{\left(\frac{\pi}{2(u+2)}\right)} = 1 \times \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi}
\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 2} (x-2) \tan \frac{\pi}{x} = \frac{4}{\pi}}$

5. លីមីតនៃអនុគមន៍អិចស្ប៉ូណង់ស្យែល

ក. អនុគមន៍អិចស្ប៉ូណង់ស្យែល

❖ $y = a^x, (a > 0, a \neq 1, x \in \mathbb{R})$ ហៅថាអនុគមន៍អិចស្ប៉ូណង់ស្យែលគោល a ។

❖ បើ $a = e$ នោះ: $y = e^x$ ហៅថាអនុគមន៍អិចស្ប៉ូណង់ស្យែលគោល e ឬ អនុគមន៍អិចស្ប៉ូណង់ស្យែលធម្មជាតិ។ ($e = 2.7$)

ខ. លីមីតនៃអនុគមន៍អិចស្ប៉ូណង់ស្យែល

❶ $\lim_{x \rightarrow -\infty} e^x = 0$

❷ $\lim_{x \rightarrow +\infty} e^x = +\infty$

❸ $\lim_{x \rightarrow +\infty} \frac{e^x}{x^n} = +\infty$

❹ $\lim_{x \rightarrow +\infty} \frac{x^n}{e^x} = 0$

❺ $\lim_{x \rightarrow -\infty} x^n e^x = 0$

លំហាត់

1) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow +\infty} e^{-x}$

b). $\lim_{x \rightarrow -\infty} e^{x^2}$

c). $\lim_{x \rightarrow -\infty} e^{2-3x}$

d). $\lim_{x \rightarrow +\infty} (e^x - 7x + 6)$

e). $\lim_{x \rightarrow -\infty} (e^{-x} - 5)$

f). $\lim_{x \rightarrow -\infty} (e^x + 3)$

g). $\lim_{x \rightarrow -\infty} (2e^x - x^2 + 1)$

h). $\lim_{x \rightarrow +\infty} (x - 2 + xe^x)$

i). $\lim_{x \rightarrow 0} (e^x + 2x^2 - 6)$

j). $\lim_{x \rightarrow -\infty} e^x (x^3 + 1)$

2) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow +\infty} (x^3 + x)e^{-x}$

b). $\lim_{x \rightarrow +\infty} (e^x - e^{-x})$

c). $\lim_{x \rightarrow -\infty} \frac{e^{2x} + 1}{e^x + 3}$

d). $\lim_{x \rightarrow +\infty} \frac{xe^x + 2}{x^2 - 1}$

e). $\lim_{x \rightarrow +\infty} \frac{3e^x - 2}{5e^x + 3}$

f). $\lim_{x \rightarrow -\infty} xe^x$

g). $\lim_{x \rightarrow +\infty} \frac{2e^x - 1}{x^2 + x - 3}$

h). $\lim_{x \rightarrow 0} \frac{e^x + x + 1}{2e^x + 9}$

i). $\lim_{x \rightarrow -\infty} (xe^x + e^x + 2)$

j). $\lim_{x \rightarrow +\infty} \frac{2e^x + 3}{x^2 + x + 1}$

3) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow +\infty} \frac{x^3 - e^x}{2e^x + 3x + 1}$

b). $\lim_{x \rightarrow +\infty} \frac{2e^x - x}{1 + e^x}$

c). $\lim_{x \rightarrow +\infty} \frac{x - e^x}{2e^x + 3}$

d). $\lim_{x \rightarrow +\infty} \frac{e^{2x} - 3e^x + 2}{2e^{2x} + 5e^x - 7}$

e). $\lim_{x \rightarrow -\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$

f). $\lim_{x \rightarrow +\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$

g). $\lim_{x \rightarrow +\infty} \frac{3e^{x-2}}{x^3}$

h). $\lim_{x \rightarrow -\infty} \frac{1}{1 + e^x}$

i). $\lim_{x \rightarrow +\infty} (x+1)e^{-x}$

j). $\lim_{x \rightarrow +\infty} \frac{1 + e^x - xe^{2x}}{x^2 e^x + e^{2x}}$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow +\infty} e^{-x} = e^{-(+\infty)} = e^{-\infty} = 0$

b). $\lim_{x \rightarrow -\infty} e^{x^2} = e^{(-\infty)^2} = e^{+\infty} = +\infty$

c). $\lim_{x \rightarrow -\infty} e^{2-3x} = e^{2-3(-\infty)} = e^{+\infty} = +\infty$

d). $\lim_{x \rightarrow +\infty} (e^x - 7x + 6) = \lim_{x \rightarrow +\infty} e^x \left(1 - \frac{7x}{e^x} + \frac{6}{e^x} \right) = +\infty$

$$e). \lim_{x \rightarrow -\infty} (e^{-x} - 5) = e^{-(-\infty)} - 5 = +\infty$$

$$f). \lim_{x \rightarrow -\infty} (e^x + 3) = 0 + 3 = 3$$

$$g). \lim_{x \rightarrow -\infty} (2e^x - x^2 + 1) = 0 - \infty + 1 = -\infty$$

$$h). \lim_{x \rightarrow +\infty} (x - 2 + xe^x) = \lim_{x \rightarrow +\infty} xe^x \left(\frac{1}{e^x} - \frac{2}{xe^x} + 1 \right) = 0 - 0 + 1 = 1$$

$$i). \lim_{x \rightarrow 0} (e^x + 2x^2 - 6) = 1 + 0 - 6 = -5$$

$$j). \lim_{x \rightarrow -\infty} e^x (x^3 + 1) = \lim_{x \rightarrow -\infty} x^3 e^x \left(1 + \frac{1}{x^3} \right) = 0(1 + 0) = 0$$

2) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow +\infty} (x^3 + x)e^{-x} \text{ មានរាង } (\infty \times 0)$$

$$= \lim_{x \rightarrow +\infty} \frac{x^3}{e^x} \left(1 + \frac{1}{x^2} \right) = 0(1 + 0) = 0$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} (x^3 + x)e^{-x} = 0}$$

$$b). \lim_{x \rightarrow +\infty} (e^x - e^{-x}) = +\infty - 0 = +\infty$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} (e^x - e^{-x}) = +\infty}$$

$$c). \lim_{x \rightarrow -\infty} \frac{e^{2x} + 1}{e^x + 3} = \frac{0 + 1}{0 + 3} = \frac{1}{3}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow -\infty} \frac{e^{2x} + 1}{e^x + 3} = \frac{1}{3}}$$

$$d). \lim_{x \rightarrow +\infty} \frac{xe^x + 2}{x^2 - 1} \text{ មានរាង } \frac{\infty}{\infty}$$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} \frac{xe^x \left(1 + \frac{2}{xe^x}\right)}{x^2 \left(1 - \frac{1}{x^2}\right)} = \lim_{x \rightarrow +\infty} \frac{e^x \left(1 + \frac{2}{xe^x}\right)}{x \left(1 - \frac{1}{x^2}\right)} \\
&= \lim_{x \rightarrow +\infty} \frac{e^x}{x} \cdot \frac{\left(1 + \frac{2}{xe^x}\right)}{\left(1 - \frac{1}{x^2}\right)} = (+\infty) \cdot \frac{1+0}{1-0} = +\infty
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{xe^x + 2}{x^2 - 1} = +\infty$

e). $\lim_{x \rightarrow +\infty} \frac{3e^x - 2}{5e^x + 3}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{e^x \left(3 - \frac{2}{e^x}\right)}{e^x \left(5 + \frac{3}{e^x}\right)} = \lim_{x \rightarrow +\infty} \frac{3 - \frac{2}{e^x}}{5 + \frac{3}{e^x}} = \frac{3-0}{5+0} = \frac{3}{5}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{3e^x - 2}{5e^x + 3} = \frac{3}{5}$

f). $\lim_{x \rightarrow -\infty} xe^x = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} = 0$

ដូចនេះ: $\lim_{x \rightarrow -\infty} xe^x = 0$

g). $\lim_{x \rightarrow +\infty} \frac{2e^x - 1}{x^2 + x - 3}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{e^x}{x^2} \cdot \frac{2 - \frac{1}{e^x}}{1 + \frac{1}{x} - \frac{3}{x^2}} = (+\infty) \cdot \frac{2-0}{1+0-0} = +\infty$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{2e^x - 1}{x^2 + x - 3} = +\infty$

$$h). \lim_{x \rightarrow 0} \frac{e^x + x + 1}{2e^x + 9} = \frac{1+0+1}{2+9} = \frac{2}{11}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{e^x + x + 1}{2e^x + 9} = \frac{2}{11}}$$

$$i). \lim_{x \rightarrow -\infty} (xe^x + e^x + 2) = 0 + 0 + 2 = 2$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow -\infty} (xe^x + e^x + 2) = 2}$$

$$j). \lim_{x \rightarrow +\infty} \frac{2e^x + 3}{x^2 + x + 1} \text{ មានរាង } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{e^x}{x^2} \cdot \frac{2 + \frac{3}{e^x}}{1 + \frac{1}{x} + \frac{1}{x^2}} = (+\infty) \cdot \frac{2+0}{1+0+0} = +\infty$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{2e^x + 3}{x^2 + x + 1} = +\infty}$$

3) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow +\infty} \frac{x^3 - e^x}{2e^x + 3x + 1} \text{ មានរាង } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{e^x \left(\frac{x^3}{e^x} - 1 \right)}{e^x \left(2 + \frac{3x}{e^x} + \frac{1}{e^x} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{x^3}{e^x} - 1 \right)}{\left(2 + \frac{3x}{e^x} + \frac{1}{e^x} \right)} = \frac{0-1}{2+0+0} = -\frac{1}{2}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{x^3 - e^x}{2e^x + 3x + 1} = -\frac{1}{2}}$$

$$b). \lim_{x \rightarrow +\infty} \frac{2e^x - x}{1 + e^x} \text{ មានរាង } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{e^x \left(2 - \frac{x}{e^x} \right)}{e^x \left(\frac{1}{e^x} + 1 \right)} = \lim_{x \rightarrow +\infty} \frac{\left(2 - \frac{x}{e^x} \right)}{\left(\frac{1}{e^x} + 1 \right)} = \frac{2-0}{0+1} = 2$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{2e^x - x}{1 + e^x} = 2$

c). $\lim_{x \rightarrow +\infty} \frac{x - e^x}{2e^x + 3}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{e^x \left(\frac{x}{e^x} - 1 \right)}{e^x \left(2 + \frac{3}{e^x} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{x}{e^x} - 1 \right)}{\left(2 + \frac{3}{e^x} \right)} = \frac{0-1}{2+0} = -\frac{1}{2}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{x - e^x}{2e^x + 3} = -\frac{1}{2}$

d). $\lim_{x \rightarrow +\infty} \frac{e^{2x} - 3e^x + 2}{2e^{2x} + 5e^x - 7}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{e^{2x} \left(1 - \frac{3}{e^x} + \frac{2}{e^{2x}} \right)}{e^{2x} \left(2 + \frac{5}{e^x} - \frac{7}{e^{2x}} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(1 - \frac{3}{e^x} + \frac{2}{e^{2x}} \right)}{\left(2 + \frac{5}{e^x} - \frac{7}{e^{2x}} \right)} = \frac{1-0+0}{2+0-0} = \frac{1}{2}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{e^{2x} - 3e^x + 2}{2e^{2x} + 5e^x - 7} = \frac{1}{2}$

e). $\lim_{x \rightarrow -\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow -\infty} \frac{e^{-x} (e^{2x} - 1)}{e^{-x} (e^{2x} + 1)} = \lim_{x \rightarrow -\infty} \frac{(e^{2x} - 1)}{(e^{2x} + 1)} = \frac{0-1}{0+1} = -1$$

ដូចនេះ: $\lim_{x \rightarrow -\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = -1$

$$\begin{aligned} \text{f). } \lim_{x \rightarrow +\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} & \text{ មានរាង } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow +\infty} \frac{e^x(1 - e^{-2x})}{e^x(1 + e^{-2x})} = \lim_{x \rightarrow +\infty} \frac{(1 - e^{-2x})}{(1 + e^{-2x})} = \frac{1 - 0}{1 + 0} = 1 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = 1}$$

$$\begin{aligned} \text{g). } \lim_{x \rightarrow +\infty} \frac{3e^{x-2}}{x^3} & \text{ មានរាង } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow +\infty} \frac{3e^x \cdot e^{-2}}{x^3} = \lim_{x \rightarrow +\infty} (3e^{-2}) \times \frac{e^x}{x^3} = (3e^{-2})(+\infty) = +\infty \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{3e^{x-2}}{x^3} = +\infty}$$

$$\text{h). } \lim_{x \rightarrow -\infty} \frac{1}{1 + e^x} = \frac{1}{1 + 0} = 1$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow -\infty} \frac{1}{1 + e^x} = 1}$$

$$\text{i). } \lim_{x \rightarrow +\infty} (x+1)e^{-x} \text{ មានរាង } (\infty \times 0)$$

$$= \lim_{x \rightarrow +\infty} \frac{x+1}{e^x} = \lim_{x \rightarrow +\infty} \left(\frac{x}{e^x} + \frac{1}{e^x} \right) = 0 + 0 = 0$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} (x+1)e^{-x} = 0}$$

$$\begin{aligned} \text{j). } \lim_{x \rightarrow +\infty} \frac{1 + e^x - xe^{2x}}{x^2 e^x + e^{2x}} & \text{ មានរាង } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow +\infty} \frac{xe^{2x} \left(\frac{1}{xe^{2x}} + \frac{1}{xe^x} - 1 \right)}{xe^{2x} \left(\frac{x}{e^x} + \frac{1}{x} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{1}{xe^{2x}} + \frac{1}{xe^x} - 1 \right)}{\left(\frac{x}{e^x} + \frac{1}{x} \right)} \\ &= \frac{0 + 0 - 1}{0^+} = -\infty \end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow +\infty} \frac{1+e^x - xe^{2x}}{x^2 e^x + e^{2x}} = -\infty$$

គ. ធានាលីមីតដោយប្រើរូបមន្ត ៖

❶ $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$

❷ $\lim_{x \rightarrow 0} \frac{a^{kx} - 1}{kx} = \ln a$

❸ $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

❹ $\lim_{x \rightarrow 0} \frac{e^{kx} - 1}{kx} = 1$

លំហាត់

1) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{x}$

b). $\lim_{x \rightarrow 0} \frac{e^{2x} - \sin 3x - 1}{x}$

c). $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^{3x} - 1}$

d). $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x}$

e). $\lim_{x \rightarrow 0} \frac{e^{\sin x} + \sin 2x - 1}{\sin x}$

f). $\lim_{x \rightarrow 0} \frac{(e^{2x} - 1)(1 - \cos 2x)}{x^3}$

g). $\lim_{x \rightarrow 0} \frac{(e^{-3x+2} - e^2) \sin \pi x}{4x^2}$

h). $\lim_{x \rightarrow 0} \left(\frac{e^x}{x} - \frac{1}{x} + \frac{\sin 2x}{x} \right)$

i). $\lim_{x \rightarrow 0} \frac{\sin x + e^x - 1}{x^2 + x}$

j). $\lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x}}{\sin 2x}$

2) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 1} \frac{1-x}{e^x - e}$

b). $\lim_{x \rightarrow 1} \frac{e^{x^2} - e^x}{x^2 - 1}$

c). $\lim_{x \rightarrow 0} \frac{5^x - 4^x}{x^2 - x}$

d). $\lim_{x \rightarrow 0} \frac{8^x - 7^x}{6^x - 5^x}$

e). $\lim_{x \rightarrow 0} \frac{2 - (e^x + e^{-x})}{x^2}$

f). $\lim_{x \rightarrow 0} \frac{e^{3x} - 2x - 1}{\sin^2 5x}$

g). $\lim_{x \rightarrow 0} \frac{e^{\sin x} - e^{\sin 3x}}{\sin 2x}$

h). $\lim_{x \rightarrow 1} \frac{e^{2x^2-1} - e^x}{x-1}$

$$i). \lim_{x \rightarrow 0} \frac{e^{ax} - e^{bx}}{x}$$

$$j). \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\sin x}}{x}$$

3) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 0} \frac{e^{\sin^2 2x} - \cos(\sin 3x)}{x^2}$$

$$b). \lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{\sin^3 x}$$

$$c). \lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2}$$

$$d). \lim_{x \rightarrow 0} \frac{e^{3x} - e^{2x}}{\sin 3x - \sin 2x}$$

$$e). \lim_{x \rightarrow 0} \frac{e^{x+1} - e}{\sin \pi x}$$

$$f). \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{e^x - e}$$

$$g). \lim_{x \rightarrow \pi} \frac{\sin 2x}{e^{x+\pi} - e^{2\pi}}$$

$$h). \lim_{x \rightarrow 0} \frac{\sin 2016x}{\sqrt{e^{\pi x}} - 1}$$

$$i). \lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{1 - \sqrt{x+1}}$$

$$j). \lim_{x \rightarrow 0} \frac{1 - \sin(\sin x) - e^{\sin x}}{\sin x - \sin 2x}$$

$$k). \lim_{x \rightarrow 0} \frac{(e^x - 1)(1 - \cos 2x) \sin 7x}{x^4}$$

$$l). \lim_{x \rightarrow 0} \frac{xe^x - x + \sin^2 x}{1 - \sqrt{\cos 2x}}$$

$$m). \lim_{x \rightarrow 0} \frac{x^2}{e^{x^2} - 4x^2}$$

$$n). \lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{e^{x-1} + \cos 2(x-1) - 2}$$

ជំនួសប្រភេទ

1) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{x} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1 + \sin x}{x} = \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + \frac{\sin x}{x} \right) = 1 + 1 = 2$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{x} = 2}$$

$$b). \lim_{x \rightarrow 0} \frac{e^{2x} - \sin 3x - 1}{x} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{x} - \frac{\sin 3x}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{2x} \times 2 - \frac{\sin 3x}{3x} \times 3 \right)$$

$$= 2 - 3 = -1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{2x} - \sin 3x - 1}{x} = -1$

c). $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^{3x} - 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\frac{e^{2x} - 1}{x}}{\frac{e^{3x} - 1}{x}} = \lim_{x \rightarrow 0} \frac{\frac{e^{2x} - 1}{x} \times 2}{\frac{e^{3x} - 1}{3x} \times 3} = \frac{1 \times 2}{1 \times 3} = \frac{2}{3}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^{3x} - 1} = \frac{2}{3}$

d). $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x} = 1 \times 1 = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x} = 1$

e). $\lim_{x \rightarrow 0} \frac{e^{\sin x} + \sin 2x - 1}{\sin x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{\sin x} - 1}{\sin x} + \frac{\sin 2x}{\sin x} \right) = \lim_{x \rightarrow 0} \left(\frac{e^{\sin x} - 1}{\sin x} + \frac{2 \sin x \cos x}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{\sin x} - 1}{\sin x} + 2 \cos x \right) = 1 + 2 = 3$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{\sin x} + \sin 2x - 1}{\sin x} = 3$

$$\text{f). } \lim_{x \rightarrow 0} \frac{(e^{2x} - 1)(1 - \cos 2x)}{x^3} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \times \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 \times \frac{2\sin^2 x}{x^2} = 1 \times 2 \times 2 = 4$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0} \frac{(e^{2x} - 1)(1 - \cos 2x)}{x^3} = 4}$$

$$\text{g). } \lim_{x \rightarrow 0} \frac{(e^{-3x+2} - e^2)\sin \pi x}{4x^2} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{e^2(e^{-3x} - 1)}{4x} \times \frac{\sin \pi x}{x} = \lim_{x \rightarrow 0} \frac{e^{-3x} - 1}{-3x} \times \frac{-3e^2}{4} \times \frac{\sin \pi x}{\pi x} \times \pi$$

$$= 1 \times \frac{-3e^2}{4} \times 1 \times \pi = -\frac{3\pi e^2}{4}$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0} \frac{(e^{-3x+2} - e^2)\sin \pi x}{4x^2} = -\frac{3\pi e^2}{4}}$$

$$\text{h). } \lim_{x \rightarrow 0} \left(\frac{e^x}{x} - \frac{1}{x} + \frac{\sin 2x}{x} \right) \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + \frac{\sin 2x}{2x} \times 2 \right) = 1 + 1 \times 2 = 1 + 2 = 3$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0} \left(\frac{e^x}{x} - \frac{1}{x} + \frac{\sin 2x}{x} \right) = 3}$$

$$\text{i). } \lim_{x \rightarrow 0} \frac{\sin x + e^x - 1}{x^2 + x} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x + e^x - 1}{x(x+1)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} + \frac{e^x - 1}{x} \right) \times \frac{1}{x+1} = (1+1) \times \frac{1}{0+1} = 2$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0} \frac{\sin x + e^x - 1}{x^2 + x} = 2}$$

$$\begin{aligned} \text{j). } \lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x}}{\sin 2x} & \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{e^{-2x}(e^{4x} - 1)}{\sin 2x} = \lim_{x \rightarrow 0} \frac{e^{4x} - 1}{4x} \times \frac{2x}{\sin 2x} \times \frac{4e^{-2x}}{2} = 1 \times 1 \times \frac{4}{2} = 2 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x}}{\sin 2x} = 2}$$

2) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow 1} \frac{1-x}{e^x - e} \text{ មានរាង } \frac{0}{0}$$

តាង $u = x - 1$ នោះ $x = u + 1$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 1} \frac{1-x}{e^x - e} &= \lim_{u \rightarrow 0} \frac{1-(u+1)}{e^{u+1} - e} = \lim_{u \rightarrow 0} \frac{-u}{e(e^u - 1)} \\ &= -\frac{1}{e} \lim_{u \rightarrow 0} \frac{u}{e^u - 1} = -\frac{1}{e} \times 1 = -\frac{1}{e} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{1-x}{e^x - e} = -\frac{1}{e}}$$

$$\text{b). } \lim_{x \rightarrow 1} \frac{e^{x^2} - e^x}{x^2 - 1} \text{ មានរាង } \frac{0}{0}$$

តាង $u = x - 1$ នោះ $x = u + 1$

ពេល $x \rightarrow 1$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 1} \frac{e^{x^2} - e^x}{x^2 - 1} &= \lim_{u \rightarrow 0} \frac{e^{(u+1)^2} - e^{u+1}}{(u+1)^2 - 1} \\ &= \lim_{u \rightarrow 0} \frac{e^{u^2+2u+1} - e^{u+1}}{u^2 + 2u + 1 - 1} = \lim_{u \rightarrow 0} \frac{e(e^{u^2+2u} - e^u)}{u^2 + 2u} \\ &= e \lim_{u \rightarrow 0} \frac{(e^{u^2+2u} - 1) - (e^u - 1)}{u^2 + 2u} = e \lim_{u \rightarrow 0} \left(\frac{e^{u^2+2u} - 1}{u^2 + 2u} - \frac{e^u - 1}{u^2 + 2u} \right) \end{aligned}$$

$$= e \lim_{u \rightarrow 0} \left(\frac{e^{u^2+2u} - 1}{u^2 + 2u} - \frac{e^u - 1}{u} \times \frac{1}{u+2} \right) = e(1 - 1 \times \frac{1}{2}) = \frac{1}{2}e$$

ដូច្នេះ: $\lim_{x \rightarrow 1} \frac{e^{x^2} - e^x}{x^2 - 1} = \frac{1}{2}e$

c). $\lim_{x \rightarrow 0} \frac{5^x - 4^x}{x^2 - x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(5^x - 1) - (4^x - 1)}{x(x-1)} = \lim_{x \rightarrow 0} \left(\frac{5^x - 1}{x} - \frac{4^x - 1}{x} \right) \times \frac{1}{x-1}$$

$$= (\ln 5 - \ln 4)(-1) = -\ln \frac{5}{4}$$

ដូច្នេះ: $\lim_{x \rightarrow 0} \frac{5^x - 4^x}{x^2 - x} = -\ln \frac{5}{4}$

d). $\lim_{x \rightarrow 0} \frac{8^x - 7^x}{6^x - 5^x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(8^x - 1) - (7^x - 1)}{(6^x - 1) - (5^x - 1)} = \lim_{x \rightarrow 0} \frac{\frac{8^x - 1}{x} - \frac{7^x - 1}{x}}{\frac{6^x - 1}{x} - \frac{5^x - 1}{x}}$$

$$= \frac{\ln 8 - \ln 7}{\ln 6 - \ln 5} = \frac{\ln(8/7)}{\ln(6/5)}$$

ដូច្នេះ: $\lim_{x \rightarrow 0} \frac{8^x - 7^x}{6^x - 5^x} = \frac{\ln(8/7)}{\ln(6/5)}$

e). $\lim_{x \rightarrow 0} \frac{2 - (e^x + e^{-x})}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{2 - e^x - e^{-x}}{x^2} = \lim_{x \rightarrow 0} \frac{2e^x - e^{2x} - 1}{x^2 e^x} = \lim_{x \rightarrow 0} \frac{-(e^{2x} - 2e^x + 1)}{x^2 e^x}$$

$$= - \lim_{x \rightarrow 0} \frac{e^{2x} - 2e^x + 1}{x^2 e^x} = - \lim_{x \rightarrow 0} \frac{(e^x - 1)^2}{x^2 e^x} = - \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right)^2 \cdot \frac{1}{e^x}$$

$$= -(1)^2(1) = -1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{2 - (e^x + e^{-x})}{x^2} = -1$

f). $\lim_{x \rightarrow 0} \frac{e^{3x} - 2x - 1}{\sin^2 5x}$ មានរាង $\frac{0}{0}$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 2x}{\sin^2 5x} = \lim_{x \rightarrow 0} \frac{\frac{e^{3x} - 1}{x} - \frac{2x}{x}}{\frac{\sin^2 5x}{x}} = \lim_{x \rightarrow 0} \frac{\frac{e^{3x} - 1}{x} \times 3 - 2}{\frac{\sin 5x}{5x} \times 5 \sin 5x} \\ &= \frac{1 \times 3 - 2}{1 \times 0} = \frac{1}{0} = \pm \infty \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{3x} - 2x - 1}{\sin^2 5x} = \pm \infty$

g). $\lim_{x \rightarrow 0} \frac{e^{\sin x} - e^{\sin 3x}}{\sin 2x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 - e^{\sin 3x} + 1}{\sin 2x} = \lim_{x \rightarrow 0} \frac{\frac{e^{\sin x} - 1}{x} - \frac{e^{\sin 3x} - 1}{x}}{\frac{\sin 2x}{x}}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x} \right) - \left(\frac{e^{\sin 3x} - 1}{\sin 3x} \times \frac{\sin 3x}{3x} \times 3 \right)}{\frac{\sin 2x}{2x} \times 2}$$

$$= \frac{(1 \times 1) - (1 \times 1 \times 3)}{1 \times 2} = \frac{1 - 3}{2} = \frac{-2}{2} = -1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{\sin x} - e^{\sin 3x}}{\sin 2x} = -1$

h). $\lim_{x \rightarrow 1} \frac{e^{2x^2 - 1} - e^x}{x - 1}$ មានរាង $\frac{0}{0}$

តាង $u = x - 1$ នោះ $x = u + 1$

ពេល $x \rightarrow 1$ នោះ $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow 1} \frac{e^{2x^2-1} - e^x}{x-1} &= \lim_{u \rightarrow 0} \frac{e^{2(u+1)^2-1} - e^{u+1}}{u} \\
 &= \lim_{u \rightarrow 0} \frac{e^{2(u^2+2u+1)-1} - e^{u+1}}{u} = \lim_{u \rightarrow 0} \frac{e^{2u^2+4u+1} - e^{u+1}}{u} \\
 &= e \lim_{u \rightarrow 0} \frac{e^{2u^2+4u} - e^u}{u} = e \lim_{u \rightarrow 0} \frac{(e^{u(2u+4)} - 1) - (e^u - 1)}{u} \\
 &= e \lim_{u \rightarrow 0} \left[\frac{e^{u(2u+4)} - 1}{u(2u+4)} \cdot (2u+4) - \frac{e^u - 1}{u} \right] = e(1 \times 4 - 1) = 3e
 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{e^{2x^2-1} - e^x}{x-1} = 3e}$$

$$\begin{aligned}
 \text{i). } \lim_{x \rightarrow 0} \frac{e^{ax} - e^{bx}}{x} &\text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(e^{ax} - 1) - (e^{bx} - 1)}{x} = \lim_{x \rightarrow 0} \left(\frac{e^{ax} - 1}{x} - \frac{e^{bx} - 1}{x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{ax} - 1}{ax} \times a - \frac{e^{bx} - 1}{bx} \times b \right) = a - b
 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{e^{ax} - e^{bx}}{x} = a - b}$$

$$\begin{aligned}
 \text{j). } \lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\sin x}}{x} &\text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(e^{\sin 2x} - 1) - (e^{\sin x} - 1)}{x} = \lim_{x \rightarrow 0} \left(\frac{e^{\sin 2x} - 1}{x} - \frac{e^{\sin x} - 1}{x} \right) \\
 &= \lim_{x \rightarrow 0} \left(\frac{e^{\sin 2x} - 1}{\sin 2x} \times \frac{\sin 2x}{2x} \times 2 - \frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x} \right) = 2 - 1 = 1
 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\sin x}}{x} = 1}$$

3) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow 0} \frac{e^{\sin^2 2x} - \cos(\sin 3x)}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{\sin^2 2x} - 1 + 1 - \cos(\sin 3x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \left[\frac{e^{\sin^2 2x} - 1}{x^2} + \frac{2 \sin^2 \left(\frac{\sin 3x}{2} \right)}{x^2} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{e^{\sin^2 2x} - 1}{\sin^2 2x} \times \frac{\sin^2 2x}{(2x)^2} \times 4 + \frac{2 \sin^2 \left(\frac{\sin 3x}{2} \right)}{\left(\frac{\sin 3x}{2} \right)^2} \times \frac{1}{4} \frac{(\sin 3x)^2}{(3x)^2} \times 9 \right]$$

$$= 1 \times 1 \times 4 + 2 \times \frac{1}{4} \times 9 = 4 + \frac{9}{2} = \frac{8+9}{2} = \frac{17}{2}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{\sin^2 2x} - \cos(\sin 3x)}{x^2} = \frac{17}{2}$

b). $\lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{\sin^3 x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{x^3} \times \frac{x^3}{\sin^3 x} = 1 \times 1 = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{\sin^3 x} = 1$

c). $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{x^2} - 1 + 1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \left(\frac{e^{x^2} - 1}{x^2} + \frac{2 \sin^2 \frac{x}{2}}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{e^{x^2} - 1}{x^2} + \frac{2 \sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{1}{4} \right) = 1 + 2 \times \frac{1}{4} = 1 + \frac{1}{2} = \frac{3}{2}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2} = \frac{3}{2}$

d). $\lim_{x \rightarrow 0} \frac{e^{3x} - e^{2x}}{\sin 3x - \sin 2x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e^{3x} - 1 + 1 - e^{2x}}{\sin 3x - \sin 2x} = \lim_{x \rightarrow 0} \frac{(e^{3x} - 1) - (e^{2x} - 1)}{\sin 3x - \sin 2x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{e^{3x} - 1}{x} - \frac{e^{2x} - 1}{x}}{\frac{\sin 3x}{x} - \frac{\sin 2x}{x}} = \lim_{x \rightarrow 0} \frac{\frac{e^{3x} - 1}{3x} \times 3 - \frac{e^{2x} - 1}{2x} \times 2}{\frac{\sin 3x}{3x} \times 3 - \frac{\sin 2x}{2x} \times 2} = \frac{3 - 2}{3 - 2} = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{3x} - e^{2x}}{\sin 3x - \sin 2x} = 1$

e). $\lim_{x \rightarrow 0} \frac{e^{x+1} - e}{\sin \pi x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{e(e^x - 1)}{\sin \pi x} = e \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \frac{x}{\sin \pi x}$$

$$= e \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \frac{\pi x}{\sin \pi x} \times \frac{1}{\pi} = e \times 1 \times 1 \times \frac{1}{\pi} = \frac{e}{\pi}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{x+1} - e}{\sin \pi x} = \frac{e}{\pi}$

f). $\lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{e^x - e}$ មានរាង $\frac{0}{0}$

តាង $u = x - 1$ នោះ: $x = u + 1$

ពេល $x \rightarrow 1$ នោះ: $u \rightarrow 0$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{e^x - e} &= \lim_{u \rightarrow 0} \frac{\cos \frac{\pi}{2} (u+1)}{e^{u+1} - e} = \lim_{u \rightarrow 0} \frac{\cos \left(\frac{\pi}{2} + \frac{\pi}{2} u \right)}{e(e^u - 1)} \\
 &= \frac{1}{e} \lim_{u \rightarrow 0} \frac{-\sin \left(\frac{\pi}{2} u \right)}{e^u - 1} = -\frac{1}{e} \lim_{u \rightarrow 0} \frac{\sin \left(\frac{\pi}{2} u \right)}{\left(\frac{\pi}{2} u \right)} \times \frac{u}{e^u - 1} \times \frac{\pi}{2} \\
 &= -\frac{1}{e} \times 1 \times 1 \times \frac{\pi}{2} = -\frac{\pi}{2e}
 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{e^x - e} = -\frac{\pi}{2e}}$$

$$\text{g). } \lim_{x \rightarrow \pi} \frac{\sin 2x}{e^{x+\pi} - e^{2\pi}} \text{ មានរាង } \frac{0}{0}$$

$$\text{តាង } u = x - \pi \text{ នៅ: } x = \pi + u$$

$$\text{ពេល } x \rightarrow \pi \text{ នៅ: } u \rightarrow 0$$

$$\begin{aligned}
 \text{គេបាន } \lim_{x \rightarrow \pi} \frac{\sin 2x}{e^{x+\pi} - e^{2\pi}} &= \lim_{u \rightarrow 0} \frac{\sin 2(\pi + u)}{e^{\pi+u+\pi} - e^{2\pi}} \\
 &= \lim_{u \rightarrow 0} \frac{\sin(2\pi + 2u)}{e^{u+2\pi} - e^{2\pi}} = \frac{1}{e^{2\pi}} \lim_{u \rightarrow 0} \frac{\sin 2u}{e^u - 1} \\
 &= \frac{1}{e^{2\pi}} \lim_{u \rightarrow 0} \frac{\sin 2u}{2u} \times \frac{u}{e^u - 1} \times 2 = \frac{1}{e^{2\pi}} \times 1 \times 1 \times 2 = \frac{2}{e^{2\pi}}
 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \pi} \frac{\sin 2x}{e^{x+\pi} - e^{2\pi}} = \frac{2}{e^{2\pi}}}$$

$$\text{h). } \lim_{x \rightarrow 0} \frac{\sin 2016x}{\sqrt{e^{\pi x}} - 1} \text{ មានរាង } \frac{0}{0}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{(\sin 2016x)(\sqrt{e^{\pi x}} + 1)}{(\sqrt{e^{\pi x}} - 1)(\sqrt{e^{\pi x}} + 1)} = \lim_{x \rightarrow 0} \frac{(\sin 2016x)(\sqrt{e^{\pi x}} + 1)}{(e^{\pi x} - 1)} \\
 &= \lim_{x \rightarrow 0} \frac{(\sin 2016x)}{2016x} \times \frac{\pi x}{e^{\pi x} - 1} \times \frac{2016}{\pi} \times (\sqrt{e^{\pi x}} + 1)
 \end{aligned}$$

$$= 1 \times 1 \times \frac{2016}{\pi} \times (1+1) = \frac{2016 \times 2}{\pi} = \frac{4032}{\pi}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sin 2016x}{\sqrt{e^{\pi x}} - 1} = \frac{4032}{\pi}$

i). $\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{1 - \sqrt{x+1}}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(e^x - 1 + \sin x)(1 + \sqrt{x+1})}{(1 - \sqrt{x+1})(1 + \sqrt{x+1})}$$

$$= \lim_{x \rightarrow 0} \frac{(e^x - 1 + \sin x)(1 + \sqrt{x+1})}{1 - (x+1)} = \lim_{x \rightarrow 0} \frac{(e^x - 1 + \sin x)(1 + \sqrt{x+1})}{-x}$$

$$= - \lim_{x \rightarrow 0} \frac{(e^x - 1 + \sin x)(1 + \sqrt{x+1})}{x}$$

$$= - \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + \frac{\sin x}{x} \right) (1 + \sqrt{x+1}) = -(1+1)(1+1) = -4$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{1 - \sqrt{x+1}} = -4$

j). $\lim_{x \rightarrow 0} \frac{1 - \sin(\sin x) - e^{\sin x}}{\sin x - \sin 2x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{-\sin(\sin x) - e^{\sin x} + 1}{\sin x - \sin 2x} = - \lim_{x \rightarrow 0} \frac{\sin(\sin x) + (e^{\sin x} - 1)}{\sin x - \sin 2x}$$

$$= - \lim_{x \rightarrow 0} \frac{\frac{\sin(\sin x)}{x} + \frac{e^{\sin x} - 1}{x}}{\frac{\sin x}{x} - \frac{\sin 2x}{x}}$$

$$= - \lim_{x \rightarrow 0} \frac{\frac{\sin(\sin x)}{\sin x} \times \frac{\sin x}{x} + \frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x}}{\frac{\sin x}{x} - \frac{\sin 2x}{2x} \times 2}$$

$$= - \frac{1 \times 1 + 1 \times 1}{1 - 2} = - \frac{2}{-1} = 2$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{1 - \sin(\sin x) - e^{\sin x}}{\sin x - \sin 2x} = 2}$$

$$\begin{aligned} \text{k). } \lim_{x \rightarrow 0} \frac{(e^x - 1)(1 - \cos 2x) \sin 7x}{x^4} & \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{x} \times \frac{2 \sin^2 x}{x^2} \times \frac{\sin 7x}{7x} \times 7 = 1 \times 2 \times 1 \times 7 = 14 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{(e^x - 1)(1 - \cos 2x) \sin 7x}{x^4} = 4}$$

$$\begin{aligned} \text{l). } \lim_{x \rightarrow 0} \frac{xe^x - x + \sin^2 x}{1 - \sqrt{\cos 2x}} & \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{(xe^x - x + \sin^2 x)(1 + \sqrt{\cos 2x})}{(1 - \sqrt{\cos 2x})(1 + \sqrt{\cos 2x})} \\ &= \lim_{x \rightarrow 0} \frac{(xe^x - x + \sin^2 x)(1 + \sqrt{\cos 2x})}{(1 - \cos 2x)} \\ &= \lim_{x \rightarrow 0} \frac{(xe^x - x + \sin^2 x)(1 + \sqrt{\cos 2x})}{2 \sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{\left(\frac{xe^x - x}{x^2} + \frac{\sin^2 x}{x^2} \right) (1 + \sqrt{\cos 2x})}{\frac{2 \sin^2 x}{x^2}} \\ &= \lim_{x \rightarrow 0} \frac{\left(\frac{e^x - 1}{x} + \frac{\sin^2 x}{x^2} \right) (1 + \sqrt{\cos 2x})}{2 \cdot \frac{\sin^2 x}{x^2}} = \frac{(1+1)(1+1)}{2} = 2 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{xe^x - x + \sin^2 x}{1 - \sqrt{\cos 2x}} = 2}$$

$$\text{m). } \lim_{x \rightarrow 0} \frac{x^2}{e^{x^2} - 4x^2} \text{ មានរាង } \frac{0}{0}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{x^2}{e^{x^2} - 1 - 4x^2 + 1} = \lim_{x \rightarrow 0} \frac{x^2}{(e^{x^2} - 1) - (4x^2 - 1)} \\
&= \lim_{x \rightarrow 0} \frac{\frac{x^2}{x^2}}{\left(\frac{e^{x^2} - 1}{x^2}\right) - \left(\frac{4x^2 - 1}{x^2}\right)} = \lim_{x \rightarrow 0} \frac{1}{\left(\frac{e^{x^2} - 1}{x^2}\right) - \left(\frac{4x^2 - 1}{x^2}\right)} = \frac{1}{1 - \ln 4}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{x^2}{e^{x^2} - 4x^2} = \frac{1}{1 - \ln 4}$

$$\text{n). } \lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{e^{x-1} + \cos 2(x-1) - 2} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+4)}{(e^{x-1} - 1) - [1 - \cos 2(x-1)]}$$

តាង $u = x - 1$ នោះ: $x = u + 1$

ពេល $x \rightarrow 1$ នោះ: $u \rightarrow 0$

$$\text{គេបាន } \lim_{x \rightarrow 1} \frac{(x-1)(x+4)}{(e^{x-1} - 1) - [1 - \cos 2(x-1)]} = \lim_{u \rightarrow 0} \frac{u(u+5)}{(e^u - 1) - (1 - \cos 2u)}$$

$$= \lim_{u \rightarrow 0} \frac{u(u+5)}{(e^u - 1) - 2\sin^2 u} = \lim_{u \rightarrow 0} \frac{u(u+5)}{\left(\frac{e^u - 1}{u}\right) - 2 \cdot \frac{\sin^2 u}{u}}$$

$$= \lim_{u \rightarrow 0} \frac{u+5}{\left(\frac{e^u - 1}{u}\right) - 2 \cdot \frac{\sin^2 u}{u}} = \frac{5}{1 - 2 \cdot 0} = \frac{5}{1} = 5$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{e^{x-1} + \cos 2(x-1) - 2} = 5$

ប. លីមីតរាងមិនកំណត់ 1^∞

ដើម្បីគណនាលីមីតរាងមិនកំណត់ 1^∞ គេត្រូវប្រើរូបមន្ត៖

❶ $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$

❷ $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{kx}\right)^{kx} = e$

❸ $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

❹ $\lim_{x \rightarrow 0} (1+kx)^{\frac{1}{kx}} = e$

លំហាត់

1) គណនាលីមីតខាងក្រោម៖

a). $\lim_{x \rightarrow +\infty} \left(\frac{x+1}{x}\right)^{x+5}$

b). $\lim_{x \rightarrow 0} (1+2x)^{\frac{5}{x}}$

c). $\lim_{x \rightarrow 0^+} (1+2x)^{\cot x}$

d). $\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{mx}$

e). $\lim_{x \rightarrow 0} (1+\sin x)^{\frac{1}{x}}$

f). $\lim_{x \rightarrow 0} (2-e^x)^{\frac{1}{x}}$

g). $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$

h). $\lim_{x \rightarrow \infty} \left(\frac{x+3}{x+1}\right)^{x+1}$

i). $\lim_{x \rightarrow \infty} \left(\frac{x-2}{x+3}\right)^x$

j). $\lim_{x \rightarrow \infty} \left(\frac{4x+3}{4x-3}\right)^{2x+1}$

2) គណនាលីមីតខាងក្រោម៖

a). $\lim_{x \rightarrow \infty} \left(\frac{x}{1+x}\right)^x$

b). $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{\frac{x+1}{x}}$

c). $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x-2}\right)^{2x-1}$

d). $\lim_{x \rightarrow \infty} \left(\frac{3x-4}{3x+2}\right)^{\frac{x+1}{3}}$

e). $\lim_{x \rightarrow 0} (1+\sin x)^{\cot x}$

f). $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$

$$g). \lim_{x \rightarrow \infty} \left(\frac{2x+3}{2x+1} \right)^{x+1}$$

$$h). \lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin x}}$$

$$i). \lim_{x \rightarrow 0} (1 + 3 \tan^2 x)^{\cot^2 x}$$

$$j). \lim_{x \rightarrow 0} \frac{\sin x}{\sqrt{1+x}}$$

3) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 0} \left(\frac{2x+3}{x+3} \right)^{\frac{1}{\sin 3x}}$$

$$b). \lim_{x \rightarrow 0} (x + 2^x)^{\frac{1}{x}}$$

$$c). \lim_{x \rightarrow 0} (\cos x + \sin x)^{\frac{1}{x}}$$

$$d). \lim_{x \rightarrow 0} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}}$$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow +\infty} \left(\frac{x+1}{x} \right)^{x+5} \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow +\infty} \left(\frac{x}{x} + \frac{1}{x} \right)^{x+5} = \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{\frac{1}{x}(x+5)}$$

$$= \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{(1+\frac{5}{x})} = e^{(1+0)} = e$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \left(\frac{x+1}{x} \right)^{x+5} = e}$$

$$b). \lim_{x \rightarrow 0} (1 + 2x)^{\frac{5}{x}} \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left[(1 + 2x)^{\frac{1}{2x}} \right]^{2x \times \frac{5}{x}} = \lim_{x \rightarrow 0} \left[(1 + 2x)^{\frac{1}{2x}} \right]^{10} = e^{10}$$

ដូចនេះ: $\lim_{x \rightarrow 0} (1+2x)^{\frac{5}{x}} = e^{10}$

c). $\lim_{x \rightarrow 0^+} (1+2x)^{\cot x}$ មានរាង 1^∞

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} (1+2x)^{\cot x} = \lim_{x \rightarrow 0^+} \left[(1+2x)^{\frac{1}{2x}} \right]^{2x \cdot \cot x} \\ &= \lim_{x \rightarrow 0^+} \left[(1+2x)^{\frac{1}{2x}} \right]^{2x \times \frac{\cos x}{\sin x}} = \lim_{x \rightarrow 0^+} \left[(1+2x)^{\frac{1}{2x}} \right]^{\frac{x}{\sin x} \times 2 \cos x} \\ &= e^{1 \times 2 \times 1} = e^2 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0^+} (1+2x)^{\cot x} = e^2$

d). $\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{mx}$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{k}{x}\right)^{\frac{x}{k}} \right]^{\frac{k}{x} \times mx} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{k}{x}\right)^{\frac{x}{k}} \right]^{km} = e^{km}$$

ដូចនេះ: $\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$

e). $\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$ មានរាង 1^∞

$$= \lim_{x \rightarrow 0} \left[(1 + \sin x)^{\frac{1}{\sin x}} \right]^{(\sin x) \frac{1}{x}} = \lim_{x \rightarrow 0} \left[(1 + \sin x)^{\frac{1}{\sin x}} \right]^{\frac{\sin x}{x}} = e^1 = e$$

ដូចនេះ: $\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}} = e$

f). $\lim_{x \rightarrow 0} (2 - e^x)^{\frac{1}{x}}$ មានរាង 1^∞

$$= \lim_{x \rightarrow 0} \left((1 + 1 - e^x)^{\frac{1}{1 - e^x}} \right)^{(1 - e^x) \frac{1}{x}} = \lim_{x \rightarrow 0} \left((1 + 1 - e^x)^{\frac{1}{1 - e^x}} \right)^{\frac{1 - e^x}{x}}$$

$$= \lim_{x \rightarrow 0} \left((1 + 1 - e^x)^{\frac{1}{1 - e^x}} \right)^{\frac{-(e^x - 1)}{x}} = e^{-1} = \frac{1}{e}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} (2 - e^x)^{\frac{1}{x}} = \frac{1}{e}}$

g). $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} \right)^x$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} \right)^x = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{3}{x} \right)^{\frac{x}{3}} \right]^{\frac{3}{x} \times x} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{3}{x} \right)^{\frac{x}{3}} \right]^3 = e^3$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x} \right)^x = e^3}$

h). $\lim_{x \rightarrow \infty} \left(\frac{x+3}{x+1} \right)^{x+1}$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left(\frac{x+3}{x+1} \right)^{x+1} = \lim_{x \rightarrow \infty} \left(\frac{x+1+2}{x+1} \right)^{x+1} = \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x+1} \right)^{x+1}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x+1} \right)^{\frac{x+1}{2}} \right]^2 = e^2$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(\frac{x+3}{x+1} \right)^{x+1} = e^2}$

$$i). \lim_{x \rightarrow \infty} \left(\frac{x-2}{x+3} \right)^x \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow \infty} \left(\frac{x+3-5}{x+3} \right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{-5}{x+3} \right)^x$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-5}{x+3} \right)^{\frac{x+3}{-5}} \right]^{\frac{-5x}{x(1+\frac{3}{x})}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-5}{x+3} \right)^{\frac{x+3}{-5}} \right]^{\frac{-5}{(1+\frac{3}{x})}}$$

$$= e^{\frac{-5}{1+0}} = e^{-5}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \infty} \left(\frac{x-2}{x+3} \right)^x = e^{-5}}$$

$$j). \lim_{x \rightarrow \infty} \left(\frac{4x+3}{4x-3} \right)^{2x+1} \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow \infty} \left(\frac{4x-3+6}{4x-3} \right)^{2x+1} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{6}{4x-3} \right)^{\frac{4x-3}{6}} \right]^{\frac{6}{4x-3}(2x+1)}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{6}{4x-3} \right)^{\frac{4x-3}{6}} \right]^{\frac{x6(2+\frac{1}{x})}{x(4-\frac{3}{x})}}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{6}{4x-3} \right)^{\frac{4x-3}{6}} \right]^{\frac{6(2+\frac{1}{x})}{(4-\frac{3}{x})}} = e^{\frac{6(2+0)}{4-0}} = e^{\frac{12}{4}} = e^3$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow \infty} \left(\frac{4x+3}{4x-3} \right)^{2x+1} = e^3}$$

2) គណនាលីមីតខាងក្រោម:

a). $\lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x$ មានរាង 1^∞

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \left(\frac{1+x-1}{1+x} \right)^x = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-1}{1+x} \right)^{-(1+x)} \right]^{\frac{-1}{1+x} \times x} \\
 &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-1}{1+x} \right)^{-(1+x)} \right]^{\frac{-x}{1+x}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-1}{1+x} \right)^{-(1+x)} \right]^{\frac{-1}{\frac{1}{x}+1}} \\
 &= e^{\frac{-1}{0+1}} = e^{-1}
 \end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x = e^{-1}}$

b). $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^{\frac{x+1}{x}}$ មានរាង 1^∞

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{\frac{1}{x} \times \frac{x+1}{x}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{\frac{x+1}{x^2}} \\
 &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{\frac{1}{x} + \frac{1}{x^2}} = e^{0+0} = e^0 = 1
 \end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^{\frac{x+1}{x}} = 1}$

c). $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x-2} \right)^{2x-1}$ មានរាង 1^∞

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \left(\frac{x-2+3}{x-2} \right)^{2x-1} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{3}{x-2} \right)^{\frac{x-2}{3}} \right]^{\frac{3}{x-2} (2x-1)}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{3}{x-2} \right)^{\frac{x-2}{3}} \right]^{\frac{3(2x-1)}{x-2}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{3}{x-2} \right)^{\frac{x-2}{3}} \right]^{\frac{3(2-\frac{1}{x})}{1-\frac{2}{x}}} \\
&= e^{\frac{3(2-0)}{1-0}} = e^6
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x-2} \right)^{2x-1} = e^6$

d). $\lim_{x \rightarrow \infty} \left(\frac{3x-4}{3x+2} \right)^{\frac{x+1}{3}}$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left(\frac{3x+2-6}{3x+2} \right)^{\frac{x+1}{3}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-6}{3x+2} \right)^{\frac{3x+2}{-6}} \right]^{\frac{-6}{3x+2} \times \frac{x+1}{3}}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-6}{3x+2} \right)^{\frac{3x+2}{-6}} \right]^{\frac{-2(x+1)}{3x+2}}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-6}{3x+2} \right)^{\frac{3x+2}{-6}} \right]^{\frac{-2(1+\frac{1}{x})}{3+\frac{2}{x}}} = e^{\frac{-2}{3}} = \frac{1}{e^{2/3}} = \sqrt[3]{e^2}$$

ដូចនេះ: $\lim_{x \rightarrow \infty} \left(\frac{3x-4}{3x+2} \right)^{\frac{x+1}{3}} = \sqrt[3]{e^2}$

e). $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$ មានរាង 1^∞

$$= \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{\cos x}{\sin x}} = \lim_{x \rightarrow 0} \left[(1 + \sin x)^{\frac{1}{\sin x}} \right]^{(\sin x) \cdot \frac{\cos x}{\sin x}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + \sin x)^{\frac{1}{\sin x}} \right]^{(\cos x)} = e^1 = e$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x} = e}$

f). $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x} \right)^x$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x} \right)^{\frac{x}{2}} \right]^{\frac{2}{x} \times x} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x} \right)^{\frac{x}{2}} \right]^2 = e^2$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x} \right)^x = e^2}$

g). $\lim_{x \rightarrow \infty} \left(\frac{2x+3}{2x+1} \right)^{x+1}$ មានរាង 1^∞

$$= \lim_{x \rightarrow \infty} \left(\frac{2x+1+2}{2x+1} \right)^{x+1} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{2x+1} \right)^{\frac{2x+1}{2}} \right]^{\frac{2}{2x+1} (x+1)}$$

$$= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{2x+1} \right)^{\frac{2x+1}{2}} \right]^{\frac{2(x+1)}{2x+1}} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{2x+1} \right)^{\frac{2x+1}{2}} \right]^{\frac{2(1+\frac{1}{x})}{2+\frac{1}{x}}}$$

$$= e^{\frac{2(1+0)}{2+0}} = e^1 = e$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow \infty} \left(\frac{2x+3}{2x+1} \right)^{x+1} = e}$

h). $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin x}}$ មានរាង 1^∞

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left[(1 - 1 + \cos x)^{-1 + \cos x} \right]^{(-1 + \cos x) \cdot \frac{1}{\sin x}} \\
&= \lim_{x \rightarrow 0} \left[(1 - 1 + \cos x)^{-1 + \cos x} \right]^{\frac{-2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}}} \\
&= \lim_{x \rightarrow 0} \left[(1 - 1 + \cos x)^{-1 + \cos x} \right]^{\frac{-\sin \frac{x}{2}}{\cos \frac{x}{2}}} = e^0 = 1
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} (\cos x)^{\sin x} = 1$

i). $\lim_{x \rightarrow 0} (1 + 3 \tan^2 x)^{\cot^2 x}$ មានរាង 1^∞

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left[(1 + 3 \tan^2 x)^{\frac{1}{3 \tan^2 x}} \right]^{3 \tan^2 x \cot^2 x} \\
&= \lim_{x \rightarrow 0} \left[(1 + 3 \tan^2 x)^{\frac{1}{3 \tan^2 x}} \right]^3 = e^3
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} (1 + 3 \tan^2 x)^{\cot^2 x} = e^3$

j). $\lim_{x \rightarrow 0} \sqrt[\sin x]{1 + x}$

$$= \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{\sin x}} = \lim_{x \rightarrow 0} \left[(1 + x)^{\frac{1}{x}} \right]^{\frac{x}{\sin x}} = e^1 = e$$

ដូចនេះ: $\lim_{x \rightarrow 0} \sin^x \sqrt{1+x} = e$

3) គណនាលីមីតខាងក្រោម:

$$\begin{aligned} \text{a). } \lim_{x \rightarrow 0} \left(\frac{2x+3}{x+3} \right)^{\frac{1}{\sin 3x}} & \text{ មានរាង } 1^\infty \\ &= \lim_{x \rightarrow 0} \left(\frac{2x+3}{x+3} \right)^{\frac{1}{\sin 3x}} = \lim_{x \rightarrow 0} \left(1 + \frac{2x+3}{x+3} - 1 \right)^{\frac{1}{\sin 3x}} \\ &= \lim_{x \rightarrow 0} \left(1 + \frac{(2x+3) - (x+3)}{x+3} \right)^{\frac{1}{\sin 3x}} \\ &= \lim_{x \rightarrow 0} \left(1 + \frac{x}{x+3} \right)^{\frac{1}{\sin 3x}} = \lim_{x \rightarrow 0} \left[\left(1 + \frac{x}{x+3} \right)^{\frac{x+3}{x}} \right]^{\frac{x}{x+3} \times \frac{1}{\sin 3x}} \\ &= \lim_{x \rightarrow 0} \left[\left(1 + \frac{x}{x+3} \right)^{\frac{x+3}{x}} \right]^{\frac{1}{3(x+3)} \times \frac{3x}{\sin 3x}} = e^{\frac{1}{3} \times 1} = e^{\frac{1}{3}} = \sqrt[3]{e} \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \left(\frac{2x+3}{x+3} \right)^{\frac{1}{\sin 3x}} = \sqrt[3]{e}$

$$\begin{aligned} \text{b). } \lim_{x \rightarrow 0} (x+2^x)^{\frac{1}{x}} & \text{ មានរាង } 1^\infty \\ &= \lim_{x \rightarrow 0} \left[(1+x+2^x-1)^{\frac{1}{x+2^x-1}} \right]^{(x+2^x-1) \frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \left[(1+x+2^x-1)^{\frac{1}{x+2^x-1}} \right]^{(1+\frac{2^x-1}{x})} = e^{1+\ln 2} = e \cdot e^{\ln 2} \end{aligned}$$

$$= e(2) = 2e$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} (x + 2^x)^{\frac{1}{x}} = 2e}$$

$$\text{c). } \lim_{x \rightarrow 0} (\cos x + \sin x)^{\frac{1}{x}} \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow 0} (1 + \cos x + \sin x - 1)^{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + \cos x + \sin x - 1)^{\frac{1}{\cos x + \sin x - 1}} \right]^{(\cos x + \sin x - 1) \frac{1}{x}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + \cos x + \sin x - 1)^{\frac{1}{\cos x + \sin x - 1}} \right]^{(\frac{\cos x - 1}{x} + \frac{\sin x}{x})} = e^{0+1} = e$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} (\cos x + \sin x)^{\frac{1}{x}} = e}$$

$$\text{d). } \lim_{x \rightarrow 0} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}} \text{ មានរាង } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left[(1 + \tan^2 \sqrt{x})^{\frac{1}{\tan^2 \sqrt{x}}} \right]^{\tan^2 \sqrt{x} \times \frac{1}{2x}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + \tan^2 \sqrt{x})^{\frac{1}{\tan^2 \sqrt{x}}} \right]^{\frac{\tan^2 \sqrt{x}}{(\sqrt{x})^2} \times \frac{1}{2}} = e^{1 \times \frac{1}{2}} = e^{\frac{1}{2}} = \sqrt{e}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} (1 + \tan^2 \sqrt{x})^{\frac{1}{2x}} = \sqrt{e}}$$

6. លីមីតនៃអនុគមន៍លោការីត

ក. អនុគមន៍លោការីត

❖ អនុគមន៍ $y = \log_a x$ ដែល ($a > 0, a \neq 1, x > 0$) ហៅថាអនុគមន៍លោការីត គោល a នៃ x ។

❖ បើ $a = e$ នោះ $y = \log_e x = \ln x$ ហៅថាអនុគមន៍លោការីតនេពែ ។

ខ. លក្ខណៈនៃលោការីតនេពែ

ចំពោះគ្រប់ចំនួនពិតវិជ្ជមាន a និង b គេបាន៖

❶ $\ln ab = \ln a + \ln b$

❷ $\ln \frac{a}{b} = \ln a - \ln b$

❸ $\ln a^p = p \ln a$

❹ $\ln \sqrt{a} = \frac{1}{2} \ln a$

❺ $\ln \frac{1}{a} = -\ln a$

❻ $e^{\ln a} = a$

គ. លីមីតនៃអនុគមន៍លោការីតនេពែ

❶ $\lim_{x \rightarrow +\infty} \ln x = +\infty$

❷ $\lim_{x \rightarrow 0^+} \ln x = -\infty$

❸ $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^n} = 0, (n > 0)$

❹ $\lim_{x \rightarrow +\infty} \frac{x^n}{\ln x} = +\infty, (n > 0)$

❺ $\lim_{x \rightarrow 0^+} x^n \ln x = 0, (n > 0)$

លំហាត់

1) គណនាលីមីតខាងក្រោម៖

a). $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2 - x}$

b). $\lim_{x \rightarrow 0^+} \frac{\ln x}{\cos x}$

c). $\lim_{x \rightarrow -\infty} \ln(e^x + 1)$

d). $\lim_{x \rightarrow +\infty} \ln(\ln x)$

e). $\lim_{x \rightarrow 0^+} \ln\left(\frac{x}{x+1}\right)$

f). $\lim_{x \rightarrow +\infty} \ln\left(\frac{x}{x+1}\right)$

$$g). \lim_{x \rightarrow \pm\infty} x \ln(x^2 + 1)$$

$$i). \lim_{x \rightarrow +\infty} (x \ln x - x^2 + 5)$$

$$h). \lim_{x \rightarrow +\infty} \frac{x^2 + \ln x}{3x^2 - 1}$$

$$j). \lim_{x \rightarrow 0} x^x$$

2) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 1^-} x \ln(4 - 3x - x^2)$$

$$c). \lim_{x \rightarrow +\infty} x[\ln(x+1) - \ln x]$$

$$e). \lim_{x \rightarrow 0^+} (x \ln x - x^2 + 2)$$

$$g). \lim_{x \rightarrow 0^+} \frac{x^2 \ln x + 2}{\ln x + 1}$$

$$i). \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{\ln x}{x} \right)$$

$$b). \lim_{x \rightarrow -4} x \ln(4 - 3x - x^2)$$

$$d). \lim_{x \rightarrow +\infty} \frac{\ln x + 1}{\ln x - 1}$$

$$f). \lim_{x \rightarrow 0^+} \frac{\ln^2 x + 2 \ln x + 3}{\ln x + 2}$$

$$h). \lim_{x \rightarrow +\infty} \ln \left(\frac{x^2 + x - 2}{x^2 + 1} \right)$$

$$j). \lim_{x \rightarrow +\infty} \frac{3x \ln x}{x + 2}$$

3) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow +\infty} \frac{e^x \ln x + 1}{x^2}$$

$$c). \lim_{x \rightarrow 0} \frac{\ln(1+x)}{1-e^x}$$

$$e). \lim_{x \rightarrow 1} \frac{x^2 - 1 + \ln x}{e^x - e}$$

$$g). \lim_{x \rightarrow +\infty} [x + 3 - 3 \ln(e^x + 1)]$$

$$i). \lim_{x \rightarrow 0} \frac{\ln(1+x)}{e^{2x} - 1}$$

$$k). \lim_{x \rightarrow +\infty} \left(x + \frac{\ln x}{2x+1} \right)$$

$$b). \lim_{x \rightarrow +\infty} \frac{\ln x^{2015} - 2015}{x^{2015}}$$

$$d). \lim_{x \rightarrow 0} \frac{\sin 3x - \sin x}{\ln(x+1)}$$

$$f). \lim_{x \rightarrow 0} \frac{\ln(x+2) - \ln 2}{x}$$

$$h). \lim_{x \rightarrow 0} \frac{x - \ln(1+3x)}{x + \sin 3x}$$

$$j). \lim_{x \rightarrow 0} \frac{\ln(1+2x)}{\sin 2x}$$

$$l). \lim_{x \rightarrow +\infty} \frac{\ln(3x-2)}{\sqrt[3]{x}}$$

ដំណោះស្រាយ

1) គណនាលីមីតខាងក្រោម:

$$\text{a). } \lim_{x \rightarrow +\infty} \frac{\ln x}{x^2 - x} = \lim_{x \rightarrow +\infty} \frac{\ln x}{x(x-1)} = \lim_{x \rightarrow +\infty} \frac{\ln x}{x} \times \frac{1}{x-1} = 0 \times 0 = 0$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2 - x} = 0}$$

$$\text{b). } \lim_{x \rightarrow 0^+} \frac{\ln x}{\cos x} = \frac{-\infty}{1} = -\infty$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0^+} \frac{\ln x}{\cos x} = -\infty}$$

$$\text{c). } \lim_{x \rightarrow -\infty} \ln(e^x + 1) = \ln(0 + 1) = \ln 1 = 0$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow -\infty} (e^x + 1) = 0}$$

$$\text{d). } \lim_{x \rightarrow +\infty} \ln(\ln x) = \ln(\ln(+\infty)) = \ln(+\infty) = +\infty$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow +\infty} \ln(\ln x) = +\infty}$$

$$\text{e). } \lim_{x \rightarrow 0^+} \ln\left(\frac{x}{x+1}\right) = \ln 0^+ = -\infty$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow 0^+} \ln\left(\frac{x}{x+1}\right) = -\infty}$$

$$\text{f). } \lim_{x \rightarrow +\infty} \ln\left(\frac{x}{x+1}\right) = \lim_{x \rightarrow +\infty} \ln\left(\frac{x}{x(1 + \frac{1}{x})}\right)$$

$$= \lim_{x \rightarrow +\infty} \ln\left(\frac{1}{1 + \frac{1}{x}}\right) = \ln 1 = 0$$

$$\text{ដូច្នេះ: } \boxed{\lim_{x \rightarrow +\infty} \ln\left(\frac{x}{x+1}\right) = 0}$$

$$\text{g). } \lim_{x \rightarrow \pm\infty} x \ln(x^2 + 1) = (\pm\infty) \ln(+\infty) = \pm\infty$$

ដូចនេះ: $\lim_{x \rightarrow \pm\infty} x \ln(x^2 + 1) = \pm\infty$

$$h). \lim_{x \rightarrow +\infty} \frac{x^2 + \ln x}{3x^2 - 1} = \lim_{x \rightarrow +\infty} \frac{x^2(1 + \frac{\ln x}{x^2})}{x^2(3 - \frac{1}{x^2})}$$

$$= \lim_{x \rightarrow +\infty} \frac{(1 + \frac{\ln x}{x^2})}{(3 - \frac{1}{x^2})} = \frac{1+0}{3-0} = \frac{1}{3}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{x^2 + \ln x}{3x^2 - 1} = \frac{1}{3}$

$$i). \lim_{x \rightarrow +\infty} (x \ln x - x^2 + 5) = \lim_{x \rightarrow +\infty} x^2 \left(\frac{\ln x}{x} - 1 + \frac{5}{x^2} \right)$$

$$= \lim_{x \rightarrow +\infty} x^2 \left(\frac{\ln x}{x} - 1 + \frac{5}{x^2} \right) = (+\infty)(0 - 1 + 0) = -\infty$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} (x \ln x - x^2 + 5) = -\infty$

$$j). \lim_{x \rightarrow 0} x^x$$

តាង $y = x^x$ នោះ $\ln y = \ln x^x = x \ln x$

គេបាន $\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} x \ln x = 0$ នោះ $\lim_{x \rightarrow 0} \ln y = 0$

នាំឲ្យ $\ln(\lim_{x \rightarrow 0} y) = 0$ ឬ $\ln(\lim_{x \rightarrow 0} x^x) = 0$ គេបាន $\lim_{x \rightarrow 0} x^x = 1$

ដូចនេះ: $\lim_{x \rightarrow 0} x^x = 1$

2) គណនាលីមីតខាងក្រោម:

$$a). \lim_{x \rightarrow 1^-} x \ln(4 - 3x - x^2) = \lim_{x \rightarrow 1^-} x \ln(1 - x)(x + 4)$$

$$= (1) \ln(0^+)(0 + 4) = \ln(0^+) = -\infty$$

ដូច្នេះ: $\lim_{x \rightarrow 1^-} x \ln(4 - 3x - x^2) = -\infty$

b). $\lim_{x \rightarrow -4} x \ln(4 - 3x - x^2) = \lim_{x \rightarrow -4} x \ln(1 - x)(x + 4)$

ចំពោះ $x \rightarrow -4^+$ នោះ $x + 4 \rightarrow 0^+$

គេបាន $\lim_{x \rightarrow -4^+} x \ln(4 - 3x - x^2) = \lim_{x \rightarrow -4^+} x \ln(1 - x)(x + 4)$

$= (-4) \ln(1 + 4)(0^+) = -4 \ln 0^+ = -4(-\infty) = +\infty$

ដូច្នេះ: $\lim_{x \rightarrow -4^+} x \ln(4 - 3x - x^2) = +\infty$

c). $\lim_{x \rightarrow +\infty} x[\ln(x+1) - \ln x] = \lim_{x \rightarrow +\infty} x \ln\left(\frac{x+1}{x}\right)$

$= \lim_{x \rightarrow +\infty} x \ln\left(1 + \frac{1}{x}\right) = \lim_{x \rightarrow +\infty} \ln\left(1 + \frac{1}{x}\right)^x = \ln e = 1$

ដូច្នេះ: $\lim_{x \rightarrow +\infty} x[\ln(x+1) - \ln x] = 1$

d). $\lim_{x \rightarrow +\infty} \frac{\ln x + 1}{\ln x - 1} = \lim_{x \rightarrow +\infty} \frac{\ln x(1 + \frac{1}{\ln x})}{\ln x(1 - \frac{1}{\ln x})} = \lim_{x \rightarrow +\infty} \frac{(1 + \frac{1}{\ln x})}{(1 - \frac{1}{\ln x})} = 1$

ដូច្នេះ: $\lim_{x \rightarrow +\infty} \frac{\ln x + 1}{\ln x - 1} = 1$

e). $\lim_{x \rightarrow 0^+} (x \ln x - x^2 + 2) = 0 - 0 + 2 = 2$

ដូច្នេះ: $\lim_{x \rightarrow 0^+} (x \ln x - x^2 + 2) = 2$

f). $\lim_{x \rightarrow 0^+} \frac{\ln^2 x + 2 \ln x + 3}{\ln x + 2} = \lim_{x \rightarrow 0^+} \frac{\ln x(\ln x + 2 + \frac{3}{\ln x})}{\ln x(1 + \frac{2}{\ln x})}$

$$= \lim_{x \rightarrow 0^+} \frac{(\ln x + 2 + \frac{3}{\ln x})}{(1 + \frac{2}{\ln x})} = \frac{-\infty + 2 + 0}{1 + 0} = -\infty$$

ដូចនេះ: $\lim_{x \rightarrow 0^+} \frac{\ln^2 x + 2 \ln x + 3}{\ln x + 2} = -\infty$

g). $\lim_{x \rightarrow 0^+} \frac{x^2 \ln x + 2}{\ln x + 1} = \frac{0 + 2}{-\infty + 1} = \frac{2}{-\infty} = 0$

ដូចនេះ: $\lim_{x \rightarrow 0^+} \frac{x^2 \ln x + 2}{\ln x + 1} = 0$

h). $\lim_{x \rightarrow +\infty} \ln \left(\frac{x^2 + x - 2}{x^2 + 1} \right) = \lim_{x \rightarrow +\infty} \ln \left(\frac{x^2(1 + \frac{1}{x} - \frac{2}{x^2})}{x^2(1 + \frac{1}{x^2})} \right)$

$$= \lim_{x \rightarrow +\infty} \ln \left(\frac{1 + \frac{1}{x} - \frac{2}{x^2}}{1 + \frac{1}{x^2}} \right) = \ln \frac{1 + 0 - 0}{1 + 0} = \ln 1 = 0$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \ln \left(\frac{x^2 + x - 2}{x^2 + 1} \right) = 0$

i). $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{\ln x}{x} \right) = \lim_{x \rightarrow 0^+} \left(\frac{1 - \ln x}{x} \right) = \frac{1 - (-\infty)}{0^+} = \frac{+\infty}{0^+} = +\infty$

ដូចនេះ: $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{\ln x}{x} \right) = +\infty$

j). $\lim_{x \rightarrow +\infty} \frac{3x \ln x}{x + 2} = \lim_{x \rightarrow +\infty} \frac{3x \ln x}{x(1 + \frac{2}{x})} = \lim_{x \rightarrow +\infty} \frac{3 \ln x}{1 + \frac{2}{x}} = \frac{+\infty}{1 + 0} = +\infty$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{3x \ln x}{x + 2} = +\infty$

3) គណនាលីមីតខាងក្រោម:

$$\begin{aligned} \text{a). } \lim_{x \rightarrow +\infty} \frac{e^x \ln x + 1}{x^2} & \text{ មានរាង } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow +\infty} \frac{e^x (\ln x + \frac{1}{e^x})}{x^2} = \lim_{x \rightarrow +\infty} \frac{e^x}{x^2} (\ln x + \frac{1}{e^x}) = (+\infty)(+\infty + 0) = +\infty \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{e^x \ln x + 1}{x^2} = +\infty}$$

$$\begin{aligned} \text{b). } \lim_{x \rightarrow +\infty} \frac{\ln x^{2015} - 2015}{x^{2015}} & \text{ មានរាង } \frac{\infty}{\infty} \\ &= \lim_{x \rightarrow +\infty} \frac{2015 \ln x - 2015}{x^{2015}} = \lim_{x \rightarrow +\infty} 2015 \left(\frac{\ln x}{x^{2015}} - \frac{1}{x^{2015}} \right) \\ &= 2015(0 - 0) = 0 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow +\infty} \frac{\ln x^{2015} - 2015}{x^{2015}} = 0}$$

$$\begin{aligned} \text{c). } \lim_{x \rightarrow 0} \frac{\ln(1+x)}{1-e^x} & \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\ln(1+x)}{x}}{\frac{1-e^x}{x}} = \lim_{x \rightarrow 0} \frac{\frac{1}{x} \ln(1+x)}{\frac{e^x - 1}{x}} = - \lim_{x \rightarrow 0} \frac{\ln(1+x)^{\frac{1}{x}}}{\frac{e^x - 1}{x}} \\ &= - \frac{\ln e}{1} = - \frac{1}{1} = -1 \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{\ln(1+x)}{1-e^x} = -1}$$

$$\begin{aligned} \text{d). } \lim_{x \rightarrow 0} \frac{\sin 3x - \sin x}{\ln(x+1)} & \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{x} - \frac{\sin x}{x}}{\frac{\ln(x+1)}{x}} = \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{3x} \times 3 - \frac{\sin x}{x}}{\frac{1}{\ln(1+x)^x}} = \frac{3-1}{\ln e} = \frac{2}{1} = 2 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sin 3x - \sin x}{\ln(x+1)} = 2$

e). $\lim_{x \rightarrow 1} \frac{x^2 - 1 + \ln x}{e^x - e}$ មានរាង $\frac{0}{0}$

តាង $u = x - 1$ នោះ: $x = 1 + u$

បើ $x \rightarrow 1$ នោះ: $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន } \lim_{x \rightarrow 1} \frac{x^2 - 1 + \ln x}{e^x - e} &= \lim_{u \rightarrow 0} \frac{(1+u)^2 - 1 + \ln(1+u)}{e^{1+u} - e} \\ &= \lim_{u \rightarrow 0} \frac{1 + 2u + u^2 - 1 + \ln(1+u)}{e(e^u - 1)} = \frac{1}{e} \lim_{u \rightarrow 0} \frac{2u + u^2 + \ln(1+u)}{e^u - 1} \\ &= \frac{1}{e} \lim_{u \rightarrow 0} \frac{\frac{2u}{u} + \frac{u^2}{u} + \frac{\ln(1+u)}{u}}{\frac{e^u - 1}{u}} = \frac{1}{e} \lim_{u \rightarrow 0} \frac{2 + u + \ln(1+u)^u}{\frac{e^u - 1}{u}} \\ &= \frac{1}{e} \cdot \frac{2+0+\ln e}{1} = \frac{1}{e} \cdot \frac{2+1}{1} = \frac{3}{e} \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^2 - 1 + \ln x}{e^x - e} = \frac{3}{e}$

f). $\lim_{x \rightarrow 0} \frac{\ln(x+2) - \ln 2}{x}$ មានរាង $\frac{0}{0}$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\ln \frac{x+2}{2}}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \ln\left(1 + \frac{x}{2}\right) = \lim_{x \rightarrow 0} \ln\left(1 + \frac{x}{2}\right)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \ln \left[\left(1 + \frac{x}{2}\right)^{\frac{2}{x}} \right]^{\frac{x}{2} \cdot \frac{1}{x}} = \lim_{x \rightarrow 0} \ln \left[\left(1 + \frac{x}{2}\right)^{\frac{2}{x}} \right]^{\frac{1}{2}} = \ln e^{\frac{1}{2}} = \frac{1}{2} \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\ln(x+2) - \ln 2}{x} = \frac{1}{2}$

g). $\lim_{x \rightarrow +\infty} [x + 3 - 3\ln(e^x + 1)]$ មានរាង $(\infty - \infty)$

$$\begin{aligned}
&= \lim_{x \rightarrow +\infty} [x + 3 - 3\ln e^x (1 + \frac{1}{e^x})] = \lim_{x \rightarrow +\infty} [x + 3 - 3\ln e^x - 3\ln(1 + \frac{1}{e^x})] \\
&= \lim_{x \rightarrow +\infty} [x + 3 - 3x - 3\ln(1 + \frac{1}{e^x})] = \lim_{x \rightarrow +\infty} [-2x + 3 - 3\ln(1 + \frac{1}{e^x})] \\
&= [-\infty + 3 - 3\ln(1 + 0)] = -\infty + 3 - 0 = -\infty
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} [x + 3 - 3\ln(e^x + 1)] = -\infty$

h). $\lim_{x \rightarrow 0} \frac{x - \ln(1 + 3x)}{x + \sin 3x}$ មានរាង $\frac{0}{0}$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\frac{x}{x} - \frac{\ln(1 + 3x)}{x}}{\frac{x}{x} + \frac{\sin 3x}{x}} = \lim_{x \rightarrow 0} \frac{1 - \ln(1 + 3x)^{\frac{1}{x}}}{1 + \frac{\sin 3x}{3x} \times 3} \\
&= \lim_{x \rightarrow 0} \frac{1 - \ln \left[(1 + 3x)^{\frac{1}{3x}} \right]^{3x \cdot \frac{1}{x}}}{1 + \frac{\sin 3x}{3x} \times 3} = \lim_{x \rightarrow 0} \frac{1 - \ln \left[(1 + 3x)^{\frac{1}{3x}} \right]^3}{1 + \frac{\sin 3x}{3x} \times 3} \\
&= \frac{1 - \ln e^3}{1 + 3} = \frac{1 - 3}{4} = -\frac{2}{4} = -\frac{1}{2}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{x - \ln(1 + 3x)}{x + \sin 3x} = -\frac{1}{2}$

i). $\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{e^{2x} - 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\frac{\ln(1 + x)}{x}}{\frac{e^{2x} - 1}{x}} = \lim_{x \rightarrow 0} \frac{\ln(1 + x)^{\frac{1}{x}}}{\frac{e^{2x} - 1}{2x} \times 2} = \frac{\ln e}{1 \times 2} = \frac{1}{2}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{e^{2x} - 1} = \frac{1}{2}$

j). $\lim_{x \rightarrow 0} \frac{\ln(1 + 2x)}{\sin 2x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\frac{\ln(1+2x)}{2x}}{\frac{\sin 2x}{2x}} = \lim_{x \rightarrow 0} \frac{\ln(1+2x)^{\frac{1}{2x}}}{\frac{\sin 2x}{2x}} = \frac{\ln e}{1} = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\ln(1+2x)}{\sin 2x} = 1$

ក). $\lim_{x \rightarrow +\infty} \left(x + \frac{\ln x}{2x+1} \right)$

$$= \lim_{x \rightarrow +\infty} x \left(1 + \frac{\ln x}{x} \times \frac{1}{2x+1} \right) = (+\infty)(1+0 \times 1) = +\infty$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \left(x + \frac{\ln x}{2x+1} \right) = +\infty$

ឈ). $\lim_{x \rightarrow +\infty} \frac{\ln(3x-2)}{\sqrt[3]{x}}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{\ln x(3-\frac{2}{x})}{\sqrt[3]{x}} = \lim_{x \rightarrow +\infty} \frac{\ln x + \ln(3-\frac{2}{x})}{x^{1/3}}$$

$$= \lim_{x \rightarrow +\infty} \left[\frac{\ln x}{x^{1/3}} + \frac{\ln(3-\frac{2}{x})}{x^{1/3}} \right] = 0 + \frac{\ln 3}{+\infty} = 0$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{\ln(3x-2)}{\sqrt[3]{x}} = 0$

លំហាត់សារឡើងវិញ

គណនាលីមីតខាងក្រោម៖

1). $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

2). $\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x - 1}$

3). $\lim_{x \rightarrow -1} \frac{x + 1}{x^2 + 2x + 1}$

4). $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 - 5x + 4}$

5). $\lim_{x \rightarrow 1} \frac{3x^3 - 3x^2 + 2x - 2}{(x - 1)^3}$

6). $\lim_{x \rightarrow 2} \frac{x^3 - x^2 + 2x - 8}{x - 2}$

7). $\lim_{x \rightarrow 3} \frac{x^3 - 5x - 12}{x - 3}$

8). $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 - 4}{x^3 - 1}$

9). $\lim_{x \rightarrow 1} \frac{x^4 - 2x^2 + 1}{x^2 - 8x + 7}$

10). $\lim_{x \rightarrow 1} \frac{3x^4 - 4x^3 + 1}{(x - 1)^2}$

11). $\lim_{x \rightarrow -3} \frac{x^4 - 6x^2 - 27}{x^3 + 3x^2 + x + 3}$

12). $\lim_{x \rightarrow 1} \frac{x^9 - 3x + 2}{x^6 + 5x - 6}$

13). $\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^2 - x}$

14). $\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6}$

15). $\lim_{x \rightarrow 1} \frac{(x - 1)\sqrt{2 - x}}{x^2 - 1}$

16). $\lim_{x \rightarrow 1/2} \frac{8x^3 - 1}{6x^2 - 5x + 1}$

17). $\lim_{x \rightarrow 1} \frac{x^3 + x - 2}{x^3 - x^2 - x + 1}$

18). $\lim_{x \rightarrow 1} \left(\frac{1}{1 - x} - \frac{3}{1 - x^2} \right)$

19). $\lim_{x \rightarrow 0} \frac{\sqrt{1 + x^2} - 1}{x}$

20). $\lim_{x \rightarrow 0} \frac{\sqrt{1 + x} - 1}{x^2}$

21). $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 1} - 1}{\sqrt{x^2 + 16} - 4}$

22). $\lim_{x \rightarrow 5} \frac{\sqrt{x - 1} - 2}{x - 5}$

23). $\lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$

24). $\lim_{h \rightarrow 0} \frac{\sqrt{x + h} - \sqrt{x}}{h}$

- 25). $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1}{x^2}$
- 26). $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x}$
- 27). $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$
- 28). $\lim_{x \rightarrow a} \frac{\sqrt{x-b} - \sqrt{a-b}}{x^2 - a^2}, (a > b)$
- 29). $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^2} - \sqrt{3+x^2}}{x-1}$
- 30). $\lim_{x \rightarrow 1} \frac{\sqrt[n]{x} - 1}{\sqrt[m]{x} - 1} \quad (n, m \text{ ជាចំនួនគត់វិជ្ជមាន})$
- 31). $\lim_{x \rightarrow 0} \frac{\sqrt{a} - \sqrt{a-x}}{x}$
- 32). $\lim_{x \rightarrow 8} \frac{\sqrt[3]{x} - 2}{\sqrt[3]{x+19} - 3}$
- 33). $\lim_{x \rightarrow 4} \frac{\sqrt[3]{3x-4} - 2}{\sqrt{4x+9} - 5}$
- 34). $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+2x} - \sqrt[4]{1+3x}}{x}$
- 35). $\lim_{x \rightarrow 0} \left(\frac{1}{x(x+1)} - \frac{1}{x} \right)$
- 36). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt[3]{1+x}}{x}$
- 37). $\lim_{x \rightarrow 0} \frac{\sqrt[5]{x+1} - 1}{\sqrt[3]{x+1} - 1}$
- 38). $\lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{x}} - \frac{x+1}{\sqrt{x}} \right)$
- 39). $\lim_{x \rightarrow 0} \frac{x^2 + |x| + x}{x^2 - |x| + 3x}$
- 40). $\lim_{x \rightarrow \infty} (\sqrt{x^2+1} - \sqrt{x^2-1})$
- 41). $\lim_{x \rightarrow \pm\infty} (\sqrt{x^2+1} - x)$
- 42). $\lim_{x \rightarrow \pm\infty} x(\sqrt{x^2+1} - x)$
- 43). $\lim_{x \rightarrow \pm\infty} (\sqrt{(x+a)(x+b)} - x)$
- 44). $\lim_{x \rightarrow \pm\infty} (\sqrt{x^2-2x-1} - \sqrt{x^2-7x+3})$
- 45). $\lim_{x \rightarrow \infty} (\sqrt[3]{(x+1)^2} - \sqrt[3]{(x-1)^2})$
- 46). $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x + \sqrt{2x + \sqrt{x}}}}{\sqrt{\frac{1}{5}x - 7}}$
- 47). $\lim_{x \rightarrow -\infty} \frac{-2x^2 - 2}{x^2 + 3x - 1}$
- 48). $\lim_{x \rightarrow +\infty} \frac{x^2 - 2x - 2}{-3x^2 + 4}$
- 49). $\lim_{x \rightarrow +\infty} \frac{x^{100} + x^{10} + 1}{100x^{100} - 10x^{10} - 1}$
- 50). $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + x + 3}}{x}$

$$51). \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sqrt{1+x} - 1}$$

$$53). \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x}$$

$$55). \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$$

$$57). \lim_{x \rightarrow -1} \frac{x^3 + 1}{\sin(x+1)}$$

$$59). \lim_{x \rightarrow 0} \frac{\cos 2x - e^{x^2}}{x^2}$$

$$61). \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \cos x}{x^2}$$

$$63). \lim_{x \rightarrow 0} \frac{\sqrt{1-x^3} - 1}{\sin x^2}$$

$$65). \lim_{x \rightarrow -\frac{\pi}{2}} \frac{\sin^2 x - 1}{1 + \sin x}$$

$$67). \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos 2x}}{x^2}$$

$$69). \lim_{x \rightarrow 0} \frac{\tan kx}{x}$$

$$71). \lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} \quad (n, m \text{ ជាចំនួនគត់វិជ្ជមាន})$$

$$73). \lim_{x \rightarrow \infty} x(e^{\frac{1}{x}} - 1)$$

$$75). \lim_{x \rightarrow \pm \infty} \left(1 + \frac{1}{x} \right)^{x^2}$$

$$52). \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x}$$

$$54). \lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + x\right) - \frac{1}{2}}{x}$$

$$56). \lim_{x \rightarrow 1} \frac{\sin(1-x)}{\sqrt{x} - 1}$$

$$58). \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\tan x - \cot x}$$

$$60). \lim_{x \rightarrow 0} \frac{e^{-x^2} \cos 4x - 1}{x^2}$$

$$62). \lim_{x \rightarrow 1} \frac{x^3 + x^2 - 2}{\sin(x-1)}$$

$$64). \lim_{x \rightarrow 0} \frac{\sin \alpha x}{\sin \beta x}$$

$$66). \lim_{x \rightarrow 0} \frac{2 \tan x + \sin x}{x}$$

$$68). \lim_{x \rightarrow +\infty} \frac{(x+1) \sin x}{x^2 + 3}$$

$$70). \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{(\pi - 2x)^2}$$

$$72). \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$$

$$74). \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x} \right)^x$$

$$76). \lim_{x \rightarrow 0} \frac{\ln(1+kx)}{x}$$

$$77). \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}}$$

$$78). \lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$$

$$79). \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}}$$

$$80). \lim_{n \rightarrow \infty} \left(\frac{1 + 2 + 3 + \dots + n}{n + 2} - \frac{n}{2} \right)$$

$$81). \lim_{n \rightarrow \infty} \left(\frac{1 - 2 + 3 - 4 + \dots - 2n}{\sqrt{n^2 + 1}} \right)$$

$$82). \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right)$$

$$83). \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$$

84). កំណត់អនុគមន៍ដ៏ក្រៃទីពីរ $f(x)$ ដែលបំពេញលក្ខខណ្ឌលីមីតទាំងពីរខាងក្រោម:

$$\text{ក). } \lim_{x \rightarrow +\infty} \frac{f(x)}{x^2 - 1} = 1$$

$$\text{ខ). } \lim_{x \rightarrow 1} \frac{f(x)}{x^2 - 1} = -1$$

85). កំណត់តម្លៃនៃចំនួនថេរ a ដើម្បីឲ្យលីមីតខាងក្រោមជាលីមីតនៃចំនួនថេរ ហើយកំណត់លីមីតនេះផង:

$$\text{ក/. } \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - a}{x-1}$$

$$\text{ខ/. } \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} + a}{x}$$

$$\text{គ/. } \lim_{x \rightarrow 2} \frac{\sqrt{x+a} - 1}{x-2}$$

$$\text{ឃ/. } \lim_{x \rightarrow -1} \frac{\sqrt{x^2 + ax} - 1}{x^2 - 1}$$

ដំណោះស្រាយ

$$1). \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} (x+2) = 2+2 = 4$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4}$

2). $\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x - 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 1} \frac{(x-1)^2}{x-1} = \lim_{x \rightarrow 1} (x-1) = 1-1 = 0$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x - 1} = 0}$

3). $\lim_{x \rightarrow -1} \frac{x+1}{x^2 + 2x + 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow -1} \frac{x+1}{(x+1)^2} = \lim_{x \rightarrow -1} \frac{1}{x+1} = \frac{1}{-1+1} = \frac{1}{0} = \pm\infty$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow -1} \frac{x+1}{x^2 + 2x + 1} = \pm\infty}$

4). $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 - 5x + 4}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+7)}{(x-1)(x-4)} = \lim_{x \rightarrow 1} \frac{x+7}{x-4} = \frac{1+7}{1-4} = \frac{8}{-3} = -\frac{8}{3}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 - 5x + 4} = -\frac{8}{3}}$

5). $\lim_{x \rightarrow 1} \frac{3x^3 - 3x^2 + 2x - 2}{(x-1)^3}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 1} \frac{(3x^3 - 3x^2) + (2x - 2)}{(x-1)^3} = \lim_{x \rightarrow 1} \frac{3x^2(x-1) + 2(x-1)}{(x-1)^3}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(3x^2 + 2)}{(x-1)^3} = \lim_{x \rightarrow 1} \frac{(3x^2 + 2)}{(x-1)^2} = \frac{5}{0^+} = +\infty$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{3x^3 - 3x^2 + 2x - 2}{(x-1)^3} = +\infty$

6). $\lim_{x \rightarrow 2} \frac{x^3 - x^2 + 2x - 8}{x - 2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 2} \frac{(x^3 - 8) - (x^2 - 2x)}{x - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4) - x(x-2)}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4 - x)}{x - 2} = \lim_{x \rightarrow 2} (x^2 + x + 4) = 10$$

ដូចនេះ: $\lim_{x \rightarrow 2} \frac{x^3 - x^2 + 2x - 8}{x - 2} = 10$

7). $\lim_{x \rightarrow 3} \frac{x^3 - 5x - 12}{x - 3}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 3} \frac{x^3 - 27 - 5x + 15}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9) - 5(x-3)}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9 - 5)}{x - 3} = \lim_{x \rightarrow 3} (x^2 + 3x + 4) = 22$$

ដូចនេះ: $\lim_{x \rightarrow 3} \frac{x^3 - 5x - 12}{x - 3} = 22$

8). $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 - 4}{x^3 - 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 1} \frac{x^3 - 1 + 3x^2 - 3}{x^3 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 1) + 3(x-1)(x+1)}{(x-1)(x^2 + x + 1)}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)[(x^2 + x + 1) + 3(x+1)]}{(x-1)(x^2 + x + 1)} = \lim_{x \rightarrow 1} \frac{(x^2 + x + 1 + 3x + 3)}{(x^2 + x + 1)}$$

$$= \lim_{x \rightarrow 1} \frac{(x^2 + 4x + 4)}{(x^2 + x + 1)} = \frac{1 + 4 + 4}{1 + 1 + 1} = \frac{9}{3} = 3$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 - 4}{x^3 - 1} = 3$

$$\begin{aligned}
 9). \quad & \lim_{x \rightarrow 1} \frac{x^4 - 2x^2 + 1}{x^2 - 8x + 7} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 1} \frac{x^4 - 1 - 2x^2 + 2}{x^2 - 8x + 7} = \lim_{x \rightarrow 1} \frac{(x-1)(x^3 + x^2 + x + 1) - 2(x-1)(x+1)}{(x-1)(x-7)} \\
 &= \lim_{x \rightarrow 1} \frac{(x-1)[(x^3 + x^2 + x + 1) - 2(x+1)]}{(x-1)(x-7)} \\
 &= \lim_{x \rightarrow 1} \frac{(x^3 + x^2 + x + 1 - 2x - 2)}{(x-7)} = \lim_{x \rightarrow 1} \frac{x^3 + x^2 - x - 1}{x-7} = \frac{0}{-7} = 0
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^4 - 2x^2 + 1}{x^2 - 8x + 7} = 0$

$$\begin{aligned}
 10). \quad & \lim_{x \rightarrow 1} \frac{3x^4 - 4x^3 + 1}{(x-1)^2} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 1} \frac{3x^4 - 3 - 4x^3 + 4}{(x-1)^2} = \lim_{x \rightarrow 1} \frac{3(x^4 - 1) - 4(x^3 - 1)}{(x-1)^2} \\
 &= \lim_{x \rightarrow 1} \frac{3(x-1)(x^3 + x^2 + x + 1) - 4(x-1)(x^2 + x + 1)}{(x-1)^2} \\
 &= \lim_{x \rightarrow 1} \frac{3(x^3 + x^2 + x + 1) - 4(x^2 + x + 1)}{(x-1)} = \lim_{x \rightarrow 1} \frac{3x^3 - x^2 - x - 1}{x-1} \\
 &= \lim_{x \rightarrow 1} \frac{(3x^3 - 3) - (x^2 + x - 2)}{x-1} = \lim_{x \rightarrow 1} \frac{3(x-1)(x^2 + x + 1) - (x-1)(x+2)}{x-1} \\
 &= \lim_{x \rightarrow 1} [3(x^2 + x + 1) - (x+2)] = 3(3) - (3) = 9 - 3 = 6
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{3x^4 - 4x^3 + 1}{(x-1)^2} = 6$

$$\begin{aligned}
 11). \quad & \lim_{x \rightarrow -3} \frac{x^4 - 6x^2 - 27}{x^3 + 3x^2 + x + 3} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow -3} \frac{(x^2 - 3)^2 - 36}{x^2(x+3) + (x+3)} = \lim_{x \rightarrow -3} \frac{(x^2 - 3 - 6)(x^2 - 3 + 6)}{(x+3)(x^2 + 1)}
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow -3} \frac{(x^2 - 9)(x^2 + 3)}{(x + 3)(x^2 + 1)} = \lim_{x \rightarrow -3} \frac{(x - 3)(x + 3)(x^2 + 3)}{(x + 3)(x^2 + 1)} \\
&= \lim_{x \rightarrow -3} \frac{(x - 3)(x^2 + 3)}{(x^2 + 1)} = \frac{(-3 - 3)(9 + 3)}{9 + 1} = \frac{(-6)(12)}{10} = -\frac{36}{5}
\end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow -3} \frac{x^4 - 6x^2 - 27}{x^3 + 3x^2 + x + 3} = -\frac{36}{5}$$

$$\begin{aligned}
12). \quad &\lim_{x \rightarrow 1} \frac{x^9 - 3x + 2}{x^6 + 5x - 6} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 1} \frac{x^9 - 1 - 3x + 3}{x^6 - 1 + 5x - 5} = \lim_{x \rightarrow 1} \frac{(x - 1)(x^8 + x^7 + x + \dots + 1) - 3(x - 1)}{(x - 1)(x^5 + x^4 + \dots + 1) + 5(x - 1)} \\
&= \lim_{x \rightarrow 1} \frac{(x^8 + x^7 + x + \dots + 1) - 3}{(x^5 + x^4 + \dots + 1) + 5} = \frac{9 - 3}{6 + 5} = \frac{6}{11}
\end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow 1} \frac{x^9 - 3x + 2}{x^6 + 5x - 6} = \frac{6}{11}$$

$$\begin{aligned}
13). \quad &\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^2 - x} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 1} \frac{(x - 1)^2}{x(x - 1)} = \lim_{x \rightarrow 1} \frac{x - 1}{x} = \frac{1 - 1}{1} = 0
\end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^2 - x} = 0$$

$$\begin{aligned}
14). \quad &\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow -2} \frac{x(x^2 + 3x + 2)}{x^2 - x - 6} = \lim_{x \rightarrow -2} \frac{x(x + 1)(x + 2)}{(x - 3)(x + 2)} = \lim_{x \rightarrow -2} \frac{x(x + 1)}{(x - 3)} \\
&= \frac{(-2)(-2 + 1)}{(-2 - 3)} = \frac{2}{-5} = -\frac{2}{5}
\end{aligned}$$

ដូចនេះ:
$$\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6} = -\frac{2}{5}$$

$$15). \lim_{x \rightarrow 1} \frac{(x-1)\sqrt{2-x}}{x^2-1} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)\sqrt{2-x}}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{\sqrt{2-x}}{(x+1)} = \frac{\sqrt{2-1}}{1+1} = \frac{1}{2}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1} \frac{(x-1)\sqrt{2-x}}{x^2-1} = \frac{1}{2}}$

$$16). \lim_{x \rightarrow 1/2} \frac{8x^3-1}{6x^2-5x+1} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 1/2} \frac{(2x)^3-1}{6x^2-5x+1} = \lim_{x \rightarrow 1/2} \frac{(2x-1)(4x^2+2x+1)}{(2x-1)(3x-1)}$$

$$= \lim_{x \rightarrow 1/2} \frac{(4x^2+2x+1)}{(3x-1)} = \frac{4\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 1}{3\left(\frac{1}{2}\right) - 1} = \frac{1+1+1}{\frac{3}{2}-1} = \frac{3 \times 2}{3-2} = 6$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1/2} \frac{8x^3-1}{6x^2-5x+1} = 6}$

$$17). \lim_{x \rightarrow 1} \frac{x^3+x-2}{x^3-x^2-x+1} \text{ មានរាង } \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{(x^3-1)+(x-1)}{(x^3-x^2)-(x-1)} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)+(x-1)}{x^2(x-1)-(x-1)}$$

$$= \lim_{x \rightarrow 1} \frac{(x^2+x+1)+1}{x^2-1} = \frac{(1+1+1)+1}{1-1} = \frac{4}{0} = \pm \infty$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 1} \frac{x^3+x-2}{x^3-x^2-x+1} = \pm \infty}$

$$18). \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^2} \right) \text{ មានរាង } (\infty - \infty)$$

$$= \lim_{x \rightarrow 1} \left(\frac{1+x}{(1-x)(1+x)} - \frac{3}{1-x^2} \right) = \lim_{x \rightarrow 1} \frac{1+x-3}{1-x^2}$$

$$= \frac{1+1-3}{1-1} = \frac{-1}{0} = \pm\infty$$

ដូចនេះ: $\lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^2} \right) = \pm\infty$

19). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}-1}{x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x^2}-1)(\sqrt{1+x^2}+1)}{x(\sqrt{1+x^2}+1)} = \lim_{x \rightarrow 0} \frac{(1+x^2-1)}{x(\sqrt{1+x^2}+1)}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{x(\sqrt{1+x^2}+1)} = \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x^2}+1} = 0$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2}-1}{x} = 0$

20). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x}-1)(\sqrt{1+x}+1)}{x^2(\sqrt{1+x}+1)} = \lim_{x \rightarrow 0} \frac{1+x-1}{x^2(\sqrt{1+x}+1)}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x^2(\sqrt{1+x}+1)} = \lim_{x \rightarrow 0} \frac{1}{x(\sqrt{1+x}+1)} = \frac{1}{0} = \pm\infty$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x^2} = \pm\infty$

21). $\lim_{x \rightarrow 0} \frac{\sqrt{x^2+1}-1}{\sqrt{x^2+16}-4}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(x^2+1-1)(\sqrt{x^2+16}+4)}{(x^2+16-16)(\sqrt{x^2+1}+1)} = \lim_{x \rightarrow 0} \frac{(x^2)(\sqrt{x^2+16}+4)}{(x^2)(\sqrt{x^2+1}+1)}$$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{x^2+16}+4)}{(\sqrt{x^2+1}+1)} = \frac{\sqrt{16}+4}{\sqrt{1}+1} = \frac{4+4}{1+1} = \frac{8}{2} = 4$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 1} - 1}{\sqrt{x^2 + 16} - 4} = 4$

22). $\lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x-5}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 5} \frac{x-1-4}{(x-5)(\sqrt{x-1}+2)} = \lim_{x \rightarrow 5} \frac{x-5}{(x-5)(\sqrt{x-1}+2)}$$

$$= \lim_{x \rightarrow 5} \frac{1}{(\sqrt{x-1}+2)} = \frac{1}{2+2} = \frac{1}{4}$$

ដូចនេះ: $\lim_{x \rightarrow 5} \frac{\sqrt{x-1} - 2}{x-5} = \frac{1}{4}$

23). $\lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 1} \frac{(x^4 - x)(\sqrt{x} + 1)}{(x-1)(x^2 + \sqrt{x})} = \lim_{x \rightarrow 1} \frac{x(x^3 - 1)(\sqrt{x} + 1)}{(x-1)(x^2 + \sqrt{x})}$$

$$= \lim_{x \rightarrow 1} \frac{x(x-1)(x^2 + x + 1)(\sqrt{x} + 1)}{(x-1)(x^2 + \sqrt{x})} = \lim_{x \rightarrow 1} \frac{x(x^2 + x + 1)(\sqrt{x} + 1)}{(x^2 + \sqrt{x})}$$

$$= \frac{(1)(1+1+1)(1+1)}{(1+1)} = \frac{3 \times 2}{2} = 3$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^2 - \sqrt{x}}{\sqrt{x} - 1} = 3$

24). $\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$ មានរាង $\frac{0}{0}$

$$= \lim_{h \rightarrow 0} \frac{(x+h-x)}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}} = \frac{\sqrt{x}}{2x}$$

ដូចនេះ: $\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{\sqrt{x}}{2x}$

$$\begin{aligned}
 25). \quad & \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1}{x^2} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(1+x^2) - 1}{x^2 (\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1)} \\
 &= \lim_{x \rightarrow 0} \frac{x^2}{x^2 (\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1)} \\
 &= \lim_{x \rightarrow 0} \frac{1}{(\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1)} = \frac{1}{1+1+1} = \frac{1}{3}
 \end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1}{x^2} = \frac{1}{3}}$

$$\begin{aligned}
 26). \quad & \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{x (\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x} \sqrt[3]{1-x} + \sqrt[3]{(1-x)^2})} \\
 &= \lim_{x \rightarrow 0} \frac{2x}{x (\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x} \sqrt[3]{1-x} + \sqrt[3]{(1-x)^2})} \\
 &= \lim_{x \rightarrow 0} \frac{2}{\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x} \sqrt[3]{1-x} + \sqrt[3]{(1-x)^2}} = \frac{2}{1+1+1} = \frac{2}{3}
 \end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x} = \frac{2}{3}}$

$$\begin{aligned}
 27). \quad & \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2} \text{ មានរាង } \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1 - \sqrt[4]{1-2x} + 1}{x+x^2} = \lim_{x \rightarrow 0} \frac{(\sqrt[3]{1+x^2} - 1) - (\sqrt[4]{1-2x} - 1)}{x+x^2} \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sqrt[3]{1+x^2} - 1}{x+x^2} - \frac{\sqrt[4]{1-2x} - 1}{x+x^2} \right)
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{1+x^2-1}{(x+x^2)[\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1]} \\
&- \lim_{x \rightarrow 0} \frac{1-2x-1}{(x+x^2)[\sqrt[4]{(1-2x)^3} + \sqrt[4]{(1-2x)^2} + \sqrt[4]{1-2x} + 1]} \\
&= \lim_{x \rightarrow 0} \frac{x^2}{x(1+x)[\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1]} \\
&- \lim_{x \rightarrow 0} \frac{-2x}{x(1+x)[\sqrt[4]{(1-2x)^3} + \sqrt[4]{(1-2x)^2} + \sqrt[4]{1-2x} + 1]} \\
&= \lim_{x \rightarrow 0} \frac{x}{(1+x)[\sqrt[3]{(1+x^2)^2} + \sqrt[3]{1+x^2} + 1]} \\
&+ \lim_{x \rightarrow 0} \frac{2}{(1+x)[\sqrt[4]{(1-2x)^3} + \sqrt[4]{(1-2x)^2} + \sqrt[4]{1-2x} + 1]} \\
&= \frac{0}{(1)(1+1+1)} + \frac{2}{(1)(1+1+1+1)} = 0 + \frac{2}{4} = \frac{1}{2}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2} = \frac{1}{2}$

28). $\lim_{x \rightarrow a} \frac{\sqrt{x-b} - \sqrt{a-b}}{x^2 - a^2}, (a > b)$ មានរាង $\frac{0}{0}$

$$\begin{aligned}
&= \lim_{x \rightarrow a} \frac{(x-b) - (a-b)}{(x-a)(x+a)(\sqrt{x-b} + \sqrt{a-b})} \\
&= \lim_{x \rightarrow a} \frac{x-a}{(x-a)(x+a)(\sqrt{x-b} + \sqrt{a-b})} \\
&= \lim_{x \rightarrow a} \frac{1}{(x+a)(\sqrt{x-b} + \sqrt{a-b})} = \frac{1}{(2a)(\sqrt{a-b} + \sqrt{a-b})} \\
&= \frac{1}{2a(2\sqrt{a-b})} = \frac{1}{4a\sqrt{a-b}}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow a} \frac{\sqrt{x-b} - \sqrt{a-b}}{x^2 - a^2} = \frac{1}{4a\sqrt{a-b}}$

$$\begin{aligned}
29). \lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^2} - \sqrt{3+x^2}}{x-1} & \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 1} \frac{\left(\sqrt[3]{7+x^2} - 2\right) - \left(\sqrt{3+x^2} - 2\right)}{x-1} \\
&= \lim_{x \rightarrow 1} \left(\frac{\sqrt[3]{7+x^2} - 2}{x-1} - \frac{\sqrt{3+x^2} - 2}{x-1} \right) \\
&= \lim_{x \rightarrow 1} \frac{(7+x^2-8)}{(x-1)[\sqrt[3]{(7+x^2)^2} + 2\sqrt[3]{7+x^2} + 4]} - \lim_{x \rightarrow 1} \frac{3+x^2-4}{(x-1)(\sqrt{3+x^2} + 2)} \\
&= \lim_{x \rightarrow 1} \frac{x^2-1}{(x-1)[\sqrt[3]{(7+x^2)^2} + 2\sqrt[3]{7+x^2} + 4]} - \lim_{x \rightarrow 1} \frac{x^2-1}{(x-1)(\sqrt{3+x^2} + 2)} \\
&= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)[\sqrt[3]{(7+x^2)^2} + 2\sqrt[3]{7+x^2} + 4]} - \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(\sqrt{3+x^2} + 2)} \\
&= \lim_{x \rightarrow 1} \frac{(x+1)}{\sqrt[3]{(7+x^2)^2} + 2\sqrt[3]{7+x^2} + 4} - \lim_{x \rightarrow 1} \frac{(x+1)}{(\sqrt{3+x^2} + 2)} \\
&= \frac{2}{4+4+4} - \frac{2}{2+2} = \frac{2}{12} - \frac{2}{4} = \frac{1}{6} - \frac{1}{2} = \frac{1-3}{6} = -\frac{2}{6} = -\frac{1}{3}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{\sqrt[3]{7+x^2} - \sqrt{3+x^2}}{x-1} = -\frac{1}{3}$

$$\begin{aligned}
30). \lim_{x \rightarrow 1} \frac{\sqrt[n]{x} - 1}{\sqrt[m]{x} - 1} \quad (n, m \text{ ជាចំនួនគត់វិជ្ជមាន}) & \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 1} \frac{(x-1)(\sqrt[m]{x^{m-1}} + \sqrt[m]{x^{m-2}} + \dots + 1)}{(x-1)(\sqrt[n]{x^{n-1}} + \sqrt[n]{x^{n-2}} + \dots + 1)} \\
&= \lim_{x \rightarrow 1} \frac{(\sqrt[m]{x^{m-1}} + \sqrt[m]{x^{m-2}} + \dots + 1)}{(\sqrt[n]{x^{n-1}} + \sqrt[n]{x^{n-2}} + \dots + 1)} = \frac{\underbrace{1+1+\dots+1}_{m \text{ ឺ}}}{\underbrace{1+1+\dots+1}_{n \text{ ឺ}}} = \frac{m}{n}
\end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 1} \frac{\sqrt[n]{x} - 1}{\sqrt[m]{x} - 1} = \frac{m}{n}}$$

$$\begin{aligned} 31). \quad & \lim_{x \rightarrow 0} \frac{\sqrt{a} - \sqrt{a-x}}{x} \quad \text{មានរាង} \quad \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{a - (a-x)}{x(\sqrt{a} + \sqrt{a-x})} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{a} + \sqrt{a-x})} = \lim_{x \rightarrow 0} \frac{1}{(\sqrt{a} + \sqrt{a-x})} \\ &= \frac{1}{\sqrt{a} + \sqrt{a}} = \frac{1}{2\sqrt{a}} = \frac{\sqrt{a}}{2a} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{\sqrt{a} - \sqrt{a-x}}{x} = \frac{\sqrt{a}}{2a}}$$

$$\begin{aligned} 32). \quad & \lim_{x \rightarrow 8} \frac{\sqrt[3]{x} - 2}{\sqrt[3]{x+19} - 3} \quad \text{មានរាង} \quad \frac{0}{0} \\ &= \lim_{x \rightarrow 8} \frac{(x-2^3)[\sqrt[3]{(x+19)^2} + 3\sqrt[3]{x+19} + 3^2]}{(x+19-3^3)(\sqrt[3]{x^2} + 2\sqrt[3]{x} + 2^2)} \\ &= \lim_{x \rightarrow 8} \frac{(x-8)[\sqrt[3]{(x+19)^2} + 3\sqrt[3]{x+19} + 9]}{(x-8)(\sqrt[3]{x^2} + 2\sqrt[3]{x} + 4)} \\ &= \lim_{x \rightarrow 8} \frac{\sqrt[3]{(x+19)^2} + 3\sqrt[3]{x+19} + 9}{\sqrt[3]{x^2} + 2\sqrt[3]{x} + 4} = \frac{9+9+9}{4+4+4} = \frac{3 \times 9}{3 \times 4} = \frac{9}{4} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 8} \frac{\sqrt[3]{x} - 2}{\sqrt[3]{x+19} - 3} = \frac{9}{4}}$$

$$\begin{aligned} 33). \quad & \lim_{x \rightarrow 4} \frac{\sqrt[3]{3x-4} - 2}{\sqrt{4x+9} - 5} \quad \text{មានរាង} \quad \frac{0}{0} \\ &= \lim_{x \rightarrow 4} \frac{(3x-4-2^3)(\sqrt{4x+9} + 5)}{(4x+9-5^2)[\sqrt[3]{(3x-4)^2} + 2\sqrt[3]{3x-4} + 2^2]} \\ &= \lim_{x \rightarrow 4} \frac{(3x-4-8)(\sqrt{4x+9} + 5)}{(4x+9-25)[\sqrt[3]{(3x-4)^2} + 2\sqrt[3]{3x-4} + 4]} \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 4} \frac{(3x-12)(\sqrt{4x+9}+5)}{(4x-16)[\sqrt[3]{(3x-4)^2} + 2\sqrt[3]{3x-4} + 4]} \\
&= \lim_{x \rightarrow 4} \frac{3(x-4)(\sqrt{4x+9}+5)}{4(x-4)[\sqrt[3]{(3x-4)^2} + 2\sqrt[3]{3x-4} + 4]} \\
&= \lim_{x \rightarrow 4} \frac{3(\sqrt{4x+9}+5)}{4[\sqrt[3]{(3x-4)^2} + 2\sqrt[3]{3x-4} + 4]} = \frac{3(5+5)}{4(4+4+4)} = \frac{30}{48} = \frac{5}{8}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 4} \frac{\sqrt[3]{3x-4}-2}{\sqrt{4x+9}-5} = \frac{5}{8}$

$$\begin{aligned}
34). \quad &\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+2x} - \sqrt[4]{1+3x}}{x} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \frac{(\sqrt[3]{1+2x}-1) - (\sqrt[4]{1+3x}-1)}{x} \\
&= \lim_{x \rightarrow 0} \left(\frac{\sqrt[3]{1+2x}-1}{x} - \frac{\sqrt[4]{1+3x}-1}{x} \right) = \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+2x}-1}{x} - \lim_{x \rightarrow 0} \frac{\sqrt[4]{1+3x}-1}{x} \\
&= \lim_{x \rightarrow 0} \frac{1+2x-1}{x[\sqrt[3]{(1+2x)^2} + \sqrt[3]{1+2x}+1]} \\
&\quad - \lim_{x \rightarrow 0} \frac{1+3x-1}{x[\sqrt[4]{(1+3x)^3} + \sqrt[4]{(1+3x)^2} + \sqrt[4]{1+3x}+1]} \\
&= \lim_{x \rightarrow 0} \frac{2}{\sqrt[3]{(1+2x)^2} + \sqrt[3]{1+2x}+1} \\
&\quad - \lim_{x \rightarrow 0} \frac{3}{\sqrt[4]{(1+3x)^3} + \sqrt[4]{(1+3x)^2} + \sqrt[4]{1+3x}+1} \\
&= \frac{2}{1+1+1} - \frac{3}{1+1+1+1} = \frac{2}{3} - \frac{3}{4} = \frac{8-9}{12} = -\frac{1}{12}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+2x} - \sqrt[4]{1+3x}}{x} = -\frac{1}{12}$

$$35). \quad \lim_{x \rightarrow 0} \left(\frac{1}{x(x+1)} - \frac{1}{x} \right) \text{ មានរាង } (\infty - \infty)$$

$$= \lim_{x \rightarrow 0} \left(\frac{1 - (x+1)}{x(x+1)} \right) = \lim_{x \rightarrow 0} \frac{-x}{x(x+1)} = \lim_{x \rightarrow 0} \frac{-1}{x+1} = \frac{-1}{1} = -1$$

ដូច្នេះ: $\lim_{x \rightarrow 0} \left(\frac{1}{x(x+1)} - \frac{1}{x} \right) = -1$

$$\begin{aligned} 36). \quad & \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt[3]{1+x}}{x} \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - 1) - (\sqrt[3]{1+x} - 1)}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} - \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - 1}{x} \\ &= \lim_{x \rightarrow 0} \frac{1+x-1}{x(\sqrt{1+x}+1)} - \lim_{x \rightarrow 0} \frac{1+x-1}{x[\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x}+1]} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{1+x}+1)} - \lim_{x \rightarrow 0} \frac{x}{x[\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x}+1]} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x}+1} - \lim_{x \rightarrow 0} \frac{1}{\sqrt[3]{(1+x)^2} + \sqrt[3]{1+x}+1} \\ &= \frac{1}{1+1} - \frac{1}{1+1+1} = \frac{1}{2} - \frac{1}{3} = \frac{3-2}{6} = \frac{1}{6} \end{aligned}$$

ដូច្នេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt[3]{1+x}}{x} = \frac{1}{6}$

$$\begin{aligned} 37). \quad & \lim_{x \rightarrow 0} \frac{\sqrt[5]{x+1} - 1}{\sqrt[3]{x+1} - 1} \text{ មានរាង } \frac{0}{0} \\ &= \lim_{x \rightarrow 0} \frac{(x+1-1)[\sqrt[3]{(x+1)^2} + \sqrt[3]{x+1}+1]}{(x+1-1)[\sqrt[5]{(x+1)^4} + \sqrt[5]{(x+1)^3} + \sqrt[5]{(x+1)^2} + \sqrt[5]{x+1}+1]} \\ &= \lim_{x \rightarrow 0} \frac{\sqrt[3]{(x+1)^2} + \sqrt[3]{x+1}+1}{\sqrt[5]{(x+1)^4} + \sqrt[5]{(x+1)^3} + \sqrt[5]{(x+1)^2} + \sqrt[5]{x+1}+1} \\ &= \frac{1+1+1}{1+1+1+1+1} = \frac{3}{5} \end{aligned}$$

ដូច្នេះ: $\lim_{x \rightarrow 0} \frac{\sqrt[5]{x+1} - 1}{\sqrt[3]{x+1} - 1} = \frac{3}{5}$

$$38). \lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{x}} - \frac{x+1}{\sqrt{x}} \right) \text{ មានរាង } (\infty - \infty)$$

$$= \lim_{x \rightarrow 0} \frac{1 - (x+1)}{\sqrt{x}} = \lim_{x \rightarrow 0} \frac{-x}{\sqrt{x}} = \lim_{x \rightarrow 0} \frac{-x\sqrt{x}}{x} = \lim_{x \rightarrow 0} (-\sqrt{x}) = 0$$

ដូចនេះ: $\lim_{x \rightarrow 0} \left(\frac{1}{\sqrt{x}} - \frac{x+1}{\sqrt{x}} \right) = 0$

$$39). \lim_{x \rightarrow 0} \frac{x^2 + |x| + x}{x^2 - |x| + 3x} \text{ មានរាង } \frac{0}{0}$$

❖ ករណី $x \rightarrow 0^+$ នោះ $|x| = x$ គេបាន

$$\lim_{x \rightarrow 0^+} \frac{x^2 + |x| + x}{x^2 - |x| + 3x} = \lim_{x \rightarrow 0^+} \frac{x^2 + x + x}{x^2 - x + 3x} = \lim_{x \rightarrow 0^+} \frac{x^2 + 2x}{x^2 + 2x} = \lim_{x \rightarrow 0^+} (1) = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0^+} \frac{x^2 + |x| + x}{x^2 - |x| + 3x} = 1$

❖ ករណី $x \rightarrow 0^-$ នោះ $|x| = -x$

គេបាន $\lim_{x \rightarrow 0^-} \frac{x^2 + |x| + x}{x^2 - |x| + 3x} = \lim_{x \rightarrow 0^-} \frac{x^2 - x + x}{x^2 + x + 3x} = \lim_{x \rightarrow 0^-} \frac{x^2}{x^2 + 4x}$

$$= \lim_{x \rightarrow 0^-} \frac{x}{x+4} = \frac{0}{0+4} = 0$$

ដូចនេះ: $\lim_{x \rightarrow 0^-} \frac{x^2 + |x| + x}{x^2 - |x| + 3x} = 0$

$$40). \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - \sqrt{x^2 - 1}) \text{ មានរាង } (\infty - \infty)$$

$$= \lim_{x \rightarrow \infty} \frac{(x^2 + 1) - (x^2 - 1)}{\sqrt{x^2 + 1} + \sqrt{x^2 - 1}} = \lim_{x \rightarrow \infty} \frac{2}{|x| \sqrt{1 + \frac{1}{x^2}} + |x| \sqrt{1 - \frac{1}{x^2}}}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

គេបាន $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 1} - \sqrt{x^2 - 1}) = \lim_{x \rightarrow -\infty} \frac{2}{-x \sqrt{1 + \frac{1}{x^2}} - x \sqrt{1 - \frac{1}{x^2}}}$

$$= \lim_{x \rightarrow -\infty} \frac{2}{-x \left(\sqrt{1 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x^2}} \right)} = \frac{2}{(+\infty)(1+1)} = \frac{1}{(+\infty)} = 0$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

$$\text{គេបាន } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - \sqrt{x^2 - 1}) = \lim_{x \rightarrow +\infty} \frac{2}{x \sqrt{1 + \frac{1}{x^2}} + x \sqrt{1 - \frac{1}{x^2}}}$$

$$= \lim_{x \rightarrow +\infty} \frac{2}{x \left(\sqrt{1 + \frac{1}{x^2}} + \sqrt{1 - \frac{1}{x^2}} \right)} = \frac{2}{(+\infty)(1+1)} = \frac{1}{(+\infty)} = 0$$

ដូចនេះ: $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - \sqrt{x^2 - 1}) = 0$

41). $\lim_{x \rightarrow \pm\infty} (\sqrt{x^2 + 1} - x)$ មានរាង $(\infty - \infty)$

❖ ករណី $x \rightarrow -\infty$ គេបាន $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 1} - x) = +\infty - (-\infty) = +\infty$

ដូចនេះ: $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 1} - x) = +\infty$

❖ ករណី $x \rightarrow +\infty$

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - x) = \lim_{x \rightarrow +\infty} \frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1} + x} = \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{x^2 + 1} + x} = 0$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - x) = 0$

42). $\lim_{x \rightarrow \pm\infty} x(\sqrt{x^2 + 1} - x)$ មានរាង $(\infty - \infty)$

❖ ករណី $x \rightarrow -\infty$ គេបាន $\lim_{x \rightarrow -\infty} x(\sqrt{x^2 + 1} - x) = (-\infty)(+\infty + \infty) = -\infty$

ដូចនេះ: $\lim_{x \rightarrow -\infty} x(\sqrt{x^2 + 1} - x) = -\infty$

❖ ករណី $x \rightarrow +\infty$

$$\begin{aligned}\lim_{x \rightarrow +\infty} x(\sqrt{x^2+1}-x) &= \lim_{x \rightarrow +\infty} \frac{x(x^2+1-x^2)}{\sqrt{x^2+1}+x} = \lim_{x \rightarrow +\infty} \frac{x}{x\sqrt{1+\frac{1}{x^2}}+x} \\ &= \lim_{x \rightarrow +\infty} \frac{x}{x\left(\sqrt{1+\frac{1}{x^2}}+1\right)} = \lim_{x \rightarrow +\infty} \frac{1}{\left(\sqrt{1+\frac{1}{x^2}}+1\right)} = \frac{1}{2}\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow +\infty} x(\sqrt{x^2+1}-x) = \frac{1}{2}}$

43). $\lim_{x \rightarrow \pm\infty} (\sqrt{(x+a)(x+b)}-x)$ មានរាង $(\infty-\infty)$

❖ ករណី $x \rightarrow -\infty$ គេបាន $\lim_{x \rightarrow -\infty} (\sqrt{(x+a)(x+b)}-x) = (+\infty+\infty) = +\infty$

ដូចនេះ: $\boxed{\lim_{x \rightarrow -\infty} (\sqrt{(x+a)(x+b)}-x) = +\infty}$

❖ ករណី $x \rightarrow +\infty$

$$\begin{aligned}\text{គេបាន } \lim_{x \rightarrow +\infty} (\sqrt{(x+a)(x+b)}-x) &= \lim_{x \rightarrow +\infty} \frac{(x+a)(x+b)-x^2}{\sqrt{(x+a)(x+b)}+x} \\ &= \lim_{x \rightarrow +\infty} \frac{x^2+(a+b)x+ab-x^2}{\sqrt{x^2+(a+b)x+ab}+x} = \lim_{x \rightarrow +\infty} \frac{(a+b)x+ab}{x\sqrt{1+\frac{a+b}{x}+\frac{ab}{x^2}}+x} \\ &= \lim_{x \rightarrow +\infty} \frac{x\left(a+b+\frac{ab}{x}\right)}{x\left(\sqrt{1+\frac{a+b}{x}+\frac{ab}{x^2}}+1\right)} = \lim_{x \rightarrow +\infty} \frac{(a+b)+\frac{ab}{x}}{\sqrt{1+\frac{a+b}{x}+\frac{ab}{x^2}}+1} \\ &= \frac{a+b+0}{1+1} = \frac{a+b}{2}\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow +\infty} (\sqrt{(x+a)(x+b)}-x) = \frac{a+b}{2}}$

44). $\lim_{x \rightarrow \pm\infty} (\sqrt{x^2-2x-1}-\sqrt{x^2-7x+3})$ មានរាង $(\infty-\infty)$

$$\begin{aligned}
&= \lim_{x \rightarrow \pm\infty} \frac{(x^2 - 2x - 1) - (x^2 - 7x + 3)}{\sqrt{x^2 - 2x - 1} + \sqrt{x^2 - 7x + 3}} \\
&= \lim_{x \rightarrow \pm\infty} \frac{5x - 4}{|x| \sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + |x| \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}}} \\
&= \lim_{x \rightarrow \pm\infty} \frac{5x - 4}{|x| \left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)}
\end{aligned}$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

$$\begin{aligned}
&\text{គេបាន } \lim_{x \rightarrow +\infty} \frac{5x - 4}{|x| \left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)} \\
&= \lim_{x \rightarrow +\infty} \frac{x \left(5 - \frac{4}{x} \right)}{x \left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)} \\
&= \lim_{x \rightarrow +\infty} \frac{\left(5 - \frac{4}{x} \right)}{\left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)} = \frac{5}{1+1} = \frac{5}{2}
\end{aligned}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

$$\begin{aligned}
&\text{គេបាន } \lim_{x \rightarrow -\infty} \frac{5x - 4}{|x| \left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)} \\
&= \lim_{x \rightarrow -\infty} \frac{x \left(5 - \frac{4}{x} \right)}{-x \left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}} \right)}
\end{aligned}$$

$$= \lim_{x \rightarrow -\infty} \frac{\left(5 - \frac{4}{x}\right)}{-\left(\sqrt{1 - \frac{2}{x} - \frac{1}{x^2}} + \sqrt{1 - \frac{7}{x} + \frac{3}{x^2}}\right)} = -\frac{5}{1+1} = -\frac{5}{2}$$

ដូចនេះ: $\lim_{x \rightarrow \pm\infty} (\sqrt{x^2 - 2x - 1} - \sqrt{x^2 - 7x + 3}) = \pm \frac{5}{2}$

45). $\lim_{x \rightarrow \infty} (\sqrt[3]{(x+1)^2} - \sqrt[3]{(x-1)^2})$ មានរាង $(\infty - \infty)$

$$= \lim_{x \rightarrow \infty} \frac{(x+1)^2 - (x-1)^2}{\sqrt[3]{[(x+1)^2]^2} + \sqrt[3]{(x+1)^2} \sqrt[3]{(x-1)^2} + \sqrt[3]{[(x-1)^2]^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{(x^2 + 2x + 1) - (x^2 - 2x + 1)}{\sqrt[3]{(x+1)^4} + \sqrt[3]{(x+1)^2(x-1)^2} + \sqrt[3]{(x-1)^4}}$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{\sqrt[3]{(x+1)^4} + \sqrt[3]{(x+1)^2(x-1)^2} + \sqrt[3]{(x-1)^4}}$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{\sqrt[3]{(x+1)^4} + \sqrt[3]{(x^2-1)^2} + \sqrt[3]{(x-1)^4}}$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{\sqrt[3]{x^4(1+\frac{1}{x})^4} + \sqrt[3]{x^4(1-\frac{1}{x^2})^2} + \sqrt[3]{x^4(1-\frac{1}{x})^4}}$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{\sqrt[3]{x^4} \left(\sqrt[3]{(1+\frac{1}{x})^4} + \sqrt[3]{(1-\frac{1}{x^2})^2} + \sqrt[3]{(1-\frac{1}{x})^4} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{x \sqrt[3]{x} \left(\sqrt[3]{(1+\frac{1}{x})^4} + \sqrt[3]{(1-\frac{1}{x^2})^2} + \sqrt[3]{(1-\frac{1}{x})^4} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{4}{\sqrt[3]{x}} \times \frac{1}{\left(\sqrt[3]{(1+\frac{1}{x})^4} + \sqrt[3]{(1-\frac{1}{x^2})^2} + \sqrt[3]{(1-\frac{1}{x})^4} \right)} = 0 \times \frac{1}{1+1+1} = 0$$

ដូចនេះ: $\lim_{x \rightarrow \infty} (\sqrt[3]{(x+1)^2} - \sqrt[3]{(x-1)^2}) = 0$

46). $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x + \sqrt{2x + \sqrt{x}}}}{\sqrt{\frac{1}{5}x - 7}}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt{x} \left(\sqrt{3 + \frac{\sqrt{2x + \sqrt{x}}}{x}} \right)}{\sqrt{x} \sqrt{\frac{1}{5} - \frac{7}{x}}} = \lim_{x \rightarrow +\infty} \frac{\left(\sqrt{3 + \sqrt{\frac{2}{x} + \sqrt{\frac{1}{x^3}}}} \right)}{\sqrt{\frac{1}{5} - \frac{7}{x}}}$$

$$= \frac{\sqrt{3}}{\sqrt{\frac{1}{5}}} = \frac{\sqrt{3}}{\frac{1}{\sqrt{5}}} = \sqrt{3} \times \sqrt{5} = \sqrt{15}$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \frac{\sqrt{3x + \sqrt{2x + \sqrt{x}}}}{\sqrt{\frac{1}{5}x - 7}} = \sqrt{15}$

47). $\lim_{x \rightarrow -\infty} \frac{-2x^2 - 2}{x^2 + 3x - 1}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow -\infty} \frac{x^2 \left(-2 - \frac{2}{x^2} \right)}{x^2 \left(1 + \frac{3}{x} - \frac{1}{x^2} \right)} = \lim_{x \rightarrow -\infty} \frac{\left(-2 - \frac{2}{x^2} \right)}{\left(1 + \frac{3}{x} - \frac{1}{x^2} \right)} = \frac{-2 - 0}{1 + 0 - 0} = -2$$

ដូចនេះ: $\lim_{x \rightarrow -\infty} \frac{-2x^2 - 2}{x^2 + 3x - 1} = -2$

48). $\lim_{x \rightarrow +\infty} \frac{x^2 - 2x - 2}{-3x^2 + 4}$ មានរាង $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{x^2 \left(1 - \frac{2}{x} - \frac{2}{x^2} \right)}{x^2 \left(-3 + \frac{4}{x^2} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(1 - \frac{2}{x} - \frac{2}{x^2} \right)}{\left(-3 + \frac{4}{x^2} \right)} = \frac{1 - 0 - 0}{-3 + 0} = -\frac{1}{3}$$

ដូចនេះ:
$$\lim_{x \rightarrow +\infty} \frac{x^2 - 2x - 2}{-3x^2 + 4} = -\frac{1}{3}$$

49).
$$\lim_{x \rightarrow +\infty} \frac{x^{100} + x^{10} + 1}{100x^{100} - 10x^{10} - 1} \text{ មានរាង } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{x^{100} \left(1 + \frac{1}{x^{90}} + \frac{1}{x^{100}} \right)}{x^{100} \left(100 - \frac{10}{x^{90}} - \frac{1}{x^{100}} \right)} = \lim_{x \rightarrow +\infty} \frac{\left(1 + \frac{1}{x^{90}} + \frac{1}{x^{100}} \right)}{\left(100 - \frac{10}{x^{90}} - \frac{1}{x^{100}} \right)} = \frac{1}{100}$$

ដូចនេះ:
$$\lim_{x \rightarrow +\infty} \frac{x^{100} + x^{10} + 1}{100x^{100} - 10x^{10} - 1} = \frac{1}{100}$$

50).
$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + x + 3}}{x} \text{ មានរាង } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 \left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x} = \lim_{x \rightarrow \infty} \frac{|x| \sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x}$$

❖ ករណី $x \rightarrow -\infty$ នោះ $|x| = -x$

គេបាន
$$\lim_{x \rightarrow -\infty} \frac{|x| \sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x} = \lim_{x \rightarrow -\infty} \frac{-x \sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x}$$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{1} = -\sqrt{4 + 0 + 0} = -2$$

❖ ករណី $x \rightarrow +\infty$ នោះ $|x| = x$

គេបាន
$$\lim_{x \rightarrow +\infty} \frac{|x| \sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x} = \lim_{x \rightarrow +\infty} \frac{x \sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{x}$$

$$= \lim_{x \rightarrow +\infty} \frac{\sqrt{\left(4 + \frac{1}{x} + \frac{3}{x^2} \right)}}{1} = \sqrt{4 + 0 + 0} = 2$$

ដូចនេះ: $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + x + 3}}{x} = \pm 2$

$$\begin{aligned}
 51). \quad & \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sqrt{1+x} - 1} \quad \text{មានរាង} \quad \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(2 \sin^2 \frac{x}{2})(\sqrt{1+x} - 1)}{(1+x) - 1} = \lim_{x \rightarrow 0} \frac{(2 \sin^2 \frac{x}{2})(\sqrt{1+x} + 1)}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\left(\frac{x}{2}\right)^2} \times \frac{2(\sqrt{1+x} + 1)}{4} = 1 \times \frac{2(1+1)}{2} = \frac{4}{4} = 1
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sqrt{1+x} - 1} = 1$

$$\begin{aligned}
 52). \quad & \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x} \quad \text{មានរាង} \quad \frac{0}{0} \\
 &= \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x + \cos^2 x)}{x(2 \sin x \cos x)} = \lim_{x \rightarrow 0} \frac{(2 \sin^2 \frac{x}{2})(1 + \cos x + \cos^2 x)}{2x(2 \sin \frac{x}{2} \cos \frac{x}{2}) \cos x} \\
 &= \lim_{x \rightarrow 0} \frac{(\sin \frac{x}{2})(1 + \cos x + \cos^2 x)}{x(2 \cos \frac{x}{2}) \cos x} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \times \frac{(1 + \cos x + \cos^2 x)}{2(2 \cos \frac{x}{2}) \cos x} \\
 &= 1 \times \frac{1+1+1}{2(2)} = \frac{3}{4}
 \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin 2x} = \frac{3}{4}$

$$\begin{aligned}
 53). \quad & \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x} = \lim_{x \rightarrow 0} \left(\frac{x}{\sin x}\right) \cdot \left(x \sin \frac{1}{x}\right) \\
 & \text{ដោយ } -1 \leq \sin \frac{1}{x} \leq 1 \text{ នាំឲ្យ } -x \leq x \sin \frac{1}{x} \leq x
 \end{aligned}$$

$$\text{គេបាន } \lim_{x \rightarrow 0} (-x) \leq \lim_{x \rightarrow 0} x \sin \frac{1}{x} \leq \lim_{x \rightarrow 0} (x)$$

$$\text{ឬ } 0 \leq \lim_{x \rightarrow 0} x \sin \frac{1}{x} \leq 0 \text{ នាំឱ្យ } \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$$\text{គេបាន } \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x} = \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right) \cdot \left(x \sin \frac{1}{x} \right) = 1 \times 0 = 0$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x} = 0}$$

$$54). \lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + x\right) - \frac{1}{2}}{x} \text{ មានរាង } \frac{0}{0}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\sin \frac{\pi}{6} \cos x + \cos \frac{\pi}{6} \sin x - \frac{1}{2}}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2} \cos x + \frac{\sqrt{3}}{2} \sin x - \frac{1}{2}}{x} \\ &= \lim_{x \rightarrow 0} \frac{-\frac{1}{2}(1 - \cos x) + \frac{\sqrt{3}}{2} \sin x}{x} = -\frac{1}{2} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} + \frac{\sqrt{3}}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} \\ &= -\frac{1}{2}(0) + \frac{\sqrt{3}}{2}(1) = \frac{\sqrt{3}}{2} \end{aligned}$$

$$\text{ដូចនេះ: } \boxed{\lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} + x\right) - \frac{1}{2}}{x} = \frac{\sqrt{3}}{2}}$$

$$55). \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right) = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sin x} - \cot x \right)$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right) = \lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1 - \cos x}{\sin x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1}{x} \frac{2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{x \cos \frac{x}{2}} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \times \frac{1}{2 \cos \frac{x}{2}} = 1 \times \frac{1}{2} = \frac{1}{2} \end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right) = \frac{1}{2}$

56). $\lim_{x \rightarrow 1} \frac{\sin(1-x)}{\sqrt{x}-1}$ មានរាង $\frac{0}{0}$

តាង $u = 1 - x$ នៅ: $x = 1 - u$

ពេល $x \rightarrow 1$ នៅ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow 1} \frac{\sin(1-x)}{\sqrt{x}-1} = \lim_{u \rightarrow 0} \frac{\sin u}{\sqrt{1-u}-1} = \lim_{u \rightarrow 0} \frac{(\sin u)(\sqrt{1-u}+1)}{(1-u)-1}$
 $= \lim_{u \rightarrow 0} \frac{(\sin u)(\sqrt{1-u}+1)}{-u} = - \lim_{u \rightarrow 0} \frac{\sin u}{u} \times (\sqrt{1-u}+1) = -1(1+1) = -2$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{\sin(1-x)}{\sqrt{x}-1} = -2$

57). $\lim_{x \rightarrow -1} \frac{x^3+1}{\sin(x+1)}$ មានរាង $\frac{0}{0}$

តាង $u = 1 + x$ នៅ: $x = u - 1$

ពេល $x \rightarrow -1$ នៅ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow -1} \frac{x^3+1}{\sin(x+1)} = \lim_{u \rightarrow 0} \frac{(u-1)^3+1}{\sin u}$
 $= \lim_{u \rightarrow 0} \frac{(u^3-3u^2+3u-1)+1}{\sin u} = \lim_{u \rightarrow 0} \frac{u^3-3u^2+3u}{\sin u}$
 $= \lim_{u \rightarrow 0} \frac{u}{\sin u} \times (u^2-3u+3) = 1(0-0+3) = 3$

ដូចនេះ: $\lim_{x \rightarrow -1} \frac{x^3+1}{\sin(x+1)} = 3$

58). $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\tan x - \cot x}$ មានរាង $\frac{0}{0}$

$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\frac{\sin x}{\cos x} - \frac{\cos x}{\sin x}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x \cos x (\sin x - \cos x)}{\sin^2 x - \cos^2 x}$

$$\begin{aligned}
&= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x \cos x (\sin x - \cos x)}{(\sin x + \cos x)(\sin x - \cos x)} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x \cos x}{(\sin x + \cos x)} \\
&= \frac{\frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}} = \frac{\frac{2}{4}}{2 \frac{\sqrt{2}}{2}} = \frac{\frac{1}{2}}{\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{2 \times 2} = \frac{\sqrt{2}}{4}
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\tan x - \cot x} = \frac{\sqrt{2}}{4}$

$$\begin{aligned}
59). \quad &\lim_{x \rightarrow 0} \frac{\cos 2x - e^{x^2}}{x^2} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \frac{\cos 2x - 1 - e^{x^2} + 1}{x^2} = - \lim_{x \rightarrow 0} \frac{(1 - \cos 2x) + (e^{x^2} - 1)}{x^2} \\
&= - \lim_{x \rightarrow 0} \frac{2\sin^2 x + e^{x^2} - 1}{x^2} = - \lim_{x \rightarrow 0} \left(\frac{2\sin^2 x}{x^2} + \frac{e^{x^2} - 1}{x^2} \right) \\
&= -(2+1) = -3
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\cos 2x - e^{x^2}}{x^2} = -3$

$$\begin{aligned}
60). \quad &\lim_{x \rightarrow 0} \frac{e^{-x^2} \cos 4x - 1}{x^2} \text{ មានរាង } \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \frac{e^{-x^2} (\cos 4x - 1) + e^{-x^2} - 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^{-x^2} (-2\sin^2 2x) + e^{-x^2} - 1}{x^2} \\
&= - \lim_{x \rightarrow 0} \frac{e^{-x^2} (2\sin^2 2x) - (e^{-x^2} - 1)}{x^2} \\
&= - \lim_{x \rightarrow 0} \left(\frac{\sin^2 2x}{(2x)^2} \times 4 \times 2e^{-x^2} + \frac{e^{-x^2} - 1}{-x^2} \right) = -(1 \times 4 \times 2 + 1) = -9
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{e^{-x^2} \cos 4x - 1}{x^2} = -9$

61). $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \cos x}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{1+x^2 - \cos^2 x}{x^2(\sqrt{1+x^2} + \cos x)} = \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x) + x^2}{x^2(\sqrt{1+x^2} + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{1+x^2 - \cos^2 x}{x^2(\sqrt{1+x^2} + \cos x)} = \lim_{x \rightarrow 0} \frac{\sin^2 x + x^2}{x^2(\sqrt{1+x^2} + \cos x)}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} + 1 \right) \times \frac{1}{(\sqrt{1+x^2} + \cos x)} = (1+1) \times \frac{1}{1+1} = 2 \times \frac{1}{2} = 1$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \cos x}{x^2} = 1$

62). $\lim_{x \rightarrow 1} \frac{x^3 + x^2 - 2}{\sin(x-1)}$ មានរាង $\frac{0}{0}$

តាង $u = x - 1$ នោះ: $x = u + 1$

ពេល $x \rightarrow 1$ នោះ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow 1} \frac{x^3 + x^2 - 2}{\sin(x-1)} = \lim_{u \rightarrow 0} \frac{(u+1)^3 + (u+1)^2 - 2}{\sin u}$

$$= \lim_{u \rightarrow 0} \frac{(u^3 + 3u^2 + 3u + 1) + (u^2 + 2u + 1) - 2}{\sin u}$$

$$= \lim_{u \rightarrow 0} \frac{u^3 + 4u^2 + 5u}{\sin u} = \lim_{u \rightarrow 0} \frac{u}{\sin u} \times (u^2 + 4u + 5) = 1 \times (0 + 0 + 5) = 5$$

ដូចនេះ: $\lim_{x \rightarrow 1} \frac{x^3 + x^2 - 2}{\sin(x-1)} = 5$

63). $\lim_{x \rightarrow 0} \frac{\sqrt{1-x^3} - 1}{\sin x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{(1-x^3)-1}{(\sqrt{1-x^3}+1)\sin x^2} = \lim_{x \rightarrow 0} \frac{-x^3}{(\sqrt{1-x^3}+1)\sin x^2}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{\sin x^2} \times \frac{-x}{(\sqrt{1-x^3}+1)} = 1 \times \frac{0}{1+1} = 1 \times 0 = 0$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sqrt{1-x^3}-1}{\sin x^2} = 0$

64). $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\sin \beta x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\sin \alpha x}{\alpha x} \times \frac{\beta x}{\sin \beta x} \times \frac{\alpha}{\beta} = 1 \times 1 \times \frac{\alpha}{\beta} = \frac{\alpha}{\beta}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\sin \beta x} = \frac{\alpha}{\beta}$

65). $\lim_{x \rightarrow -\frac{\pi}{2}} \frac{\sin^2 x - 1}{1 + \sin x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow -\frac{\pi}{2}} \frac{(\sin x - 1)(\sin x + 1)}{(1 + \sin x)} = \lim_{x \rightarrow -\frac{\pi}{2}} (\sin x - 1) = -1 - 1 = -2$$

ដូចនេះ: $\lim_{x \rightarrow -\frac{\pi}{2}} \frac{\sin^2 x - 1}{1 + \sin x} = -2$

66). $\lim_{x \rightarrow 0} \frac{2 \tan x + \sin x}{x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \left(\frac{2 \tan x}{x} + \frac{\sin x}{x} \right) = 2 + 1 = 3$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{2 \tan x + \sin x}{x} = 3$

67). $\lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos 2x}}{x^2}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2(1 + \sqrt{\cos 2x})} = \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2} \times \frac{1}{(1 + \sqrt{\cos 2x})}$$

$$= 2 \times \frac{1}{1+1} = 2 \times \frac{1}{2} = 1$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos 2x}}{x^2} = 1}$

68). $\lim_{x \rightarrow +\infty} \frac{(x+1)\sin x}{x^2 + 3}$

ដោយ $-\leq \sin x \leq 1 \Rightarrow -(x+1) \leq (x+1)\sin x \leq (x+1)$

នាំឲ្យ $-\frac{(x+1)}{x^2 + 3} \leq \frac{(x+1)\sin x}{x^2 + 3} \leq \frac{(x+1)}{x^2 + 3}$

នាំឲ្យ $-\frac{x\left(1 + \frac{1}{x}\right)}{x^2\left(1 + \frac{3}{x^2}\right)} \leq \frac{(x+1)\sin x}{x^2 + 3} \leq \frac{x\left(1 + \frac{1}{x}\right)}{x^2\left(1 + \frac{3}{x^2}\right)}$

នាំឲ្យ $-\frac{\left(1 + \frac{1}{x}\right)}{x\left(1 + \frac{3}{x^2}\right)} \leq \frac{(x+1)\sin x}{x^2 + 3} \leq \frac{\left(1 + \frac{1}{x}\right)}{x^2\left(1 + \frac{3}{x^2}\right)}$

គេបាន $\lim_{x \rightarrow +\infty} -\frac{\left(1 + \frac{1}{x}\right)}{x\left(1 + \frac{3}{x^2}\right)} \leq \lim_{x \rightarrow +\infty} \frac{(x+1)\sin x}{x^2 + 3} \leq \lim_{x \rightarrow +\infty} \frac{\left(1 + \frac{1}{x}\right)}{x^2\left(1 + \frac{3}{x^2}\right)}$

នាំឲ្យ $0 \leq \lim_{x \rightarrow +\infty} \frac{(x+1)\sin x}{x^2 + 3} \leq 0$ នាំឲ្យ $\lim_{x \rightarrow +\infty} \frac{(x+1)\sin x}{x^2 + 3} = 0$

ដូចនេះ: $\boxed{\lim_{x \rightarrow +\infty} \frac{(x+1)\sin x}{x^2 + 3} = 0}$

69). $\lim_{x \rightarrow 0} \frac{\tan kx}{x}$ មានរាង $\frac{0}{0}$

$= \lim_{x \rightarrow 0} \frac{\tan kx}{kx} \times k = 1 \times k = k$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\tan kx}{x} = k$

70). $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{(\pi - 2x)^2}$ មានរាង $\frac{0}{0}$

តាង $u = x - \frac{\pi}{2}$ នោះ: $x = \frac{\pi}{2} + u$

ពេល $x \rightarrow \frac{\pi}{2}$ នោះ: $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{(\pi - 2x)^2} = \lim_{u \rightarrow 0} \frac{\ln[\sin(\frac{\pi}{2} + u)]}{[\pi - 2(\frac{\pi}{2} + u)]^2}$

$$= \lim_{u \rightarrow 0} \frac{\ln(\cos u)}{(-2u)^2} = \lim_{u \rightarrow 0} \frac{\ln(1 + \cos u - 1)}{4u^2} = \frac{1}{4} \lim_{u \rightarrow 0} \frac{1}{u^2} \ln(1 - 2\sin^2 \frac{x}{2})$$

$$= \frac{1}{4} \lim_{u \rightarrow 0} \ln(1 - 2\sin^2 \frac{u}{2}) u^{\frac{1}{2}}$$

$$= \frac{1}{4} \lim_{u \rightarrow 0} \ln \left(\left[1 + (-2\sin^2 \frac{u}{2}) \right]^{\frac{1}{(-2\sin^2 \frac{u}{2})}} \right)^{\frac{(-2\sin^2 \frac{u}{2})}{u^2}}$$

$$= \frac{1}{4} \lim_{u \rightarrow 0} \ln \left(\left[1 + (-2\sin^2 \frac{u}{2}) \right]^{\frac{1}{(-2\sin^2 \frac{u}{2})}} \right)^{\frac{(\sin^2 \frac{u}{2})}{(\frac{u}{2})^2} \times \frac{-2}{4}}$$

$$= \frac{1}{4} \ln e^{-\frac{2}{4}} = \left(\frac{1}{4} \right) \left(\frac{-2}{4} \right) \ln e = \left(\frac{1}{4} \right) \left(-\frac{1}{2} \right) = -\frac{1}{8}$$

ដូចនេះ:
$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{(\pi - 2x)^2} = -\frac{1}{8}$$

71). $\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m}$ (n, m ជាចំនួនគត់វិជ្ជមាន)

❖ ករណី $m = n$

គេបាន $\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = \lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^n} = \lim_{x \rightarrow 0} \frac{\sin(x^n)}{x^n} \times \frac{x^n}{(\sin x)^n} = 1$

ដូចនេះ:
$$\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = 1$$

❖ ករណី $n > m$ នៅ: $n - m > 0$

គេបាន $\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = \lim_{x \rightarrow 0} \frac{\sin(x^n)}{x^n} \times \frac{x^m}{(\sin x)^m} \times \frac{x^n}{x^m}$
 $= \lim_{x \rightarrow 0} \frac{\sin(x^n)}{x^n} \times \frac{x^m}{(\sin x)^m} \times x^{n-m} = 1 \times 1 \times 0 = 0$

ដូចនេះ:
$$\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = 0$$

❖ ករណី $n < m$ នៅ: $m - n > 0$

គេបាន $\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = \lim_{x \rightarrow 0} \frac{\sin(x^n)}{x^n} \times \frac{x^m}{(\sin x)^m} \times \frac{x^n}{x^m}$
 $= \lim_{x \rightarrow 0} \frac{\sin(x^n)}{x^n} \times \frac{x^m}{(\sin x)^m} \times \frac{1}{x^{m-n}} = 1 \times 1 \times \frac{1}{0} = \pm \infty$

ដូចនេះ:
$$\lim_{x \rightarrow 0} \frac{\sin(x^n)}{(\sin x)^m} = \pm \infty$$

72). $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right)$ មានរាង $(\infty - \infty)$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right) = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \\
&= \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} = \frac{0}{1} = 0
\end{aligned}$$

ដូចនេះ: $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{\tan x} \right) = 0$

73). $\lim_{x \rightarrow \infty} x(e^x - 1)$ មានរាង $(\infty \times 0)$

តាង $u = \frac{1}{x}$ នោះ $x = \frac{1}{u}$

ពេល $x \rightarrow \infty$ នោះ $u \rightarrow 0$

គេបាន $\lim_{x \rightarrow \infty} x(e^x - 1) = \lim_{u \rightarrow 0} \frac{1}{u}(e^u - 1) = \lim_{u \rightarrow 0} \frac{e^u - 1}{u} = 1$

ដូចនេះ: $\lim_{x \rightarrow \infty} x(e^{\frac{1}{x}} - 1) = 1$

74). $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x} \right)^x$ មានរាង (1^∞)

$$= \lim_{x \rightarrow \infty} \left[\left(1 - \frac{1}{x} \right)^{(-x)} \right]^{(-1)} = e^{-1} = \frac{1}{e}$$

ដូចនេះ: $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x} \right)^x = \frac{1}{e}$

75). $\lim_{x \rightarrow \pm \infty} \left(1 + \frac{1}{x} \right)^{x^2}$ មានរាង (1^∞)

❖ ករណី $x \rightarrow +\infty$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^{x^2} = \lim_{x \rightarrow +\infty} \left[\left(1 + \frac{1}{x}\right)^x \right]^x = e^{+\infty} = +\infty$$

ដូចនេះ: $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^{x^2} = +\infty$

❖ ករណី $x \rightarrow -\infty$

$$\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^{x^2} = \lim_{x \rightarrow -\infty} \left[\left(1 + \frac{1}{x}\right)^x \right]^x = e^{-\infty} = 0$$

ដូចនេះ: $\lim_{x \rightarrow -\infty} \left(1 + \frac{1}{x}\right)^{x^2} = 0$

76). $\lim_{x \rightarrow 0} \frac{\ln(1+kx)}{x}$ មានរាង $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+kx) = \lim_{x \rightarrow 0} \ln(1+kx)^{\frac{1}{x}} = \lim_{x \rightarrow 0} \ln \left[(1+kx)^{\frac{1}{kx}} \right]^k$$

$$= \ln e^k = k \ln e = k$$

ដូចនេះ: $\lim_{x \rightarrow 0} \frac{\ln(1+kx)}{x} = k$

77). $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}}$ មានរាង (1^∞)

$$= \lim_{x \rightarrow 0} \left(1 + \frac{\sin x}{x} - 1 \right)^{\frac{\sin x}{x - \sin x}} = \lim_{x \rightarrow 0} \left(1 + \frac{\sin x}{x} - 1 \right)^{\frac{\sin x}{x - \sin x}}$$

$$= \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{\sin x}{x} - 1 \right) \right]^{\frac{1}{\left(\frac{\sin x}{x} - 1 \right)}} \right\}^{\left(\frac{\sin x}{x} - 1 \right) \left(\frac{\sin x}{x - \sin x} \right)}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{\sin x}{x} - 1 \right) \right]^{\left(\frac{\frac{\sin x}{x}}{\frac{\sin x}{x} - 1} \right)} \right\}^{\left(\frac{\sin x - x}{x} \right) \left(\frac{\sin x}{x - \sin x} \right)} \\
&= \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{\sin x}{x} - 1 \right) \right]^{\left(\frac{\frac{\sin x}{x}}{\frac{\sin x}{x} - 1} \right)} \right\}^{-\left(\frac{x - \sin x}{x} \right) \left(\frac{\sin x}{x - \sin x} \right)} \\
&= \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{\sin x}{x} - 1 \right) \right]^{\left(\frac{\frac{\sin x}{x}}{\frac{\sin x}{x} - 1} \right)} \right\}^{-\left(\frac{\sin x}{x} \right)} = e^{-1} = \frac{1}{e}
\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} = \frac{1}{e}}$

78). $\lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n)$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \frac{1}{n^2} \times \frac{n(n+1)}{2} = \lim_{n \rightarrow \infty} \frac{1}{n^2} \times \frac{n^2 \left(1 + \frac{1}{n} \right)}{2} \\
&= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n} \right)}{2} = \frac{1+0}{2} = \frac{1}{2}
\end{aligned}$$

ដូចនេះ: $\boxed{\lim_{n \rightarrow \infty} \frac{1}{n^2} (1 + 2 + 3 + \dots + n) = \frac{1}{2}}$

79). $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}}$

ឥឡូវ $1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \left(\frac{1}{2}\right)} = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{\left(\frac{1}{2}\right)}$

$$1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n} = \frac{1 - \left(\frac{1}{3}\right)^{n+1}}{1 - \left(\frac{1}{3}\right)} = \frac{1 - \left(\frac{1}{3}\right)^{n+1}}{\left(\frac{2}{3}\right)}$$

ឥឡូវ $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}} = \lim_{n \rightarrow \infty} \frac{\frac{1 - \left(\frac{1}{2}\right)^{n+1}}{\left(\frac{1}{2}\right)}}{\frac{1 - \left(\frac{1}{3}\right)^{n+1}}{\left(\frac{2}{3}\right)}} = \frac{\frac{(1-0)}{\left(\frac{1}{2}\right)}}{\frac{(1-0)}{\left(\frac{2}{3}\right)}} = \frac{\frac{2}{1}}{\frac{3}{2}} = \frac{2}{3} \cdot \frac{2}{1} = \frac{4}{3}$

ដូចនេះ: $\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}}{1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^n}} = \frac{4}{3}$

80). $\lim_{n \rightarrow \infty} \left(\frac{1 + 2 + 3 + \dots + n}{n + 2} - \frac{n}{2} \right)$

$$= \lim_{n \rightarrow \infty} \left(\frac{\frac{n(1+n)}{2}}{n + 2} - \frac{n}{2} \right) = \lim_{n \rightarrow \infty} \left(\frac{n(1+n)}{2(n+2)} - \frac{n}{2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n(1+n) - n(n+2)}{2(n+2)} \right) = \lim_{n \rightarrow \infty} \frac{n + n^2 - n^2 - 2n}{2(n+2)}$$

$$= \lim_{n \rightarrow \infty} \frac{-n}{2(n+2)} = \lim_{n \rightarrow \infty} \frac{-n}{2n \left(1 + \frac{2}{n}\right)} = \lim_{n \rightarrow \infty} \frac{-1}{2 \left(1 + \frac{2}{n}\right)} = -\frac{1}{2}$$

$$\text{ដូចនេះ: } \lim_{n \rightarrow \infty} \left(\frac{1+2+3+\dots+n}{n+2} - \frac{n}{2} \right) = -\frac{1}{2}$$

$$\begin{aligned} 81). \quad & \lim_{n \rightarrow \infty} \left(\frac{1-2+3-4+\dots-2n}{\sqrt{n^2+1}} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1+3+\dots+(2n-1)-(2+4+\dots+2n)}{\sqrt{n^2+1}} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{\frac{n(1+2n-1)}{2} - \frac{n(2+2n)}{2}}{\sqrt{n^2+1}} \right) = \lim_{n \rightarrow \infty} \left(\frac{\frac{n(2n)}{2} - \frac{n(2+2n)}{2}}{\sqrt{n^2+1}} \right) \\ &= \lim_{n \rightarrow \infty} \frac{n^2 - n(1+n)}{\sqrt{n^2+1}} = \lim_{n \rightarrow \infty} \frac{n^2 - n - n^2}{\sqrt{n^2+1}} = \lim_{n \rightarrow \infty} \frac{-n}{\sqrt{n^2+1}} \\ &= \lim_{n \rightarrow \infty} \frac{-n}{n\sqrt{1+\frac{1}{n^2}}} = \lim_{n \rightarrow \infty} \frac{-1}{\sqrt{1+\frac{1}{n^2}}} = -1 \end{aligned}$$

$$\text{ដូចនេះ: } \lim_{n \rightarrow \infty} \left(\frac{1-2+3-4+\dots-2n}{\sqrt{n^2+1}} \right) = -1$$

$$82). \quad \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right)$$

$$\text{គេមាន } \frac{1}{(k-1)k} = \frac{1}{k-1} - \frac{1}{k}$$

$$\text{បើ } k=2 \text{ នោះ: } \frac{1}{1 \cdot 2} = \frac{1}{1} - \frac{1}{2}$$

$$\text{បើ } k=3 \text{ នោះ: } \frac{1}{2 \cdot 3} = \frac{1}{2} - \frac{1}{3}$$

$$\text{បើ } k=4 \text{ នោះ: } \frac{1}{3 \cdot 4} = \frac{1}{3} - \frac{1}{4}$$

.....

$$\text{បើ } k=n \text{ នោះ: } \frac{1}{(n-1)n} = \frac{1}{n-1} - \frac{1}{n}$$

បូកអង្គនិងអង្គគេបាន:

$$\left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right) = \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n-1} - \frac{1}{n} \right)$$

$$\left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right) = \left(\frac{1}{1} - \frac{1}{n} \right) = 1 - \frac{1}{n}$$

$$\text{គេបាន } \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n} \right) = 1 - 0 = 1$$

$$\text{ដូចនេះ: } \boxed{\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1)n} \right) = 1}$$

$$83). \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right)$$

$$\text{គេមាន } \frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right)$$

$$\text{បើ } k=1 \text{ នោះ: } \frac{1}{1 \cdot 3} = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} \right)$$

$$\text{បើ } k=2 \text{ នោះ: } \frac{1}{3 \cdot 5} = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$\text{បើ } k=3 \text{ នោះ: } \frac{1}{5 \cdot 7} = \frac{1}{2} \left(\frac{1}{5} - \frac{1}{7} \right)$$

.....

$$\text{បើ } k=n \text{ នោះ: } \frac{1}{(2n-1)(2n+1)} = \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

បូកអង្គនិងអង្គគេបាន:

$$\left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right) = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

$$\left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right) = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{2n+1} \right)$$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right) = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{1}{1} - \frac{1}{2n+1} \right) = \frac{1}{2}$$

$$\text{ដូចនេះ: } \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} \right) = \frac{1}{2}$$

84). កំណត់អនុគមន៍ដឺក្រេទីពីរ $f(x)$

តាង $f(x) = ax^2 + bx + c$ ដែល $a \neq 0$

ក). $\lim_{x \rightarrow +\infty} \frac{f(x)}{x^2 - 1} = 1$ គេបាន:

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{ax^2 + bx + c}{x^2 - 1} &= \lim_{x \rightarrow +\infty} \frac{x^2 \left(a + \frac{b}{x} + \frac{c}{x^2} \right)}{x^2 \left(1 - \frac{1}{x^2} \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{\left(a + \frac{b}{x} + \frac{c}{x^2} \right)}{\left(1 - \frac{1}{x^2} \right)} = \frac{a + 0 + 0}{1 - 0} = a \text{ នាំឲ្យ } a = 1 \end{aligned}$$

$$\text{ខ/}. \lim_{x \rightarrow 1} \frac{f(x)}{x^2 - 1} = -1 \text{ គេបាន } \lim_{x \rightarrow 1} \frac{x^2 + bx + c}{x^2 - 1} = -1$$

ដោយ $\lim_{x \rightarrow 1} (x^2 - 1) = 0$ ដូច្នេះដើម្បីឲ្យ $\lim_{x \rightarrow 1} \frac{x^2 + bx + c}{x^2 - 1}$ ជាលីមីតនៃចំនួនថេរ

លុះត្រាតែ $\lim_{x \rightarrow 1} (x^2 + bx + c) = 0$ ឬ $(1 + b + c) = 0$ នាំឲ្យ $c = -b - 1$

$$\text{គេបាន } \lim_{x \rightarrow 1} \frac{x^2 + bx + c}{x^2 - 1} = -1 \Leftrightarrow \lim_{x \rightarrow 1} \frac{x^2 + bx - b - 1}{x^2 - 1} = -1$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1) + b(x-1)}{(x-1)(x+1)} = -1 \Leftrightarrow \lim_{x \rightarrow 1} \frac{(x+1) + b}{x+1} = -1$$

$$\frac{1+1+b}{1+1} = -1 \Leftrightarrow \frac{b+2}{2} = -1 \Leftrightarrow b+2 = -2 \Rightarrow b = -4$$

$$\text{នាំឲ្យ } c = -b - 1 = 4 - 1 = 3$$

$$\text{ដូចនេះ: } f(x) = x^2 - 4x + 3 \text{ ។}$$

85). កំណត់តម្លៃនៃចំនួនថេរ a ដើម្បីឲ្យលីមីតខាងក្រោមជាលីមីតនៃចំនួនថេរ ហើយកំណត់លីមីតនេះផង:

$$\text{ក/} \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - a}{x-1}$$

$$\text{ដោយ } \lim_{x \rightarrow 1} (x-1) = 1-1=0 \text{ នាំឲ្យ } \lim_{x \rightarrow 1} (\sqrt{x+3} - a) = 0$$

$$\text{គេបាន } \sqrt{1+3} - a = 0 \Leftrightarrow 2 - a = 0 \Rightarrow a = 2$$

$$\diamond \text{ ចំពោះ } a = 2$$

$$\text{គេបាន } \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x-1} = \lim_{x \rightarrow 1} \frac{x+3-4}{(x-1)(\sqrt{x+3}+2)}$$

$$= \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(\sqrt{x+3}+2)} = \lim_{x \rightarrow 1} \frac{1}{(\sqrt{x+3}+2)} = \frac{1}{2+2} = \frac{1}{4}$$

$$\text{ដូចនេះ: } \boxed{a = 2, \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x-1} = \frac{1}{4}}$$

$$\text{ខ/} \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} + a}{x}$$

$$\text{ដោយ } \lim_{x \rightarrow 0} (x) = 0 \text{ នាំឲ្យ } \lim_{x \rightarrow 0} (\sqrt{1+3x} + a) = 0$$

$$\text{នាំឲ្យ } \sqrt{1+0} + a = 0 \Leftrightarrow 1 + a = 0 \Rightarrow a = -1$$

$$\diamond \text{ ចំពោះ } a = -1$$

$$\text{គេបាន } \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} - 1}{x} = \lim_{x \rightarrow 0} \frac{1+3x-1}{x(\sqrt{1+3x}+1)}$$

$$= \lim_{x \rightarrow 0} \frac{3x}{x(\sqrt{1+3x}+1)} = \lim_{x \rightarrow 0} \frac{3}{(\sqrt{1+3x}+1)} = \frac{3}{1+1} = \frac{3}{2}$$

$$\text{ដូចនេះ: } \boxed{a = -1, \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} - 1}{x} = \frac{3}{2}}$$

$$\text{គ/} \lim_{x \rightarrow 2} \frac{\sqrt{x+a} - 1}{x-2}$$

$$\text{ដោយ } \lim_{x \rightarrow 2} (x-2) = 0 \text{ នាំឲ្យ } \lim_{x \rightarrow 2} (\sqrt{x+a} - 1) = 0$$

$$\text{គេបាន } \sqrt{2+a} - 1 = 0 \Leftrightarrow \sqrt{2+a} = 1 \Leftrightarrow 2+a = 1 \Rightarrow a = -1$$

$$\diamond \text{ ចំពោះ } a = -1$$

$$\begin{aligned}\text{គេបាន } \lim_{x \rightarrow 2} \frac{\sqrt{x-1}-1}{x-2} &= \lim_{x \rightarrow 2} \frac{x-1-1}{(x-2)(\sqrt{x-1}+1)} \\ &= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(\sqrt{x-1}+1)} = \lim_{x \rightarrow 2} \frac{1}{\sqrt{x-1}+1} = \frac{1}{1+1} = \frac{1}{2}\end{aligned}$$

$$\text{ដូចនេះ: } \boxed{a = -1, \lim_{x \rightarrow 2} \frac{\sqrt{x-1}-1}{x-2} = \frac{1}{2}}$$

$$\text{ឃ/. } \lim_{x \rightarrow -1} \frac{\sqrt{x^2+ax}-1}{x^2-1}$$

$$\text{ដោយ } \lim_{x \rightarrow -1} (x^2-1) = 1-1=0 \text{ នាំឲ្យ } \lim_{x \rightarrow -1} (\sqrt{x^2+ax}-1) = 0$$

$$\sqrt{1-a}-1=0 \Leftrightarrow \sqrt{1-a}=1 \Leftrightarrow 1-a=1 \Rightarrow a=0$$

❖ ចំពោះ $a=0$

$$\text{គេបាន } \lim_{x \rightarrow -1} \frac{\sqrt{x^2}-1}{x^2-1} = \lim_{x \rightarrow -1} \frac{(x^2-1)}{(x^2-1)(\sqrt{x^2}+1)} = \lim_{x \rightarrow -1} \frac{1}{\sqrt{x^2}+1} = \frac{1}{2}$$

$$\text{ដូចនេះ: } \boxed{a=0, \lim_{x \rightarrow -1} \frac{\sqrt{x^2}-1}{x^2-1} = \frac{1}{2}}$$

